
A cyclic-coloring obstruction to maximal binary DFAO reversal

A proposed solution to Davies's nonattainment conjecture,
with two independent proofs, constructive certificates,
exhaustive small-state enumeration, and finite optimality bounds

ChatGPT

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Abstract

An n -state deterministic finite automaton with k output values admits a reversal construction with k^n states. We investigate whether this bound can be attained with only two input letters when $3 \leq k \leq n$. We give a self-contained proposed proof that it cannot. More precisely, for arbitrary transformations a, b of an n -element set and arbitrary k -valued output map τ , we prove

$$|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n,$$

where $g(k)$ is the maximum order of a permutation of k objects. The central construction is the graph formed by translating one collision pair under a permutation. This graph is always 3-colorable. Its proper colorings can only be reached through permutation powers, whereas three suitable output relabelings cannot lie in a single cyclic orbit.

Two independent proofs of nonattainment are given: a short qualitative one, which uses no counting and no Landau function, and the quantitative one that yields the displayed additive deficit. We also give a polynomial-time algorithm producing an explicit missing coloring together with two independently checkable certificate formats, extend the obstruction to an abelian group of permutation letters and one singular letter, and derive two sharper upper bounds from chromatic polynomials: an instance-specific one that is exact for the worked examples, and a weaker one that abstracts into a universal structural bound. A complete C++ enumeration settles all 462 066 reduced instances with $n \leq 4$, and exact arithmetic verifies agreement with Davies's proposed binary optimum for all 372 pairs $7 \leq n \leq 30$, $3 \leq k < n$. This finite optimality statement is distinct from the unrestricted theorem. Source, verification programs, explicit automata, certificates, and numerical records accompany the article.

Scope. The primary target is the nonattainment question, not the exact maximum number of reverse states for every pair (n, k) . The latter remains undetermined by the methods of this article outside the finite range checked in section 10. [Added 28 September 2026, batch 39: for $k = 3$, Part II determines it for every $n \geq 7$ (theorem 13.1).]

Keywords. Finite automata with output; reversal; state complexity; transformation monoids; graph coloring; cyclic actions; certificates.

Research status. The target is explicitly posed in Davies's 2017 paper. A literature search on 20 September 2026 found no later resolution, but was not an exhaustive bibliographic review, a systematic citation-index review, a survey of theses, or correspondence with the author; no conclusion about priority should be drawn from a missing search hit. The proofs below have not been independently refereed or checked by a proof assistant. The status of the main inequality is therefore a *candidate theorem with a self-contained proposed proof*, not an independently certified result. The principal obstruction theorem does not depend on the computations.

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Reading routes. The shared core is sections 2 and 3. Two independent proofs of nonattainment then follow: the short qualitative one in section 4, which a referee can check first, and the quantitative one in section 5. Section 6 makes both constructive. Sections 9 and 10 contain the stronger numerical bounds and their finite verification, and section 11 the computational audit. Section C proves one

orbit size by hand, with no program; section D lists the places where the argument could go wrong. The archive's `README.md` gives build and reproduction commands. Part II (section 13 onwards, added 28 September 2026) proves the exact three-output maximum $R_2(n, 3)$ for every $n \geq 7$. Part III (section 28 onwards, added 29 September 2026) proves it for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ beyond an explicit threshold. Part IV (section 48 onwards, added 29 September 2026) gives a second proof of the case $k = 4$, analytic from $n = 26$, with stability theorems, the missing-orbit inventory and a minimal recurrence.

1 The problem and the result

A deterministic finite automaton normally gives a binary answer: accept or reject. A deterministic finite automaton with output, or DFAO, instead gives one value from a finite output alphabet. The output is attached to the final state reached after the entire input word has been read. Thus a DFAO computes a function $f : \Sigma^* \rightarrow \Delta$, not a stream of output symbols.

Reversing the direction of input means computing

$$f^R(w) = f(w^R),$$

where w^R is the word written backwards. The number of states can increase substantially. If the original machine has n states and k output values, the natural reversal construction has a potential state for each function from the original state set to the output alphabet. There are k^n such functions.

Davies explicitly asks whether this bound is attainable with two input letters and $k \geq 3$, and conjectures that it is not [1, Section 5, Problem 1, printed page 17]. That is the problem selected here. The diagonal case $k = n$ was already settled in that source; the remaining range was $3 \leq k < n$. Her second problem concerns the exact binary maximum; we treat it separately. The binary restriction is on the *input* alphabet. The output alphabet has k symbols throughout.

1.1 Algebraic formulation

Let Q and Δ be finite sets of sizes n and k . A transformation is a function $Q \rightarrow Q$. Write $T(Q)$ for the monoid of all transformations of Q , with usual function composition:

$$(u \circ v)(q) = u(v(q)).$$

If $M \subseteq T(Q)$ and $\tau : Q \rightarrow \Delta$, define

$$\tau M = \{\tau \circ t : t \in M\} \subseteq \Delta^Q.$$

The monoid $\langle a, b \rangle$ includes the identity transformation. The automata question reduces to whether

$$\tau \langle a, b \rangle = \Delta^Q \tag{1.1}$$

can hold. We prove that it cannot when $3 \leq k \leq n$.

Theorem 1.1 (Binary obstruction and quantitative gap). *Let $3 \leq k \leq n$, let $a, b : Q \rightarrow Q$, and let $\tau : Q \rightarrow \Delta$ be arbitrary. Define*

$$g(k) = \max\{\text{ord}(\sigma) : \sigma \in \text{Sym}(\Delta)\}.$$

Then

$$|\tau \langle a, b \rangle| \leq k^n - k! + g(k) < k^n. \tag{1.2}$$

Consequently, the state complexity of the reversal of any n -state binary DFAO with k output symbols is strictly less than k^n .

The function g is often called Landau's function; the standard terminology is used, for example, in [4]. Only the finite-group definition is needed, and no estimate for its growth: a cyclic subgroup cannot equal the nonabelian group $\text{Sym}(\Delta)$ when $k \geq 3$, so $g(k) < k!$.

The conclusion is deliberately stronger than necessary for minimal automata: it applies to every transformation pair and every output map, surjective or not. Accessibility, distinguishability, and the initial state enter only when translating the orbit size into exact automaton state complexity.

Corollary 1.2 (Automaton interpretation and the alphabet threshold). *If a function f is computed by an n -state DFAO with at most two input letters and $3 \leq k \leq n$ output labels, then*

$$\text{sc}(f^R) \leq k^n - k! + g(k) < k^n.$$

Moreover, for each $n \geq k \geq 3$, the least number of input letters needed by a minimal n -state DFAO to attain k^n reverse states is exactly three; see remark 7.2 for $k = 2$.

The first assertion follows from the reversal construction of section 2. The second combines the obstruction with the three-letter construction of section 7; a one-letter machine is covered by duplicating its transformation as a second letter.

For $k = n$, the numerical bound specializes to the classical transformation-monoid bound $n^n - n! + g(n)$ of Holzer and König [2, Theorem 13]. The substantive extension here is to $k < n$: counting omitted permutations of Q does not directly survive the projection $t \mapsto \tau \circ t$. The proof instead makes the symmetric group act on the k output labels.

The proof has three cases. If both letters are permutations, they preserve the number of used output colors. If both are singular, two collision pairs already obstruct reachability. The interesting case is one permutation and one singular transformation. A single collision then generates a small chromatic obstruction under repeated rotation.

1.2 Size of the guaranteed deficit

A permutation with disjoint cycle lengths $\lambda_1, \dots, \lambda_r$ has order $\text{lcm}(\lambda_1, \dots, \lambda_r)$, and every partition of k occurs as a cycle type. Consequently

$$g(k) = \max_{\lambda_1 + \dots + \lambda_r = k} \text{lcm}(\lambda_1, \dots, \lambda_r), \quad (1.3)$$

a small exact computation over integer partitions; the bundled implementation uses no asymptotic approximation. Exact values of $g(k)$ and of the deficit $k! - g(k)$ for $3 \leq k \leq 10$ are tabulated in table 1; the machine-readable file `results/landau_gaps.json` extends the same data to $k = 12$.

For $k = 3$ the theorem gives $3^n - 3$. At $n = 3$ this is sharp, as the word certificate of section C shows by hand. At $n = 4$ the uniform bound gives 78, whereas the graph bound of section 9 gives 67 for a maximizing witness, which is also the true value. The uniform theorem therefore answers the attainment question without purporting to solve the entire optimization problem.

1.3 Provenance of this report

This article merges two independently prepared research packages, both dated 20 September 2026 and both targeting the same question.

`dfao-reversal-cyclic-coloring` proved the shared theorem by the three-case argument reproduced in section 5, packaging the relabeling bound as a subgroup $K \leq \mathbb{Z}$ of exponents with a homomorphism $\phi : K \rightarrow \text{Sym}(\Delta)$ (lemma 5.1); it supplied in addition the independent qualitative proof of section 4, the abelian extension of section 8, the universal structural bound $U(n, k)$ of section 9, the 372-pair finite optimality result of section 10, the correction to the source’s printed Table 2, the Holzer–König attribution, and the independent C++ breadth-first search.

`dfao-reversal-coloring-bound` proved the same theorem by the same three-case argument with the same collision-orbit graph, packaging the relabeling bound instead as the *setwise* stabilizer $H \leq \langle p \rangle$ of the fiber partition of a surjective coloring, with induced homomorphism $\rho : H \rightarrow \text{Sym}(\Delta)$ (remark 5.2); it supplied in addition the sharper instance-specific chromatic bound proposition 9.1, the complete $n \leq 4$

enumeration of section 11, the standalone certificate checker with its pairwise-compatibility Chinese remainder criterion, the hand-checkable 24-word certificate of section C, and the research-status and literature-search notes retained in the archive.

The two proofs of the shared inequality are the same mathematics in different dress, and it is proved here exactly once, in section 5. Everything built on top of that core was complementary, and is kept. Where the two packages stated the status of a source or of the literature differently, the more cautious statement is the one retained. [Added 28 September 2026, batch 39: this paragraph describes Part I. Part II was added from a third, later manuscript; its provenance is in sections 13.1 and 27.]

1.4 What is and is not established

The main theorem is unrestricted in n and k within $3 \leq k \leq n$. It applies even without accessibility, minimality, or surjectivity assumptions on the supplied machine. The article also proves an extension to commuting permutation letters, and gives a deterministic algorithm for constructing one unreachable coloring.

The exact maximum number of binary reversal states is a different question. We derive an explicit universal upper bound $U(n, k)$ by classifying the collision graphs. Exact evaluation gives the same number as the published coprime-block lower bound for $7 \leq n \leq 30$ and $3 \leq k < n$. This is a finite, computer-assisted optimality result. We do not prove equality with that lower bound for all n . [Added 28 September 2026, batch 39: the preceding sentences describe the report at the state fbba58593 against which Part II was written. For $k = 3$, Part II proves equality with the lower bound for every $n \geq 7$ (theorem 13.1), and with it $U(n, 3) = L(n, 3)$ (remark 18.3). For $k \geq 4$ nothing changes, and the list below still holds for general n, k .] [Added 29 September 2026, batches 40–41: for $k \geq 4$, Part III proves equality with the lower bound, and $U(n, k) = L(n, k)$, for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$ (theorem 28.1 and remark 33.6). The exact optimum for $k \geq 8$ below $N(k)$, beyond the finite ranges already certified, remains open.]

The following are *not* claimed. No proof of the exact optimal binary reversal complexity for arbitrary n, k , and no resolution of Davies’s other open questions. No novelty for the orbit reduction, the ternary witness, the standard Landau function, the elementary group facts, the previously reported small maxima, the diagonal specialization $k = n$ to the classical bound $n^n - n! + g(n)$ of Holzer and König, or graph coloring and chromatic polynomials as a method in the transformation-monoid setting. No claim that the absence of a located later solution proves the problem remained open until this draft. No Lean, Isabelle, Coq, or other proof-assistant certification. No claim that random tests establish a universal theorem. No claim that all improper graph colorings are reachable in arbitrary cases. The archive’s `source_audit.md` records the search scope, the locations actually inspected in the source, and the attribution boundaries of this merged report in full. `RESEARCH_STATUS.md` and `LITERATURE_SEARCH.md` are kept verbatim from one of the two source reports and describe that report rather than this one; each carries a note saying so.

2 Why reversal is an orbit of colorings

2.1 The reversal construction

Write a DFAO as

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, \Delta, \tau).$$

Each input letter x induces a transformation $t_x(q) = \delta(q, x)$. For a word $w = x_1 \cdots x_r$, the forward transformation is

$$t_w = t_{x_r} \circ \cdots \circ t_{x_1}.$$

The order here matters. The machine reads x_1 first, whereas function composition applies its rightmost factor first. The function computed by $\mathcal{A}$ is $f(w) = \tau(t_w(q_0))$, and we write $w^R = x_r \cdots x_1$ and $f^R(w) = f(w^R)$.

An automaton is *accessible* if every state is reachable from q_0 , and two states are *distinguishable* if some continuation word gives different outputs. An accessible automaton whose distinct states are pairwise distinguishable is *minimal*: two distinguishable states cannot be identified in any deterministic machine with the same continuation behaviour. Accordingly $\text{sc}(f)$ denotes the *state complexity* of f , the least number of states in a DFAO computing it.

Define a new DFAO on the set Δ^Q . Its initial state is τ , its transition on x sends a coloring c to $c \circ t_x$, and its output at c is $c(q_0)$. After reading $x_1 \cdots x_r$, its state is

$$\tau \circ t_{x_1} \circ \cdots \circ t_{x_r} = \tau \circ t_{w^R}.$$

Its output is therefore $\tau(t_{w^R}(q_0)) = f(w^R)$, as required.

If M is the transition monoid generated by the original letters, the reachable state set of this reversal construction is exactly τM . It follows immediately that

$$\text{sc}(f^R) \leq |\tau M| \leq k^n.$$

Proposition 2.1 (Reachable colorings are distinguishable). *If every state of the original DFAO is reachable from q_0 , then*

$$\text{sc}(f^R) = |\tau M|.$$

Proof. Let $c, d \in \tau M$ be distinct. Choose q with $c(q) \neq d(q)$. By accessibility, some word u satisfies $t_u(q_0) = q$. Read u^R in the reversal construction starting from c or from d . The resulting outputs are respectively

$$c(t_u(q_0)) = c(q), \quad d(t_u(q_0)) = d(q).$$

They differ, so c and d cannot be identified by minimization. Every reachable state is therefore necessary. $\square$

This orbit characterization is also established by Davies [1, Proposition 4]; the proof is included to fix all conventions and to make the later argument independent of a black-box automata theorem.

2.2 Rank, fibers, and output relabeling

The rank of a coloring $c : Q \rightarrow \Delta$ is $|c(Q)|$. Precomposition by a permutation preserves this rank and, more strongly, preserves the size of every labeled color class. Precomposition by a singular transformation can merge distinctions: if $t(x) = t(y)$, then

$$(c \circ t)(x) = (c \circ t)(y)$$

for every coloring c .

There is a second action, on the opposite side. An output permutation $\sigma \in \text{Sym}(\Delta)$ acts by relabeling colors:

$$c \mapsto \sigma \circ c.$$

The two actions commute, in the precise sense

$$(\sigma \circ c) \circ t = \sigma \circ (c \circ t). \tag{2.1}$$

This elementary associativity identity is the algebraic core of the proof.

If c is surjective, its output-relabeling orbit

$$\mathcal{R}(c) = \{\sigma \circ c : \sigma \in \text{Sym}(\Delta)\}$$

has exactly $k!$ elements. Indeed, $\sigma \circ c = \rho \circ c$ implies $\sigma = \rho$ by evaluating at a preimage of each output value.

3 The collision-orbit graph

3.1 Definition and low chromatic number

Suppose p is a permutation of Q and s is a singular transformation. Choose one collision pair

$$x \neq y, \quad s(x) = s(y).$$

Define a simple undirected graph $G = G(p; x, y)$ with vertex set Q and edge set

$$E(G) = \{\{p^j(x), p^j(y)\} : j \in \mathbb{Z}\}. \quad (3.1)$$

Repeated edges are retained only once. There are no loops because p is injective. The graph is invariant under both p and p^{-1} .

Lemma 3.1. *Every graph $G(p; x, y)$ is 3-colorable.*

Proof. If x and y lie in different cycles of p , every edge runs between those two cycles. They form the two parts of a bipartite graph, and all other vertices are isolated.

Suppose instead that x and y lie on the same cycle of length m . Identify that cycle with $\mathbb{Z}/m\mathbb{Z}$, with $p(i) = i + 1$, and write the pair as $\{0, h\}$, where $h \not\equiv 0 \pmod{m}$. Its translates are $\{i, i + h\}$. Each vertex has at most two neighbors, $i + h$ and $i - h$. Hence the nontrivial components are cycles or single edges. Such a graph is 3-colorable. Isolated vertices cause no restriction. $\square$

Lemma 3.2 (Refinement to a surjective coloring). *If a graph on n vertices is r -colorable and $r \leq k \leq n$, then it has a proper coloring using exactly k colors.*

Proof. Start with a proper coloring using $j \leq r$ nonempty color classes. If $j < k$, then $j < n$, so some class contains at least two vertices. Give one of those vertices a fresh color. Splitting a color class cannot create a monochromatic edge. Repeating the operation reaches exactly k nonempty classes. $\square$

In particular, for every $3 \leq k \leq n$ the graph G has a proper surjective k -coloring. Every output relabeling of that coloring is still proper. The same refinement applies to the two-edge graph used in the case of two singular generators: a simple graph with at most two edges contains no cycle, hence no odd cycle, and is bipartite.

3.2 A singular letter cannot reach a proper coloring

Lemma 3.3 (Rightmost singular factor). *Every coloring in $\tau\langle p, s \rangle$ obtained from a product containing s is improper on $G(p; x, y)$. Consequently,*

$$\tau\langle p, s \rangle \cap \text{Col}_k(G) \subseteq \{\tau \circ p^j : j \in \mathbb{Z}\}, \quad (3.2)$$

where $\text{Col}_k(G)$ is the set of all proper k -colorings, not necessarily surjective.

Proof. Take a product containing s . Locate its rightmost occurrence of s in usual composition notation. All factors to its right are powers of p , so the product has the form

$$t = u \circ s \circ p^j$$

for some $u \in \langle p, s \rangle$ and $j \geq 0$. Put $x' = p^{-j}(x)$ and $y' = p^{-j}(y)$. Then

$$t(x') = u(s(x)) = u(s(y)) = t(y').$$

The pair $\{x', y'\}$ is an edge of G . Thus $(\tau \circ t)(x') = (\tau \circ t)(y')$, which makes the coloring improper. Products with no s are powers of p , proving the inclusion. $\square$

A proper coloring remains proper under precomposition by p . Therefore, if τ itself is proper, the reachable proper colorings form its whole cyclic orbit; if τ is improper, that set is empty. We will usually need only the weaker inclusion (3.2).

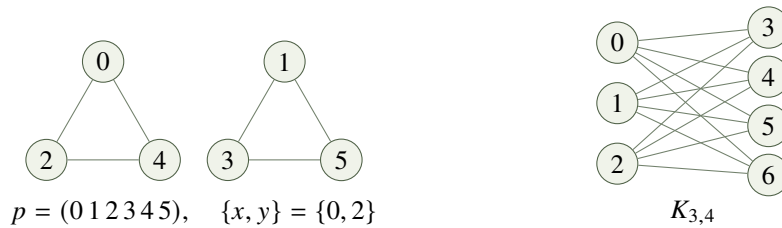


Figure 1: The two basic mechanisms: a same-cycle pair produces disjoint cycles; a pair from different permutation cycles produces a bipartite graph. These diagrams illustrate the construction, not an additional assumption.

4 First proof of nonattainment: a short qualitative route

Before counting missing states, we prove that at least one must be missing. The useful ingredient is a small obstruction to fitting output symmetries into a cyclic input action.

First of two independent proofs. This section uses no counting, no Landau function, and no refinement beyond one surjective proper base coloring. What it buys: it is the shorter statement a referee can check first, it is logically independent of the counting argument of section 5, and it is what makes the constructive algorithm of section 6 cheap, needing only three candidate colorings instead of the six relabelings that the counting route suggests.

Lemma 4.1 (Three relabelings cannot share a cyclic orbit). *Let $c : Q \rightarrow \Delta$ be surjective with $k \geq 3$, and let p be a permutation of Q . Choose three distinct output symbols, written 0, 1, 2, and let $\sigma = (0\ 1)$ and $\rho = (1\ 2)$ fix all other symbols. No orbit of precomposition by p contains all three colorings*

$$c, \quad \sigma \circ c, \quad \rho \circ c.$$

Proof. Suppose one orbit contains all three. Taking c as its base point, there exist integers r, t such that

$$\sigma \circ c = c \circ p^r, \quad \rho \circ c = c \circ p^t.$$

Using (2.1),

$$\begin{aligned} \sigma \circ \rho \circ c &= \sigma \circ c \circ p^t = c \circ p^{r+t}, \\ \rho \circ \sigma \circ c &= \rho \circ c \circ p^r = c \circ p^{t+r}. \end{aligned}$$

The two expressions are equal. Surjectivity of c then forces $\sigma\rho = \rho\sigma$, contrary to the chosen transpositions. For example, they send the symbol 0 to different symbols in the two orders. $\square$

Theorem 4.2 (Qualitative binary obstruction). *For $3 \leq k \leq n$ and arbitrary a, b, τ , equality (1.1) is impossible.*

Proof. If τ is not surjective, every coloring in its orbit omits at least one of the original missing labels. Thus assume τ is surjective.

Both a and b are permutations. Every orbit coloring has rank k , so no constant coloring is reachable.

Both a and b are singular. Choose one collision pair for each and form a graph H consisting of the resulting one or two edges. This graph is 3-colorable, so lemma 3.2 gives a proper surjective k -coloring. Every nonempty product ends on the right in a or b and collapses the chosen pair for that factor. Hence the only possibly reachable proper coloring is the initial coloring τ . There are at least two distinct proper colorings, obtained by swapping two used output colors, so one is missing.

Exactly one generator is a permutation. Write it as p and the singular one as s . Choose a proper surjective coloring c of $G(p; x, y)$. All three colorings in lemma 4.1 are proper. If every coloring were reachable, lemma 3.3 would put all three in the single cyclic orbit of τ under p . This contradicts lemma 4.1. $\square$

The argument does not estimate the size of the transformation monoid. In particular, merely knowing that two transformations cannot generate all of $T(Q)$ would not have sufficed: a proper submonoid could in principle still meet every fiber of the map $t \mapsto \tau \circ t$. The graph argument excludes this possibility directly in the output-coloring space.

5 Second proof of nonattainment: counting the missing colorings

Second of two independent proofs. This section proves the quantitative inequality (1.2). What it buys over section 4: a numerical additive deficit $k! - g(k)$ valid uniformly in n , which the qualitative route does not produce, and the lemma that generalizes to arbitrary permutation groups in proposition 8.1. This is the only place where the shared core theorem is proved; every later bound refines it rather than reproving it.

5.1 A cyclic orbit sees only a cyclic subgroup of relabelings

Lemma 5.1 (Relabeling intersection bound). *Let $c : Q \rightarrow \Delta$ be surjective. If O is any orbit under precomposition by a permutation p , then*

$$|O \cap \mathcal{R}(c)| \leq g(k).$$

Proof. If the intersection is empty, there is nothing to prove. Otherwise choose $d \in O \cap \mathcal{R}(c)$. Then $O = \{d \circ p^j : j \in \mathbb{Z}\}$ and $\mathcal{R}(d) = \mathcal{R}(c)$. Define

$$K = \{j \in \mathbb{Z} : d \circ p^j = \sigma \circ d \text{ for some } \sigma \in \text{Sym}(\Delta)\}.$$

For each $j \in K$, surjectivity makes σ unique; denote it by $\phi(j)$. We claim that K is a subgroup of $\mathbb{Z}$ and that $\phi : K \rightarrow \text{Sym}(\Delta)$ is a homomorphism.

The exponent 0 is in K . If $r, t \in K$, then

$$d \circ p^{r+t} = \phi(r) \circ d \circ p^t = \phi(r) \circ \phi(t) \circ d.$$

Hence $r + t \in K$ and $\phi(r + t) = \phi(r)\phi(t)$. Also, $d \circ p^r = \phi(r) \circ d$ implies $d \circ p^{-r} = \phi(r)^{-1} \circ d$, so $-r \in K$.

Every subgroup of $\mathbb{Z}$ is cyclic. Its image $\phi(K)$ is therefore a cyclic subgroup of $\text{Sym}(\Delta)$ and has at most $g(k)$ elements. The colorings of $O \cap \mathcal{R}(d)$ correspond one-to-one to these image elements. Thus the desired intersection has size at most $g(k)$. $\square$

The statement is about distinct output relabelings, not about the number of exponents realizing them. Many powers of p may induce the same coloring. This is why the argument uses the image of a subgroup rather than the order of p itself.

It is convenient to name the cyclic orbit of an arbitrary output map:

$$\text{Orb}_\tau = \{\tau \circ p^j : j \in \mathbb{Z}\}. \quad (5.1)$$

With this notation lemma 5.1 reads $|\text{Orb}_\tau \cap \mathcal{R}(c)| \leq g(k)$ for every surjective c .

Remark 5.2 (A second packaging of the same bound: the setwise stabilizer). The same inequality can be obtained inside the finite cyclic group $A = \langle p \rangle$ rather than inside $\mathbb{Z}$. Take $\tau = c$ first. The nonempty fibers of c form an unordered partition $\mathcal{B}_c = \{c^{-1}(d) : d \in \Delta\}$. Let $H \leq A$ be its *setwise* stabilizer: those elements of A that permute the blocks, without necessarily preserving each block individually. For $h \in H$ there is a unique $\rho(h) \in \text{Sym}(\Delta)$ with

$$c \circ h = \rho(h) \circ c, \quad (5.2)$$

uniqueness coming from surjectivity of c ; conversely any $h \in A$ satisfying (5.2) for some color permutation lies in H . The same composition computation as above makes $\rho : H \rightarrow \text{Sym}(\Delta)$ a homomorphism, and $|\text{Orb}_c \cap \mathcal{R}(c)| = |\rho(H)|$. A subgroup of a cyclic group is cyclic, so $\rho(H)$ is a cyclic subgroup of $\text{Sym}(\Delta)$ and has at most $g(k)$ elements. For general τ , if the intersection is nonempty choose $c' = \pi_0 \circ c$ in it; then $\text{Orb}_\tau = \text{Orb}_{c'}$, and relabeling by π_0 bijects Orb_c with $\text{Orb}_{c'}$ while mapping $\mathcal{R}(c)$ onto itself, so the two intersections have equal size.

What this packaging buys over lemma 5.1: it is the form that generalizes verbatim to an arbitrary permutation group in proposition 8.1, and it is the form whose characteristic failure mode (the partition is stabilized setwise, not block by block) is flagged in section D. The $\mathbb{Z}$ -version is kept as the working lemma because choosing the base point d inside the intersection absorbs the reduction from a general τ that the finite-group version has to perform separately. The two are the same mathematics in different dress, and the different dress does different work.

Lemma 5.3. *For $k \geq 3$, $g(k) < k!$.*

Proof. A permutation of order $k!$ would generate a cyclic subgroup of the same size as $\text{Sym}(\Delta)$, hence all of it. But $\text{Sym}(\Delta)$ is not cyclic for $k \geq 3$: the transpositions of the first two and of the last two of three distinct symbols do not commute. $\square$

5.2 Proof of the quantitative theorem

Proof of theorem 1.1. We bound the number of missing colorings in each case.

If τ is not surjective, choose any surjective coloring $c : Q \rightarrow \Delta$. All $k!$ members of $\mathcal{R}(c)$ are absent from $\tau\langle a, b \rangle$.

Suppose τ is surjective and both generators are permutations. Choose a coloring c using exactly $k - 1$ labels. None of its relabelings is reachable, since rank is preserved. Its stabilizer in $\text{Sym}(\Delta)$ is trivial: a permutation fixing the $k - 1$ used symbols must fix the one remaining symbol as well. Thus again at least $k!$ colorings are missing.

If both generators are singular, use the graph H from the qualitative proof and a proper surjective coloring c . All $k!$ colorings in $\mathcal{R}(c)$ are proper, but at most the initial coloring τ is reachable. Hence at least $k! - 1$ are missing.

Finally, in the mixed case choose a proper surjective coloring c of the collision-orbit graph. Its $k!$ relabelings are proper. By lemmas 3.3 and 5.1, at most $g(k)$ of them belong to the reachable set. At least $k! - g(k)$ are therefore missing.

Every case gives at least $k! - g(k)$ missing colorings, since $g(k) \geq 1$. This proves the first inequality in (1.2). The strict inequality is lemma 5.3. Applying the reversal construction proves the automata assertion. $\square$

k	$k!$	$g(k)$	Guaranteed missing colorings $k! - g(k)$
3	6	3	3
4	24	4	20
5	120	6	114
6	720	6	714
7	5 040	12	5 028
8	40 320	15	40 305
9	362 880	20	362 860
10	3 628 800	30	3 628 770

Table 1: An additive deficit valid for every $n \geq k$, computed exactly from (1.3). It is not asserted to be the optimal deficit except at $(n, k) = (3, 3)$. The accompanying file `results/landau_gaps.json` continues the table to $k = 12$, where $g(11) = 30$ with deficit 39 916 770 and $g(12) = 60$ with deficit 479 001 540.

The universal gap is deliberately simple rather than usually sharp. It ignores all but one relabeling orbit of proper colorings. Counting the entire proper-coloring set will give substantially stronger bounds in section 9.

Remark 5.4 (Why three output labels matter). For two colors the counting deficit vanishes, since $g(2) = 2! = 2$. Independently, some collision-orbit graphs are odd cycles and have no proper 2-coloring. Both halves of the obstruction genuinely change at three colors. This is not a hidden argument that would incorrectly prohibit binary-alphabet reversal witnesses for ordinary languages; section 7.1 exhibits a $k = 2$ machine reaching all 2^n colorings.

Finally, a caution about what has been bounded. The proof does not classify two-generated monoids, estimate their orders, or enumerate their elements. Its bound concerns the action on output maps, which may be substantially smaller than the monoid itself. The distinction is especially relevant when $k < n$, where different transformations can have exactly the same composition with τ .

6 Constructing a missing reversal state

The qualitative proof yields a useful algorithm. Its input is two explicit transition arrays and an output array. Its output is a particular coloring that cannot be a reversal state, together with a short explanation that can be checked without exploring all k^n colorings.

6.1 The case split

If an output label does not occur in τ , return the constant coloring with that label. If both letters are permutations, every reachable coloring has the same number of occurrences of each label as τ , so any

constant coloring with a different label-count vector is unreachable; deterministically, return the constant zero coloring, unless τ is already constantly zero, in which case return the constant one coloring.

If both letters are singular, find one collision pair for each. Properly color their one- or two-edge graph, split classes until k labels are used, and choose this coloring unless it equals τ . In that exceptional case, swap two output labels. The resulting proper coloring is not the initial state, and no nonempty word can reach it.

In the mixed case, form $G(p; x, y)$ by iterating the unordered pair under p . At most $\binom{n}{2}$ different pairs can occur, so the graph is built in polynomial time even when p has very large order. Construct a proper surjective coloring c : try a bipartite coloring first, and if the graph is not bipartite, the proof of lemma 3.1 shows that every nonisolated component is a cycle and every vertex has degree at most two, so greedy coloring uses at most three colors. Then consider

$$c_0 = c, \quad c_1 = (0\ 1) \circ c, \quad c_2 = (1\ 2) \circ c.$$

Test these colorings for membership in the cyclic orbit of τ . At least one test must fail by lemma 4.1. Since each candidate is proper, a failed cyclic-orbit test certifies that it is unreachable by any word.

Remark 6.1 (The counting route gives a six-candidate variant). If one starts instead from lemma 5.1 rather than from lemma 4.1, the natural search tests all six relabelings of a proper coloring that uses exactly the three labels 0, 1, 2: at most $g(3) = 3$ of them can lie in Orb_τ , so at most six tests are needed. More explicitly, if $\text{im}(\tau)$ is not exactly $\{0, 1, 2\}$ then the cyclic orbit contains none of the six, and otherwise the lemma applies directly. Both variants are implemented and both are correct; the three-candidate version is the cheaper one, and it is available only because section 4 was proved separately. This is one concrete payoff of keeping the two proofs apart.

The only potentially nontrivial computational issue is deciding

$$c = \tau \circ p^j \quad \text{for some integer } j \tag{6.1}$$

without iterating through all powers of p . The next construction resolves it with modular arithmetic.

6.2 A rotation constraint for each permutation cycle

Decompose p into cycles. For one cycle

$$C = (q_0, q_1, \dots, q_{m-1}), \quad p(q_i) = q_{i+1 \bmod m},$$

form the two cyclic words

$$U_i = \tau(q_i), \quad V_i = c(q_i).$$

Equation (6.1) requires

$$V_i = U_{i+j \bmod m} \quad (0 \leq i < m).$$

If V is not a cyclic rotation of U , the candidate is immediately rejected. Otherwise find one working shift r , and let d be the least positive cyclic period of U :

$$U_{i+d \bmod m} = U_i \quad \text{for all } i.$$

Here $d \mid m$. Every working shift satisfies exactly the congruence

$$j \equiv r \pmod{d}. \tag{6.2}$$

Lemma 6.2. *The shifts fixing a cyclic word of length m form a subgroup of $\mathbb{Z}/m\mathbb{Z}$. If a target word is one rotation of the source, the shifts taking the source to the target form a coset of this subgroup. Equivalently, they are precisely one residue class modulo the least cyclic period.*

Proof. The identity shift fixes the word; the composition and inverse of fixing shifts also fix it. Thus the fixing shifts form a subgroup. If r takes the source to the target, another shift j does so exactly when $j - r$ fixes the source. Subgroups of $\mathbb{Z}/m\mathbb{Z}$ consist of the multiples of a divisor of m , and the least positive period is that divisor. $\square$

For example, the word 010101 has least cyclic period 2, not 6. A target 101010 contributes the constraint $j \equiv 1 \pmod{2}$. Using the whole cycle length as the modulus would incorrectly discard valid shifts.

6.3 Combining the constraints

Maintain a congruence $j \equiv R \pmod{M}$, initially $R = 0$, $M = 1$. To add $j \equiv r \pmod{d}$, put $g = \gcd(M, d)$. The two congruences are compatible if and only if

$$r - R \equiv 0 \pmod{g}.$$

When they are compatible, write $j = R + Mt$. Dividing by g gives

$$\frac{M}{g}t \equiv \frac{r - R}{g} \pmod{d/g}.$$

The coefficient M/g is invertible modulo d/g , so this determines a unique class of t and hence one class of j modulo $\text{lcm}(M, d)$. Modulus 1 is handled as a vacuous congruence.

A rejection certificate is either a cycle whose target is not a rotation of the source, or a pair of incompatible accumulated congruences. A successful test returns an exponent representing the solution class. At most three such tests are needed to produce an unreachable proper coloring.

6.4 A second, deliberately different membership criterion

The incremental merge just described is not the only way to decide the system. Suppose the cycles contribute congruences $j \equiv r_i \pmod{p_i}$ for $i = 1, \dots, v$. The generalized Chinese remainder theorem says that the system is solvable exactly when every pair is compatible:

$$r_i \equiv r_{i'} \pmod{\gcd(p_i, p_{i'})} \quad \text{for all } i, i'. \quad (6.3)$$

Sufficiency follows by prime powers. For each prime, select a congruence whose modulus carries the largest exponent of that prime. Pairwise compatibility makes its residue agree with every weaker requirement at that prime. The selected prime-power congruences have coprime moduli and combine by the ordinary Chinese remainder theorem, and their common solution satisfies the original system.

What this buys over the incremental merge: nothing mathematically, and that is the point. The two criteria are used by two programs that share no code, so a disagreement between them exposes an implementation mistake in one of them. The producer code/dfao.py merges congruences iteratively; the standalone verifier code/certificate_checker.py uses (6.3). This deliberate divergence of method is the substance of the trust separation described in section 6.7, and the two implementations are not unified.

Proposition 6.3 (Polynomial-time constructive obstruction). *For explicit arrays a, b, τ with $3 \leq k \leq n$, one can construct a coloring outside $\tau\langle a, b \rangle$ in time polynomial in the input size, without enumerating the coloring orbit or the transformation monoid. The construction also admits a polynomial-size certificate that can be checked in polynomial time.*

Proof. Finding collisions and decomposing a permutation into cycles are elementary finite-array operations. The graph has at most $\binom{n}{2}$ edges and can be built in polynomial time. It is either bipartite or has maximum

degree at most two, so the initial proper coloring is obtained by a breadth-first bipartite coloring or a greedy 3-coloring. Splitting color classes takes at most k further steps.

The simple implementation tests all rotations of each cycle word, requiring $O(\sum m_i^2) \leq O(n^2)$ symbol comparisons for one cyclic-orbit test. The least periods can be found within the same bound. The moduli in the Chinese remainder calculation divide $\text{lcm}(1, 2, \dots, n)$ and therefore have at most $O(n \log n)$ bits, using the elementary bound by $n!$. Euclidean arithmetic and modular inverses on these integers take polynomial time. Only three candidate tests are needed.

For the certificate clause, a certificate consists of the input arrays and the target, together with either a differing label-count vector, or two collision pairs, or one collision pair with its orbital graph and evidence of cyclic nonmembership. In the last case the evidence is a cycle whose target word is no rotation of the source word, or a pair of incompatible period congruences; a checker may instead recompute all congruences and reject membership directly. The graph has at most $\binom{n}{2}$ edges and every tuple has length at most n , so both the certificate size and the checking time are polynomial. $\square$

6.5 A concrete certificate

Consider the 8-state, 5-output machine whose arrays are

$$\begin{aligned} p &= (1, 2, 0, 4, 5, 6, 7, 3), \\ s &= (3, 2, 1, 3, 4, 5, 6, 0), \\ \tau &= (0, 1, 2, 3, 4, 4, 4, 4). \end{aligned}$$

Array notation means, for example, $p(0) = 1$. The collision $s(0) = s(3)$ generates $K_{3,5}$ between $\{0, 1, 2\}$ and $\{3, 4, 5, 6, 7\}$. The algorithm returns

$$c = (2, 3, 0, 4, 1, 1, 1, 1).$$

Its two block palettes, $\{0, 2, 3\}$ and $\{1, 4\}$, are disjoint, so it is proper. On the first cycle, however, $(2, 3, 0)$ is not a rotation of the source word $(0, 1, 2)$. Thus c is not a power-rotation of τ , and no word containing s can reach a proper coloring. This proves that c is missing. The full machine-readable certificate is in `data/example_certificate.json`. Its schema is

```
{n, k, a, b, τ, unreachable{case, target, permutation, singular,
                                collision, edges, candidate_tests[...]}},
```

recording the 15 edges of $K_{3,5}$ and, for each of the three candidates, the outcome of its cyclic-orbit test with reason `nonrotation`.

6.6 Two certificate formats and two generators

The archive ships a second, differently shaped certificate family, produced by a different search and validated by a different program. Its schema is

```
{n, k, a, b, τ, case, permutation_letter, pair, edges, target, cyclic_nonmembership{...}},
```

in `results/certificates.json`, where the nonmembership block records `member`, `reason`, `cycle` and `constraints`. Four such certificates are supplied, all in the one-permutation case: at $(n, k) = (3, 3)$ with target $[0, 2, 1]$; at $(4, 3)$ and $(4, 4)$ with target $[2, 1, 0, 1]$; and at $(5, 3)$ with target $[2, 0, 1, 1, 1]$. All four carry the reason `no_rotation`.

The two formats are kept side by side and neither schema is normalized onto the other. What each buys: the `results/certificates.json` family is produced by the six-relabeling search of remark 6.1 and is checked by the import-free verifier described below, which is what makes it useful as a trust artifact, and its structural variant scales to $n \leq 50$ without any orbit enumeration; the `data/example_certificate.json` family is produced by the three-candidate search driven by lemma 4.1, records the candidate-by-candidate test log that the cheaper search generates, and describes a single large machine, the 8-state, 5-output $K_{3,5}$ example whose orbit is computed exactly in section 10. Different search strategies and different scales make agreement between them informative.

6.7 Implementation and trust separation

The file `code/dfao.py` constructs the certificates of the first family. The separate file `code/certificate_checker.py` imports no code from that module. It recomputes the collision conditions, the entire pair orbit, properness, and cyclic nonmembership, using the pairwise criterion (6.3) rather than the producer's iterative merge, and exits with a nonzero status if any supplied certificate is rejected.

This independence is limited, and is stated precisely. Both programs implement the same mathematical argument, so they are not an independent proof of the theorem. They are independent checks of finite certificates under that argument. The archive also includes randomized comparisons with explicit cyclic-orbit enumeration and direct comparisons with complete reverse-state searches.

7 Why three input letters suffice

The binary obstruction is an exact alphabet-size threshold, not an obstruction to maximal reversal in general. We include a construction to make this distinction explicit.

Theorem 7.1 (Ternary attainability). *For every $3 \leq k \leq n$, some minimal n -state DFAO with three input letters and k output values has reversal state complexity k^n .*

Proof. Let $Q = \{0, \dots, n-1\}$. Use three transformations:

$$p = (0\ 1 \cdots n-1), \quad t = (0\ 1),$$

and the map e that sends 1 to 0 and fixes every other point. Conjugating t by powers of p yields the adjacent transpositions, so p and t generate $\text{Sym}(Q)$.

Conjugates of e by permutations give every elementary collapse that sends one chosen point to another and fixes all remaining points. These collapses and the permutations generate every transformation of Q . To see this, let $f : Q \rightarrow Q$ be arbitrary. Choose one representative $r_y \in f^{-1}(y)$ for each $y \in \text{im}(f)$. Collapse every other point of $f^{-1}(y)$ onto r_y . The resulting transformation h satisfies $h(q) = r_{f(q)}$. The bijection $r_y \mapsto y$ between two equally sized subsets extends to a permutation u of Q . Then $f = u \circ h$.

Choose any surjective output map $\tau : Q \rightarrow \Delta$. Given any coloring $c : Q \rightarrow \Delta$, choose a representative $r_d \in \tau^{-1}(d)$ for every output symbol d , and define $f_c(q) = r_{c(q)}$. Then $\tau \circ f_c = c$. Since the transition monoid is all of $T(Q)$, every coloring is reachable in the reversal construction. The original machine is accessible because it contains the n -cycle p . Hence proposition 2.1 gives k^n distinguishable reversal states.

The original machine is also minimal. If $q \neq q'$, choose states u, v with different outputs. A transformation sending q to u and q' to v is realized by an input word, which distinguishes q from q' . $\square$

Combining this theorem with theorem 1.1 proves the second assertion of corollary 1.2: the least input-alphabet size that permits all k^n reversal states is exactly three for $3 \leq k \leq n$. A one-letter machine is covered by duplicating its transformation.

Remark 7.2 (The two-value case). One of the two source reports stated this attainability for every $n \geq k \geq 2$. Its construction uses only that p and t generate $\text{Sym}(Q)$ and that elementary mergers together with permutations generate all of $T(Q)$, neither of which needs $k \geq 3$, so the case $k = 2$ is covered on the same grounds. The deficit of [theorem 1.1](#) is a separate matter and is claimed only for $k \geq 3$.

Ternary attainability was already established in the target paper [1, Theorem 1 and Corollary 2]; the verification above only fixes conventions, and the new obstruction supplies the missing binary exclusion.

7.1 Why two output values are different

The proof genuinely needs three output values. A collision-orbit graph can contain an odd cycle and then has no proper 2-coloring. Independently, $\text{Sym}(\{0, 1\})$ is cyclic, so the noncommuting-relabellings argument is unavailable even when a proper binary coloring exists.

A small explicit example uses

$$p = (1, 2, 0), \quad s = (0, 0, 2), \quad \tau = (0, 0, 1).$$

The orbit contains all eight binary colorings of three states. Powers of p provide the three colorings with one 1. Applying s to a suitable one provides a coloring with two 1s, whose rotations give the other two. The same collapse produces 000 from a one-1 coloring and 111 from a two-1 coloring. Thus the small example already shows that the conclusion cannot simply be extended to $k = 2$.

The cases $k = 1$ and $k > n$ are outside the interesting range. The former gives a constant function; in the latter, no output map can use all k labels, and missing output values already prevent a full coloring orbit.

8 An extension to abelian permutation groups

The cyclic nature of the single permutation letter can be weakened. The same mechanism applies to any number of commuting permutation letters, provided there is only one singular letter.

8.1 The group-theoretic form of the intersection lemma

Proposition 8.1 (Relabelings induced by a permutation group). *Let $H \leq \text{Sym}(Q)$ and let $c : Q \rightarrow \Delta$ be surjective. If O is an H -orbit of colorings, then the nonempty intersection $O \cap \mathcal{R}(c)$ has size $|J|$ for a subgroup $J \leq \text{Sym}(\Delta)$ that is a homomorphic image of a subgroup of H .*

Proof. Choose $d \in O \cap \mathcal{R}(c)$, so $O = dH$ and $\mathcal{R}(d) = \mathcal{R}(c)$. Define

$$K = \{h \in H : d \circ h = \sigma \circ d \text{ for some } \sigma \in \text{Sym}(\Delta)\}.$$

The output permutation is unique. The same composition and inverse calculation as in [lemma 5.1](#) shows that K is a subgroup of H and that the assignment $h \mapsto \sigma$ is a homomorphism. Its image J parametrizes exactly the colorings in the intersection. $\square$

Thus a full output-relabeling orbit would force $\text{Sym}(\Delta)$ to be a quotient of a subgroup of H . This is a general obstruction, although the graph colorability condition must still be checked.

Remark 8.2 (Several singular generators). Nothing above needs a single singular letter. Let $H \leq \text{Sym}(Q)$ be generated by any set of permutation letters, let $b_1, \dots, b_r$ be singular letters, and choose one collision pair e_i for each b_i . Put

$$E = \bigcup_{i=1}^r \{h(e_i) : h \in H\}. \tag{8.1}$$

A composition containing a singular letter has a rightmost such letter and a suffix in H , so exactly as in lemma 3.3 its output coloring is improper for this graph, and the reachable proper colorings lie in τH . If that graph has a proper surjective k -coloring c , taking H to be the setwise stabilizer of the fiber partition of c as in remark 5.2 shows that full reachability forces $\rho(H) = \text{Sym}(\Delta)$.

The colorability hypothesis cannot simply be dropped. With many collision pairs or a large permutation group, the graph of (8.1) can require more than k colors. Moreover, a nonabelian permutation group can genuinely have a symmetric-group quotient. These two escape routes are exactly why the extension does not contradict the three-letter full-transformation witness of theorem 7.1.

8.2 Orbit graphs for abelian groups

For a collision pair $s(x) = s(y)$, define

$$E(G_H) = \{\{h(x), h(y)\} : h \in H\}.$$

A word containing s can again be written $u \circ s \circ h$ with $h \in H$. It collapses an edge of G_H , so reachable proper colorings must lie in the single H -orbit τH .

Lemma 8.3. *If H is abelian, the graph G_H is 3-colorable.*

Proof. If x and y lie in different H -orbits, all edges join those two orbits, so the graph is bipartite. Otherwise write $y = h_0(x)$ for some $h_0 \in H$. For a vertex $z = h(x)$, commutativity gives

$$h(y) = h(h_0(x)) = h_0(h(x)) = h_0(z).$$

Thus the edges on the orbit of x are exactly $\{z, h_0(z)\}$. Every vertex has at most two neighbors. All other vertices are isolated, so three colors suffice. $\square$

Let $A(k)$ denote the largest order of an abelian subgroup of $\text{Sym}(k)$. There is a simple exact expression:

$$A(k) = \max_{\lambda \vdash k} \prod_i \lambda_i. \quad (8.2)$$

To prove the upper bound, decompose an abelian permutation group into its orbits of sizes λ_i . On one orbit, an element fixing one point fixes every point, because it commutes with the transitive action. The faithful image on that orbit is therefore regular, of order λ_i . The original group embeds into the product of these images, so its order is at most $\prod_i \lambda_i$. Conversely, disjoint cyclic groups of orders λ_i attain that product.

For $k \geq 2$, the usual elementary product maximization gives

$$A(k) = \begin{cases} 3^{k/3}, & k \equiv 0 \pmod{3}, \\ 4 \cdot 3^{(k-4)/3}, & k \equiv 1 \pmod{3}, k \geq 4, \\ 2 \cdot 3^{(k-2)/3}, & k \equiv 2 \pmod{3}. \end{cases}$$

Indeed, a part at least 5 is improved by replacing it with 3 and the remaining part; parts 1 are eliminated in a maximizing partition for $k \geq 2$; and three parts 2 are improved by two parts 3. A leftover 4 may be kept as 4 or split into $2 + 2$.

Theorem 8.4 (Abelian-permutation obstruction). *Let $H \leq \text{Sym}(Q)$ be abelian, let $s : Q \rightarrow Q$ be singular, and let $3 \leq k \leq n$. Then, for every $\tau : Q \rightarrow \Delta$,*

$$|\tau\langle H, s \rangle| \leq k^n - k! + A(k) < k^n.$$

Proof. A nonsurjective τ omits all $k!$ relabelings of any surjective coloring. Otherwise, lemmas 3.2 and 8.3 gives a proper surjective k -coloring of G_H . Its reachable relabelings must lie in τH . By proposition 8.1, those relabelings form a set whose size is the order of an abelian subgroup of $\text{Sym}(\Delta)$, and so is at most $A(k)$. The remaining $k! - A(k)$ relabelings are missing. Since $\text{Sym}(k)$ is nonabelian for $k \geq 3$, $A(k) < k!$. $\square$

This does not forbid three-letter maximal examples in general: the two permutation letters in theorem 7.1 generate a nonabelian group. It instead identifies a structural requirement on any such construction with one singular letter.

9 Sharper bounds from all proper colorings

The graph-coloring viewpoint has antecedents in Holzer and König’s work on transformation monoids [2, Sections 4–5]; the coloring count is not presented here as a newly invented technique. The factorial gap counts one free output-relabeling orbit. We now count all proper colorings of a collision-orbit graph. The resulting bound retains information about the cycle structure of the permutation and is often sharp.

9.1 The basic chromatic bound

Let $P_G(k)$ be the number of proper maps $Q \rightarrow \Delta$. In the mixed case, lemma 3.3 immediately gives

$$|\tau\langle p, s \rangle| \leq k^n - P_G(k) + \text{ord}(p). \quad (9.1)$$

More generally, $\text{ord}(p)$ may be replaced by any bound on the p -orbit length of a proper coloring. If τ is improper, no proper coloring is reachable at all; using the larger term remains a valid upper bound.

Replacing $\text{ord}(p)$ by the actual orbit size, and splitting on properness rather than absorbing it, gives a strictly sharper statement for a given machine.

Proposition 9.1 (Chromatic deficit, instance-specific form). *Let p be a permutation, let s be singular, fix a collision pair $\{x, y\}$ of s , and put $G = G(p; x, y)$ and $M = \langle p, s \rangle$. Then*

$$|\tau M| \leq k^n - P_G(k) + \begin{cases} |\text{Orb}_\tau|, & \tau \text{ is proper for } G, \\ 0, & \tau \text{ is not proper for } G. \end{cases} \quad (9.2)$$

Proof. The permutation p is a graph automorphism of G , since $E(G)$ was defined in (3.1) as a p -invariant set of pairs. Its pullback action therefore preserves properness, so the orbit Orb_τ of (5.1) is wholly proper or wholly improper according to whether τ is proper. By lemma 3.3, every reachable proper coloring lies in that orbit. There are $k^n - P_G(k)$ improper colorings in total, and at most the stated number of reachable proper ones. Adding the two counts gives (9.2). $\square$

Remark 9.2 (Two instance-specific chromatic bounds, and what each buys). Inequalities (9.1) and (9.2) are both kept, and neither substitutes for the other. Proposition 9.1 is sharper for any fixed machine, because $|\text{Orb}_\tau|$ divides $\text{ord}(p)$ and is usually much smaller, and because an improper τ contributes nothing at all; it is what yields the exact values 24, 67, 176 and 215 in section 9.5. Inequality (9.1) is weaker for a given instance but is stated in terms of $\text{ord}(p)$ alone, and that is precisely what allows it to be maximized over all collision graphs without reference to τ ; it is the form that abstracts into the residual-order function $H_r(t)$, the period improvement $T_{r,k}(a, b)$, and hence the universal bound $U(n, k)$ of theorem 9.4. The instance-specific version cannot play that role, since $|\text{Orb}_\tau|$ is not a function of the graph parameters.

Neither inequality asserts that all improper colorings are reachable. That additional saturation property must be proved or checked separately for any claimed equality witness, which is why a graph bound alone does not solve the exact state-complexity optimization problem.

There are only two structural forms for $G(p; x, y)$.

9.2 A pair in one permutation cycle

Suppose the pair lies on a cycle of length m , with displacement h . Put

$$d = \gcd(m, h), \quad L = m/d, \quad r = n - m.$$

The graph consists of d disjoint cycles of length L , together with r isolated vertices. When $L = 2$, each component is one edge rather than a cycle with two parallel edges; the coloring count is unchanged.

For $L \geq 2$, the appropriate cycle factor is

$$C_L(k) = (k - 1)^L + (-1)^L(k - 1). \quad (9.3)$$

One direct derivation uses the $k \times k$ matrix B with zeros on the diagonal and ones elsewhere. A proper coloring of a labeled cyclic sequence is a closed walk, so the count is $\text{tr}(B^L)$. The constant vectors have eigenvalue $k - 1$, and the subspace of coordinate sum zero has eigenvalue -1 and dimension $k - 1$. This gives (9.3). At $L = 2$ the formula is $k(k - 1)$, the count for one edge.

Therefore

$$P_G(k) = C_L(k)^d k^r. \quad (9.4)$$

Every divisor $d \mid m$ with $d < m$ is possible: take displacement $h = d$.

9.3 A pair in different permutation cycles

Let the cycle lengths be a, b , and put

$$d = \gcd(a, b), \quad u = a/d, \quad v = b/d, \quad r = n - a - b.$$

Label the two cycles by $\mathbb{Z}/a\mathbb{Z}$ and $\mathbb{Z}/b\mathbb{Z}$, putting the chosen endpoints at their respective origins. An edge joins i to j exactly when a common power of p carries the origin pair to (i, j) . The generalized Chinese remainder criterion says this occurs precisely when $i \equiv j \pmod{d}$. Hence

$$G \cong dK_{u,v} \sqcup rK_1.$$

The proper-coloring count for $K_{u,v}$ is

$$B_k(u, v) = \sum_{i=1}^{\min(u, k)} (k)_i \begin{Bmatrix} u \\ i \end{Bmatrix} (k - i)^v. \quad (9.5)$$

To derive it, let the left side use exactly i colors. Partition its u labeled vertices into i nonempty color classes in $\begin{Bmatrix} u \\ i \end{Bmatrix}$ ways, assign distinct output labels to the classes in $(k)_i = k!/(k - i)!$ ways, and color each right-side vertex with one of the remaining $k - i$ labels. Thus

$$P_G(k) = B_k(u, v)^d k^r. \quad (9.6)$$

This derivation also proves the symmetry $B_k(u, v) = B_k(v, u)$, even though (9.5) does not display it symmetrically.

For three colors the expression simplifies considerably:

$$B_3(u, v) = 3(2^u + 2^v - 2), \quad u, v \geq 1. \quad (9.7)$$

Only $i = 1, 2$ contribute in (9.5), and $\begin{Bmatrix} u \\ 2 \end{Bmatrix} = 2^{u-1} - 1$.

9.4 Choosing a collision pair

A singular map may have many collision pairs. Each one gives a valid bound of the form (9.2), so taking the minimum over all collision pairs of s is a legitimate improvement. Alternatively, the union of all the corresponding orbital graphs is still a valid obstruction graph, because a word containing s collapses one of its edges whichever pair is chosen. That union may, however, cease to be three-colorable, and its chromatic polynomial may be harder to compute. The uniform proof deliberately uses a single pair: that choice is what preserves the elementary three-colorability argument of lemma 3.1 for every permutation, and hence the guaranteed supply of surjective proper colorings.

9.5 Worked instances

Tuples list map values in the order $0, 1, \dots, n-1$. All orbit counts in this subsection are computed by exact integer-state breadth-first search.

Sharpness at three states and three outputs. Take

$$a = (1, 2, 0), \quad b = (0, 0, 2), \quad \tau = (0, 1, 2). \quad (9.8)$$

The collision pair $\{0, 1\}$ produces a triangle under the three-cycle a . There are six proper three-colorings, and the pure- a orbit of τ is $(0, 1, 2), (1, 2, 0), (2, 0, 1)$, of size three. Thus proposition 9.1 gives at most $27 - 6 + 3 = 24$ reachable states. The word table of section C exhibits 24 distinct colorings, so this automaton has exactly 24 reverse states and the three missing ones are

$$(0, 2, 1), \quad (1, 0, 2), \quad (2, 1, 0).$$

The original machine is accessible and minimal: the cycle reaches every state, and τ already separates states without a continuation. Hence equality holds in theorem 1.1 at $n = k = 3$.

Four-state examples. Use $a = (1, 2, 3, 0)$, $b = (0, 0, 3, 2)$, again with the collision pair $\{0, 1\}$, whose orbital graph is the four-cycle C_4 . For three colors take $\tau = (0, 1, 0, 2)$: it is proper, its cyclic orbit has size four, and $C_4(3) = 2^4 + 2 = 18$ by (9.3), so the bound is $81 - 18 + 4 = 67$. For four colors take $\tau = (0, 1, 2, 3)$: now $C_4(4) = 3^4 + 3 = 84$ and the bound is $256 - 84 + 4 = 176$. Exact search attains both. Both machines are accessible and minimal.

A complete-bipartite example. Let

$$a = (1, 0, 3, 4, 2), \quad b = (2, 1, 2, 3, 0), \quad \tau = (0, 1, 2, 2, 2).$$

The permutation has cycles $(0\ 1)$ and $(2\ 3\ 4)$, while b identifies 0 and 2, so the orbital graph is $K_{2,3}$. By (9.5),

$$B_3(2, 3) = 3 \cdot 2^3 + 6 \cdot 1^3 = 30,$$

in agreement with (9.7). The coloring τ is proper with orbit size two under a , so proposition 9.1 gives $3^5 - 30 + 2 = 215$, which the included search attains.

Remark 9.3 (Two different $n = 5$, $k = 3$ examples). The value 215 just obtained is an *instance-specific* equality for this particular $K_{2,3}$ machine; it is not a claim that 215 is the global five-state, three-output maximum. Section A lists a different five-state machine, $p = (1, 2, 3, 4, 0)$, $s = (0, 0, 3, 4, 2)$, $\tau = (0, 1, 0, 1, 2)$, whose orbit has size $218 = U(5, 3)$ and which realizes the small-case maximum of section 10. The two examples illustrate different claims, the first that the cross-cycle formula is exactly attained and the second that the structural bound is met, and $215 < 218$ is a consequence of the first machine not being extremal. They are not in conflict, and neither replaces the other.

n	k	k^n	Uniform bound	Graph bound	Actual witness orbit
3	3	27	24	24	24
4	3	81	78	67	67
4	4	256	236	176	176
5	3	243	240	215	215

Table 2: The uniform estimate of theorem 1.1 is designed to prove nonattainment; the instance-specific graph estimate of proposition 9.1 can be much stronger for a particular pair of generators. The last row is the $K_{2,3}$ machine of remark 9.3, not the extremal five-state one.

The values 24, 67 and 176 are not new numerical records: they occur among the previously verified maxima in [1, Table 3]. Here they serve as independent regression tests and as examples where the collision graph explains the missing states. For every example the archive records the input maps, the exact orbit size, a minimality check, and an explicit unreachable coloring; the certificate checker verifies nonreachability structurally, without relying on the breadth-first search that produced the orbit counts.

9.6 Accounting for the remaining permutation cycles

Let O_r be the set of orders of permutations on r points, with $O_0 = \{1\}$. Define

$$H_r(t) = \max_{o \in O_r} \text{lcm}(t, o). \quad (9.9)$$

For a same-cycle pair the distinguished cycle contributes order m , so

$$\text{ord}(p) \leq H_r(m).$$

For a different-cycle pair, the two distinguished cycles contribute $\text{lcm}(a, b)$, giving

$$\text{ord}(p) \leq H_r(\text{lcm}(a, b)).$$

The exact set of possible residual orders matters. In general one must not replace $H_r(t)$ by $\text{lcm}(t, g(r))$: the residual order maximizing an ordinary integer magnitude need not maximize its least common multiple with t .

For example, $g(5) = 6$. If $t = 6$, then $\text{lcm}(t, g(5)) = 6$, whereas a residual 5-cycle gives $\text{lcm}(6, 5) = 30$. The definition (9.9) handles such interactions correctly.

9.7 The three-color period improvement

When $k = 3$ and $\gcd(a, b) = 1$, the nonisolated part is the connected graph $K_{a,b}$. In a proper coloring the palettes on the two sides are disjoint. Since only three labels are available and both palettes are nonempty, at least one side is monochromatic. Its rotation contributes period 1. Thus the proper-coloring period on the two distinguished cycles divides a or divides b , rather than necessarily requiring $\text{lcm}(a, b)$. Including the isolated vertices gives the improved bound

$$T_{r,k}(a, b) = \begin{cases} \max\{H_r(a), H_r(b)\}, & k = 3, \gcd(a, b) = 1, \\ H_r(\text{lcm}(a, b)), & \text{otherwise.} \end{cases} \quad (9.10)$$

The connectedness condition is important. We do not apply the improvement component by component when $\gcd(a, b) > 1$, because the permutation may rotate the components as well as the colors within them.

9.8 A universal finite formula

The two unmixed generator cases also admit easy bounds. If both letters are permutations, the orbit lies in one fixed color histogram. The largest possible histogram class is the balanced multinomial coefficient

$$M(n, k) = \frac{n!}{(q!)^{k-z}((q+1)!)^z}, \quad n = qk + z, \quad 0 \leq z < k.$$

Balancing is optimal because replacing two counts $a \geq b + 2$ by $a - 1, b + 1$ decreases the product of their factorials.

If both letters are singular, their two chosen collision edges have at least $(k - 1)^2 k^{n-2}$ proper colorings. For two distinct edges the count is exactly this whether they share a vertex or are disjoint; for a repeated edge the count is larger. At most the initial coloring is reachable among them. Thus the missing-coloring count is at least

$$(k - 1)^2 k^{n-2} - 1.$$

Define $D(n, k)$ to be the minimum of the following finite collection:

$$k^n - (k - 1)^n, \tag{9.11}$$

$$k^n - M(n, k), \tag{9.12}$$

$$(k - 1)^2 k^{n-2} - 1, \tag{9.13}$$

$$C_{m/d}(k)^d k^{n-m} - H_{n-m}(m), \tag{9.14}$$

$$B_k(a/d, b/d)^d k^{n-a-b} - T_{n-a-b,k}(a, b). \tag{9.15}$$

In (9.14), the parameters range over $2 \leq m \leq n$ and positive divisors $d \mid m$ with $d < m$. In (9.15), they range over $1 \leq a \leq b, a + b \leq n$, with $d = \gcd(a, b)$. The first entry handles nonsurjective output maps. It is unnecessary when surjectivity is part of the definition of the extremal problem, but makes the bound valid for arbitrary supplied output arrays.

Theorem 9.4 (Structural upper bound). *For every $3 \leq k \leq n$ and arbitrary a, b, τ ,*

$$|\tau\langle a, b \rangle| \leq U(n, k) := k^n - D(n, k). \tag{9.16}$$

It is always valid to combine this with theorem 1.1 by taking the smaller of $U(n, k)$ and $k^n - k! + g(k)$.

Proof. Every machine belongs to one of the nonsurjective, two-permutation, two-singular, or mixed cases. The first three give (9.11)–(9.13). In the mixed case choose any collision pair. If its endpoints are on the same permutation cycle, use (9.4) and the order bound $H_{n-m}(m)$. Otherwise use (9.6) and (9.10). These are exactly the entries (9.14) and (9.15). The actual machine therefore misses at least one entry of this collection, and hence at least its minimum. $\square$

The bound is obtained without enumerating transformations. There are only quadratically many choices of the displayed graph parameters up to divisor bookkeeping. Computing the residual order sets involves integer partitions; we do not claim that this part is a polynomial-time algorithm in n . It is nevertheless a much smaller exact computation than enumerating all n^{2n} ordered pairs of transformations, let alone their output orbits.

10 Finite optimality results and explicit witnesses

10.1 Comparison with the published lower bound

For $3 \leq k < n$, define

$$L(n, k) = \max_{\substack{1 < \ell < m, \ell+m=n \\ \gcd(\ell, m)=1}} [k^n - B_k(\ell, m) + G_k(\ell, m)], \quad (10.1)$$

where

$$G_k(\ell, m) = \begin{cases} m, & k = 3, \\ \ell m, & k \geq 4. \end{cases}$$

Davies's Corollary 3 provides accessible binary DFAO witnesses attaining this lower bound [1]. We use that result as a literature dependency for the finite optimality comparison, not for the principal obstruction theorem.

Let $R_2(n, k)$ denote the largest reversal complexity of an accessible n -state binary DFAO using k output values. Then

$$L(n, k) \leq R_2(n, k) \leq U(n, k).$$

For the displayed range $n \geq 7$, admissible coprime splits always exist. For example, take the neighboring halves for odd n ; for $n \equiv 0 \pmod{4}$ take $n/2 - 1, n/2 + 1$; and for $n \equiv 2 \pmod{4}$, $n \geq 10$, take $n/2 - 2, n/2 + 2$.

Proposition 10.1 (Computer-assisted finite equality). *Exact evaluation of (10.1) and (9.16) gives*

$$L(n, k) = U(n, k) \quad (7 \leq n \leq 30, 3 \leq k < n).$$

Consequently these values equal $R_2(n, k)$ throughout that finite range. There are $\sum_{n=7}^{30} (n-3) = 372$ parameter pairs.

[Added 28 September 2026, batch 39: for $k = 3$ the equality $L(n, 3) = U(n, 3) = R_2(n, 3)$ holds for every $n \geq 7$ (theorem 13.1 and remark 18.3), and the splits listed above are the optimal ones (lemma 18.1). Part II's closed form reproduces all 24 rows $k = 3$ of the table described next (section 18.2).] [Added 29 September 2026, batches 40–41: for $4 \leq k \leq 7$ the equality $L(n, k) = U(n, k) = R_2(n, k)$ holds for every $n \geq \max(7, k+1)$, and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$ (theorem 28.1 and remark 33.6); Part III's finite certificates extend the equality $R_2(n, k) = L(n, k)$ to $7 \leq n \leq 100$, $3 \leq k \leq \min(12, n-1)$ (proposition 35.1). All 372 rows of the table described next agree with Part III's formula (section 35.5).]

The complete exact-integer table is supplied as `data/finite_range_bounds.csv`. Each row gives the upper and lower values, a minimizing graph type, a maximizing coprime split, and the number of upper-bound candidates examined. The program compares the integers directly; there is no floating-point tolerance or probabilistic component. An independent checker reproduces all 372 pairs using an unrestricted composition recurrence for permutation orders, inclusion–exclusion for surjections, and nonzero cycle displacements rather than the primary implementation's divisor enumeration.

The witnesses also have minimal original state count n . Indeed, the complete bipartite graph for an admissible split contains two disjoint edges, so

$$B_k(\ell, m) \leq (k-1)^2 k^{n-2}.$$

Thus the witness reversal size is strictly larger than

$$k^n - (k-1)^2 k^{n-2} = (2k-1)k^{n-2} > k^{n-1}.$$

If its function had an original DFAO with fewer than n states, the general reversal construction would give at most k^{n-1} reversal states, a contradiction. Therefore the finite maxima apply equally when the original function is required to have state complexity exactly n .

n	k	Exact maximum in the checked range	k^n	Split (ℓ, m)
7	3	2 125	2 187	(3, 4)
7	4	15 472	16 384	(3, 4)
7	5	71 037	78 125	(3, 4)
8	3	6 452	6 561	(3, 5)
8	4	63 403	65 536	(3, 5)
8	5	369 020	390 625	(3, 5)
8	6	1 539 561	1 679 616	(3, 5)
9	3	19 550	19 683	(4, 5)
9	4	258 360	262 144	(4, 5)
10	3	58 654	59 049	(3, 7)

10.2 Small cases and the limitation at the diagonal

The structural upper bound also meets explicit breadth-first-search witnesses in nine of the ten pairs $3 \leq k \leq n \leq 6$.

n	k	Witness orbit size	Structural upper bound	Outcome
3	3	24	24	Equality
4	3	67	67	Equality
4	4	176	176	Equality
5	3	218	218	Equality
5	4	826	826	Equality
5	5	2 110	2 271	Bound not sharp
6	3	699	699	Equality
6	4	3 526	3 526	Equality
6	5	12 031	12 031	Equality
6	6	32 262	32 262	Equality

These are finite verifications: the upper bounds follow from a proved formula, and each listed lower witness is checked by exhaustive exploration of its coloring orbit. This is not exhaustive enumeration of all automata. At $(n, k) = (5, 5)$ the collision graph loses information and leaves a gap. The diagonal case $k = n$ is closely tied to the size of a two-generated transformation monoid, because a surjective output map is then bijective. For background on that monoid problem, see Holzer and König [2]. The graph bound alone does not recover all restrictions on such a monoid.

For example, at $n = 6$ the minimizing same-cycle graph consists of two triangles. Therefore its proper-coloring count is $[k(k-1)(k-2)]^2$, and the full-cycle order is 6. For $k = 3, 4, 5, 6$ the bound is

$$k^6 - [k(k-1)(k-2)]^2 + 6,$$

which gives exactly 699, 3526, 12031, and 32262.

10.3 An arithmetic correction to the source table

The $n = 8, k = 5$ entry in Table 2 of arXiv version 2 is printed as 368020 [1, p. 13]. Its own formula, evaluated at $(\ell, m) = (3, 5)$, gives instead

$$\begin{aligned}
B_5(3, 5) &= 5 \cdot 4^5 + (5 \cdot 4) \cdot 3 \cdot 3^5 + (5 \cdot 4 \cdot 3) \cdot 2^5 \\
&= 5120 + 14580 + 1920 = 21620, \\
5^8 - B_5(3, 5) + 15 &= 390625 - 21620 + 15 = 369020.
\end{aligned}$$

This is an arithmetic discrepancy in the displayed table, not a counterexample to the formula or to the lower-bound conjecture. The printed PDF page was inspected to distinguish it from an HTML extraction error.

The explicit 8-state witness in section 6 has exactly 369020 reachable colorings. Two independent breadth-first-search implementations, one storing Python tuples and one encoding colorings as base-5 integers in C++, return that same count. As additional comparison checks, the sums and bitwise exclusive-or values of all reached integer encodings are respectively

$$72076669650 \quad \text{and} \quad 128342.$$

These checksums are auxiliary consistency checks, not substitutes for the mathematical bound or the reachability computations.

11 Verification record and reproducibility

All computations described here were executed for this draft. The programs use exact integers, finite sets, and deterministic searches. The only randomness is the fixed-seed selection of test instances; no random sampling is used to establish the exhaustive maxima below.

11.1 Exhaustive small-state enumeration

For each $2 \leq n \leq 4$ and $2 \leq k \leq n$, a C++17 program enumerates all n^n transformations, every unordered pair of transformations with repetition, and one representative of every surjective output map up to renaming colors. The representatives are restricted-growth strings with exactly k blocks, of which there are $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$. The number of searched instances is therefore exactly

$$\binom{n^n + 1}{2} \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}. \quad (11.1)$$

Only two reductions are used: interchanging the names of the two input letters, which does not change the generated monoid, and relabeling all output colors, which bijects reverse orbits because it commutes with every pullback. No restriction to one permutation generator, no state-conjugacy representative, no accessibility or minimality requirement, and no conjecturally maximal monoid is imposed. Every orbit is explored to closure by breadth-first search, with colorings encoded as base- k integers and pullbacks precomputed.

n	k	Output partitions	Searched instances	Maximum orbit
2	2	1	10	4
3	2	3	1 134	8
3	3	1	378	24
4	2	7	230 272	16
4	3	6	197 376	67
4	4	1	32 896	176

Table 3: Complete searches with only input-letter interchange and output-color relabeling removed, 462 066 reduced instances in total. The $k = 2$ rows are boundary checks lying outside the hypotheses of theorem 1.1.

The Python implementation independently recalculates the orbit of each maximizing witness emitted by the C++ program, and verifies that some initial state makes the original automaton accessible and

minimal; those witnesses therefore realize the listed orbit values as genuine reversal state complexities. The unrestricted maxima themselves rest on the C++ enumeration, which Python does not repeat for every four-state candidate. The per-row counts of maximizers in the reduced search, together with one explicit maximizing triple (a, b, τ) per row, are recorded in `results/exhaustive.json`.

11.2 What was actually checked

The archive includes executable source and the exact output records from the following checks.

Check	Instances
All ordered pairs of transformations and all output maps for $n = k = 3$	19 683
Cyclic membership: CRT algorithm versus literal orbit search, $n = 3, 4$ with $k = 3$	161 838
Random and mixed-case orbit searches, checking constructed missing states and both upper bounds	2 550
Collision graphs: proper coloring, closed count, direct enumeration, and inclusion–exclusion	1 362
Permutation orders from cycle partitions versus all permutations, $0 \leq n \leq 8$	46 234
Upper versus lower bound for $7 \leq n \leq 30$, $3 \leq k < n$	372
Explicit automaton witnesses checked by independent C++ orbit search	20

The exhaustive $n = k = 3$ test uses 3^3 choices for each of a , b , and τ , giving $27^3 = 19683$ triples. Its largest coloring orbit has size 24, agreeing with [theorem 1.1](#) exactly. Nonsurjective output maps are included, so that branch of the constructive algorithm is also exercised.

For the graph tests, the code enumerates permutations and collision pairs for $3 \leq n \leq 5$, constructs the actual edge set, and compares three ways of counting proper 3-colorings: direct enumeration, inclusion–exclusion over edges, and the formulas in [section 9](#). The two general counting procedures do not use the graph classification.

The random tests use the recorded seed 20260920. They are regression tests, not evidence for an unrestricted quantifier. The unrestricted statements are supported by the proofs; the tests are intended to expose orientation, period, and implementation mistakes.

11.3 A second audit suite at a different reduction level

A separately written suite repeats and extends these checks with different conventions, and its counts are recorded separately rather than merged into the table above.

Remark 11.1 (Two exhaustive $n = k = 3$ audits, at different reduction levels). The first row of [table 4](#) covers all $27^2 \cdot 6 = 4374$ ordered transformation pairs with all six output bijections, checking each against the bound 24 and verifying that each constructed certificate names a state genuinely missing from the complete orbit; the corresponding row of [table 3](#) uses the symmetry-reduced count 378. The first row of the table in the previous subsection instead uses the fully unreduced $27^3 = 19683$ triples, including nonsurjective output maps, which exercises the constructive algorithm’s nonsurjective branch as well. All three counts are kept, because the reduction assumptions are themselves part of what is being checked: a mistake in the letter-interchange or color-relabeling reduction would show up as a disagreement between them, and they agree on the maximum 24.

For the chromatic-formula audit, all permutations and all unordered pairs are tested for $n = 3, 4, 5$, and every coloring is counted for each of $k = 3, 4$. Testing $k = 4$ on three vertices is legitimate for those

Audit	Cases
All <i>ordered</i> three-state map pairs, all six output bijections	4 374
Two cyclic-membership implementations versus explicit rotation	1 000
Cyclic orbit/relabeling intersection versus $g(k)$	252
Orbital chromatic formulas versus all colorings, through $n = 5$	2 724
Random exact reverse-orbit and missing-certificate checks, $n \leq 8$	330
Larger structural certificate checks, $3 \leq n \leq 50$	1 440
C++ maximizing witnesses recomputed and checked in Python	6

Table 4: All listed checks passed. The larger certificate tests do not enumerate their reverse orbits.

chromatic identities, although it lies outside the $k \leq n$ convention of [theorem 1.1](#). The random reverse-orbit checks use $k = 3, 4$ where applicable and balance the zero-, one-, and two-permutation cases. The larger certificate tests allow k up to eight and include nonsurjective output maps.

There are also ten focused unit tests, covering invalid inputs, small Landau values, the sharp three-state example, cyclic-shift obstructions, consistent and inconsistent Chinese remainder data, the cross-cycle and odd-cycle coloring formulas, certificate validation and certificate tampering, and the finite-search safety limit. One design rule is tested explicitly: a breadth-first search that reaches its configured state limit raises an error rather than returning an incomplete orbit as if it were exact.

11.4 What the experiments establish

The searches give exact finite cardinalities for their explicitly stated ranges and validate the implementations on additional instances. They provide useful evidence against composition-direction mistakes, forgotten constant colorings, improper treatment of repeated edges, or incompatible cycle rotations. They do not establish the universal theorem by induction, prove priority, or certify the correctness of arbitrary source code. The proof of [theorem 1.1](#) is independent of all of them.

11.5 Files in the archive

The main implementation is `code/reversal.py`. It contains ordinary function composition, coloring actions, literal orbit search, accessibility and minimality checks, the missing-coloring algorithm, cyclic membership via generalized CRT, chromatic counts, and the structural upper-bound evaluator. It uses only the Python standard library and supports Python 3.9 or later.

`code/run_checks.py` reproduces the recorded Python verification suite and the finite-range CSV. `code/orbit_count.cpp` is an independent C++17 implementation of coloring-orbit search. It consumes the plain-text fixtures in `code/bfs_cases.txt`. `code/check_bound_certificate.py` independently reevaluates the finite upper and lower bounds without importing the primary implementation; `code/check_large_example.py` repeats the 369020-state example using Python tuples. The C++ representation encodes c as $\sum_i c(i)k^i$, so it does not depend on Python tuple hashing or on any graph theorem.

The second implementation family is rooted at `code/dfao.py`, which contains its own validation, pullback, cycle and Landau routines, an exact reverse-orbit search that refuses to return a partial orbit when its `max_states` limit is reached, orbital-edge and properness tests, three-coloring and refinement, cyclic membership, missing-certificate generation, Stirling numbers, the bipartite and orbital chromatic counts, and an accessibility-and-distinguishability check. `code/certificate_checker.py` is the standalone verifier of [section 6.7](#), and imports nothing from `dfao.py`. `code/run_experiments.py` runs the audits of [table 4](#) and `code/test_dfao.py` the ten unit tests.

Remark 11.2 (Two independent C++ programs). The archive contains two C++17 programs, which do different jobs and are both kept. `code/exhaustive.cpp` performs the complete enumeration of all transformation pairs and output-partition representatives for $n \leq 4$, emitting the maxima and maximizers of table 3; `code/orbit_count.cpp` performs single-instance breadth-first search over the fixtures of `code/bfs_cases.txt` with base- k integer encoding and sum and exclusive-or checksums, its largest case visiting 1 539 561 states. The first answers “what is the maximum over all small machines”; the second answers “exactly how large is the orbit of this one machine”, at sizes far beyond the first program’s reach. Their outputs are `results/exhaustive.json` and `data/cpp_orbit_counts.csv`.

Remark 11.3 (Two independent arithmetic paths, and two independent confirmations of 369020). The finite bounds of section 10 are computed twice by different arithmetic. `reversal.py`’s `structural_bound` uses divisor enumeration, the Stirling recurrence, and order sets built from nondecreasing cycle parts; `check_bound_certificate.py` re-derives all 372 upper and lower values using an unrestricted composition recurrence for permutation orders, inclusion–exclusion for surjections, and direct enumeration of all nonzero displacements. The independence of method is the evidence, so both programs and both result files, `data/finite_range_bounds.csv` and `data/independent_bounds_check.json`, are kept. In the same spirit the 369020-state example is confirmed twice, by the integer-encoded C++ search and by `check_large_example.py`’s Python tuple-set search, agreeing on the count and on the checksums 72076669650 and 128342; both runs and both records, `data/cpp_orbit_counts.csv` and `data/independent_python_bfs.json`, are what turn the correction of section 10.3 from an assertion into a verified one.

`data/small_witnesses.json` and `data/bfs_witnesses.json` contain explicit arrays. The data directory also contains the finite-range bounds, the constructive certificate, and verification summaries. The results directory contains the second suite’s records: `certificates.json`, `exhaustive.json`, `examples.json`, `sharp_n3_orbit.json` (the word table of section C in machine-readable form), `landau_gaps.json`, `audit_summary.json`, which distinguishes exact orbit checks from larger certificate-only tests, the test and reproduction logs, `environment.txt`, and `pdf_checks.json`. The last is a rendering audit recording page count, absence of LaTeX warnings and of unresolved reference markers, absence of text outside the page box, enumerated external links, verified word certificates, rendering at 140 dpi, checked `cleveref` names, and full-page visual inspection; it was performed on the 19-page predecessor article from which the certificate and enumeration material is drawn, and has not been rerun against the merged text. Two provenance records are kept, and both are needed because they document different things. `source_audit.md` lists the seven numbered locations actually inspected in the source, the Holzer–König attribution points, and the dependency boundary of the proofs; `LITERATURE_SEARCH.md` together with `RESEARCH_STATUS.md` lists the inspected URLs, the conference DOI, the representative search queries, and the seven-item statement of what is not claimed. The first says what was read in the sources; the second says what was searched for and what is being asserted.

To reproduce the main checks from the archive root:

```
python code/run_checks.py --max-n 30
python code/check_bound_certificate.py
c++ -std=c++17 -O2 code/orbit_count.cpp -o orbit_count
./orbit_count < code/bfs_cases.txt
c++ -std=c++17 -O3 code/exhaustive.cpp -o exhaustive
./exhaustive 4 > results/exhaustive.json
python code/run_experiments.py
python code/certificate_checker.py results/certificates.json
python -m unittest discover -s code -p 'test_*.py' -v
```

The certificate checker prints `ACCEPTED 4 certificate(s)` and exits nonzero on any rejection. Do not invoke Python with `-O` or with `PYTHONOPTIMIZE` set: the audit scripts use assertions deliberately. On

Windows, the compiled executable can be invoked as `.\orbit_count.exe`, with the input file supplied by the shell or an equivalent process-redirection command. The mathematics does not require a particular operating system.

To rebuild the article, run `latexmk -pdf -interaction=nonstopmode article.tex`, or `pdflatex article.tex` at least twice, or use the included Makefile. No bibliography processor is needed; the bibliography is contained in the TeX source. No checksum manifest accompanies the archive: re-running verification rewrites some generated files, including recorded elapsed times, so fixed hashes would be misleading.

11.6 Limits of the verification

No Lean, Coq, Isabelle, or other proof-assistant formalization is supplied. The code does not prove the general theorem by enumeration, and the finite comparison does not settle the exact-optimum conjecture beyond its checked range. The proof of the lower-bound existence theorem used in [proposition 10.1](#) is a cited literature dependency. Conversely, none of those qualifications changes the quantifiers of the independently proved upper bounds: their proofs apply to every allowed n , k , and pair of transformations.

12 Further questions and conclusions

12.1 The remaining extremal inequality

The finite results suggest a sharply defined next problem. Prove, or refute,

$$U(n, k) = L(n, k) \quad (n \geq 7, 3 \leq k < n).$$

This is a sufficient route to the exact binary optimum. It is an extremal inequality over chromatic counts and residual permutation orders, rather than a search over all automata.

A proof would need to show that no same-cycle graph, disconnected bipartite graph, or graph with isolated vertices yields a smaller proper-coloring loss than a suitable connected $K_{\ell, m}$ with coprime block sizes. The residual-order term $H_r(t)$ and the special three-color period bound are part of the statement; discarding either can obscure the sharp value.

Failure of this numerical equality would not automatically disprove the original exact-optimum conjecture. The graph bound is only an upper bound; a minimizing graph need not be realized by a two-generated monoid that reaches every coloring not excluded by the graph. The gap at $(5, 5)$ already illustrates this distinction. [Added 28 September 2026, batch 39: for $k = 3$ this equality is now proved for every $n \geq 7$ ([remark 18.3](#)), with the eliminations described above ([sections 16 and 17](#)), and every extremizer is shown to reach all improper colorings ([theorem 19.1](#)). For $k \geq 4$ the problem remains open ([question 23.1](#)).] [Added 29 September 2026, batches 40–41: for $k \geq 4$ the equality holds for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$ ([remark 33.6](#)); it remains open for $k \geq 8$ below $N(k)$ outside the certified finite ranges ([question 44.1](#)).]

12.2 Using more than one collision

The proof selects only one collision of the singular generator. A map of rank substantially below $n - 1$ has many collisions. Taking the union of all their permutation translates may exclude many more colorings. Its proper colorings would still be unreachable through words containing the singular letter. However, the union need not remain 3-colorable. This creates a tradeoff: additional collision information strengthens counting but can remove the convenient supply of surjective proper colorings.

For a fixed output size, a useful refinement would optimize over subsets of collision pairs whose orbit graph remains k -colorable. Such a refinement may improve the diagonal case $k = n$, or provide sharper instance-specific certificates even when it does not improve the universal maximum. [Added 29 September 2026, batches 40–41: Part III’s matching hierarchy (theorem 40.1) uses every kernel pair of the singular letter for two coprime cycles.]

12.3 What is bounded, and what is not

The yes-or-no target is settled within this draft: the complete k^n coloring space cannot be reached by two generators when $k \geq 3$. That conclusion is conditional only on the correctness of the proof, not on a monoid-classification conjecture and not on the experimental data. Acceptance of the result as a new theorem nevertheless requires external scrutiny and a more comprehensive bibliographic check.

The next quantity is $R_2(n, k)$, the maximum of $|\tau\langle a, b \rangle|$ over transition pairs and output maps of minimal n -state DFAOs using k labels, as in section 10. We have proved universal upper bounds, not a formula for $R_2(n, k)$, and outside the finite range of proposition 10.1 this article does not establish the optimality of previously proposed lower-bound families or settle the remaining open questions in [1, Section 5]. [Added 28 September 2026, batch 39: this paragraph describes the report at the state fbba58593 against which Part II was written. For $k = 3$, Part II proves the formula $R_2(n, 3) = 3^n - 3(2^{A_n} + 2^{B_n} - 2) + B_n$ for every $n \geq 7$, where (A_n, B_n) is the closest coprime split of n (theorem 13.1), using Davies’s lower construction for attainment; this settles the $k = 3$ slice of Davies’s Section 5, Problem 2. For $k \geq 4$ the paragraph stands, and Davies’s other open questions remain open.] [Added 29 September 2026, batches 40–41: for $k \geq 4$, Part III proves $R_2(n, k) = k^n - B_k(a_n, b_n) + a_n b_n$, the optimality of the lower-bound family, for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$ and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$ (theorem 28.1); the slice $k \geq 8$ below $N(k)$ and Davies’s other open questions remain open.]

The chromatic estimate suggests a concrete refinement program in three steps: optimize over collision pairs, count proper colorings for their unions, and analyze when all improper colorings are in fact reachable. The first two steps produce obstructions; the third is a constructive saturation problem of a different nature. Confusing them would turn a valid upper bound into an unjustified equality claim. [Added 28 September 2026, batch 39: Part II carries out the second and third steps partly, for $k = 3$: two cross collisions cost at least $6A_n B_n$ further states (proposition 19.2), and equality forces saturation (theorem 19.1), which is necessary but not sufficient (example 19.3).] [Added 29 September 2026, batches 40–41: Part III does the same for $k \geq 4$ in its exactness range (theorems 39.1 and 40.1 and example 39.3).]

A different question is whether the uniform additive deficit can be strengthened in n as well as in k . At fixed k the quantity $k! - g(k)$ is independent of n , whereas a collision-orbit graph can exclude a number of colorings growing exponentially in n ; isolated vertices make that growth particularly transparent, since each one multiplies the proper-coloring count by k . Extracting the best universal bound from all possible cycle structures is a separate task from proving nonattainment, and is what section 9 begins.

Finally, a proof-assistant formalization would be valuable. The argument uses only finite functions, elementary graph coloring, and elementary group actions, and its main logical steps are already isolated as explicit lemmas.

12.4 Group structure beyond cyclicity

The subgroup-image statement in proposition 8.1 separates two independent requirements for full reachability: enough proper colorings must exist, and the permutation group must induce enough output symmetries on them. For an abelian permutation group, the second requirement already fails at three outputs. For a general group, one can ask which output symmetric groups occur as quotients of subgroups and how that restriction interacts with the chromatic number of its collision-orbit graphs.

This viewpoint is not limited to the numerical gap $k! - g(k)$. It supplies an obstruction to a whole family of colorings being in one orbit and identifies why noncommuting input permutations are effective in the ternary witness.

12.5 Conclusion

The central result is the uniform inequality

$$|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n \quad (3 \leq k \leq n).$$

Its proof has a concrete mechanism. A singular letter collapses a pair; the permutation letter moves that pair around a 3-colorable graph; proper colorings can then be reached only by permutation powers. Output relabeling creates symmetries that a cyclic orbit cannot realize. This gives both a quantitative proposed resolution of the selected problem and a polynomial-time procedure for exhibiting a missing reversal state.

Counting the full graph-coloring obstruction, rather than one relabeling orbit, gives the stronger structural bound. That bound establishes the reported finite optimality results when combined with the published lower construction. The unrestricted exact binary maximum remains a separate question beyond what is proved here. [Added 28 September 2026, batch 39: for three outputs Part II answers it for every $n \geq 7$ (theorem 13.1); for $k \geq 4$ it remains open.] [Added 29 September 2026, batches 40–41: for $k \geq 4$ Part III answers it for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$ (theorem 28.1).]

Part II. The exact three-output maximum for all $n \geq 7$

Sections 1–12 and Appendices A–D form the original report (Part I) of 20 September 2026, unchanged except for dated pointers to this addition and a layout fix of its title page. Part II was added on 28 September 2026 in batch 39 of ProveIt’s incoming-report intake, from one manuscript written against the state of Part I printed above. It answers, for three outputs, the question Part I left open in section 12: it determines $R_2(n, 3)$ for every $n \geq 7$, where Part I had established it only for $7 \leq n \leq 30$ (proposition 10.1). Section 13 records the provenance and fixes the notation of the addition.

13 Provenance, conventions and the exact answer

13.1 One manuscript, one addition

Part II is batch-39 manuscript 06, *Exact Binary Reversal with Three Outputs: An all-state solution and rigidity of extremizers*, an AI-assisted research article prepared for Vladimir Reshetnikov and dated 28 September 2026 (19 pages as delivered). It arrived as the archive `ProveIt_Three_Output_Reversal.zip` (inner directory `three_output_reversal/`) in ProveIt commit `74f7f5bdb` and was placed in `e2b1f016a`. It was written against ProveIt commit `fbba58593`, at which Part I’s `article.tex` had the Git blob `e6dbd47b`; that blob is unchanged at the placement commit, so every statement the manuscript makes about Part I refers to the text printed above. Its code, data and source audit are shipped beside Part I’s files with the prefix `02-three-output-` (Part I’s unprefix files count as 01); its manuscript, README and PDF are not shipped and survive in the arrival commit.

The manuscript contributes, in the order printed here: an all-state upper bound for three outputs, obtained by eliminating every collision graph except the complete bipartite graph on the optimal coprime two cycles (sections 14 to 17); the optimization of the surviving graph and, with Davies’s published lower construction, the exact value of $R_2(n, 3)$ for every $n \geq 7$ (section 18); the necessary structure of every extremizer (section 19); the orbit inventory of the missing colorings (section 20); asymptotics and recurrences (section 21); its finite checks and their limits (section 22); eight research questions (section 23); its conclusion (section 24); the fully displayed finite base table (section 25); a formalization plan (section 26); and its source audit (section 27). Every result, proof, example, remark, limitation and question of the manuscript is printed. The manuscript re-derives the coloring-orbit description, the collision-coloring bound, and the two graph counts of Part I; those are printed once, in Part I, and credited where Part II uses them (section 14.1). Its different proof of the three-color biclique count is kept as a second proof. Remark 18.3 and the finite comparison of section 18.2 were added in the merge.

Status (manuscript 06). This is a research draft with complete conventional proofs of its new upper-bound and rigidity statements. It has not been independently refereed or checked in a proof assistant. Novelty is assessed relative to the sources inspected, not asserted as an established priority claim. The result concerns exactly three outputs; it does not settle the full variable-output conjecture or the largest-two-generated-monoid problem. Nothing in Part II is formalized in any proof assistant.

13.2 Notation of Part II

Part I’s symbols keep their meanings: Q , Δ , τ , q_0 , M , τM , sc , $G(p; x, y)$, $\text{Col}_k(G)$, $P_G(k)$, $C_L(k)$, $B_k(u, v)$, $H_r(t)$, $T_{r,k}(a, b)$, $D(n, k)$, $U(n, k)$, $L(n, k)$, $G_k(\ell, m)$, $M(n, k)$ and $R_2(n, k)$. Part II works at

$k = 3$ throughout, except in question 23.1 and section 23, and its new symbols, and the ones renamed from the manuscript, are listed in table 5.

Table 5: Symbols of Part II. “Renamed” gives the manuscript’s own symbol. The last column prints the tempting false reading.

Symbol	Meaning in Part II	Not to be confused with
(A_n, B_n) , or (A, B)	the closest coprime split of n , (13.1)	Part I’s split (ℓ, m) of (10.1); at $k = 3$ the optimal (ℓ, m) is (A_n, B_n) . B_n (one index, an integer) is not a chromatic count.
$\mathcal{B}(u, v)$	$3(2^u + 2^v - 2)$, the proper three-colorings of $K_{u,v}$	equal to Part I’s $B_3(u, v)$ of (9.7); not B_n .
$\mathcal{B}_k(a, b)$	Part I’s $B_k(a, b)$ of (9.5), used only in question 23.1	—
D_n (one index)	$\mathcal{B}(A_n, B_n) - B_n$, the deficit $3^n - R_2(n, 3)$	Part I’s $D(n, k)$, the minimum of the collection (9.11)–(9.15). By definition they differ; that $D(n, 3) = D_n$ for $n \geq 7$ is a theorem (remark 18.3).
$C_L, C_*(m)$	$C_L = C_L(3) = 2^L + 2(-1)^L$; $C_*(m)$ its minimum in section 25	Part I’s $C_L(k)$ at general k .
P_G	$P_G(3) = \text{Col}_3(G) $	Part I’s $P_G(k)$; not $P_2(d)$.
$\text{Col}_3(G)$ (renamed from $\text{Prop}(G)$)	proper three-colorings of G	—
$\text{Orb}_p(c)$ (renamed from $O_p(c)$)	$\{c \circ p^j : j \geq 0\}$	Part I’s $\text{Orb}_\tau = \text{Orb}_p(\tau)$ of (5.1); not Part I’s order set O_r of (9.9).
$J(r)$	the largest product of the parts of a partition of r , (14.4)	Landau’s $g(r)$ (a least common multiple, $g(r) \leq J(r)$) and Part I’s $H_r(t)$; $H_r(t) \leq tJ(r)$.
E_n	the finite same-cycle minimum (16.4)	—
$P_2(d)$	primitive binary words of length d , (20.1)	P_G .
X, Y	the two cycles of an extremal permutation	—
h	a cyclic displacement (section 14) or $\lfloor n/2 \rfloor$ ((13.1) and later)	—
d	a gcd (sections 14, 16 and 17) or an orbit size (section 20)	—
s, t in theorem 13.1	two arbitrary transformations	Part I’s letters a, b of theorem 1.1; in Part II a, b are cycle lengths.

No normalization changed: every renamed symbol denotes exactly the object the manuscript denotes, and $\mathcal{B}, C_L, P_G$ are Part I’s $B_3, C_L(3), P_G(3)$ under the names the manuscript uses. As in Part I, “binary” always describes the *input* alphabet: there are two input letters but three output values.

13.3 The exact answer

A deterministic finite automaton with output (DFAO) returns one of finitely many labels after reading its input word. Reversal asks how much memory is needed to compute the same function with the input read in the opposite order. Davies’s lower construction and her question about its optimality are the published starting point [1, Corollary 3 and Section 5, Problem 2]. Part I supplies a collision-coloring upper-bound method and verifies agreement with that lower construction for $7 \leq n \leq 30, 3 \leq k < n$ (proposition 10.1). The purpose of Part II is to remove the finite-range restriction for $k = 3$, and to determine what equality forces. [Added 29 September 2026, batches 40–41: Part III does so for $k \geq 4$, and gives a second proof of theorem 13.1 (section 38).]

For $n \geq 7$, define (A_n, B_n) by

$$(A_n, B_n) = \begin{cases} (h, h+1), & n = 2h+1, \\ (h-1, h+1), & n = 2h, h \text{ even}, \\ (h-2, h+2), & n = 2h, h \text{ odd}. \end{cases} \quad (13.1)$$

The last case has $h \geq 5$. In each case $1 < A_n < B_n$ and the two integers are coprime. When n is fixed we abbreviate them to A, B . These are the splits Part I lists after (10.1) to show that admissible splits exist; lemma 18.1 shows that they are the optimal ones.

Theorem 13.1 (Exact three-output reversal). *For every integer $n \geq 7$,*

$$R_2(n, 3) = 3^n - D_n, \quad D_n = 3(2^{A_n} + 2^{B_n} - 2) - B_n. \quad (13.2)$$

The same maximum is obtained whether the original automaton is required to be accessible with n states or to be minimal with exactly n states. Moreover, for arbitrary transformations s, t of an n -element set and any three-valued map τ , even without accessibility or surjectivity,

$$|\tau\langle s, t \rangle| \leq 3^n - D_n. \quad (13.3)$$

Here $\langle s, t \rangle$ includes the identity. Thus the initially supplied output coloring is counted among the reachable reverse states. The deficit has the explicit form

$$D_n = \begin{cases} 9 \cdot 2^h - h - 7, & n = 2h+1, \\ \frac{15}{2} 2^h - h - 7, & n = 2h, h \text{ even}, \\ \frac{51}{4} 2^h - h - 8, & n = 2h, h \text{ odd}. \end{cases} \quad (13.4)$$

In particular,

$$0 < D_n < \frac{51}{4} 2^{n/2} \quad (n \geq 7). \quad (13.5)$$

Positivity follows at once from the expression for D_n in (13.2); the upper estimate follows from (13.4). This single benchmark will dispose of several nonextremal graph families.

The proof classifies the orbit graph of a collision pair, controls the period of a proper three-coloring, and excludes every graph except the complete bipartite graph on the optimal two cycles. Elementary product bounds replace enumeration of residual permutation orders. The inequalities also yield rigidity: every extremizer has one permutation letter with the specified two cycles and one rank- $(n-1)$ singular letter with exactly one cross-cycle collision. Its reachable colorings are all improper colorings of that complete bipartite graph together with one maximal cyclic orbit of proper colorings. Part II also gives the complete orbit inventory of the missing colorings and a quantitative penalty for additional collisions.

Proof map of Part II. Section 14 fixes conventions, recalls Part I's collision-graph bound, and adds the residual product bound. The all-state comparison is in sections 15 to 18. Equality is analyzed in sections 19 and 20. The verification record and the explicit finite base table are in sections 22 and 25.

13.4 What is inherited and what is proved here

The following division of responsibility is important. The lower construction and the collision-graph method are not presented as new.

Ingredient	Attribution and role
Reversal as an orbit of colorings	Davies, Proposition 4; proved in Part I (proposition 2.1) and reproved in the manuscript with the same composition convention (section 14.1).
Coprime two-cycle lower construction	Davies, Theorem 3 and Corollary 3; used as a published theorem, not reproved at the semigroup-generation level.
Collision-orbit graphs and their chromatic bound	Part I (sections 3 and 9); reproved in the manuscript and recalled in section 14.1.
All- n elimination for three outputs	The comparisons in sections 15 to 17, including the delicate one-residual-state case.
Exact formula and necessary extremal structure	Theorems 13.1 and 19.1; the all-state conclusion and equality analysis extend Part I.
Additional-collision penalty and missing-orbit inventory	Proposition 19.2 and theorem 20.1; consequences developed in Part II.

In Part I the manuscript inspected the README, the article’s structural bound and finite-optimality sections, its research-status record, and the `u_witness` implementation of `code/reversal.py`. Part I explicitly separates its finite agreement from an all- n proof. The proof in Part II needs no unproved claim elsewhere in the repository.

The numerical values 24, 67, 218, 699 at $n = 3, 4, 5, 6$ are already recorded as exact in Davies’s Table 3 [1], and Part I reproduces them (section 10). They are not new results of Part II. The threshold $n \geq 7$ is essential to the formula: at five states the coprime two-cycle construction is not the optimum.

14 Coloring orbits, collision graphs and residual products

14.1 What Part II takes from Part I

Automata, composition and the reversal construction are as in section 2: letters act by transformations $t_a : Q \rightarrow Q$, a word $w = a_1 \cdots a_m$ acts by $t_w = t_{a_m} \circ \cdots \circ t_{a_1}$, the computed function is $f(w) = \tau(t_w(q_0))$, an automaton is accessible if every state is $t_w(q_0)$ for some word w , and $f^R(w) = f(w^R)$. Here $\Delta = \{0, 1, 2\}$. The state complexity $\text{sc}(f)$ is the least number of states in a DFAO computing f . Part II defines $R_2(n, 3)$ using accessible n -state automata with two input letters whose output map uses all three labels; the equality with the convention $\text{sc}(f) = n$ is proved separately in theorem 13.1 rather than assumed. (Part I states $R_2(n, k)$ once for accessible and once for minimal n -state DFAOs, sections 10 and 12; at $k = 3$ theorem 13.1 shows the two readings agree.)

Coloring-orbit description. For $M = \langle t_a : a \in \Sigma \rangle$ the reversal has a realization with reachable state set $\tau M = \{\tau \circ t : t \in M\} \subseteq \Delta^Q$, transition $c \mapsto c \circ t_a$, initial state τ and output $c(q_0)$; if the original automaton is accessible, these reachable states are pairwise distinguishable, so $\text{sc}(f^R) = |\tau M|$. This is proposition 2.1 and the construction preceding it; the manuscript proves it again with the same argument (after reading $a_1 \cdots a_m$ the coloring is $\tau \circ t_{w^R}$; distinct colorings are separated by the reverse of a word reaching a state where they differ), and notes that removing unreachable states therefore yields a minimal reversal automaton.

Remark 14.1 (The meaning of a word). When a reverse search applies letter a and then letter b , it replaces c by $c \circ t_a \circ t_b$. This is the pullback action, not the forward state action on the same two-letter word. The order of composition in the collision lemma depends on this convention. Part I states the same convention for its word certificates (section C).

For $c : Q \rightarrow \Delta$ and a permutation p , write

$$\text{Orb}_p(c) = \{c \circ p^j : j \geq 0\}.$$

Its size is the least positive j with $c \circ p^j = c$ and divides $\text{ord}(p)$. A nonbijective transformation is called *singular*. A pair $x \neq y$ with $s(x) = s(y)$ is a *collision* of s .

Collision-coloring bound. Suppose that the two generators are a permutation p and a singular map s , choose a collision $\{x, y\}$ of s , and form the collision-orbit graph $G = G(p; x, y)$ of (3.1). A coloring is *proper* if it assigns different labels to both ends of every edge; $\text{Col}_3(G)$ is the set of proper three-colorings and $P_G = |\text{Col}_3(G)|$. Every element of $\tau\langle p, s \rangle$ represented by a word containing s is improper for G (lemma 3.3), and consequently, by proposition 9.1 at $k = 3$,

$$|\tau\langle p, s \rangle| \leq 3^n - P_G + \begin{cases} |\text{Orb}_p(\tau)|, & \tau \text{ is proper,} \\ 0, & \tau \text{ is improper.} \end{cases} \quad (14.1)$$

The manuscript reproves this with the same rightmost-singular-factor argument and the observation that p , an automorphism of G , preserves properness. As Part I stresses after remark 9.2, the inequality makes no claim that all improper colorings are reachable. Such saturation is an extra property. In Part II it will be forced by equality for an extremizer (theorem 19.1), not assumed in proving the upper bound.

The two graph types. If x, y lie in one p -cycle of length m with cyclic displacement h , put $d = \gcd(m, h)$, $L = m/d$ and $r = n - m$. The nonisolated graph is d disjoint cycles of length L (for $L = 2$ each component is a single edge) and, by (9.4) at $k = 3$,

$$P_G = C_L^d 3^r, \quad C_L = 2^L + 2(-1)^L, \quad L \geq 2. \quad (14.2)$$

If x, y lie in distinct cycles of lengths $a \leq b$, put $d = \gcd(a, b)$, $u = a/d$, $v = b/d$ and $r = n - a - b$. Then $G \cong dK_{u,v} \sqcup rK_1$ and, by (9.6) and (9.7),

$$P_G = \mathcal{B}(u, v)^d 3^r, \quad \mathcal{B}(u, v) = 3(2^u + 2^v - 2). \quad (14.3)$$

The manuscript proves both counts again. Its cycle count is Part I's (the trace of the L th power of the 3×3 matrix with zero diagonal and all other entries one, whose eigenvalues are $2, -1, -1$; at $L = 2$ the count is 6, the number for a single edge), and its edge criterion for two cycles is Part I's Chinese remainder criterion $i \equiv j \pmod{d}$. Its derivation of the biclique count differs from Part I's specialization of the Stirling sum (9.5), and is kept.

Second proof of (9.7) (manuscript 06). A proper three-coloring of a complete bipartite graph uses disjoint palettes on its two nonempty sides, so at least one side is monochromatic. There are $3 \cdot 2^v$ colorings with the left side monochromatic and $3 \cdot 2^u$ with the right side monochromatic. The six colorings with both sides monochromatic were counted twice. This gives $\mathcal{B}(u, v)$. $\square$

14.2 A safe bound on residual orders

Let $J(r)$ be the largest product of the positive parts in a partition of r , with the empty-product convention $J(0) = 1$. Then

$$J(0) = J(1) = 1, \quad J(r) = \begin{cases} 3^q, & r = 3q, \\ 4 \cdot 3^{q-1}, & r = 3q + 1 \geq 4, \\ 2 \cdot 3^q, & r = 3q + 2. \end{cases} \quad (14.4)$$

Lemma 14.2 (Residual product bound). *Every permutation on r points has order at most $J(r)$, and $J(r) \leq 3^{r/3}$. In the same-cycle case,*

$$|\text{Orb}_p(\tau)| \leq mJ(r). \quad (14.5)$$

In the two-cycle case,

$$|\text{Orb}_p(\tau)| \leq \begin{cases} bJ(r), & d = 1 \text{ and } \tau \text{ is proper,} \\ duvJ(r), & \text{in general.} \end{cases} \quad (14.6)$$

Proof. The order of a permutation is the least common multiple of its cycle lengths, which is at most their product. A maximal product partition has no part $\ell \geq 5$, since replacing it by 3 and $\ell - 3$ increases the product. A part 1 is absorbed into another part, except when $r = 1$; parts 4 may be replaced by two parts 2. Three parts 2 are inferior to two parts 3. This gives (14.4). Also every positive integer ℓ satisfies $\ell \leq 3^{\ell/3}$: check 1, 2, 3, 4, and use the same replacement for $\ell \geq 5$. Multiplying proves $J(r) \leq 3^{r/3}$.

Distinguished cycles of total order t and the residual cycles have combined order at most $tJ(r)$. This gives $mJ(r)$ and $\text{lcm}(a, b)J(r) = \text{duv}J(r)$.

If $d = 1$, the nonisolated graph is the connected $K_{a,b}$. In a proper three-coloring one side is monochromatic, so that side has period one. The period on the other side divides its length. Hence the period on the two distinguished cycles is at most b , not merely at most ab . Combining with the residual period gives the improved bound. $\square$

The last paragraph is Part I's three-color period improvement (9.10). This argument avoids a potentially invalid shortcut: the residual order maximizing an ordinary integer need not maximize its least common multiple with a fixed distinguished order (Part I makes the same point after (9.9)). A product upper bound is used, for which no such issue arises. The price is that $J(r)$ is coarser than Part I's exact order sets: $H_r(t) \leq tJ(r)$. The improvement for proper three-colorings is the reason the proof is specialized to three outputs.

15 Eliminating the unmixed generator cases

Lemma 15.1 (Large deficits without a mixed pair). *For $n \geq 7$, none of the following can attain $3^n - D_n$: a nonsurjective output map, two permutation generators, or two singular generators.*

Proof. A nonsurjective output map has an orbit of size at most 2^n , so its deficit is at least $3^n - 2^n$.

Two permutation generators preserve the number of occurrences of each color. Thus their orbit lies in a single multinomial class. Every multinomial coefficient of degree n in three variables is at most 3^{n-1} . To see this, the assertion holds in degree one; each coefficient in the next degree is the sum of at most three coefficients in the previous degree. The deficit is therefore at least $2 \cdot 3^{n-1}$.

For two singular generators choose one collision edge for each. There are at least $4 \cdot 3^{n-2}$ colorings proper for both edges. If the edges are distinct this is exact, whether they meet or are disjoint; if the edges coincide the count is $2 \cdot 3^{n-1}$, which is larger. Any nonempty composition collapses the chosen edge of its rightmost factor. Hence only the initial coloring could be reachable among the colorings proper for both edges. The deficit is at least $4 \cdot 3^{n-2} - 1$.

Each of the first two deficits is no smaller than this last one. For the first, use $5 \cdot 3^{n-2} + 1 \geq 2^n$, true for $n = 2$ and preserved on increasing n ; the second comparison is immediate. Finally,

$$4 \cdot 3^{n-2} - 1 > \frac{51}{4} 2^{n/2} \quad (n \geq 7).$$

At $n = 7$ the left side is 971, while the right side is $102\sqrt{2} < 153$. On increasing n , the left side grows by a factor greater than three and the right side by $\sqrt{2}$. The comparison with (13.5) completes the proof. $\square$

These are the entries (9.11)–(9.13) of Part I at $k = 3$, with the balanced multinomial $M(n, 3)$ replaced by the cruder bound 3^{n-1} . We can consequently assume that the output map is surjective and the generators are a permutation p and a singular transformation s . No hypothesis about the rank of s is being imposed at this stage.

16 Eliminating collisions within one cycle

Lemma 16.1 (Cycle graphs have excessive deficit). *For every $n \geq 7$, a mixed pair with any collision of s lying within one p -cycle has orbit size strictly less than $3^n - D_n$.*

Proof. Use the parameters of (14.2). For $L \geq 2$,

$$C_L = 2^L + 2(-1)^L \geq 6^{L/3}. \quad (16.1)$$

For even L , this follows from $2^L \geq 6^{L/3}$. For odd L , equality holds at $L = 3$, and $C_{L+2} = 4C_L + 6$ with $4 > 6^{2/3}$ proves the induction. Thus $P_G \geq 6^{m/3}3^r$.

For $m \geq 3$ we have $m/6^{m/3} \leq 1/2$: equality holds at three, and the ratio decreases since $(m+1)/m \leq 4/3 < 6^{1/3}$. Therefore

$$\frac{mJ(r)}{P_G} \leq \frac{1}{2} 3^{-2r/3} \leq \frac{1}{2}.$$

When $m = 2$, the exact value $C_2 = 6$ gives the same conclusion, with $1/3$ in place of $1/2$. The collision bound consequently gives a deficit at least

$$P_G - mJ(r) \geq \frac{1}{2}P_G \geq \frac{1}{2}6^{n/3}. \quad (16.2)$$

The last inequality uses $3 \geq 6^{1/3}$.

For $n \geq 13$,

$$\frac{1}{2}6^{n/3} > \frac{51}{4}2^{n/2}.$$

The ratio of the exponential terms increases with n ; its value at thirteen is certified by the integer inequality

$$9^{13}2^6 > 51^{62^{13}}, \quad (16.3)$$

which is equivalent to $(9/2)^{13} > (51/2)^6$. This proves strict exclusion for all $n \geq 13$.

For the six remaining values we use the exact finite bound

$$E_n = \min_{\substack{2 \leq m \leq n \\ d|m, d < m}} \left[(2^{m/d} + 2(-1)^{m/d})^d 3^{n-m} - mJ(n-m) \right]. \quad (16.4)$$

Its values and minimizing parameters are

n	7	8	9	10	11	12
E_n	102	250	207	639	1926	1284
(m, d)	(6, 2)	(8, 1)	(9, 3)	(9, 3)	(9, 3)	(12, 4)
D_n	62	109	133	395	276	467

Every entry has $E_n > D_n$. The complete short arithmetic reduction of this table is displayed in section 25, so it is not an inference from random tests or from an unbounded computation. $\square$

17 Eliminating disconnected graphs and residual states

We now suppose that a chosen collision joins distinct cycles. Throughout this section use $a \leq b$, $d = \gcd(a, b)$, $u = a/d$, $v = b/d$, and $r = n - a - b$.

17.1 Two elementary estimates for a biclique

Lemma 17.1. *For positive integers u, v ,*

$$\mathcal{B}(u, v) \geq 3(u + v) \geq 6\sqrt{uv}, \quad (17.1)$$

$$\mathcal{B}(u, v) \geq 3 \cdot 2^{(u+v)/2}. \quad (17.2)$$

If $u + v \geq 3$, the second estimate improves to

$$\mathcal{B}(u, v) \geq 3\sqrt{2} 2^{(u+v)/2}. \quad (17.3)$$

Proof. The inequality $2^j \geq j + 1$ gives (17.1). Writing $T = 2^{(u+v)/2} \geq 2$, arithmetic–geometric mean gives $2^u + 2^v \geq 2T$, hence

$$\frac{\mathcal{B}(u, v)}{T} \geq 3 \left(2 - \frac{2}{T} \right) \geq 3.$$

For $u + v = 3$, direct evaluation at $(1, 2)$ gives the stronger constant $3\sqrt{2}$. For $u + v \geq 4$, $T \geq 4$, and the last display is at least $9/2 > 3\sqrt{2}$. $\square$

17.2 Disconnected bicliques: the case $d \geq 2$

Lemma 17.2. *If $d \geq 2$ and $n \geq 7$, the deficit is strictly greater than D_n .*

Proof. Put $P = \mathcal{B}(u, v)^d 3^r$. By (17.1) and $12d \leq 6^d$ for $d \geq 2$,

$$\mathcal{B}(u, v)^d \geq 6^d (uv)^{d/2} \geq 12duv.$$

Since $J(r) \leq 3^r$, the general period estimate yields

$$\frac{duvJ(r)}{P} \leq \frac{1}{12}. \quad (17.4)$$

We claim also that

$$P \geq 18 \cdot 2^{n/2}. \quad (17.5)$$

When $d \geq 3$, (17.2) gives $P/2^{n/2} \geq 3^d (3/\sqrt{2})^r \geq 27$. When $d = 2$ and $u + v \geq 3$, (17.3) gives the lower bound $(3\sqrt{2})^2 (3/\sqrt{2})^r \geq 18$. The only other case is $d = 2$, $u = v = 1$. Here $n = 4 + r$, so $r \geq 3$, and $P/2^{n/2} = 9(3/\sqrt{2})^r > 18$.

Combining (17.4) and (17.5), the missing-coloring count is at least

$$P - duvJ(r) \geq \frac{11}{12}P \geq \frac{33}{2} 2^{n/2} > \frac{51}{4} 2^{n/2} > D_n.$$

$\square$

17.3 Connected bicliques with at least two residual states

Lemma 17.3. *If $d = 1$ and $r \geq 2$, the deficit is strictly greater than D_n .*

Proof. Here $P = \mathcal{B}(a, b)3^r$. From (17.1) and $J(r) \leq 3^{r/3}$,

$$\frac{bJ(r)}{P} \leq \frac{1}{3} 3^{-2r/3} < \frac{1}{9}.$$

We next show $P \geq (27/\sqrt{2})2^{n/2}$. If $a + b \geq 3$, (17.3) gives

$$\frac{P}{2^{n/2}} \geq 3\sqrt{2} \left(\frac{3}{\sqrt{2}} \right)^r \geq \frac{27}{\sqrt{2}}.$$

If $a + b = 2$, then $a = b = 1$ and $r \geq 5$ since $n \geq 7$. Now $P/2^{n/2} = 3(3/\sqrt{2})^r$, which is larger still. Therefore the deficit is at least

$$P - bJ(r) \geq \frac{8}{9}P \geq 12\sqrt{2}2^{n/2} > \frac{51}{4}2^{n/2} > D_n.$$

The numerical comparison is exact: $48\sqrt{2} > 51$, as follows on squaring. □

17.4 The delicate case of one residual state

Lemma 17.4. *If $d = 1$ and $r = 1$, the deficit is strictly greater than D_n .*

Proof. The residual permutation is the identity, so $J(1) = 1$. The missing-coloring count is at least

$$9(2^a + 2^b - 2) - b, \quad a + b = n - 1. \quad (17.6)$$

For a lower estimate we may ignore the coprimality restriction. At a fixed sum, moving (a, b) to $(a + 1, b - 1)$ when $b \geq a + 2$ decreases this expression by

$$9(2^{b-1} - 2^a) - 1 > 0.$$

It is consequently minimized by the most balanced pair.

For $n = 2h + 1$, this pair is (h, h) , and (17.6) is at least $18 \cdot 2^h - h - 18$. Subtracting the target deficit from (13.4) leaves $9 \cdot 2^h - 11 > 0$.

For $n = 2h$, the pair is $(h - 1, h)$ and the lower estimate is $(27/2)2^h - h - 18$. If h is even, its difference from D_n is $6 \cdot 2^h - 11 > 0$. If h is odd, its difference is $(3/4)2^h - 10$, which is at least fourteen since $h \geq 5$. □

The last comparison explains why a uniform but coarser exponential estimate would not be enough. The worst residue class for the target deficit has coefficient $51/4$, while the one-residual-state competitor has leading coefficient $27/2$. The proof must retain this gap rather than discard it in a loose period estimate.

18 Optimizing the surviving graph and proving the formula

After the strict exclusions, the only possible extremal collision graph is $K_{a,b}$, with $a + b = n$ and $\gcd(a, b) = 1$. There are no residual states. The graph bound gives

$$3^n - |\tau\langle p, s \rangle| \geq \mathcal{B}(a, b) - b = 3(2^a + 2^b - 2) - b. \quad (18.1)$$

Lemma 18.1 (Closest coprime split). *For $n \geq 7$, among positive $a \leq b$ with $a + b = n$ and $\gcd(a, b) = 1$, the right side of (18.1) has the unique minimum D_n , attained at (A_n, B_n) in (13.1).*

Proof. For $b \geq a + 2$, replacing (a, b) by $(a + 1, b - 1)$ decreases $3(2^a + 2^b - 2) - b$ by

$$3(2^{b-1} - 2^a) - 1 > 0.$$

Thus the optimum is the admissible pair closest to equality, and is unique. For $n = 2h + 1$, the neighboring pair $(h, h + 1)$ is coprime. For $n = 2h$, the equal pair is not coprime because $h \geq 4$. If h is even, $h - 1$ and $h + 1$ are odd and their gcd divides two, hence is one. If h is odd, $h - 1$ and $h + 1$ are both even, while $h - 2$ and $h + 2$ are odd with gcd dividing four, hence are coprime. These are exactly (13.1). □

Proposition 18.2 (Published lower construction, specialized). *For $1 < a < b$ with $\gcd(a, b) = 1$ and $n = a + b$, there is an accessible n -state DFAO with two input letters and three output values whose reversal has $3^n - \mathcal{B}(a, b) + b$ states.*

This proposition is the $k = 3$ specialization of Davies’s Theorem 3 and Corollary 3 [1], the lower bound $L(n, 3)$ of (10.1) with $G_3(\ell, m) = m$. Its existence assertion is an external theorem used here, as it is in Part I’s proposition 10.1. It is not replaced by the finite computations below. The underlying two-generated monoid construction is discussed by Holzer and König [2]. No assertion that this monoid is the largest two-generated monoid is needed.

Proof of theorem 13.1. The algebraic inequality follows from the exhaustive cases: nonsurjective outputs and the two unmixed generator cases are strictly smaller by lemma 15.1; same-cycle collisions are strictly smaller by lemma 16.1; cross-cycle collisions with $d \geq 2$ or $r \geq 1$ are strictly smaller by lemmas 17.2 to 17.4. The remaining case is bounded by lemma 18.1.

For attainability choose $a = A_n$, $b = B_n$ in proposition 18.2. All its hypotheses hold: in particular $A_n > 1$, $A_n < B_n$, and the lengths are coprime. By proposition 2.1, its reversal state complexity is $3^n - D_n$.

It remains to verify that an attaining original function really has state complexity n . Since $A, B \geq 2$, the graph $K_{A,B}$ contains two disjoint edges. Therefore

$$\mathcal{B}(A, B) \leq 4 \cdot 3^{n-2},$$

by counting colorings proper for just these two edges. The attaining reversal has more than

$$3^n - 4 \cdot 3^{n-2} = 5 \cdot 3^{n-2} > 3^{n-1}$$

states. A function representable by at most $n - 1$ original states would have a reversal with at most 3^{n-1} states by the coloring construction. This contradiction establishes original minimality, proving both conventions have the same maximum. $\square$

The minimality step is Part I’s argument after proposition 10.1 at $k = 3$.

Remark 18.3 (The $k = 3$ slice of Part I’s remaining extremal inequality; added in the merge). Part I asks in section 12 to prove or refute $U(n, k) = L(n, k)$ for $n \geq 7$, $3 \leq k < n$. For $k = 3$ it holds for every $n \geq 7$:

$$D(n, 3) = D_n, \quad U(n, 3) = L(n, 3) = R_2(n, 3) = 3^n - D_n.$$

Indeed, $L(n, 3) = 3^n - \min[\mathcal{B}(\ell, m) - m]$ over coprime $1 < \ell < m$ with $\ell + m = n$, which is $3^n - D_n$ by lemma 18.1. Part I’s collection (9.11)–(9.15) contains the entry $\mathcal{B}(A_n, B_n) - T_{0,3}(A_n, B_n) = \mathcal{B}(A_n, B_n) - B_n = D_n$, so $D(n, 3) \leq D_n$. Conversely every entry is at least the corresponding quantity bounded in Part II: (9.11) and (9.13) are the first and third deficits of lemma 15.1, and (9.12) is at least the second, since $M(n, 3) \leq 3^{n-1}$; in (9.14), $H_{n-m}(m) \leq mJ(n - m)$; in (9.15), $T_{r,3}(a, b) \leq bJ(r)$ when $d = 1$ and $T_{r,3}(a, b) = H_r(\text{lcm}(a, b)) \leq \text{div}J(r)$ when $d \geq 2$, because $\text{lcm}(t, o) \leq to$ and every permutation order on r points is at most $J(r)$ (lemma 14.2). The proofs of lemmas 15.1, 16.1 and 17.2 to 17.4 bound exactly these relaxed quantities from below by numbers exceeding D_n , and lemma 18.1 handles the remaining entries ($d = 1$, $r = 0$), so $D(n, 3) \geq D_n$. This remark is the merge’s reading of the manuscript’s comparisons against Part I’s definitions; the manuscript does not state it. For $k \geq 4$ the equality remains open. [Added 29 September 2026, batches 40–41: for $k \geq 4$ it holds on Part III’s exactness range (remark 33.6).]

18.1 Explicit witnesses and examples

Here is the zero-based parity version of the published two-cycle witness, as also implemented in Part I's `code/reversal.py` (`u_witness`). Put

$$Q = \{0, \dots, a + b - 1\}, \quad p = (0 \ 1 \ \cdots \ a - 1)(a \ a + 1 \ \cdots \ a + b - 1).$$

Start s as the identity and set

$$s(0) = a, \quad s(a + b - 1) = 0.$$

When a is odd, also set $s(1) = 2$ and $s(2) = 1$; for even a leave these two values unchanged. Take

$$\tau = 0^a \ 1 \ 2^{b-1}. \quad (18.2)$$

The initial state is zero. Powers of p visit the first cycle; applying s at zero enters the second, whose states are then visited by p . The pure- p orbit of (18.2) has size b , since the unique occurrence of label one rotates around the second cycle.

The general attainment assertion for this witness is part of the cited lower construction. The programs of section 22 independently explore its full coloring orbit for the displayed small instances; they do not prove the all-state semigroup-generation theorem.

n	(A, B)	$\mathcal{B}(A, B)$	D_n	$R_2(n, 3)$	3^n
7	(3, 4)	66	62	2125	2187
8	(3, 5)	114	109	6452	6561
9	(4, 5)	138	133	19550	19683
10	(3, 7)	402	395	58654	59049
11	(5, 6)	282	276	176871	177147
12	(5, 7)	474	467	530974	531441

For example, the seven-state arrays are

$$p = (1, 2, 0, 4, 5, 6, 3), \quad s = (3, 2, 1, 3, 4, 5, 0), \quad \tau = (0, 0, 0, 1, 2, 2, 2).$$

Exactly four proper colorings are reached, and all $3^7 - 66 = 2121$ improper colorings are reached. The total is 2125. Notice that the *deficit* is not monotone in n : $D_{10} = 395 > D_{11} = 276$. This reflects the arithmetic constraint on the closest coprime split, not an error in the exponential asymptotics. The rows $n = 7, 8, 9, 10$ are the $k = 3$ rows of Part I's table in section 10.

18.2 Comparison with Part I's finite table

In the write phase the closed form (13.2) was compared with the 24 rows $k = 3, 7 \leq n \leq 30$, of Part I's `data/finite_range_bounds.csv`. In every row it equals both the recorded upper value $U(n, 3)$ and the recorded lower value $L(n, 3)$, and the recorded maximizing split (ℓ, m) is (A_n, B_n) . The same 24 values and splits appear in the rows $7 \leq n \leq 30$ of `data/02-three-output-structural_checks.csv`. This is a consistency check of two independently written computations, not part of the proof.

19 Rigidity of every extremizer

A triple (p, s, τ) is called *extremal* if its coloring orbit has size $3^n - D_n$. This algebraic definition does not require an initial state. The next theorem therefore applies in particular to every accessible extremal DFAO.

Theorem 19.1 (Necessary extremal structure). *Let $n \geq 7$ and let a two-generator, three-valued coloring orbit be extremal. After interchanging the two generators, one generator is a permutation p and the other is singular, say s . There is a partition $Q = X \sqcup Y$ with $|X| = A_n$ and $|Y| = B_n$ such that:*

- (i) p consists of one cycle on X and one cycle on Y .
- (ii) s has rank $n - 1$ and exactly one nonsingleton fiber. That fiber has two elements, one in X and one in Y .
- (iii) τ is constant on X , and on Y it is a word in the other two labels with least rotational period exactly B_n .
- (iv) Writing $G = K_{X,Y}$, the entire reachable coloring set is

$$\tau\langle p, s \rangle = (\Delta^Q \setminus \text{Col}_3(G)) \sqcup \text{Orb}_p(\tau), \quad |\text{Orb}_p(\tau)| = B_n. \quad (19.1)$$

Proof. The strict inequalities in the proof of theorem 13.1 force the mixed case. Choose any collision of s . Its graph cannot be a same-cycle graph, a disconnected union of bicliques, or a graph with residual vertices. The only remaining possibility is a connected biclique on two coprime cycles covering Q . The strict optimization in lemma 18.1 forces the cycle lengths to be A, B . This proves (i).

The same exclusion applies to *every* collision of s . In particular, there can be no collision internal to X or internal to Y . Thus s is injective on each of these two sets. Every nonsingleton fiber has exactly two elements and is a cross pair; distinct such fibers give disjoint pairs.

Use one such pair in the graph bound. If τ were improper, its orbit would have size at most $3^n - \mathcal{B}(A, B)$, less than the claimed maximum. So τ is proper. In

$$|\tau\langle p, s \rangle| \leq 3^n - \mathcal{B}(A, B) + |\text{Orb}_p(\tau)| \leq 3^n - \mathcal{B}(A, B) + B$$

both inequalities must be equalities. Hence every improper coloring is reachable, and the proper reachable states form exactly a B -element pure orbit. This proves (iv).

At least one side of a proper three-coloring is monochromatic. If Y were monochromatic, its total period would be at most $A < B$. Therefore X is monochromatic and the word on Y has period B . It must use both of the other labels, since a constant word has period one. This proves (iii).

Suppose finally that s had at least two cross-collision pairs. Every composition containing s , considered at its rightmost occurrence, would collapse at least two distinct cross pairs, translated by the same power of p . Choose $x \in X, y \in Y$ and define a coloring by assigning zero to x, y , one to $X \setminus \{x\}$, and two to $Y \setminus \{y\}$. Its only monochromatic cross pair is $\{x, y\}$. It cannot be represented by a word containing s , and it cannot be in the pure orbit, whose colorings are proper. This contradicts saturation of all improper colorings in (iv). Thus s has exactly one double fiber, proving (ii). $\square$

The theorem is stronger than identification of a numerical maximum. It reduces the search for all extremizers from arbitrary pairs of transformations to a fixed two-cycle permutation, a rank- $(n - 1)$ map with a single specified type of kernel pair, and a primitive binary word on the larger cycle. It does not identify all possible images of the singular letter, and does not assert that these necessary conditions are sufficient. For three outputs and $n \geq 7$ it settles, for extremizers, the saturation question that Part I separates from the graph bound (sections D and 9): equality forces every improper coloring to be reached.

19.1 An exact penalty for more than one collision

Proposition 19.2 (Additional-collision penalty). *Suppose p has the optimal cycles X, Y , s is injective on each cycle but has at least two cross-collision fibers, and τ is proper for $K_{X,Y}$. Then*

$$|\tau\langle p, s \rangle| \leq R_2(n, 3) - 6A_nB_n. \quad (19.2)$$

Proof. For each $(x, y) \in X \times Y$, choose a common label for x, y in three ways and assign the two remaining labels to $X \setminus \{x\}$ and $Y \setminus \{y\}$ in two ways. These give $6AB$ distinct colorings with exactly one monochromatic cross pair. They are distinct because that unique pair recovers (x, y) and the labels then recover the remaining choices.

In fact these are all colorings with exactly one monochromatic cross pair: its common label cannot occur elsewhere on either side, and the nonempty remaining sides must use disjoint subsets of the two remaining labels, forcing one label on each side. Every such coloring is unreachable by the rightmost-singular argument, while the pure orbit is proper.

The ordinary graph bound misses at least $\mathcal{B}(A, B) - B = D_n$ proper colorings. The $6AB$ additional missing colorings are improper and hence disjoint from this set. Adding the two deficits gives (19.2). $\square$

This is a stability statement under its stated structural assumptions. It is not asserted for an arbitrary singular map and arbitrary improper output coloring, where the pure orbit can meet the additional forbidden family. It is also an instance of the refinement Part I proposes in section 12, using more than one collision of the singular letter.

19.2 Why the necessary structure is not sufficient

Example 19.3 (A rank-one defect without saturation). Let p have the optimal cycles and let s send a single $x_0 \in X$ to a single $y_0 \in Y$, fixing every other state. Take τ as in (18.2). Then (i)–(iii) of theorem 19.1 hold, and the automaton is accessible from x_0 .

Nevertheless Y is invariant under both letters. Consequently no reachable coloring takes label zero at any point of Y . In particular the constant zero coloring, which is improper, is absent. Thus (iv) fails, and the machine is not extremal. More quantitatively, its orbit is contained in $\Delta^X \times \{1, 2\}^Y$, a set of size $3^A 2^B$.

An exact classification of extremizers must therefore include an image or generation condition beyond the cycle structure, kernel pair, and output period. This is a concrete remaining problem, rather than an unproved sufficiency assertion hidden inside the rigidity theorem.

20 The complete orbit inventory of missing colorings

The forbidden proper colorings have more structure than their cardinality alone reveals. Consider two cycles of lengths a, b , with a proper three-coloring of $K_{a,b}$, and let the permutation rotate both cycles.

Define $P_2(d)$ to be the number of binary words of length d whose least rotational period is d . Equivalently,

$$P_2(d) = \sum_{e|d} \mu(d/e) 2^e, \quad (20.1)$$

where μ is the Möbius function on positive integers. This identity follows by partitioning all 2^d binary words according to their least period and applying divisor inversion. The case $d = 1$ gives $P_2(1) = 2$.

Theorem 20.1 (Proper-coloring orbit inventory). *There are exactly six proper-coloring orbits of size one. For each $d > 1$, the number of proper-coloring orbits of size d is*

$$\frac{3P_2(d)}{d} (\mathbf{1}_{d|a} + \mathbf{1}_{d|b}). \quad (20.2)$$

For an extremizer at $n \geq 7$, the missing colorings consist of this inventory with $(a, b) = (A_n, B_n)$ after deleting precisely one orbit of size B_n .

Proof. The six fixed colorings assign distinct constant labels to the two sides. Every other proper coloring has exactly one monochromatic side, since three labels cannot be split into two disjoint palettes both of size two.

Suppose the a -side is monochromatic. Choose its label in three ways. The other side is a nonconstant binary word of length b . For $d > 1$ dividing b , there are $P_2(d)$ words of least period d on that side: repeat a primitive word of length d exactly b/d times. Rotation partitions them into orbits of size d , giving $3P_2(d)/d$ orbits. The argument with the sides interchanged gives the second term. These two nonconstant families are disjoint, and exhaust all nonfixed proper colorings.

For an extremizer, theorem 19.1 says that the only reachable proper colorings form one B_n -element orbit, proving the last assertion. $\square$

As a consistency check, weighting the orbit inventory by orbit sizes gives

$$6 + 3 \sum_{\substack{d|a \\ d>1}} P_2(d) + 3 \sum_{\substack{d|b \\ d>1}} P_2(d) = 6 + 3(2^a - 2) + 3(2^b - 2) = \mathcal{B}(a, b).$$

For $(A, B) = (3, 4)$, the full proper inventory contains six orbits of size one, three of size two, six of size three, and nine of size four. The extremal machine reaches one of the size-four orbits, leaving

$$6 \cdot 1 + 3 \cdot 2 + 6 \cdot 3 + 8 \cdot 4 = 62$$

missing colorings. This describes their cyclic organization, not just the number missing.

21 Asymptotics and exact recurrences

Corollary 21.1 (Three residue-class asymptotics). *Along the indicated subsequences,*

$$\frac{3^n - R_2(n, 3)}{2^{n/2}} \longrightarrow \begin{cases} 9/\sqrt{2}, & n \text{ odd}, \\ 15/2, & n \equiv 0 \pmod{4}, \\ 51/4, & n \equiv 2 \pmod{4}. \end{cases} \quad (21.1)$$

In particular,

$$\lim_{n \rightarrow \infty} \frac{\log(3^n - R_2(n, 3))}{n} = \frac{\log 2}{2}, \quad 1 - \frac{R_2(n, 3)}{3^n} = \Theta\left(\left(\frac{\sqrt{2}}{3}\right)^n\right).$$

Proof. Divide (13.4) by $2^{n/2}$. Its linear terms tend to zero on this scale. The three positive finite limiting constants imply the uniform two-sided Θ estimate and the logarithmic limit. $\square$

Thus the maximal reversal occupies asymptotically almost every coloring, even though full occupancy is impossible and the absolute number missing grows exponentially. The residue-class constants arise from the nearest coprime split; they are not a feature of a numerical fitting procedure. Compare Part I's remark in section 12 that a collision-orbit graph can exclude a number of colorings growing exponentially in n , whereas the uniform deficit $k! - g(k)$ of theorem 1.1 is independent of n : at $k = 3$ the exact growth rate of the deficit is $2^{n/2}$.

Corollary 21.2 (Recurrence for the deficit). *For all $n \geq 7$,*

$$D_{n+4} = 4D_n + 3\lfloor n/2 \rfloor + 19 + 3\mathbf{1}_{n \equiv 2 \pmod{4}}, \quad (21.2)$$

$$D_{n+12} - 6D_{n+8} + 9D_{n+4} - 4D_n = 0. \quad (21.3)$$

Consequently the ordinary generating function of the tail $(D_n)_{n \geq 7}$ is rational, with denominator dividing $(1 - 4z^4)(1 - z^4)^2$.

Proof. The split in (13.1) changes from (A, B) to $(A + 2, B + 2)$ when n increases by four. Hence

$$D_{n+4} = 4D_n + 3B_n + 16.$$

Substituting the three formulas for B_n yields (21.2). On each residue class modulo four the inhomogeneous term is linear in the step number. Applying a second forward difference annihilates it, giving (21.3). The generating-function assertion is the usual coefficient calculation for a constant-coefficient recurrence. $\square$

22 Reproducible finite checks and their limits

Manuscript 06 ships two standard-library Python programs, `code/02-three-output-verify.py` and `code/02-three-output-independent_check.py`. They share no project imports, and neither imports any of Part I’s code. The first performs direct structural comparisons and literal breadth-first exploration of coloring orbits; the second uses a transfer recurrence for cycle colorings and a max-product composition recurrence for residual products. All comparisons use exact integers.

22.1 What was actually run

Check	Completed scope
Conservative structural upper bound and unique minimizer	$7 \leq n \leq 200$
Distinct values of n in that comparison	194
Structural candidates evaluated	752291
Full witness-orbit searches	$7 \leq n \leq 11$
Largest fully explored witness orbit	176871 states
Proper-orbit inventory comparisons	$(3, 4), (3, 5), (4, 5)$
Unique-cross-pair forbidden colorings checked	72
Independent closest-coprime formula comparisons	$7 \leq n \leq 1000$
Independent recurrence comparisons	982

All checks passed. For every structural comparison, the minimum deficit was D_n , and the unique minimizing graph parameters were $(a, b, d, r) = (A_n, B_n, 1, 0)$. These comparisons use $J(r)$, a coarser estimate than exact residual permutation-order sets; agreement therefore checks the specific relaxed inequalities used in this proof. (Part I’s `data/finite_range_bounds.csv` evaluates the exact order sets for $n \leq 30$; see section 18.2.)

For each breadth-first witness search, the program also enumerated the full three-coloring universe and checked that every missing coloring is proper, that every improper coloring is reached, and that the proper reached set is exactly the pure permutation orbit. Thus those searches check (19.1), not merely a total count.

The additional-collision experiment used the $n = 7$ optimal permutation and a map with two cross collisions. It checked all $6 \cdot 3 \cdot 4 = 72$ unique-cross-pair colorings and found every one missing. This is a finite regression check of proposition 19.2, not its proof.

22.2 Commands and files

The two programs are shipped byte-identical to the delivery and still use the delivered layout: each computes the report root from its own location and reads or writes `data/verification.json`, `data/structural_checks.csv` and `data/independent_check.json`. Run in this report’s directory, the

first would therefore write two stray unprefixed files beside Part I’s data, and the second fails unless the first has run there. The report’s README gives a recipe that restores the delivered names in a scratch copy; there the commands are

```
python code/verify.py --max-n 200 --bfs-max-n 11
python code/independent_check.py
```

The first command writes `data/structural_checks.csv` and `data/verification.json`. The second reads the small table from that JSON record only to compare outputs; it imports no implementation from the first program and writes `data/independent_check.json`. The recorded outputs are shipped as `data/02-three-output-structural_checks.csv`, `data/02-three-output-verification.json` and `data/02-three-output-independent_check.json`, and the recorded console output as `data/02-three-output-console.txt` and `data/02-three-output-independent_console.txt`.

The breadth-first search has an explicit state limit and raises an error rather than returning a partial orbit count. Validation uses explicit exceptions, not Python `assert` statements that disappear under optimization. The supplied parameter limits guard against accidental exponential searches much larger than those reported.

22.3 What the checks do not establish

No finite enumeration proves the universal quantifier in theorem 13.1. That quantifier is supplied by the analytical eliminations and the exact coprime optimization. The six small same-cycle cases are ordinary finite integer calculations fully displayed in section 25. The larger computer checks are supplementary.

No Lean, Rocq, Isabelle, or other proof-assistant verification was carried out for Part II. The published lower theorem is cited, not formally reconstructed. The independent scripts are independent implementations of particular arithmetic tasks, not independent mathematical proofs and not proof-assistant certificates.

23 Further research questions and concrete next steps

Question 23.1 (Four outputs, then arbitrary output count). For $n \geq 7$ and $4 \leq k < n$, is

$$R_2(n, k) = \max_{\substack{1 < a < b, a+b=n \\ \gcd(a,b)=1}} [k^n - \mathcal{B}_k(a, b) + ab],$$

where

$$\mathcal{B}_k(a, b) = \sum_{i=1}^{\min(a,k)} \frac{k!}{(k-i)!} \binom{a}{i} (k-i)^b?$$

This is the remaining range of the same published optimality question, and the right side is Part I’s $L(n, k)$ of (10.1). A first target is $k = 4$: both sides of a proper biclique coloring can now be nonconstant, so the useful period contribution becomes ab rather than b . More importantly, there is no longer the two-term expression $3(2^a + 2^b - 2)$ to control all graph types. A promising route is to derive uniform palette-size inequalities for $\mathcal{B}_k$ before attempting the full structural minimization. Theorem 13.1 does not supply those inequalities. Part I’s finite equality $U(n, k) = L(n, k)$ for $7 \leq n \leq 30$ (proposition 10.1) remains the only evidence for $k \geq 4$ in this report. [Added 29 September 2026, batches 40–41: the preceding sentences describe the report before Part III. Theorem 28.1 answers this question affirmatively for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$ (computer-assisted below the threshold), and for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$, with Davies’s construction for attainment; it uses strict palette-balancing and

explicit palette estimates, the route suggested above. The question is re-scoped, not closed: it remains open for $k \geq 8$ and $\max(7, k + 1) \leq n < N(k)$, apart from $n \leq 30$ (proposition 10.1) and, for $k \leq 12$, $n \leq 100$ (proposition 35.1); for example $k = 8$, $101 \leq n \leq 2687$. See question 44.1.]

Question 23.2 (Classify the singular letter). For the fixed optimal two-cycle permutation p , which rank- $(n - 1)$ maps s with one cross-cycle double fiber, together with which primitive output words on the larger cycle, satisfy the saturation identity (19.1)?

Example 19.3 shows that kernel data alone do not answer this. The useful additional information should concern the image omitted by s and the permutations induced on the noncollapsed part. A necessary-and-sufficient criterion here would classify all numerical extremizers, not just one construction. It should distinguish coloring saturation from generation of a particular large transformation monoid.

Question 23.3 (Recognizing saturation efficiently). Can saturation for the reduced class in question 23.2 be decided in time polynomial in n , or verified by a polynomial-size certificate?

Literal exploration has an exponential state space. The rigidity theorem removes almost all arbitrary transformation parameters, making an algebraic criterion plausible. A negative complexity result would also be valuable, but would require a reduction that preserves the optimal two-cycle and rank- $(n - 1)$ restrictions.

Question 23.4 (Second-best values and stability). What is the largest attainable reversal size strictly below $R_2(n, 3)$? Can a near-extremal orbit be forced to have nearly the structure in theorem 19.1?

The exclusions in the proof give quantitative gaps for several graph types, and proposition 19.2 supplies an exact $6AB$ penalty under fixed-cycle proper-output assumptions. These bounds do not identify the second-best value: a rank- $(n - 1)$ map with incomplete saturation may lose fewer states than a different graph type. The first step is to compare these mechanisms rather than equate the smallest displayed bound with an attainable second maximum.

Question 23.5 (The near-maximum spectrum). For fixed n , which integers between a useful lower threshold and $R_2(n, 3)$ occur as reversal complexities of minimal n -state binary DFAOs?

The missing-orbit inventory suggests divisibility restrictions when the only losses are proper permutation orbits. Additional losses among improper colorings need not respect the same inventory. Separating these two kinds of loss may clarify the near-maximum “magic number” problem mentioned by Davies, without presupposing that every intermediate integer is attainable.

Question 23.6 (Growing output alphabets). As $k = k(n)$ grows, when do the extremizing cycle split and the extremizing collision graph cease to resemble the fixed-three-output answer?

The diagonal $k = n$ encodes the size of a two-generated transformation monoid when the output map is bijective (compare Part I, section 10). That objective is not the same as maximizing a coloring orbit for fixed small k . The present proof gives an exact fixed- k anchor point from which a genuine transition in behavior can be investigated. [Added 29 September 2026, batches 40–41: partly touched by Part III: for every $4 \leq k \leq (n/3)^{1/4}$ the extremizing split and collision graph are those of the fixed-output answer (corollary 33.5 and theorem 39.1); larger k , in particular k a positive proportion of n , remains open (question 44.14).]

Question 23.7 (Formal certification). Can the upper bound and rigidity theorem be formalized with no admitted combinatorial lemmas, while keeping the lower construction as a separately visible dependency until its semigroup-generation proof is available?

The proof breaks naturally into finite-set counting, permutation-cycle analysis, elementary exponential inequalities, and the six explicit base cases. A proof assistant should first certify the universal algebraic inequality (13.3); importing a lower bound is not necessary for that milestone. Section 26 gives a more precise dependency plan.

Question 23.8 (Beyond one cyclic permutation group). Which permutation groups and collision configurations admit an analogous sharp coloring-deficit theorem, rather than only nonattainment of the full coloring space?

Part I already develops qualitative extensions with abelian permutation groups (section 8). A sharper theory would need to replace the two-cycle classification by an analysis of orbital graphs and proper-coloring periods under a general group action. Even restricted families could connect exact state complexity with finite permutation-group structure.

24 Conclusion of Part II

The three-output case permits an exact answer because proper colorings of a connected biclique have one monochromatic side. This reduces the allowable permutation contribution to the longer cycle length. Once that period is kept in the estimate, disconnected collision graphs, unused states, and same-cycle collisions all have strictly larger deficits. The only surviving optimization is the closest coprime split, yielding (13.2) for all $n \geq 7$.

Equality is informative: it forces the singular letter to have a single cross collision and forces saturation of every improper coloring. The missing states form a completely described family of cyclic orbits. These conclusions extend Part I's finite-range result while leaving a clear distinction between the solved three-output problem, the published lower-bound dependency, and the genuinely remaining questions.

25 The finite same-cycle base table

This section exposes all ingredients in the small arithmetic comparison of lemma 16.1, so that the six base cases can be checked without accepting the output of a large search. Define

$$C_*(m) = \min_{\substack{d|m \\ d < m}} (2^{m/d} + 2(-1)^{m/d})^d.$$

Direct evaluation gives

m	2	3	4	5	6	7	8	9	10	11	12
$C_*(m)$	6	6	18	30	36	126	258	216	900	2046	1296
A minimizing d	1	1	1	1	2	1	1	3	2	1	4

For instance, at $m = 6$ the possible counts are $C_6 = 66$, $C_3^2 = 36$, and $C_2^3 = 216$, so the minimum is 36. At $m = 12$, the five possibilities for $d = 1, 2, 3, 4, 6$ are

$$4098, \quad 4356, \quad 5832, \quad 1296, \quad 46656,$$

so the minimum is 1296. Prime values of m have only $d = 1$. The remaining composite values 4, 8, 9, 10 require respectively two, three, two, and three divisor evaluations.

The required residual products are

$$(J(0), \dots, J(10)) = (1, 1, 2, 3, 4, 6, 9, 12, 18, 27, 36).$$

For each $n = 7, \dots, 12$, calculate

$$E_n = \min_{2 \leq m \leq n} [C_*(m)3^{n-m} - mJ(n-m)]. \quad (25.1)$$

For clarity, here is every entry in that last minimization. A dash means $m > n$.

$m \backslash n$	7	8	9	10	11	12
2	1446	4356	13098	39330	118044	354222
3	474	1440	4347	13086	39312	118017
4	474	1442	4350	13086	39318	118026
5	260	795	2410	7260	21825	65550
6	102	312	954	2892	8712	26190
7	119	371	1120	3381	10178	30576
8	–	250	766	2306	6942	20866
9	–	–	207	639	1926	5805
10	–	–	–	890	2690	8080
11	–	–	–	–	2035	6127
12	–	–	–	–	–	1284

Taking the minimum in each column gives 102, 250, 207, 639, 1926, 1284, respectively, all strictly above the target deficits in lemma 16.1.

The program `code/02-three-output-independent_check.py` recomputes the cycle counts without the closed exponential formula. With the first color fixed, let S_j, T_j count paths of j vertices ending in that color or one of the other colors. Then

$$(S_1, T_1) = (1, 0), \quad S_{j+1} = T_j, \quad T_{j+1} = 2S_j + T_j,$$

and closing the cycle gives $C_j = 3T_j$. It computes J by the independent composition recurrence

$$J(0) = 1, \quad J(r) = \max_{1 \leq t \leq r} tJ(r-t).$$

It then enumerates all nonzero displacements rather than all divisors. These differences make the agreement a useful implementation check, while (25.1) remains the transparent mathematical calculation. (The local S_j, T_j here are unrelated to Part I's $T_{r,k}$.)

26 Formalization and trust-boundary plan

A formal development can be staged without treating executable searches as mathematical axioms. The following modules describe a dependency plan; they are not claims that corresponding files have been written or checked, and none has been started in ProveIt.

Finite coloring semantics. For a finite type Q , define transformations, generated monoids, and the right action on $Q \rightarrow \text{Fin}(3)$ by composition. Prove the reversal realization and accessibility-based distinguishability lemma. This module has no asymptotics or graph theory.

Collision graphs. Define the orbit of an unordered pair under the cyclic subgroup generated by a permutation. Prove the rightmost-singular factorization by induction on words. Establish that every singular-containing word produces an improper coloring. This is the principal semantic lemma connecting graph counts to an automaton theorem.

Graph classification and counting. Use permutation cycles to show that the graph is either a union of cycles with isolates or a union of complete bipartite graphs with isolates. Prove the two exact coloring

counts and the monochromatic-side period bound. The treatment of a two-vertex cycle as a simple edge should be explicit (compare section D).

Arithmetic comparison. Prove $J(r) \leq 3^{r/3}$ and the linear/exponential estimates on C_L and $\mathcal{B}(u, v)$. Real powers in the displayed exposition can be removed from a formal proof by raising positive quantities to suitable integer powers. The threshold (16.3) and the finite base table can then be checked by kernel arithmetic. Separate the one-residual-state case rather than hide it in a numerical bound.

Universal upper bound and equality. Assemble the exhaustive generator cases to prove $|\tau\langle s, t \rangle| \leq 3^n - D_n$ for arbitrary arrays. Track strictness in every excluded case. Equality then yields the two-cycle structure, proper period, saturation, and rank- $(n - 1)$ conclusion. The unique-cross-pair coloring gives a small finite witness inside a universally quantified proof.

Lower bound as a separate obligation. Formalize the particular two-generated monoid and the saturation theorem behind Davies’s lower construction. Until that development is complete, expose it as a named hypothesis of the *equality* theorem, not as an axiom silently attached to the upper bound. The conventional article uses the published result; a future kernel-certified equality theorem must account for its proof explicitly.

No certification by executable coincidence. The CSV of bounds, the breadth-first counts, and the Python inventory checks are useful test vectors. They do not establish the semantic graph lemma or universal inequalities. A formal trust ledger should distinguish accepted finite certificates, proved checkers, and external theorems, rather than label all three simply “verified.”

27 Sources, dependency boundaries and provenance

This section follows the manuscript’s source audit, shipped verbatim as `02-three-output-SOURCE_AUDIT.md`, dated 28 September 2026 (Pacific time).

27.1 Repository evidence the manuscript inspected

The manuscript read Part I at ProveIt commit

`fbba58593dc0622aa914972896150d4848f935b5`

in the directory

`SetTheory/Cardinals/docs/reports/automata-and-formal-languages/
dfao-reversal-coloring-obstruction/`

through the GitHub connector: the README (explicitly retrieved), the article, the research-status record and the relevant implementation. It checked the following distinctions. Part I’s headline $k^n - k! + g(k)$ theorem is a nonattainment result, not an exact all- n optimization formula. Its graph bound distinguishes proper and improper initial colorings. Its structural formula accounts for same-cycle and cross-cycle collision graphs, isolated residual states, and residual permutation orders. Its proper-period improvement for $k = 3$ applies to a connected biclique. Its equality with Davies’s lower bound is reported for $7 \leq n \leq 30$ and $3 \leq k < n$, 372 finite parameter pairs, and it explicitly distinguishes that statement from an all- n theorem. The function `u_witness` in `code/reversal.py` gives the zero-based parity-version arrays used for the independent orbit experiments. The broad repository tree was inspected for topic selection; the manuscript makes no claim to have audited every document or every Lean proof in the repository. Its main upper-bound proof is self-contained and does not rely on assuming Part I’s candidate theorems are correct.

27.2 Primary published source

For Davies [1] (the abstract page, the version-2 HTML and the PDF) the manuscript inspected the coloring-orbit formulation (Proposition 4), the coprime two-cycle construction and parity generators, Theorem 3 and Corollary 3, the three-output period contribution, Table 3, and Section 5’s optimality question (Problem 2). The PDF pages containing Corollary 3 (printed page 12) and the open question and small-case table (printed page 17) were inspected as page images in addition to parsed text. The exact lower-bound existence theorem is a published dependency. Part II proves the new universal upper bound and equality restrictions, then uses the lower theorem for attainability. It does not claim a new proof of the two-generator monoid-generation theorem. Finite breadth-first tests do not replace that dependency.

Holzer and König [2] is background; the manuscript checked its publisher metadata. The operative lower-bound theorem is cited through Davies, rather than attributed to an uninspected proof in the journal paper.

27.3 Literature-search scope

The searches included the exact Davies title; combinations of DFAO, reversal, three outputs, optimality, and 2026; and the exact Holzer–König title. The broad acronym queries produced substantial unrelated material, which was not used as mathematical evidence. The primary Davies text and Part I’s explicit status statement were the substantive sources. No later complete solution was located by these searches. This is not an exhaustive citation-index review, a survey of theses or unpublished work, or correspondence with the author. It is not evidence sufficient to establish priority or to assert that no earlier solution exists. Novelty claims are limited to the inspected sources, in particular the concrete extension of Part I’s finite result to all n for three outputs.

27.4 Claims and nonclaims

The principal new contribution is the all- n comparison proving the exact three-output formula and forcing the optimal two-cycle graph. The rank rigidity, $6AB$ collision penalty, and missing-orbit inventory are developed as further results. Theorem 13.1 settles the complete $k = 3$ slice of Davies’s Problem 2, not the full problem for variable k , and it does not settle the largest-two-generated-transformation-monoid conjecture.

The lower witness, the reversal-orbit reduction, the graph classification method, standard primitive-word counting, and the small known values are not claimed as inventions. The missing-orbit inventory applies primitive-word counting to the extremal forbidden family. Example 19.3, showing that necessary conditions are not sufficient, is included to prevent an overstatement of the rigidity theorem.

Part II is not externally refereed and has no proof-assistant certification. The all- n proof is a conventional proof with a fully displayed finite base table. The exact finite experiments are implementation checks and regression tests. They are not an independently certified proof of the infinite theorem.

27.5 What remains open

Four and more outputs [Added 29 September 2026, batches 40–41: now settled for $4 \leq k \leq 7$ and, beyond an explicit threshold, for every $k \geq 4$; see Part III, section 44]; a necessary-and-sufficient classification of the singular letter for saturation; efficient saturation recognition; second-best values and stability; the near-maximum attainable spectrum; output counts growing with n ; proof-assistant formalization; sharp general-group analogues. Section 23 formulates the eight corresponding research questions with specific obstacles and next steps. Davies’s Problem 1, the nonattainment question of Part I, is not affected by Part II.

27.6 Where the merge had to choose

- *Placement.* Part II follows Part I's last main section and precedes Part I's appendices, which contain no numbered equations, tables, figures or theorem-like items; no number of Part I changed.
- *Duplicates.* The coloring-orbit description, the collision-coloring bound, the two graph counts and the Chinese remainder edge criterion (the manuscript's Proposition 2.1, Lemmas 3.1–3.3) are printed once, in Part I, and recalled with credit in section 14.1; the manuscript's proofs of them are Part I's arguments, except its biclique count, which is kept as a second proof. The manuscript's Remark 2.2 is kept as remark 14.1. The manuscript's minimality argument is printed in the proof of theorem 13.1, with a pointer to Part I's identical argument.
- *Order.* The manuscript's two appendices are printed as sections 25 and 26, after its conclusion, and its source audit as this section; its title-page status box is the status box of section 13.1.
- *Notation.* $\text{Prop}(G) \rightarrow \text{Col}_3(G)$ and $\mathcal{O}_p(c) \rightarrow \text{Orb}_p(c)$, to use Part I's notation and keep $\mathcal{O}_r$ for Part I's order sets; s, t keep their meaning (table 5). No normalization changed.
- *References to Part I.* The manuscript cites Part I as a repository report; here those citations are internal cross-references, and its bibliography entries for Davies and for Holzer and König are Part I's [1, 2].
- *Added in the merge.* Remark 18.3 (the $k = 3$ case of $U(n, k) = L(n, k)$), section 18.2, the sentences relating Part II's statements to Part I's (after lemma 15.1, theorem 19.1, proposition 19.2, and corollary 21.1 and in questions 23.1 and 23.8), and the dated pointers in Part I's title page, reading-routes box and sections 1, 10 and 12. Part I's title page was also respaced (smaller gaps, the text after the abstract in a smaller size) because it already overflowed onto a second page before this addition; its wording is unchanged apart from the pointer.
- *Cross-reference names.* Part I declared its lemmas, propositions, corollaries, remarks and other theorem-like environments on the shared theorem counter, and `cleveref` named every one of them “theorem” (“theorem 3.3” for lemma 3.3). The preamble now declares them through alias counters, as the manuscript did, so each reference carries its own name; the shared numbering, and every number of Part I, is unchanged.

Part III. The exact maximum for four and more outputs

Part III was added on 29 September 2026 in batches 40 and 41 of ProveIt's incoming-report intake, from three manuscripts, all written before Part II was printed. It answers, for four through seven outputs and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ once n exceeds an explicit threshold $N(k) \leq 3k^4$, the question Part II leaves open in question 23.1: $R_2(n, k)$ equals Part I's lower bound $L(n, k)$. It also gives two further, non-effective routes to eventual exactness, a second proof of Part II's three-output theorem, and sharp asymptotics. Section 28 records the provenance and fixes the notation, which differs from the manuscripts' own in several places.

28 Provenance, conventions and the answer

28.1 Three manuscripts, one addition

Part III merges three AI-assisted research manuscripts prepared for Vladimir Reshetnikov and dated 28 September 2026. They prove overlapping theorems on one spine, the exactness of the closest-coprime-split construction beyond three outputs, so they form one part. In this report's local file sequence (Part I's unprefixed files count as 01, Part II's are 02) they are 03, 04 and 05.

- *Manuscript 06 of batch 41, the base; file prefix 05-quartic-threshold-. Eventual Exactness in Binary DFAO Reversal: Balanced coprime cycles, a uniform quartic threshold, and rank-sensitive obstructions, 22 pages as delivered. Archive ProveIt_Binary_Reversal_Exactness.zip (inner directory binary_reversal_eventual_exactness/), arrived in 9754e8360, placed in eaf787d50. Pinned to ProveIt commit e2b1f016a (the batch-39 placement, at which Part II's files were staged but Part II was not yet written).*
- *Manuscript 01 of batch 40; prefix 03-eventual-. Eventual Optimality in Binary DFAO Reversal: Closest coprime cycles, a parity-sensitive defect law, and exact finite certificates, 22 pages as delivered. Archive ProveIt_DFAO_Eventual_Optimality.zip, arrived in 017e6c0d8, placed in afb2d1227. Pinned to ProveIt commit 2cdcd74f3, with Part I's article.tex blob e6dbd47b and README.md blob 13166e22.*
- *Manuscript 03 of batch 40; prefix 04-deficit-asymptotics-. Binary DFAO Reversal: Exact Three-Output Complexity, Odd-Output Rigidity, and Sharp Deficit Asymptotics, 21 pages as delivered. Archive ProveIt_Binary_DFAO_Reversal.zip, arrived in 017e6c0d8, placed in afb2d1227. Pinned to ProveIt commit 2cdcd74f3.*

Below, “manuscript 06” always means batch 41's manuscript 06, the base, and “manuscripts 01 and 03” mean batch 40's. Part I's text was unchanged between both pins and the placement commits, so every statement the three manuscripts make about Part I refers to the text printed above. None of them knew Part II's text: manuscripts 01 and 03 predate its placement, and manuscript 06 read only Part II's source audit (02-three-output-SOURCE_AUDIT.md), which already records the complete three-output result. The code, data and audit files of the three manuscripts are shipped with the prefixes listed above; their manuscripts, READMEs and PDFs are not shipped and survive in the arrival commits.

What each manuscript contributes. Manuscript 06: the exact value of $R_2(n, k)$ for every $k \geq 4$ and $n \geq N(k)$ with an explicit integer-defined threshold $N(k) \leq 3k^4$ (sections 31 to 33); a computer-assisted closure of the slices $4 \leq k \leq 7$ for every $n \geq \max(7, k + 1)$ (section 34); rigidity of maximizers including rank $n - 1$ (section 39); a matching hierarchy of rank-sensitive bounds (section 40); exact exponential expansions of the optimal deficit (section 41); ten questions; a verification kernel and threshold data.

Manuscript 01: a non-effective proof of eventual exactness for every fixed $k \geq 3$ (section 36); a proof of the lower witness's orbit from the Holzer–König two-generation theorem (section 30); a quantitative balancing gap; an eventual if-and-only-if extremizer criterion; sharp asymptotics; eventual recurrences and rational generating functions (section 41); 919 finite certificates for $7 \leq n \leq 100$, $3 \leq k \leq \min(12, n-1)$ (section 35); ten questions. Manuscript 03: a second proof of Part II's theorem 13.1 (section 38); sharp fixed- k asymptotics and eventual exactness for odd k by a third route, with near-optimal rigidity (section 37); the even- k interval of asymptotically optimal proportions (section 42); nine questions. Every result, proof, example, remark, limitation and question of the three manuscripts is printed; a result proved by more than one of them is printed once, with the proofs credited, and genuinely different proofs are kept as marked second routes. Section 47.7 lists where the merge had to choose.

Status (all three manuscripts). AI-assisted, unrefereed research drafts. Nothing in Part III is checked in Lean, Rocq or any other proof assistant, and no manuscript claims otherwise. Manuscript 06's theorem for $n \geq N(k)$ has a conventional proof; its extension to every $n \geq \max(7, k+1)$ for $4 \leq k \leq 7$ is computer-assisted. Manuscript 01's infinite theorem has the quantifiers $\forall k \geq 3 \exists N_{\exists}(k) \forall n \geq N_{\exists}(k)$, with no explicit threshold, and its 919 finite equalities are exact-integer certificates checked by two implementations. Manuscript 03's three-output theorem is analytic for $n \geq 32$ and computer-assisted for $7 \leq n \leq 31$; its fixed- k statements are not uniform in k . Attainment in all three rests on a published external theorem: Davies's Theorem 3 and Corollary 3 (manuscripts 06 and 03), respectively the Holzer–König two-generation theorem for $U_{a,b}$ (manuscript 01, section 30); none is reproved. No priority is claimed; every literature search was targeted, not exhaustive.

28.2 Notation of Part III

Part I's symbols keep their meanings, in particular $B_k(u, v)$ of (9.5), $P_G(k)$, $C_L(k)$ of (9.3), $H_r(t)$ and O_r of (9.9), $T_{r,k}(a, b)$ of (9.10), $D(n, k)$, $U(n, k)$, $L(n, k)$, $G_k(\ell, m)$, $M(n, k)$ and $R_2(n, k)$; so do Part II's $J(r)$ of (14.4), (A_n, B_n) and D_n . The three manuscripts use clashing names for the same objects, and the same letter for different objects. Part III therefore fixes one convention, used throughout including in the material from manuscripts 01 and 03, and table 6 lists every symbol that was renamed, with the reading to avoid. No normalization changed except the factor two in δ_n noted in the table.

Table 6: Symbols of Part III. The second column gives the manuscripts' own symbols (06, 01, 03 are the three manuscripts). The last column prints the tempting false reading.

Symbol here	Manuscripts' symbols	Meaning	Not to be confused with
$R_2(n, k)$	06: $R_2(n, k)$; 01: $M_k(n)$; 03: $B_k(n)$	the maximum binary reversal complexity (Part I's R_2)	03's $B_k(n)$ is <i>not</i> Part I's coloring count $B_k(u, v)$; 01's $M_k(n)$ is not Part I's multinomial $M(n, k)$.
$\mathcal{B}_k(a, b)$	06, 03: $P_k(a, b)$; 01: $B_k(a, b)$, $P_{K_{a,b}}(k)$	proper k -colorings of $K_{a,b}$, equal to Part I's $B_k(a, b)$; $\mathcal{B}_3 = \mathcal{B}$ of Part II	B_n , $B_k(n)$ (integers); Part I's $P_G(k)$ is the general graph count, $\mathcal{B}_k(a, b) = P_{K_{a,b}}(k)$.
$P_G(k)$	06: $\chi_G(k)$; 01, 03: $P_G(k)$	proper k -colorings of a graph G (Part I)	not a chromatic number.
(a_n, b_n)	all three	the closest coprime split, (28.1)	equals Part II's (A_n, B_n) and Part I's optimal (ℓ, m) .

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Symbol here	Manuscripts' symbols	Meaning	Not to be confused with
$G_k(a, b)$	01: $T_k(a, b)$; 03: $\gamma_k(a, b)$	b if $k = 3$, ab if $k \geq 4$ ($a \leq b$): Part I's G_k of (10.1)	Part I's $T_{r,k}(a, b)$, a residual-order bound.
$D_k(n)$	01, 03: $D_k(n)$	$k^n - R_2(n, k)$, the optimal deficit	06's $D_k(n)$, which is $D_k^*(n)$ here; Part I's $D(n, k)$; Part II's $D_n = D_3^*(n)$.
$D_k^*(n)$	06: $D_k(n)$; 01: $D_k^*(n)$	$\mathcal{B}_k(a_n, b_n) - G_k(a_n, b_n)$, the witness deficit	Theorems here prove $D_k = D_k^*$; by definition they differ.
$\text{def}(\tau; p, s)$	03: $\mathcal{D}(\tau; a, b)$; 01: D	$k^n - \tau\langle p, s \rangle $ for one machine	—
$\tau\langle a, b \rangle$	03: $\text{Orb}(\tau; a, b)$	the whole reachable coloring set	Part I's Orb_τ , the pure p -orbit.
$N(k)$	06: $N(k)$	manuscript 06's <i>explicit</i> threshold, (33.1)	$N_\exists(k)$.
$N_\exists(k)$	01, 03: $N(k)$	<i>some</i> threshold, not computed	$N(k)$; the two need not agree.
h	06, 03: h ; 01: m	$\lfloor k/2 \rfloor$	Part I's split part m ; Part II's h in (13.1).
$k - h$	03: ℓ	$\lfloor k/2 \rfloor$	Part I's split part ℓ .
λ_k	06: λ ; 01: s_k, s ; 03: α_k, α	$\sqrt{\lfloor k^2/4 \rfloor} = \sqrt{h(k-h)}$	01's λ , which is ω_h here.
ω_h, t_h	01: λ ; 03: $\log t, t$	$\omega_h = \log t_h$, $t_h = (h+1)/h$	06's matching count t .
δ_n	06, 03: δ_n ; 01: $2\delta_n$	$(b_n - a_n)/2 \in \{\frac{1}{2}, 1, 2\}$	01's $\delta_n = b_n - a_n \in \{1, 2, 4\}$; formulas from 01 are rewritten accordingly.
$\hat{H}, \hat{M}, \hat{C}, \hat{B}, \hat{L}, \hat{S}, \hat{U}, \hat{K}, \hat{\theta}$	06: $H, M, C, B, L, S, U, K, \theta$	manuscript 06's threshold constants (32.5), (33.1); the JSON keys drop the hat	Part I's $H_r, M(n, k), C_L(k), B_k, L(n, k), U(n, k)$; Stirling $S(r, i)$; 03's θ_h .
$\mathcal{L}_k(s, \delta)$	06: $L_k(s, \delta)$	leading palette term (32.7)	Part I's $L(n, k)$.
β_k	06: β ; 01: γ_k ; 03: $\beta_k, \beta = 6^{1/3}$	$(k(k-1)(k-2))^{1/3}$	—
γ_r	06: g_r	$\lfloor 3^{r/3} \rfloor$	Landau's $g(k)$ of theorem 1.1.
$J(r)$	01: $A(r)$	largest product of a partition of r (Part II)	Part II's A_n .
$D_k^{\text{cert}}(n)$	01: $L_k(n)$	01's finite certificate minimum (35.3)	Part I's $L(n, k)$ and $D(n, k)$.
ϱ_k	01: R	k/λ_k	R_2 .
$\Omega_{k,1}(a, b)$	06: $M_{k,1}(a, b)$	colorings with matching number one	$M(n, k), M_k(n)$.
ξ_1	06: ν_1	an error base in theorem 41.2	the matching statistic $\nu(c)$.
Π	03: O	bound on reachable proper colorings in the pure orbit	Part I's O_r .
$O_r, H_r(t)$	03: $O(r), H(r, t)$	Part I's (9.9)	—
Λ_n	03: A	$2^{n/2}$ (in section 38)	A_n .
Φ_n, Ψ_n	03: X_n, Y_n	plateau exponents, theorem 42.1	the cycles X, Y .
X, Y	06, 01: A, B	the two cycles (sets) of a two-cycle permutation	lengths a_n, b_n ; Part II's A_n, B_n .

Continued on next page

Symbol here	Manuscripts' symbols	Meaning	Not to be confused with
$\text{sur}_j(m)$	01: $\text{Sur}_j(m)$; 03: $\text{onto}(m, j)$	onto maps to a fixed j -set, $j! S(m, j)$	—
E	01: E	forward shift operator	Part II's E_n .

Throughout, $S(r, i) = \begin{Bmatrix} r \\ i \end{Bmatrix}$ is a Stirling number of the second kind (with $S(0, 0) = 1$ and zeros outside $0 \leq i \leq r$), $(k)_i = k!/(k-i)!$, and the output alphabet Δ is identified with $[k] = \{1, \dots, k\}$ where the manuscripts write $[k]$. Manuscript 06 extends $\mathcal{B}_k$ by $\mathcal{B}_k(0, b) = k^b$, $\mathcal{B}_k(a, 0) = k^a$ and $\mathcal{B}_k(0, 0) = 1$. Its kernel-pair count t in section 40 keeps its letter; so do the local letters q, x, z of individual proofs. As in Parts I–II, “binary” describes the *input* alphabet.

28.3 The answer

For $n \geq 7$ put

$$(a_n, b_n) = \begin{cases} ((n-1)/2, (n+1)/2), & n \text{ odd}, \\ (n/2 - 1, n/2 + 1), & n \equiv 0 \pmod{4}, \\ (n/2 - 2, n/2 + 2), & n \equiv 2 \pmod{4}, \end{cases} \quad \delta_n = \frac{b_n - a_n}{2} \in \left\{ \frac{1}{2}, 1, 2 \right\}. \quad (28.1)$$

This is Part II's split (13.1): $(a_n, b_n) = (A_n, B_n)$. The two numbers are coprime and $2 \leq a_n < b_n$. With Part I's $G_k(a, b) = b$ for $k = 3$ and ab for $k \geq 4$ ((10.1)), put

$$D_k^*(n) = \mathcal{B}_k(a_n, b_n) - G_k(a_n, b_n), \quad D_k(n) = k^n - R_2(n, k). \quad (28.2)$$

At $k = 3$, $D_3^*(n) = D_n$ of Part II.

Manuscript 06 defines $R_2(n, k)$ as the maximum of $\text{sc}(f^R)$ over functions f of state complexity n , computed over a binary input alphabet and having exactly k output values, with $2 \leq k \leq n$ and $f^R(w) = f(w^R)$; manuscript 01 maximizes over minimal n -state DFAOs with surjective output map, and manuscript 03 over accessible minimal ones. These are Part I's two readings of R_2 (sections 10 and 12); every exactness theorem below shows that the accessible and the minimal readings agree in its range, and that the unrestricted maximum of $|\tau\langle a, b \rangle|$ over all transformation pairs and output maps is the same number.

Theorem 28.1 (Exact binary reversal for $k \geq 4$; manuscript 06). *For every integer $k \geq 4$ there is an explicitly defined integer $N(k) \leq 3k^4$ such that*

$$R_2(n, k) = k^n - D_k^*(n) = k^n - \mathcal{B}_k(a_n, b_n) + a_n b_n \quad (n \geq N(k)). \quad (28.3)$$

The same formula holds for every $4 \leq k \leq 7$ and every $n \geq \max(7, k+1)$. This second assertion uses the finite exact-integer verification of section 34 below the analytic threshold.

The threshold is defined in section 33; for example $N(4) = 212$, $N(5) = 240$, $N(6) = 954$ and $N(7) = 840$. These are sufficient thresholds, not claims about the smallest possible ones, and the finite closure shows that they are very conservative in these four cases. [Added 29 September 2026, batch 43: at $k = 4$ Part IV's second route is analytic from $n = 26$ (section 50 and remark 50.6).] Two points make the statement stronger than a leading-order estimate. First, the entire integer optimum is determined, including the additive $a_n b_n$ term. Second, the range $n \geq 3k^4$ permits k to vary with n : the theorem applies simultaneously to every integer $4 \leq k \leq (n/3)^{1/4}$.

The right side of (28.3) is Part I's $L(n, k)$ of (10.1) (corollary 31.5 shows that the nearest coprime split maximizes it), so theorem 28.1 answers question 23.1 for $4 \leq k \leq 7$ and for every $k \geq 4$ once $n \geq N(k)$;

remark 33.6 adds $U(n, k) = L(n, k)$ in the same range. For $k \geq 8$ and $\max(7, k + 1) \leq n < N(k)$ the question stays open except where finite certificates already settle it (proposition 10.1 for $n \leq 30$; proposition 35.1 for $n \leq 100, k \leq 12$); for instance $k = 8$ and $101 \leq n \leq 2687$ remain open. The case $k = 3$ is Part II's theorem 13.1.

Three routes, one formula. The exact formula $R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n)$ is proved in Part III:

- for $k \geq 4, n \geq N(k)$ with $N(k) \leq 3k^4$ explicit, and for $4 \leq k \leq 7, n \geq \max(7, k + 1)$ (manuscript 06, theorem 28.1);
- for every fixed $k \geq 3$ and all $n \geq N_{\exists}(k)$, with no explicit threshold (manuscript 01, theorem 36.1); at $k = 3$ Part II already gives all $n \geq 7$, and for $k \geq 4$ manuscript 06 supersedes this headline, which remains a second, non-effective route;
- for every fixed odd $k \geq 3$ and all $n \geq N_{\exists}(k)$, and at $k = 3$ for all $n \geq 7$ (manuscript 03, theorems 37.2 and 38.1); the $k = 3$ statement is a second proof of Part II's theorem 13.1.

The finite certificates of manuscript 01 (proposition 35.1) cover $7 \leq n \leq 100, 3 \leq k \leq \min(12, n - 1)$ independently of any threshold.

28.4 What is inherited and what is proved here

Manuscript 06 divides responsibility as follows; manuscripts 01 and 03 give their own ledgers in section 47.

Item	Status in manuscript 06
Coloring-orbit formulation	Known from Davies; reproved for clarity.
Coprime-cycle lower witness	Imported from Davies's Theorem 3 and Corollary 3. Its two-generator monoid-generation theorem is not reproved.
Collision-edge graph method	Builds on Part I; the upper-bound argument used here is written out independently.
Three-output optimum	Already developed in the repository (Part II); not a new claim of manuscript 06.
Strict balancing and explicit split	Proved as the optimization tool; no assertion of priority for classical Stirling-number facts.
Uniform eventual exactness	Principal extension, with integer-defined $N(k) \leq 3k^4$.
Four through seven outputs	All- n slices obtained by the analytic theorem plus two finite exact implementations.
Rigidity, rank hierarchy, asymptotics	Additional results developed from the exact optimum and the collision mechanism.

Manuscript 06's literature search located the primary paper and the repository's explicitly stated boundary. It was not an exhaustive review of all theses, preprints, or unpublished work. Accordingly, "extension" means an extension of the inspected mathematical sources; historical priority is not certified. The proofs and their scope can be assessed independently of that priority question. The lower construction is the only substantive imported existence theorem of manuscript 06; its proof does not assume that unformalized claims elsewhere in the repository are correct, and it does not rely on a numerical search over automata for the infinite part of the result.

29 Coloring orbits, collision graphs and order bounds

29.1 What Part III takes from Parts I and II

All three manuscripts reprove the reversal-orbit description and the collision-graph machinery of Part I, with the same composition convention (in $u \circ v$ the map v acts first, and a letter acts on colorings by $c \mapsto c \circ t$). Those statements are printed once, in Part I, and recalled here with credit:

- *Orbit reduction.* For an accessible DFAO, $\text{sc}(f^R) = |\tau\langle a, b \rangle|$ (proposition 2.1; Davies's Proposition 4). Manuscripts 06, 01 and 03 each reprove it by Part I's argument (manuscript 06 as its Proposition 2.1, 01 as its Proposition 2.1, 03 in its Section 2). Manuscript 06 adds that the upper bounds below apply to arbitrary transformation pairs, accessible or not, so they remain valid when taking the extremum defining $R_2(n, k)$, and that accessibility matters because unreachable original coordinates cannot distinguish reverse states.
- *Collision obstruction.* For a permutation p , a singular s and a collision pair $\{x, y\}$ of s , every coloring reached by a word containing s is improper for $G = G(p; x, y)$ (lemma 3.3), so $|\tau\langle p, s \rangle| \leq k^n - P_G(k) + \text{ord}(p)$ ((9.1)), and more precisely the subtraction is the length of the pure p -orbit of τ if τ is proper and zero otherwise (proposition 9.1). All three manuscripts reprove this (06: Lemma 2.3, 01: Lemma 2.2 and Proposition 2.3, 03: Lemma 3.2). As manuscript 06 notes, G is p -invariant, so properness is constant on each pure orbit; if equality holds with a positive proper contribution, then τ is proper, its pure orbit has size $\text{ord}(p)$, and every improper coloring is reached.
- *Classification.* A same-cycle pair on a cycle of length m with displacement of $\text{gcd } d$ gives d cycles C_L , $L = m/d \geq 2$, and $r = n - m$ isolates, $P_G(k) = C_L(k)^d k^r$ ((9.4)); a cross pair on cycles of lengths $a = du$, $b = dv$ gives $dK_{u,v} \sqcup rK_1$, $r = n - a - b$, $P_G(k) = \mathcal{B}_k(u, v)^d k^r$ ((9.6)). All three manuscripts reprove this with Part I's trace and Chinese-remainder arguments (06: Lemma 2.4, 01: Section 2.2 with the schematic of figure 2, 03: Lemma 3.3); manuscript 06 also mentions the two-state transfer recurrence for the cycle count, which Part II's section 25 spells out.
- *Residual orders.* $H_r(t)$ and O_r are Part I's (9.9) (manuscript 03's $H(r, t)$, $O(r)$); the three-color period bound for a connected biclique is Part I's (9.10) (manuscript 03's Lemma 3.4, printed with its finite reduction in proposition 38.2); $J(r)$ is Part II's (14.4).

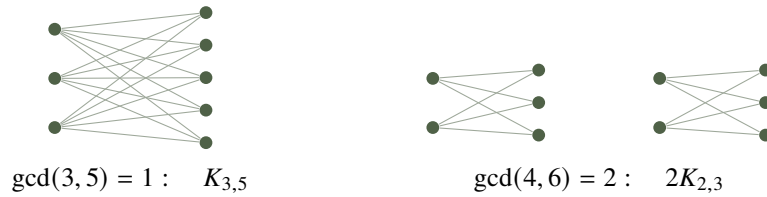


Figure 2: A single collision pair between permutation cycles produces one complete bipartite component for each common residue class (manuscript 01). These are schematics of the proved classification, not experimental graphs.

Manuscript 03 adds a small test, used for the minimality of witnesses.

Lemma 29.1 (A minimality test; manuscript 03). *If an accessible n -state DFAO with k outputs has an orbit with more than k^{n-1} colorings, the original DFAO is minimal.*

Proof. If two original states were indistinguishable, every orbit coloring would agree on those states. Only k^{n-1} colorings have that property. Accessibility and pairwise distinguishability characterize minimality. $\square$

This is the argument Part I uses after proposition 10.1 and Part II in the proof of theorem 13.1, stated once as a lemma.

29.2 The two unmixed generator cases

Lemma 29.2 (Manuscript 06; also manuscript 03). *For any $a, b : Q \rightarrow Q$ and any $\tau : Q \rightarrow [k]$, with $M = \langle a, b \rangle$:*

- (i) *if both a and b are permutations, then $|\tau M| \leq k^{n-1}$;*
- (ii) *if both are singular, then*

$$|\tau M| \leq k^n - (k-1)^2 k^{n-2} + 1.$$

The second statement is used only for $n \geq 3$.

Proof. A permutation preserves the number of occurrences of each output color. Within a family with fixed color multiplicities, restriction to any fixed $n-1$ coordinates is injective: the omitted color is determined by the multiplicities. This proves (i).

For (ii), choose collision pairs $\{x_a, y_a\}$ and $\{x_b, y_b\}$ for the two singular transformations. A coloring obtained from a nonempty word is equal on the collision pair for its rightmost generator. Hence it cannot be a proper coloring of the graph consisting of these one or two edges. Two distinct edges form a forest, with exactly $(k-1)^2 k^{n-2}$ proper colorings; this holds whether they share a vertex or are disjoint. If the selected edges coincide, there are even more proper colorings. Only the empty word, producing τ , can contribute a proper coloring. $\square$

Manuscript 03 proves the same two bounds, as deficits $\text{def} \geq (k-1)k^{n-1}$ and $\text{def} \geq (k-1)^2 k^{n-2} - 1$, by the same argument (“after specifying the first $n-1$ positions, the last one is determined if completion is possible”). Part I’s entries (9.12) and (9.13) are the same two cases; the first uses the balanced multinomial $M(n, k)$, which is at most k^{n-1} by (i).

Remark 29.3 (Manuscript 01’s version). Manuscript 01 states the unmixed cases with a weaker two-permutation bound and adds the nonsurjective case: if both generators are permutations, the coloring-orbit deficit is at least k^{n-1} (after fixing the values at $n-1$ vertices, at most $k-1$ choices at the last vertex can have the prescribed count of a chosen color, so the orbit has at most $(k-1)k^{n-1}$ colorings); if both are singular, it is at least $(k-1)^2 k^{n-2} - 1$, by the argument above; and a nonsurjective output map has deficit at least $k^n - (k-1)^n$, because all reached colorings use at most the $k-1$ colors appearing initially (Part I’s (9.11)). All three lower bounds grow faster than λ_k^n , since $k > \lambda_k$.

29.3 Bounds on permutation orders

Lemma 29.4 (Elementary permutation-order bound; manuscripts 06 and 03). *A permutation on r points has order at most $3^{r/3}$, including order one when $r = 0$. In the cross-cycle case, with the notation of (9.6),*

$$\text{ord}(p) \leq \text{dov} \, 3^{r/3}; \tag{29.1}$$

in the same-cycle case, $\text{ord}(p) \leq m \, 3^{r/3}$.

Proof. The order divides the product of the cycle lengths. Each positive integer j satisfies $j \leq 3^{j/3}$: check $j = 1, 2, 3$ and note that $\log x/x$ is decreasing for $x \geq 3$. Multiplication gives the bound. The active cycles have least common multiple $\text{lcm}(du, dv) = \text{dov}$ in the cross-cycle case, and length m in the same-cycle case. Multiply their order by the residual bound. $\square$

Second proof of $j \leq 3^{j/3}$ (manuscript 03). Check $j = 1, 2, 3$, and use $j + 3 \leq 3j$ for $j \geq 2$ to induct in steps of three, with $j = 4$ checked directly. Manuscript 03 then uses the crude forms $\Pi \leq n3^{r/3}$ (one cycle) and $\Pi \leq n^23^{r/3}$ (two cycles), which it calls intentionally crude and uses only after an exponential separation has made their precision irrelevant.

The bound $3^{r/3}$ is Part II's $J(r) \leq 3^{r/3}$ (lemma 14.2). No asymptotic estimate for the maximal order of a permutation is required; in particular manuscript 06's proof does not depend on a precise estimate for Landau's function. Manuscript 01 uses a different, asymptotic bound.

Lemma 29.5 (Subexponential order bound; manuscript 01). *Uniformly over all permutations p of n objects,*

$$\log \text{ord}(p) = O(\sqrt{n} \log n).$$

In particular, for every fixed real $\sigma > 1$, $\text{ord}(p) = o(\sigma^n)$ uniformly in p .

Proof. Put $L = \lceil \sqrt{n} \rceil$. The least common multiple of the cycle lengths at most L divides $L!$. There are at most $n/(L+1)$ longer cycles, each of length at most n . Thus

$$\text{ord}(p) \leq L! n^{n/(L+1)}.$$

Taking logarithms proves the assertion. This deliberately avoids using a sharp asymptotic formula for Landau's function. $\square$

Manuscript 01's finite certificates use instead the exact product bound: every permutation of r points has order at most $J(r)$ (Part II's (14.4), manuscript 01's $A(r)$), because the least common multiple of its cycle lengths is at most their product. Manuscript 01 proves the closed form of J as Part II does (a part at least five can be replaced by three and its remainder to increase the product; ones are removed in a maximizing partition of an integer at least two, except that $r = 1$ must be retained separately; three twos are improved by two threes; a leftover four can be retained or replaced by two twos), and its independent checker computes $J(r) = \max_{1 \leq j \leq r} jJ(r-j)$ without the formula, as Part II's does.

30 The exact lower witness from the two-generation theorem

Parts I–II and manuscripts 06 and 03 cite Davies's Theorem 3 and Corollary 3 for attainment. Manuscript 01 isolates the external algebraic ingredient more narrowly, as the Holzer–König two-generation theorem, and proves exactly which colorings the resulting monoid gives. This section is manuscript 01's. Fix $1 < a < b$, $\gcd(a, b) = 1$, and $Q = X \sqcup Y$ with $|X| = a$, $|Y| = b$; let p cycle X and Y separately.

Definition 30.1. The monoid $U_{a,b}$ consists of the powers of p , together with all transformations $t : Q \rightarrow Q$ such that

$$t(X) \cap t(Y) \neq \emptyset \quad \text{and} \quad Y \not\subseteq t(Q).$$

The published two-generator theorem says that this monoid is generated by p and one singular transformation [2, Theorem 8]; Davies records the theorem and explicit generators [1, Section 3]. The explicit generator is recorded in Davies's account of Krawetz's construction [3]. Number the states $0, \dots, a+b-1$, with $X = \{0, \dots, a-1\}$. Start with the identity array, change $s(0)$ to a and $s(a+b-1)$ to 0 , and, when a is odd, interchange $s(1)$ and $s(2)$. This is the witness of Part II's section 18.1 and of Part I's `u_witness`, and the convention of the verification code. The theorem is used as an imported theorem, not as a computational inference.

Lemma 30.2 (Projection saturates improper colorings). *Suppose $\tau : Q \rightarrow \Delta$ is surjective, its palettes on X and Y are disjoint, and two distinct points of Y have the same output. Then*

$$\tau U_{a,b} = \{c : c \text{ is improper for } K_{a,b}\} \sqcup \{\tau \circ p^j : j \geq 0\}. \quad (30.1)$$

Proof. Every nonpermutation member of $U_{a,b}$ identifies a cross pair, so its projected coloring is improper. Every power of p preserves the disjoint palettes and gives a proper coloring.

Conversely, choose a point $z \in Y$ whose output has another representative in Y . Choose one representative $q_c \neq z$ for every output color c . For any improper coloring h , define $t(q) = q_{h(q)}$. There is a cross pair with the same h -value, hence the same t -value, and $z \notin t(Q)$. Therefore $t \in U_{a,b}$ and $\tau \circ t = h$. $\square$

Lemma 30.3 (Maximum proper period). *If p has two cycles of coprime lengths $a \leq b$, the orbit length of a proper k -coloring of $K_{a,b}$ is at most $G_k(a, b)$. For $k \geq 4$, this bound is attained whenever both cycles can be given primitive words using disjoint palettes of at least two colors. For $k = 3$ and $b \geq 2$, it is attained by a constant word on the shorter cycle and a primitive two-color word on the longer.*

Proof. Every coloring orbit length divides $\text{lcm}(a, b) = ab$. With at most three colors, disjoint nonempty palettes force at least one cycle to be monochromatic. Its coloring period is one, so the full period divides the length of the other cycle and is at most b . A cyclic word in which one color occurs exactly once has its full length as period. Using such words on both cycles, when possible, gives period ab by coprimality. $\square$

Proposition 30.4 (Attainable exact formula). *For $n \geq 7$ and $3 \leq k < n$, there is a minimal binary n -state DFAO with*

$$|\tau\langle p, s \rangle| = k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n).$$

In particular, $D_k(n) \leq D_k^(n)$.*

Proof. For $k = 3$, use one color on X and two on Y , the latter in a primitive word with a repeated color. For $k \geq 4$, put

$$i = \min(a_n, k - 2), \quad j = k - i.$$

Then $2 \leq i \leq a_n$ and $2 \leq j \leq b_n - 1$. Give X exactly i colors and Y the other j , using a primitive word on each. Such a word of length L with $t \geq 2$ prescribed colors is obtained by writing each color once and filling the remaining positions with the last color; the first color occurs exactly once. The bound $j \leq b_n - 1$ ensures a repeated color on Y . Apply lemmas 30.2 and 30.3.

The automaton is accessible: p reaches every state of X from 0, and $s(0) = a_n$ enters Y , where p reaches every remaining state. It is minimal as well. For distinct q, r , color r with one color and all other states with another. This coloring separates q and r and is improper for K_{a_n, b_n} , since both parts have at least two vertices and therefore have a cross pair avoiding r . By lemma 30.2, it equals $\tau \circ t$ for a word transformation t . The corresponding continuation distinguishes q, r . $\square$

This is Davies's lower bound, Part I's $L(n, k)$ at the nearest split, with the saturation step proved and only the generation theorem imported. It also discharges part of the "lower bound as a separate obligation" of Part II's formalization plan (section 26): given generation of $U_{a,b}$, the saturation identity is elementary.

31 Strict balancing of complete bipartite colorings

A structural comparison still has infinitely many possible side sizes. The next result removes that freedom exactly rather than approximately. A leading exponential estimate cannot by itself select a unique split when k is even: a whole interval of proportions has the same leading term (section 42). Manuscripts 06 and 01 therefore both prove an exact balancing inequality, by different routes; both are printed.

31.1 Order-two total positivity of Stirling numbers

Lemma 31.1 (Manuscript 06). *For $1 \leq r \leq s$ and $1 \leq i < j$,*

$$S(r, i)S(s, j) \geq S(r, j)S(s, i). \quad (31.1)$$

Proof. We first prove log-concavity of every row. Define the Touchard polynomial $T_r(x) = \sum_i S(r, i)x^i$ (unrelated to Part I's $T_{r,k}$). The Stirling recurrence gives

$$T_{r+1}(x) = x(T_r(x) + T'_r(x)).$$

Inductively, T_r has r distinct real roots, one at zero and the others negative. The base case is $T_1 = x$. If the claim holds for T_r , the function $e^x T_r(x)$ has a critical point in each interval between successive roots. It also has one to the left of the leftmost root: it tends to zero as $x \rightarrow -\infty$, is zero at that root, and is nonzero in between, so an interior extremum exists. Thus $T_r + T'_r$ has at least r distinct negative roots. Since its degree is r , these are all its roots. Multiplication by x supplies the new simple root at zero.

For completeness, real nonpositive roots imply the needed coefficient log-concavity without invoking an unproved property of the Stirling array. Every derivative of T_r also has real nonpositive roots, by Rolle's theorem. For a polynomial g with such roots and $x > 0$,

$$g'(x)^2 - g(x)g''(x) = g(x)^2 \sum_{\rho: g(\rho)=0} \frac{1}{(x-\rho)^2} \geq 0,$$

where roots are counted with multiplicity. Apply this to $g = T_r^{(i-1)}$ and let $x \downarrow 0$. It yields

$$(i! S(r, i))^2 \geq (i-1)! S(r, i-1) (i+1)! S(r, i+1),$$

and hence ordinary log-concavity $S(r, i)^2 \geq S(r, i-1)S(r, i+1)$. On the positive support, the ratios $S(r, i-1)/S(r, i)$ are therefore nondecreasing in i .

Now compare adjacent rows using the recurrence:

$$\begin{aligned} & S(r, i)S(r+1, j) - S(r, j)S(r+1, i) \\ &= (j-i)S(r, i)S(r, j) + S(r, i)S(r, j-1) - S(r, j)S(r, i-1) \geq 0. \end{aligned}$$

The ratio observation proves the final inequality whenever the displayed entries are positive. At the support boundary, the zero entries make the same inequality immediate. Finally, for fixed $i < j$ with positive entries, these adjacent-row inequalities say that $S(r, j)/S(r, i)$ is nondecreasing as the row index increases. Iterating from r to s gives (31.1); the remaining zero cases again have the required sign. $\square$

Second proof (manuscript 01). Manuscript 01 proves row log-concavity without real-rootedness.

Lemma 31.2 (Stirling rows are log-concave; manuscript 01). *For $n \geq 1$, the positive support of $S(n, i)$ is $1 \leq i \leq n$, and $S(n, i)^2 \geq S(n, i-1)S(n, i+1)$.*

Proof. The support assertion follows from the interpretation by set partitions. For a nonnegative log-concave sequence (x_i) with no internal zeros, put $y_i = ix_i + x_{i-1}$. Extend the sequence by zero outside its support. Direct expansion gives

$$\begin{aligned} y_i^2 - y_{i-1}y_{i+1} &= (i^2 - 1)(x_i^2 - x_{i-1}x_{i+1}) + x_i^2 \\ &\quad + (i+1)(x_i x_{i-1} - x_{i-2}x_{i+1}) + (x_{i-1}^2 - x_{i-2}x_i). \end{aligned}$$

For $i \geq 1$, all terms are nonnegative. The middle product inequality follows from monotonicity of adjacent ratios on the positive support; zero boundary cases are immediate. Starting with the row $S(1, 1) = 1$, the recurrence $S(n+1, i) = iS(n, i) + S(n, i-1)$ proves the assertion by induction. $\square$

With this, manuscript 01 derives (31.1) as follows. If $j > r$, the right side is zero; if $i > r$, both sides are zero. Otherwise all relevant entries up to row s are positive. For $1 \leq t \leq m$,

$$\frac{S(m+1, t)}{S(m, t)} = t + \frac{S(m, t-1)}{S(m, t)}.$$

By lemma 31.2, the second term is nondecreasing in t , while the first is strictly increasing. Multiplying these row-to-row ratios from $m = r$ to $s-1$ shows that $S(s, t)/S(r, t)$ is nondecreasing in t , which is (31.1). When $r = s$, the assertion is equality.

31.2 Moving a vertex away from balance

Theorem 31.3 (Strict bipartite balancing; manuscript 06). *For integers $k \geq 3$ and $2 \leq a \leq b$,*

$$\mathcal{B}_k(a-1, b+1) > \mathcal{B}_k(a, b). \quad (31.2)$$

Thus, among positive side sizes with a fixed sum, $\mathcal{B}_k$ is strictly increasing with the distance from balance.

Proof. Start with $K_{r,s}$, where $r = a-1$ and $s = b$, and designate one new vertex. Attaching it to the side of size s produces $K_{a-1, b+1}$; attaching it to the side of size r produces $K_{a, b}$.

Suppose a proper coloring of the old graph uses i colors on its first side and j on its second. These color sets are disjoint. The numbers of choices for the new vertex are respectively $k-i$ and $k-j$, so the difference is $j-i$. The number of old colorings with these color-class counts is $(k)_{i+j}S(r, i)S(s, j)$. Pairing the contribution (i, j) with (j, i) gives

$$\begin{aligned} & \mathcal{B}_k(a-1, b+1) - \mathcal{B}_k(a, b) \\ &= \sum_{\substack{1 \leq i < j \\ i+j \leq k}} (j-i)(k)_{i+j} (S(r, i)S(s, j) - S(r, j)S(s, i)). \end{aligned} \quad (31.3)$$

Every term is nonnegative by lemma 31.1. The term $i=1, j=2$ is positive: $s > r \geq 1$, $S(t, 1) = 1$, and $S(t, 2) = 2^{t-1} - 1$. Its coefficient is nonzero because $k \geq 3$. $\square$

Retaining the pair $(1, 2)$ quantitatively gives manuscript 01's form; its proof is the same extension count, read from the other side.

Theorem 31.4 (Quantitative strict balancing; manuscript 01). *For $k \geq 3$ and positive integers a, b with $b \geq a+2$,*

$$\mathcal{B}_k(a, b) - \mathcal{B}_k(a+1, b-1) \geq k(k-1)(k-2)(2^{b-2} - 2^{a-1}) > 0. \quad (31.4)$$

Proof. First compare extensions of a proper coloring of $K_{a,c}$ to $K_{a+1,c}$ and $K_{a,c+1}$. If its left and right palettes have sizes i, j , the respective numbers of extensions are $k-j$ and $k-i$. The number of such colorings is $(k)_{i+j}S(a, i)S(c, j)$. Consequently

$$\mathcal{B}_k(a+1, c) - \mathcal{B}_k(a, c+1) = \sum_{i+j \leq k} (i-j)(k)_{i+j} S(a, i)S(c, j).$$

Pairing the summands with indices (i, j) and (j, i) , for $i < j$, gives

$$(i-j)(k)_{i+j} (S(a, i)S(c, j) - S(a, j)S(c, i)).$$

When $c \geq a$, every such expression is nonpositive by lemma 31.1. Take $c = b-1 > a$ and retain only the pair $(i, j) = (1, 2)$. Since $S(t, 1) = 1$ and $S(t, 2) = 2^{t-1} - 1$, reversing the comparison gives precisely (31.4). $\square$

Corollary 31.5 (Explicit optimization of the lower construction; manuscript 06). *For $n \geq 7$ and $k \geq 4$, among all a, b satisfying $1 < a < b$, $a + b = n$ and $\gcd(a, b) = 1$, the quantity*

$$k^n - \mathcal{B}_k(a, b) + ab$$

is uniquely maximized, up to interchanging sides, by (28.1). This optimization conclusion holds without any large- n hypothesis.

Proof. Moving toward balance strictly decreases $\mathcal{B}_k$ and strictly increases ab . The nearest admissible split therefore uniquely maximizes the expression. For odd n , consecutive integers give the nearest split. For $n = 2h$ with h even, the pair $h - 1, h + 1$ is coprime because its common divisor divides two and both terms are odd. For h odd, $h - 1, h + 1$ is not coprime, whereas $h - 2, h + 2$ is: the common divisor divides four and both terms are odd. The equal split is not admissible for $n \geq 7$. $\square$

(In this proof, as in Part II's lemma 18.1, h is $\lfloor n/2 \rfloor$, not $\lfloor k/2 \rfloor$.) Manuscript 06 also uses the nearest coprime split for component sizes $s \geq 2$; the same formula applies, except that $s = 2$ has split $(1, 1)$. For this auxiliary use, sides of size one are allowed.

Manuscript 01 proves the same optimization for every $k \geq 3$, with Part I's period term G_k .

Corollary 31.6 (Balancing with the period correction; manuscript 01). *For fixed n and $k \geq 3$, the quantity $\mathcal{B}_k(a, n - a) - G_k(a, n - a)$, $1 \leq a < n - a$, strictly decreases when a moves toward $n/2$ without crossing it.*

Proof. For $k \geq 4$, the product $a(n - a)$ increases under balancing, so subtracting it strengthens theorem 31.4. For $k = 3$, the subtracted value $n - a$ decreases by one, but the right side of (31.4) is at least six. Thus the net change still has the required strict sign. $\square$

Lemma 31.7 (Closest coprime split; manuscript 01). *For $n \geq 7$, (a_n, b_n) is the unique unordered coprime pair closest to equality with sum n . It minimizes $\mathcal{B}_k(a, b) - G_k(a, b)$ among coprime positive pairs $a < b$, for every $k \geq 3$.*

Proof. For odd n , the two consecutive integers $(n - 1)/2, (n + 1)/2$ are coprime. For $n = 4t$, the equally sized pair is not coprime, while $2t - 1, 2t + 1$ has greatest common divisor dividing two and consists of odd numbers. For $n = 4t + 2$, the equal pair is not coprime; the next pair $2t, 2t + 2$ is even; and $2t - 1, 2t + 3$ has greatest common divisor dividing four and consists of odd numbers. This gives the listed minimal differences. The minimization follows from corollary 31.6, even though some intervening pairs are not coprime. $\square$

Manuscript 03 gives the same admissibility argument for the nearest split, without a number-theoretic search. At $k = 3$ this is Part II's lemma 18.1, which proves it with the explicit decrement $3(2^{b-1} - 2^a) - 1$.

32 Palette expansions and uniform estimates

32.1 Exact counting formulas

Proposition 32.1 (Manuscripts 06 and 03). *For $a, b \geq 1$,*

$$\mathcal{B}_k(a, b) = \sum_{i=1}^k (k)_i S(a, i) (k - i)^b, \quad (32.1)$$

$$\mathcal{B}_k(a, b) = \sum_{\substack{i, j \geq 1 \\ i+j \leq k}} (-1)^{k-i-j} \frac{k!}{i! j! (k - i - j)!} i^a j^b. \quad (32.2)$$

Proof. For (32.1), partition the first side into its i nonempty color classes, assign i distinct colors to them, and color the second side with any of the other $k - i$ colors. (This is Part I's (9.5); manuscript 01 writes the same sum up to $i \leq \min(a, k - 1)$, the term $i = k$ being zero.)

For (32.2), split the k available colors into three labeled sets I, J, U . Allow colors in I on the first side and colors in J on the second, and assign weight $(-1)^{|U|}$. A coloring using a color on both sides is never counted. For a proper coloring, each actually used color is forced into its correct palette, whereas an unused color may be put into I, J, U with weights $1, 1, -1$. Its total contribution is therefore $1 + 1 - 1 = 1$. Summing over palette sizes gives the formula. Terms with an empty I or J contribute zero because both sides are nonempty. $\square$

Second proof of (32.2) (manuscript 03). A single color contributes $e^x + e^y - 1$ to the exponential generating function: its vertices may occur on the first side, the second side, or neither, but not both. Expanding $(e^x + e^y - 1)^k$ and taking the coefficient of $x^a y^b / (a! b!)$ proves (32.2). The zero exponents disappear because $a, b > 0$.

Manuscript 03 records two further exact forms. With $\text{sur}_j(m) = j! S(m, j) = \sum_{q=0}^j (-1)^q \binom{j}{q} (j - q)^m$ the number of onto maps from an m -set to a prescribed j -set ($m, j \geq 1$), counting the colors used on the first part gives

$$\mathcal{B}_k(a, b) = \sum_{i=1}^{k-1} \binom{k}{i} \text{sur}_i(a) (k - i)^b, \quad (32.3)$$

which is (32.1) again, and counting the exact palettes on both parts gives the nonnegative expansion

$$\mathcal{B}_k(a, b) = \sum_{\substack{i, j \geq 1 \\ i+j \leq k}} \frac{k!}{i! j! (k - i - j)!} \text{sur}_i(a) \text{sur}_j(b). \quad (32.4)$$

Manuscript 01's lemma 41.5 below is (32.3) with sur_i expanded by inclusion–exclusion. The signed formula is a finite exponential sum; in particular, its subtractions cannot be discarded in exact computations: counting disjoint *allowed* palettes without inclusion–exclusion overcounts colorings that do not use every allowed color (manuscript 06). The positive formula (32.4) is best for lower bounds, the signed formula (32.2) makes the bounded-imbalance asymptotics transparent, and using them for their respective purposes avoids a potentially invalid lower bound obtained by discarding negative terms (manuscript 03).

32.2 Constants for the effective theorem

Fix an integer $k \geq 4$ for the remainder of this subsection and the next section, and define

$$\begin{aligned} \hat{H} &= \lfloor k^2/4 \rfloor, & \lambda_k &= \sqrt{\hat{H}}, \\ \hat{M} &= \begin{cases} \hat{H} - 1, & k \text{ even}, \\ \hat{H} - 2, & k \text{ odd}, \end{cases} & \mu &= \sqrt{\hat{M}}, & \hat{\theta} &= \mu/\lambda_k < 1, \\ \hat{C} &= \begin{cases} \binom{k}{k/2}, & k \text{ even}, \\ 2\binom{k}{(k-1)/2}, & k \text{ odd}, \end{cases} & \hat{B} &= k^2 3^k. \end{aligned} \quad (32.5)$$

We always have $\hat{C} \geq 6$, $\lambda_k \geq 2$, and $k/\lambda_k \geq 2$.

Lemma 32.2 (Uniform near-balanced estimate; manuscript 06). *Let $a + b = s$, $a, b \geq 1$, and write $\delta = (b - a)/2$. If $|\delta| \leq 2$, then*

$$|\mathcal{B}_k(a, b) - \mathcal{L}_k(s, \delta)| \leq \hat{B} \mu^s, \quad (32.6)$$

where

$$\mathcal{L}_k(s, \delta) = \begin{cases} \hat{C} \lambda_k^s, & k \text{ even}, \\ \hat{C} \lambda_k^s \cosh(\delta \omega_h), & k = 2h + 1. \end{cases} \quad (32.7)$$

Proof. A term in (32.2) has exponential factor $i^a j^b = (ij)^{s/2} (j/i)^\delta$. Among positive i, j with $i + j \leq k$, the maximum of ij is $\hat{H}$. For even $k = 2h$, it is attained only at (h, h) ; its coefficient is $\hat{C}$. For odd $k = 2h + 1$, it is attained at $(h, h + 1)$ and $(h + 1, h)$, each with coefficient $\hat{C}/2$. These terms give (32.7). Every remaining product ij is at most $\hat{M}$. For even k , the next possible product is at most $h^2 - 1$. For odd $k = 2h + 1$ with $h \geq 2$, either $i + j = k$ and the next product is at most $(h - 1)(h + 2) = \hat{H} - 2$, or $i + j \leq k - 1$ and the product is at most $h^2 \leq \hat{H} - 2$. Also $|(j/i)^\delta| \leq k^2$ in the stated range. The sum of all absolute multinomial coefficients is at most 3^k , by the multinomial theorem. This proves (32.6). $\square$

Let $\hat{S} = \hat{S}(k)$ be the least integer $\hat{S} \geq 4$ satisfying

$$64\hat{B}^2\hat{M}^{\hat{S}} \leq \hat{C}^2\hat{H}^{\hat{S}}. \quad (32.8)$$

Equivalently, $\hat{B}\hat{\theta}^{\hat{S}} \leq \hat{C}/8$. The definition is effective using only integer multiplication and comparison, and such an $\hat{S}$ exists because $\hat{M} < \hat{H}$.

Corollary 32.3 (Manuscript 06). *For $s \geq \hat{S}$,*

$$\mathcal{B}_k(\lfloor s/2 \rfloor, \lceil s/2 \rceil) \geq \frac{7}{8} \hat{C} \lambda_k^s. \quad (32.9)$$

For $n \geq \max(7, \hat{S})$,

$$\mathcal{B}_k(a_n, b_n) < \frac{3}{2} \hat{C} \lambda_k^n. \quad (32.10)$$

Proof. In the balanced split, $|\delta| \leq 1/2$, and the leading expression in (32.7) is at least $\hat{C} \lambda_k^s$. Subtract the error $\hat{C} \lambda_k^s/8$.

For the nearest coprime split, $|\delta| \leq 2$. In the even-output case, the upper coefficient is at most $1 + 1/8 = 9/8$. In the odd-output case, $h \geq 2$ and

$$\cosh(2\omega_h) \leq \cosh(2 \log(3/2)) = \frac{97}{72}.$$

Adding the error coefficient gives $97/72 + 1/8 = 53/36 < 3/2$. $\square$

Lemma 32.4 (A bound with no size restriction; manuscripts 06, 01 and 03). *For every $k \geq 3$ and all $u, v \geq 1$,*

$$\mathcal{B}_k(u, v) \geq 2\lambda_k^{u+v}. \quad (32.11)$$

Proof. Fix a partition of the colors into sets of sizes $h = \lfloor k/2 \rfloor$ and $k - h$. Color the two sides from these palettes, and also count the colorings with the palettes exchanged. The two families are disjoint, since both sides are nonempty and the palettes are disjoint. Their total is $h^u(k - h)^v + (k - h)^u h^v \geq 2[h(k - h)]^{(u+v)/2}$. $\square$

All three manuscripts prove lemma 32.4 by this argument (06: Lemma 4.5, 01: Lemma 5.1, 03: first part of Lemma 7.1); manuscript 06 states it for $k \geq 4$, manuscripts 01 and 03 for $k \geq 3$.

32.3 Sharpened palette bounds for fixed k (manuscripts 01 and 03)

Manuscripts 01 and 03 need asymptotic lower bounds valid for unbounded imbalance. For fixed j , the union bound on a missing color gives

$$j^m - j(j - 1)^m \leq \text{sur}_j(m) \leq j^m; \quad (32.12)$$

for $j = 1$ and $m \geq 1$ this reads $\text{sur}_1(m) = 1$.

Lemma 32.5 (Sharpened lower bounds; manuscript 01, also manuscript 03). *Let $u, v \rightarrow \infty$, with $k \geq 3$ fixed and $h = \lfloor k/2 \rfloor$. If $k = 2h$,*

$$\mathcal{B}_k(u, v) \geq (1 - o(1)) \binom{k}{h} \lambda_k^{u+v}. \quad (32.13)$$

If $k = 2h + 1$,

$$\mathcal{B}_k(u, v) \geq (1 - o(1)) 2 \binom{k}{h} \lambda_k^{u+v} \cosh\left(\frac{(v-u)\omega_h}{2}\right). \quad (32.14)$$

The $o(1)$ tends to zero whenever $\min(u, v) \rightarrow \infty$, with no assumption on the ratio u/v .

Proof. In the even case, count colorings using exactly a chosen h -color palette on the left and exactly its complement on the right. These choices are disjoint as the chosen palette varies, so they give $\binom{k}{h} \text{sur}_h(u) \text{sur}_h(v)$. Apply (32.12).

In the odd case, use every color and the two possible palette-size pairs $(h, h+1)$ and $(h+1, h)$. Their combined count is $\binom{k}{h} (\text{sur}_h(u) \text{sur}_{h+1}(v) + \text{sur}_{h+1}(u) \text{sur}_h(v))$. Each factor has relative error tending to zero with the smaller side. Factoring out λ_k^{u+v} gives (32.14). $\square$

Manuscript 03 proves the same bounds (its Lemma 7.1, in the form $\mathcal{B}_{2h+1}(u, v) \geq (\binom{k}{h} - o(1)) \lambda_k^{u+v} (t_h^{(v-u)/2} + t_h^{-(v-u)/2})$, where $t_h^z + t_h^{-z} = 2 \cosh(z\omega_h)$) by retaining in (32.4) the term (h, h) , respectively $(h, h+1)$ and $(h+1, h)$; since $\text{sur}_j(m)/j^m \rightarrow 1$ uniformly over the finite set $1 \leq j \leq k$, the relative errors depend only on $\min(u, v)$, which proves the uniformity.

Lemma 32.6 (Near-balanced asymptotics; manuscript 01). *For $a + b = n \rightarrow \infty$ and bounded $|a - b|$, the lower bounds (32.13) and (32.14) are asymptotic equalities. Consequently*

$$\frac{D_k^*(n)}{\lambda_k^n} = \begin{cases} \binom{k}{h} + o(1), & k \text{ even}, \\ 2 \binom{k}{h} \cosh(\delta_n \omega_h) + o(1), & k \text{ odd}. \end{cases} \quad (32.15)$$

Proof. Every proper coloring admits a partition of the full k -color alphabet into a nonempty left palette and a nonempty right palette. It may admit several if some colors are unused, but this gives the upper bound

$$\mathcal{B}_k(a, b) \leq \sum_{i=1}^{k-1} \binom{k}{i} i^a (k-i)^b. \quad (32.16)$$

For bounded imbalance, each summand has exponential base $\sqrt{i(k-i)}$ in n . The unique largest product is h^2 for even k ; for odd k , the two largest products are $h(h+1)$, from $i = h, h+1$. All remaining products are strictly smaller. The sharpened lower bounds match these largest summands. Finally, $G_k(a_n, b_n) \leq n^2$ is negligible relative to λ_k^n . $\square$

(In manuscript 01's own normalization, $\delta_n \in \{1, 2, 4\}$ and the cosine argument is $\lambda \delta_n / 2$ with $\lambda = \log((m+1)/m)$; this is $\delta_n \omega_h$ in ours.)

Lemma 32.7 (Bounded-imbalance asymptotics; manuscript 03). *If $a + b = n$ and $|a - b|$ is bounded, then for some $\rho_k < \lambda_k$,*

$$\mathcal{B}_{2h}(a, b) = \binom{2h}{h} h^n + O_k(\rho_k^n), \quad (32.17)$$

$$\mathcal{B}_{2h+1}(a, b) = \binom{2h+1}{h} \lambda_k^n (t_h^{(b-a)/2} + t_h^{-(b-a)/2}) + O_k(\rho_k^n). \quad (32.18)$$

The implied constants may depend on the fixed bound on $|a - b|$.

Proof. In (32.2), the largest possible product ij under $i + j \leq k$ is $\lfloor k^2/4 \rfloor$. The maximizing pair is (h, h) for even k , and the two pairs $(h, h+1)$, $(h+1, h)$ for odd k . All their coefficients are positive and have magnitude $\binom{k}{h}$. Every other product is strictly smaller. Bounded imbalance means $i^a j^b = (ij)^{n/2} (j/i)^{(b-a)/2}$, with a bounded final factor. There are only finitely many terms, so the rest have a strictly smaller exponential base. For $k = 3$ the rest reduce to the constant -6 . $\square$

By proposition 30.4 (or Davies’s Corollary 3) and lemma 32.7, the leading terms of (32.17)–(32.18) are already upper bounds for the optimal deficit to first order. The remaining work, in manuscript 03’s words, is an upper bound on reversal complexity, equivalently a matching lower bound on the deficit, uniform over all pairs of transformations. Lemma 32.2 is the effective version of lemmas 32.6 and 32.7 for $|\delta| \leq 2$.

33 The effective eventual theorem

This section and the next are manuscript 06’s. Retain (32.5) and (32.8), with $k \geq 4$ fixed.

33.1 The complete integer definition of the threshold

Define

$$\begin{aligned}\hat{L} &= \min\{\ell \geq 0 : 2^\ell \geq 2\hat{C}\}, \\ \hat{K} &= k(k-1)(k-2), \quad \beta_k = \hat{K}^{1/3}, \\ \hat{U} &= \min\{u \geq 3 : \hat{K}^{2u} \geq (2\hat{C})^6 \hat{H}^{3u}\}, \\ N(k) &= \max\{k+1, 7, \hat{L}(\hat{S}+1), \hat{U}\}.\end{aligned}\tag{33.1}$$

The condition defining $\hat{U}$ is equivalent to $(\beta_k/\lambda_k)^{\hat{U}} \geq 2\hat{C}$. All the threshold-defining tests are exact integer inequalities. In particular, rounding a numerical logarithm is not part of the theorem’s specification. The shipped data/05-quartic-threshold-thresholds.json lists the constants for $4 \leq k \leq 16$ under the keys H, M, C, B, L, S, U, N; in the write phase they were recomputed from (32.5), (32.8) and (33.1) and agree in all 13 rows.

33.2 Separating incomplete or disconnected cross-cycle graphs

Call a cross-cycle graph *full* if it is a single $K_{a,b}$ on all n vertices, with $\gcd(a, b) = 1$; in the notation of (9.6), fullness means $d = 1$, $r = 0$.

Lemma 33.1. *If $n \geq \hat{L}(\hat{S}+1)$ and a cross-cycle graph is not full, then*

$$P_G(k) \geq \frac{7}{4}\hat{C}\lambda_k^n.\tag{33.2}$$

Proof. Write $n = d(u+v) + r$, $d \geq 1$, $r \geq 0$. First suppose $d+r \geq \hat{L}$. Lemma 32.4 gives

$$\frac{P_G(k)}{\lambda_k^n} = \left(\frac{\mathcal{B}_k(u, v)}{\lambda_k^{u+v}}\right)^d \left(\frac{k}{\lambda_k}\right)^r \geq 2^{d+r} \geq 2\hat{C}.$$

This is stronger than required.

Now suppose $d+r < \hat{L}$, and put $s = u+v$. Since $r < \hat{L}$ and $d < \hat{L}$, $n \geq \hat{L}(\hat{S}+1)$ implies $s = (n-r)/d \geq \hat{S}$. Strict balancing, without any coprimality requirement for the intermediate balanced split, and (32.9) imply

$$\mathcal{B}_k(u, v) \geq \mathcal{B}_k(\lfloor s/2 \rfloor, \lceil s/2 \rceil) \geq \frac{7}{8}\hat{C}\lambda_k^s.$$

If $d = 1$, nonfullness forces $r \geq 1$, and the normalized coloring count is at least $(7\hat{C}/8) \cdot 2 = 7\hat{C}/4$. If $d \geq 2$, it is at least $(7\hat{C}/8)^2 \geq 7\hat{C}/4$, because $\hat{C} \geq 6$. $\square$

Corollary 33.2 (A quantitative nonfull-graph loss). *For $n \geq N(k)$, any mixed pair admitting a nonfull cross-cycle collision graph has orbit size strictly less than $k^n - D_k^*(n)$, by at least*

$$\left(\frac{\hat{C}}{4} - 1\right) \lambda_k^n \quad (33.3)$$

as a real-valued lower bound on the difference.

Proof. By the collision bound and lemma 29.4, its deficit from k^n is at least $P_G(k) - 3^{n/3}$. Since $3^{n/3} < \lambda_k^n$ and $\mathcal{B}_k(a_n, b_n) < (3\hat{C}/2)\lambda_k^n$, subtracting the optimal candidate deficit $D_k^*(n) = \mathcal{B}_k(a_n, b_n) - a_n b_n$ gives a difference strictly larger than the right side of (33.3). It is positive because $\hat{C} \geq 6$. $\square$

33.3 Separating the remaining graph types

Lemma 33.3 (Manuscript 06). *For $k \geq 4$:*

- (i) every same-cycle collision graph has at least β_k^n proper colorings;
- (ii) $(k-1)^2 k^{n-2} \geq \beta_k^n$ for $n \geq 3$;
- (iii) $k^n - k^{n-1} \geq \beta_k^n$ for $n \geq 1$;
- (iv) $\beta_k > \lambda_k$.

Proof. Set $q = k - 1$. Then $\beta_k^3 = q^3 - q$, so $\beta_k < q < k$. For an even component length ℓ , its coloring count $q^\ell + q$ exceeds β_k^ℓ . For odd $\ell \geq 3$,

$$q^\ell - q = q^\ell (1 - q^{1-\ell}) \geq q^\ell (1 - q^{-2})^{\ell/3} = \beta_k^\ell.$$

Indeed $1 - q^{1-\ell} \geq 1 - q^{-2}$ and raising a number in $(0, 1)$ to a power at least one can only decrease it. Multiplication over components and the factor $k^r \geq \beta_k^r$ from isolates prove (i).

At $n = 3$, (ii) is $k(k-1)^2 \geq k(k-1)(k-2)$; increasing n multiplies the left side by k and the right side by $\beta_k < k$. For (iii), $(k-1)k^{n-1} > \beta_k^n$ follows from $k-1 > \beta_k$ and $k > \beta_k$. Finally, using $\hat{H} \leq k^2/4$ gives

$$\frac{\beta_k^3}{\lambda_k^3} = \frac{k(k-1)(k-2)}{\hat{H}^{3/2}} \geq 8 \left(1 - \frac{1}{k}\right) \left(1 - \frac{2}{k}\right) \geq 3. \quad (33.4)$$

$\square$

For $n \geq \hat{U}$, the definition of $\hat{U}$ gives $\beta_k^n \geq 2\hat{C}\lambda_k^n$. The possible correction $\text{ord}(p)$ in the same-cycle case is less than λ_k^n . The possible correction one in the two-singular case is at most λ_k^n as well. Thus every deficit in these cases exceeds $(3\hat{C}/2)\lambda_k^n$, because $\hat{C} \geq 6$. The same conclusion holds, without a correction, in the two-permutation case. Each is strictly larger than the candidate $D_k^*(n)$.

Second routes to (i) and (iv). Manuscript 01 proves (i) and (iv) for every $k \geq 3$ (its Lemma 6.3, with β_k called γ_k): for odd $L \geq 3$, $C_L(k)^{1/L} = (k-1)(1 - (k-1)^{1-L})^{1/L} \geq (k-1)(1 - (k-1)^{-2})^{1/3} = \beta_k$; for even $L \geq 4$, $C_L(k)^{1/L} \geq k-1 > \beta_k$; for $L = 2$, $[k(k-1)]^{1/2} \geq \beta_k$; isolated vertices contribute $k \geq \beta_k$; and $\beta_k > k/2 \geq \lambda_k$, since after cubing the strict inequality is $8(k-1)(k-2) > k^2$, valid for all $k \geq 3$. Manuscript 03 proves $s^L + (-1)^L s \geq (s^3 - s)^{L/3} = \beta_k^L$ for $s = k-1 \geq 2$, $L \geq 2$, by the argument of its three-output case ((38.6)), and $\beta_k > \lambda_k$ by checking $(1 - 1/k)(1 - 2/k) > 1/8$, true at $k = 3$ and increasing thereafter, because $\lambda_k \leq k/2$. At $k = 3$, $\beta_3 = 6^{1/3}$, and (i) is Part II's (16.1).

33.4 Completion of the exactness proof

Theorem 33.4 (Eventual exactness with explicit threshold; manuscript 06). *For $k \geq 4$ and $n \geq N(k)$ as defined in (33.1), (28.3) holds.*

Proof. Consider any binary transformation pair on n points. The two-permutation and two-singular cases are strictly below the candidate optimum by lemma 29.2 and the preceding estimates. In the mixed case, select any collision pair of the singular map. A same-cycle graph is strictly excluded by lemma 33.3; a nonfull cross-cycle graph is strictly excluded by corollary 33.2.

The only remaining possibility is that the permutation has exactly two cycles, of coprime lengths a, b summing to n , and the collision graph is $K_{a,b}$. Now $\text{ord}(p) = ab$, so

$$|\tau\langle p, s \rangle| \leq k^n - \mathcal{B}_k(a, b) + ab \leq k^n - \mathcal{B}_k(a_n, b_n) + a_n b_n,$$

where the last inequality follows from strict balancing. This argument also covers a cycle of length one: such a split is farther from balance than (28.1). It proves the universal upper bound.

For attainability, use the published lower-bound theorem [1, Theorem 3 and Corollary 3]: for $4 \leq k < n$ and a coprime split $1 < a < b$, there is an accessible binary DFAO with reverse complexity $k^n - \mathcal{B}_k(a, b) + ab$. Apply it to (a_n, b_n) .

We must also check the original-state minimality in the definition of $R_2(n, k)$, since accessibility alone is not minimality. Because $n \geq \hat{L}$ and $k/\lambda_k \geq 2$, we have $\hat{C}(\lambda_k/k)^n \leq \hat{C}2^{-n} \leq 1/2$. Equation (32.10) gives $\mathcal{B}_k(a_n, b_n) < 3k^n/4$, so the attained reverse complexity is greater than $k^n/4 \geq k^{n-1}$. If the original automaton minimized to fewer than n states, the elementary coloring construction would bound its reverse complexity by k^{n-1} , a contradiction. The witness therefore has original state complexity exactly n (lemma 29.1). $\square$

33.5 A uniform quartic bound

Corollary 33.5 (Manuscript 06). *For every $k \geq 4$, $N(k) \leq 3k^4$. Consequently (28.3) holds whenever $k \geq 4$ and $n \geq 3k^4$.*

Proof. First, $\hat{C} \leq 2^{k+1}$, hence $\hat{L} \leq k + 2$. Since $\hat{M} \leq \hat{H} - 1$,

$$\log(1/\hat{\theta}) = \frac{1}{2} \log(\hat{H}/\hat{M}) \geq \frac{1}{2\hat{H}}.$$

For $k \geq 4$ we have

$$\log(8\hat{B}) = \log 8 + 2 \log k + k \log 3 \leq 3k.$$

For example, the three summands are bounded respectively by $3k/4$, k , and $5k/4$; the middle inequality follows from $2 \log k \leq k$ for $k \geq 4$. Thus $S_0 = 6\hat{H}k$ satisfies $\hat{B}\hat{\theta}^{S_0} \leq 1/8 \leq \hat{C}/8$, and

$$\hat{S} \leq 6\hat{H}k \leq \frac{3}{2}k^3.$$

Therefore

$$\hat{L}(\hat{S} + 1) \leq (k + 2)(\frac{3}{2}k^3 + 1) \leq 3k^4,$$

using $k + 2 \leq 3k/2$ and $\frac{3}{2}k^3 + 1 \leq 2k^3$.

By (33.4), $\beta_k/\lambda_k \geq 3^{1/3}$. At $u = 2(k + 2)$,

$$(\beta_k/\lambda_k)^u \geq 3^{2(k+2)/3} > 2^{k+2} \geq 2\hat{C},$$

so $\hat{U} \leq 2(k + 2)$. The remaining entries $k + 1, 7$ in the maximum defining $N(k)$ are plainly at most $3k^4$. $\square$

The constants were chosen to make a fully explicit comparison easy to audit, not to optimize the threshold. In particular, the argument deliberately uses an elementary exponential order bound instead of a sharper bound on permutation orders.

Remark 33.6 (Part I's structural bound in the same range; added in the merge). Part I asks in section 12 to prove or refute $U(n, k) = L(n, k)$ for $n \geq 7$, $3 \leq k < n$; Part II's remark 18.3 proves it for $k = 3$. It also holds on manuscript 06's exactness range: for $k \geq 4$ and either $n \geq N(k)$ or $4 \leq k \leq 7$, $\max(7, k + 1) \leq n$ (with $k < n$),

$$D(n, k) = D_k^*(n), \quad U(n, k) = L(n, k) = R_2(n, k) = k^n - D_k^*(n).$$

Indeed $L(n, k) = k^n - D_k^*(n)$ by corollary 31.5, and Part I's collection contains the entry (9.15) at $(a, b) = (a_n, b_n)$, $d = 1$, $r = 0$, which is $\mathcal{B}_k(a_n, b_n) - H_0(a_n b_n) = D_k^*(n)$; so $D(n, k) \leq D_k^*(n)$. Conversely every entry is at least a quantity that manuscript 06 bounds below by $D_k^*(n)$. Entry (9.13) is lemma 29.2(ii); entry (9.12) is at least $k^n - k^{n-1}$, because $M(n, k) \leq k^{n-1}$ by lemma 29.2(i); in (9.14), $H_{n-m}(m) \leq m\gamma_{n-m} \leq m 3^{(n-m)/3}$; in (9.15), where $u = a/d$, $v = b/d$ are coprime with sum s , $T_{r,k}(a, b) = H_r(\text{lcm}(a, b)) \leq duv\gamma_r$, $\mathcal{B}_k(u, v) \geq \mathcal{B}_k(u_s, v_s)$ and $uv \leq u_s v_s$ for the nearest coprime split (u_s, v_s) of s (theorem 31.3), so the entry is at least manuscript 06's compressed quantity (34.3). For $n \geq N(k)$ the proofs of lemmas 33.1 and 33.3 and corollary 33.2 bound these quantities from below by numbers exceeding $D_k^*(n)$, except the entries with $d = 1$, $r = 0$, which strict balancing handles; below $N(k)$, for $4 \leq k \leq 7$, the finite verification of section 34 checks exactly the compressed quantities. The one entry manuscript 06 does not treat is the nonsurjective (9.11): for $n \geq N(k)$, $k^n - (k-1)^n \geq n(k-1)^{n-1} \geq (k-1)^n > \beta_k^n \geq 2\hat{C}\lambda_k^n > D_k^*(n)$ (using $n \geq k+1$ and $n \geq \hat{U}$), and for $4 \leq k \leq 7$ below $N(k)$ the inequality $k^n - (k-1)^n > D_k^*(n)$ was checked in the write phase for all 2,217 rows by exact integer arithmetic. So $D(n, k) \geq D_k^*(n)$. This remark is the merge's reading of manuscript 06's comparisons against Part I's definitions; the manuscripts do not state it, and for $n \leq 30$ it agrees with Part I's data/finite_range_bounds.csv (section 35.5).

34 Finite exact closure for four through seven outputs

34.1 What must actually be checked

The analytic theorem leaves finitely many n for each fixed k . It is unnecessary to enumerate all pairs of transformations. The graph classification and strict balancing reduce the upper-bound verification to integer inequalities indexed by graph parameters.

For any component total $s \geq 2$, let (u_s, v_s) be its nearest positive coprime split, allowing $(1, 1)$ for $s = 2$, and write $\mathcal{B}_k(u_s, v_s)$ for its coloring count (manuscript 06's $C_{k,s}$). Let $\gamma_r = \lfloor 3^{r/3} \rfloor$. For each n , set $D = D_k^*(n)$. The following list is exhaustive:

$$k^n - k^{n-1} \geq D, \tag{34.1}$$

$$(k-1)^2 k^{n-2} - 1 \geq D, \tag{34.2}$$

$$\mathcal{B}_k(u_s, v_s)^d k^r - du_s v_s \gamma_r \geq D, \quad s \geq 2, d \geq 1, r = n - ds \geq 0, \tag{34.3}$$

$$((k-1)^\ell + (-1)^\ell (k-1))^{m/\ell} k^{n-m} - m\gamma_{n-m} \geq D, \quad 2 \leq m \leq n, \ell \mid m, \ell \geq 2. \tag{34.4}$$

To justify the compression in (34.3), fix s, d, r and vary over all coprime pairs $u + v = s$. Strict balancing minimizes $\mathcal{B}_k(u, v)$ at (u_s, v_s) , while uv is maximized there. Thus $\mathcal{B}_k(u, v)^d k^r - duv\gamma_r$ is minimized at that same split. No pair is lost. Equation (34.4) includes every possible displacement class inside a cycle. The factor γ_r is a valid bound because a residual permutation order is an integer and at most $3^{r/3}$.

For $s = n$, $d = 1$, $r = 0$, (34.3) is the distinguished identity $\mathcal{B}_k(a_n, b_n) - a_n b_n = D$. All other comparisons in the verified ranges are *strict*. This extra fact extends the structural rigidity theorem (theorem 39.1) as well as the maximum-value theorem.

34.2 Two implementations with different arithmetic routes

The primary program, `code/05-quartic-threshold-theorem_checks.py`, computes $\mathcal{B}_k$ by the signed palette formula, uses the explicit residue formula for the split, and computes γ_r by integer binary search. The independent program, `code/05-quartic-threshold-independent_check.py`, does not import the primary program or read its computed tables. It instead uses the positive Stirling formula, finds the split by a descending gcd scan, and enumerates active sizes and divisors rather than component sizes and multiplicities.

The independent program also uses a different residual-order comparison. Given a coloring count X , target deficit D , and active period factor t , it first checks

$$X - D > t 3^{\lceil r/3 \rceil}.$$

If needed, it can check the sharper real-order inequality by cubing: $(X - D)^3 > t^3 3^r$ with $X - D > 0$. In the reported runs, the coarser test already succeeded for every excluded mixed case; no cubic fallback was needed. Thus the second implementation verifies inequalities at least as strong as the corresponding primary exclusions, rather than merely reproducing rounded roots.

Both programs raise exceptions on any failed comparison or unexpected equality. Neither depends on Python's optional assertion statements. The independent program also checks that the attained candidate exceeds k^{n-1} in every finite row, completing the original-state minimality argument below the analytic threshold.

k	$\hat{L}$	$\hat{S}$	$\hat{U}$	$N(k)$	finite n	rows	comparisons
4	4	52	7	212	7–211	205	180,964
5	6	39	8	240	7–239	233	238,946
6	6	158	8	954	7–953	947	5,022,662
7	8	104	10	840	8–839	832	3,804,474
Total per implementation						2,217	9,247,046

Table 7: Exact finite closure (manuscript 06). The two implementations agree on every row and on the number of structural comparisons.

There were zero failed inequalities and zero unexpected equalities. There was precisely one distinguished equality per row. The primary program's smallest strict excluded margins, taken over its entire range for each k , were respectively 1265, 9285, 41615, and 327855. The per-row CSV files (`data/05-quartic-threshold-finite_k4.csv` to `..._k7.csv`) retain the target deficit, strict margin, and closest excluded case, not only an overall success flag.

Theorem 34.1 (Computer-assisted all- n slices; manuscript 06). *For $4 \leq k \leq 7$ and $n \geq \max(7, k + 1)$,*

$$R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + a_n b_n.$$

Moreover, in these ranges every collision-graph case other than a full nearest-coprime two-cycle graph is strictly suboptimal.

Proof. For $n \geq N(k)$ this is theorem 33.4 and its strict exclusions. Below $N(k)$, the finite ranges in table 7 cover every remaining integer n under consideration. The two implementations verify (34.1)–(34.4), with strictness outside the distinguished case. Their mathematical sufficiency was proved above. The published lower witness supplies equality, and the separate k^{n-1} test ensures its original minimality. $\square$

This is a computer-assisted proof with explicit finite obligations. It is not a proof-assistant certificate, and agreement between two programs does not make them independent of the common mathematical reduction. The programs do, however, separate several potential implementation failures: counting formula, split selection, loop indexing, and order-bound arithmetic.

34.3 Examples and supplementary checks

n	(a_n, b_n)	$k = 4$	$k = 5$	$k = 6$	$k = 7$
7	(3, 4)	15,472	71,037	243,438	—
8	(3, 5)	63,403	369,020	1,539,561	5,101,594
9	(4, 5)	258,360	1,902,365	9,657,446	37,886,085
10	(3, 7)	1,035,217	9,544,826	58,260,447	267,231,328
11	(5, 6)	4,179,298	48,491,775	358,442,016	1,939,367,551

Table 8: Exact optima in the proved all- n slices (manuscript 06). The dash excludes the boundary $k = n$, which is not covered here. Values are computed from the proved formula, not by exhaustive search over automata. The shipped data/05-quartic-threshold-sample_values.csv also has the row $n = 12$.

The supplementary program, code/05-quartic-threshold-supplementary_checks.py, checks 1,183 signed-versus-positive palette identities, 1,999 nearest-split formulas, and 6,728 strict balancing inequalities. It also enumerates every permutation and collision pair on up to five vertices, checking 1,364 collision-graph coloring counts by literal three-color assignments. Twenty-seven small matching-spectrum enumerations test the rank-sensitive formulas of section 40. These are regression checks, not substitutes for the proofs of the identities. The program is not a third independent implementation of the full proof: it imports the primary chromatic routine when comparing that formula to an independently written positive formula.

35 Finite certificates for $n \leq 100$, $k \leq 12$

This section is manuscript 01's. Its infinite argument (section 36) uses a subexponential permutation-order bound and a subsequence contradiction; its finite computations use a different bound, weaker asymptotically but especially easy to check in integers. They close a large initial region directly; they do not replace the infinite proof.

35.1 A finite universal bound

For a same-cycle collision, write the support size as $m = dL$, $L \geq 2$, and put $r = n - m$. The permutation order is at most $mJ(r)$, so the deficit is at least

$$C_L(k)^d k^r - mJ(r). \quad (35.1)$$

For a collision across cycles $a \leq b$, write $d = \gcd(a, b)$ and $r = n - a - b$. The analogous lower bound is

$$\mathcal{B}_k(a/d, b/d)^d k^r - V_k(a, b, d)J(r), \quad V_k(a, b, d) = \begin{cases} b, & k = 3 \text{ and } d = 1, \\ \text{lcm}(a, b), & \text{otherwise.} \end{cases} \quad (35.2)$$

To justify the improvement when $k = 3$, $d = 1$: a proper coloring of $K_{a,b}$ has one monochromatic side, so its period on the two selected cycles is at most b . The period on the remaining vertices divides the residual permutation order, and the total period is at most the product of those bounds. When the initial coloring is improper, subtracting this possible proper period is still a valid, weaker bound. (This is Part I's (9.10) with H_r relaxed to a product with $J(r)$, as in Part II's (14.6).)

Proposition 35.1 (Finite certificate principle; manuscript 01). *Let $D_k^{\text{cert}}(n)$ be the minimum of the three bounds of remark 29.3, all values of (35.1) with $m = dL \leq n$ and $L \geq 2$, and all values of (35.2) with $1 \leq a \leq b$, $a + b \leq n$. Then every coloring orbit has defect at least $D_k^{\text{cert}}(n)$. If $n \geq 7$, $3 \leq k < n$, and*

$$D_k^{\text{cert}}(n) = D_k^*(n), \quad (35.3)$$

then $D_k(n) = D_k^*(n)$ for that particular pair (n, k) .

Proof. The minimum includes a valid bound for every case of the generator classification and every possible collision-orbit graph. This proves the universal lower bound. The explicit accessible minimal witness has defect $D_k^*(n)$ by proposition 30.4, so equality of the two integer bounds proves the second assertion. $\square$

Some terms in the minimum could be negative for other parameter ranges; this only makes the bound weaker, never invalid. No positivity assumption on the certificate expressions is used. The principle is Part I's proposition 10.1 with Part I's exact residual order sets H_r replaced by the product bound $J(r)$, and with Part I's two-permutation entry (9.12) replaced by the weaker k^{n-1} .

35.2 The certified finite region

Both programs independently obtained

$$D_k^{\text{cert}}(n) = D_k^*(n) \quad \text{for every } 7 \leq n \leq 100, \quad 3 \leq k \leq \min(12, n-1). \quad (35.4)$$

There are 919 such pairs. For every row, `data/03-eventual-finite-certificates.csv` records $n, k, a_n, b_n, D_k^*(n), D_k^{\text{cert}}(n)$ (columns `candidate_defect`, `universal_lower_bound`) and a minimizing graph type. These are finite exhaustive checks of *graph-parameter bounds*, not exhaustive enumerations of all n -state automata. The latter would be vastly larger and is unnecessary for this certificate principle.

The result extends Part I's finite range (proposition 10.1, $n \leq 30$) in the state variable, but does not establish a new record for each individual value. Some of the small values, including the example in example 35.2, were already computed in Part I or in Davies's experiments. No claim of priority for the finite table is made. For $4 \leq k \leq 7$ the region lies inside manuscript 06's closure; for $8 \leq k \leq 12$ it is the only settled part of the window below $N(k)$ ($N(8), \dots, N(12) = 2688, 1863, 5364, 3883, 10483$, table 11).

n	k	a_n	b_n	Defect $D_k(n)$	Maximum $R_2(n, k)$
7	3	3	4	62	2 125
8	3	3	5	109	6 452
9	3	4	5	133	19 550
10	3	3	7	395	58 654
12	3	5	7	467	530 974
20	3	9	11	7 663	3 486 776 738
7	4	3	4	912	15 472
8	4	3	5	2 133	63 403
10	4	3	7	13 359	1 035 217
20	4	9	11	7 047 969	1 099 504 579 807
8	5	3	5	21 605	369 020
10	6	3	7	2 205 729	58 260 447

Table 9: Selected exact finite optima certified by proposition 35.1 (manuscript 01). All entries are integer counts, not asymptotic approximations.

Example 35.2 (A worked finite optimum; manuscript 01). For $n = 8$ and $k = 5$, the selected split is $(a, b) = (3, 5)$. Direct evaluation of (32.1) gives

$$\mathcal{B}_5(3, 5) = 5 \cdot 4^5 + 20 \cdot 3 \cdot 3^5 + 60 \cdot 2^5 = 21620.$$

The two primitive cycle words return $ab = 15$ proper colorings, so the candidate defect is 21605 and the candidate maximum is $5^8 - 21605 = 369020$. The finite certificate proves that this candidate is the

actual maximum, without needing to know whether $8 \geq N_{\exists}(5)$. This value is already established in Part I (section 10.3, which corrects Davies’s printed 368020); it is an explanatory example, not a newly claimed numerical record.

35.3 Separation of the producer and checker

The producer code/`03-eventual-verify.py` evaluates Stirling numbers by their recurrence, uses the explicit maximum-product formula for J , and uses the residue formula for the proposed split. Its separate finite-verification tasks also check the two chromatic formulas, strict balancing, collision graphs, actual witness orbits, and recurrences.

The checker code/`03-eventual-check-certificates.py` imports none of the producer’s code. It computes complete-bipartite color counts by inclusion–exclusion, computes residual product bounds by an integer-partition dynamic program, and finds the closest coprime split by direct enumeration rather than by residue cases. It then enumerates the full parameter minimum independently for each CSV row. All 919 rows were accepted, with zero rejected rows.

This is implementation independence, not logical independence of the underlying theorem: both programs check the finite certificate principle of proposition 35.1. Neither is a proof assistant, and agreement between them cannot validate an incorrect mathematical statement shared by both. The proof of the principle is therefore included in full rather than hidden behind the programs.

35.4 Recorded checks and their meaning

The producer recorded 42960 successful checks, grouped as follows.

Check family	Count
Stirling row log-concavity	630
Order-two total positivity	18207
Chromatic inclusion–exclusion agreement	3610
Chromatic symmetry	3610
Quantitative strict balancing	15210
Closest coprime splits	94
Universal finite-bound comparisons	919
Witness orbit sizes, improper saturation, and proper periods	24
Direct counts for sampled collision-orbit graphs	120
Bounds against sampled complete automaton orbits	120
Candidate residue-recurrence instances	416
Total	42960

The 24 witness checks are three checks for each of eight automata:

$$(n, k) \in \{(7, 3), (8, 3), (9, 3), (10, 3), (7, 4), (8, 4), (7, 5), (8, 5)\}.$$

Their complete breadth-first searches visited 605713 coloring states in total. Each search finishes the entire orbit; none reports a truncated partial orbit as complete. The program’s explicit state limit is a guard that raises an error, not a stopping rule that manufactures a result.

The 120 graph tests and 120 automaton-bound tests use a recorded, deterministic random seed and small state spaces where literal enumeration is possible. They are regression tests, not an exhaustive classification of all automata in their size ranges. The exact infinite balancing proof and the structural exclusion argument do not depend on their success.

35.5 Comparison with Parts I and II and between the manuscripts

In the write phase the shipped tables were compared by exact integer arithmetic with each other and with $D_k^*(n)$ computed from (32.1). All 919 rows of `data/03-eventual-finite-certificates.csv` have $D_k^{\text{cert}}(n) = D_k^*(n)$ equal to the formula and the split (a_n, b_n) . All 372 rows of Part I's `data/finite-range_bounds.csv` have `upper = lower =  $k^n - D_k^*(n)$` and $(\ell, m) = (a_n, b_n)$; 219 of them ($k \leq 12$) are also rows of manuscript 01's table, and 95 ($4 \leq k \leq 7$) rows of manuscript 06's. Manuscript 06's 2,217 rows have the split (a_n, b_n) , and their `target_gap` equals $D_k^*(n)$ in all 1,025 rows with $n \leq 300$ that were recomputed; 375 of them are also rows of manuscript 01's table, with equal values. The 23 values of `data/05-quartic-threshold-sample_values.csv` equal $k^n - D_k^*(n)$. The 25 rows of manuscript 03's table (table 10) equal $D_3^*(n) = D_n$ and Part II's rows. These are consistency checks of independently written computations, not part of any proof.

36 A second route: eventual exactness for every fixed $k \geq 3$

This section is manuscript 01's, the base of its argument. It proves the exact formula for every fixed $k \geq 3$ and all sufficiently large n , without an explicit threshold. For $k \geq 4$, theorem 33.4 supersedes it with $N(k) \leq 3k^4$; for $k = 3$, Part II's theorem 13.1 gives every $n \geq 7$. It remains a genuinely different route: a compactness argument over graph parameters instead of explicit estimates, valid for every fixed $k \geq 3$ (not uniformly in k), and it does not presume that a singular generator has rank $n - 1$. Manuscript 01 stresses that Davies's two binary questions, nonattainment of k^n (Part I) and exactness of the two-cycle lower bound for $n \geq 7$, are not the same question: the second requires controlling virtually all colorings, not merely exhibiting a missing one. Manuscript 01 also does not assert that the transition monoid of every extremal DFAO must equal a particular $U_{a,b}$: it proves an exact coloring-saturation criterion (theorem 36.4), and monoid classification is a stronger question.

Theorem 36.1 (Eventual exact optimum; manuscript 01). *For each fixed integer $k \geq 3$, there is an integer $N_{\exists}(k)$ such that*

$$D_k(n) = D_k^*(n), \quad R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n) \quad (36.1)$$

for all $n \geq N_{\exists}(k)$. One may require $N_{\exists}(k) \geq \max(7, k + 1)$. The same maximum is obtained if one maximizes the coloring-orbit size over all transformation pairs and all k -valued output maps, without first requiring accessibility, minimality, or surjectivity.

The last sentence matters: the upper bound is genuinely universal, whereas the equality witness is an actual accessible minimal DFAO (proposition 30.4). No tacit passage from an abstract monoid to a realizable minimal automaton is needed. The proof of theorem 36.1 does not depend on the correctness of the repository's other mathematical projects; manuscript 01 reproves the needed graph and counting facts, and its one non-elementary imported assertion is the published fact that $U_{a,b}$ is two-generated for $1 < a < b$, $\gcd(a, b) = 1$ [2, Theorem 8].

Proposition 36.2 (Structural separation; manuscript 01). *Fix $k \geq 3$. For all sufficiently large n , any transformation pair and output map whose defect is at most $D_k^*(n)$ has, after possibly exchanging its generators, one permutation p and one singular map s . Moreover, p consists of exactly two coprime cycles. Every collision pair of s connects those two cycles.*

Proof. Suppose there are arbitrarily large exceptions, and select a sequence of them with $n \rightarrow \infty$. By lemma 32.6, their defects are $O_k(\lambda_k^n)$. The unmixed and nonsurjective cases are excluded by remark 29.3.

Choose any collision pair of the singular generator. By the collision bound and lemma 29.5, its graph satisfies

$$\frac{P_G(k)}{\lambda_k^n} \leq \frac{D_k^*(n)}{\lambda_k^n} + o(1). \quad (36.2)$$

The same-cycle case is excluded by lemma 33.3(i),(iv) in manuscript 01's form for $k \geq 3$ (section 33). We therefore have $G = dK_{u,v} \sqcup rK_1$, $n = d(u + v) + r$, with integers $d \geq 1$, $r \geq 0$. From lemma 32.4, with $\varrho_k = k/\lambda_k$,

$$\frac{P_G(k)}{\lambda_k^n} \geq 2^d \varrho_k^r. \quad (36.3)$$

Thus d and r are bounded. Pass to a subsequence on which both are constant, and, where needed, $n \bmod 4$ is constant.

Both u and v tend to infinity on this subsequence. Indeed, if one stays bounded, take a further subsequence on which it is constant. Color that part with one color and the other part freely with the remaining $k - 1$. This gives at least $k(k - 1)^{\max(u,v)}$ colorings per component. Since $k - 1 > \lambda_k$, the normalized total tends to infinity, contradicting (36.2).

If k is even, let $C = \binom{k}{k/2}$. The sharpened palette lower bound (32.13) and boundedness of d give $P_G(k)/\lambda_k^n \geq (1 - o(1))C^d 2^r$. The witness limit in (32.15) is C . Since $C \geq 6$, either $d \geq 2$ or $r \geq 1$ gives a strictly larger limiting lower bound. Therefore $d = 1$, $r = 0$.

Now suppose $k = 2h + 1$ is odd. Write $C = \binom{k}{h}$ and $c = \cosh \omega_h$. Then

$$0 < \omega_h \leq \log 2, \quad 1 < c \leq \frac{5}{4}, \quad \varrho_k = 2 \cosh(\omega_h/2).$$

By (32.14),

$$\frac{P_G(k)}{\lambda_k^n} \geq (1 - o(1))(2C)^d \varrho_k^r \cosh^d\left(\frac{(v - u)\omega_h}{2}\right). \quad (36.4)$$

The witness constant is at most $2C \cosh(2\omega_h) \leq 2C \cdot 17/8$. If $d \geq 2$, the lower coefficient is at least $(2C)^2$, larger than this because $C \geq 3$. If $r \geq 2$, it is at least $2C \varrho_k^2 \geq 8C$, also larger. It remains to exclude the single delicate case $d = 1$, $r = 1$.

If n is odd, its witness constant is $2C \cosh(\omega_h/2)$, whereas (36.4) is at least $2C \varrho_k = 4C \cosh(\omega_h/2)$, already larger. If n is even, then $u + v = n - 1$ is odd, so $|v - u| \geq 1$. The factor after $2C$ on the right of (36.4) is at least

$$\varrho_k \cosh(\omega_h/2) = 2 \cosh^2(\omega_h/2) = 1 + c.$$

For $n \equiv 0 \pmod{4}$, the witness factor is $c < 1 + c$. For $n \equiv 2 \pmod{4}$, it is $\cosh(2\omega_h) = 2c^2 - 1$, and

$$(1 + c) - (2c^2 - 1) = 2 + c - 2c^2 \geq \frac{1}{8} > 0 \quad (1 \leq c \leq 5/4). \quad (36.5)$$

Again there is a strict separation. Hence $d = 1$, $r = 0$ also for odd k .

Since $d = \gcd(a, b)$ and $r = n - a - b$, these conclusions say that p has exactly two coprime cycles. The collision pair was arbitrary. If an exceptional pair lay on one cycle for arbitrarily large n , choose that pair in the preceding argument and obtain the same-cycle contradiction. Thus every collision pair connects the two cycles, once n is sufficiently large. This contradicts the assumed sequence of exceptions and proves the proposition. $\square$

Remark 36.3 (Why the odd- k margin is necessary; manuscript 01). Merely observing that an isolated vertex multiplies a coloring count by k is not enough. Removing that vertex also changes $n \bmod 4$ and can improve the available coprime imbalance from four to one. Inequality (36.5) compares both effects on the same scale. Omitting it would leave a genuine hole in the structural argument, particularly at $k = 3$. Manuscript 06's effective proof avoids the issue by bounding every nonfull graph by $\frac{7}{4}\hat{C}\lambda_k^n$ (lemma 33.1), and manuscript 03's by the explicit gap (37.4).

Proof of theorem 36.1. For each n there are finitely many transformation pairs and output maps, so the maximum of their coloring-orbit sizes exists. By proposition 30.4, the minimum defect among them is at most $D_k^*(n)$. For fixed k and sufficiently large n , proposition 36.2 applies to an optimizer. Its permutation has exactly two coprime cycles, say of lengths $a < b$, and any collision pair produces $K_{a,b}$ on the whole state set.

The proper-period bound of lemma 30.3 sharpens the chromatic bound to

$$D \geq \mathcal{B}_k(a, b) - G_k(a, b).$$

This holds even if τ is improper, because then the true subtraction is zero and the displayed bound is weaker. By lemma 31.7,

$$D \geq \mathcal{B}_k(a_n, b_n) - G_k(a_n, b_n) = D_k^*(n).$$

The opposite inequality comes from the witness, so equality holds. Since the witness is accessible and minimal, the unrestricted algebraic maximum and the automata maximum coincide. $\square$

Theorem 36.4 (Criterion for eventual extremizers; manuscript 01). *For fixed $k \geq 3$ and sufficiently large n , a transformation pair and output map attain the unrestricted maximum if and only if, after exchanging the generators when necessary, the following hold:*

1. *The permutation generator has two cycles of lengths a_n, b_n .*
2. *The output map is proper for the corresponding K_{a_n, b_n} and its permutation orbit has length $G_k(a_n, b_n)$.*
3. *Every improper coloring of that complete bipartite graph is reachable.*

For every maximizing pair, each nontrivial kernel class of the singular generator has size two and meets both cycles.

Proof. Necessity of the two-cycle structure follows from proposition 36.2. A split other than (a_n, b_n) would make the lower defect bound strictly larger by corollary 31.6 and lemma 31.7. Equality in the chromatic bound requires both full reachability of improper colorings and the maximum allowed number of proper ones. The latter forces τ itself to be proper and have the stated period.

Conversely, the three properties give exactly $k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n)$ reached colorings, which is the maximum by theorem 36.1.

Finally, proposition 36.2 prohibits a kernel pair within either cycle. A kernel class of three or more points would contain such a pair by the pigeonhole principle. Any nontrivial class is therefore a cross pair. $\square$

Manuscript 01 notes that this criterion does not force the singular map to have exactly one collision pair: several disjoint cross pairs are not excluded by the kernel argument, and whether full improper-coloring saturation permits them is a separate algebraic problem (its question on rank and kernel structure, question 44.6). That problem is now answered: at $k = 3$, for every $n \geq 7$, by Part II's theorem 19.1(ii), and for $k \geq 4$ on manuscript 06's exactness range by theorem 39.1(ii) below, whose proof exploits exactly the full improper saturation of (3). Both show rank $n - 1$ with one cross kernel pair.

37 A third route: uniform separation and sharp constants

This section is manuscript 03's fixed- k theory. Its separation step is a second route to the exclusions of proposition 36.2, organized as a compactness lemma with explicit quantifiers; its conclusions are the sharp first-order asymptotics for every fixed $k \geq 3$ (printed once, as theorem 41.1), eventual exactness for fixed odd k , and near-optimal rigidity. Manuscript 03 distinguishes three statements that can otherwise be confused: proving $R_2(n, k) < k^n$ is nonattainment (Part I); determining the first-order behavior of

$k^n - R_2(n, k)$ is a sharp deficit asymptotic; proving an exact formula for $R_2(n, k)$ is an extremal theorem. It reaches the third level for all $n \geq 7$ when $k = 3$, eventually for each fixed odd k , and the second level for every fixed $k \geq 3$. The meaning of “fixed” is essential: none of its asymptotic or eventual statements is uniform in a number of outputs growing with n , and none settles the monoid-size problem on the diagonal $k = n$. In its words, the execution of two checkers adds reproducibility, not a second source of mathematical authority.

37.1 The cyclic correction cannot conceal a large graph count

Fix $k \geq 3$. For a two-cycle graph, (32.11) and lemma 29.4 give

$$P_G(k) \geq 2^d \lambda_k^{n-r} k^r, \quad \Pi \leq n^2 3^{r/3},$$

where Π bounds the reachable proper colorings of the pure orbit. Consequently, with $\eta = \min\{\lambda_k, k/3^{1/3}\} > 1$, we have the uniform bound

$$\frac{\Pi}{P_G(k)} \leq \frac{n^2}{2\eta^n} \longrightarrow 0. \quad (37.1)$$

No assumption that d , r , or the imbalance is bounded was used. Also

$$\frac{P_G(k)}{\lambda_k^n} \geq 2^d (k/\lambda_k)^r \geq 2^{d+r}. \quad (37.2)$$

For one-cycle graphs, $s^L + (-1)^L s \geq (s^3 - s)^{L/3} = \beta_k^L$ for $s = k - 1 \geq 2$, $L \geq 2$, and $\beta_k > \lambda_k$ (section 33), so

$$P_G(k) \geq \beta_k^{n-r} k^r \geq \beta_k^n, \quad \frac{\Pi}{P_G(k)} \leq \frac{n}{\min\{\beta_k, k/3^{1/3}\}^n} \longrightarrow 0.$$

Every one-cycle collision graph forces a deficit exponentially larger than λ_k^n . The two easy generator cases have the same exclusion property by lemma 29.2.

Lemma 37.1 (Reduction of bounded normalized deficits; manuscript 03). *Consider a sequence of binary DFAOs with fixed output size k and $n \rightarrow \infty$, whose deficits are $O(\lambda_k^n)$. For all sufficiently large n the generators are mixed. Choose any collision of the singular generator for each n . Its endpoints must lie on different permutation cycles. The associated $d + r$ is bounded. Along any subsequence on which d, r are fixed, both reduced cycle lengths $u = a/d$ and $v = b/d$ tend to infinity.*

Proof. The easy and one-cycle cases were just excluded. In a two-cycle case, (37.1) and the collision obstruction show $\text{def} \geq (1 - o(1))P_G(k)$. Thus $P_G(k) = O(\lambda_k^n)$, and (37.2) bounds $d + r$.

Suppose, after passing to a fixed- d, r subsequence, that u is bounded. Make its part monochromatic and use any of the other $k - 1$ colors on the opposite part. This gives $\mathcal{B}_k(u, v) \geq k(k - 1)^v$. Since $dv = n - r - du$ and du stays bounded,

$$\frac{P_G(k)}{\lambda_k^n} \geq \frac{k^{d+r} (k - 1)^{dv}}{\lambda_k^n} \longrightarrow \infty,$$

because $k - 1 > \lambda_k$. This is a contradiction. Interchanging the parts handles bounded v . If u or v failed to tend to infinity it would have a bounded subsequence, already excluded. $\square$

This lemma is the reason pointwise asymptotics suffice in the next step. An arbitrary sequence of prospective counterexamples has a subsequence with fixed d, r and with both reduced lengths tending to infinity. There are no hidden exceptional graphs with growing numbers of components or with a bounded side.

37.2 Even output size

Let $k = 2h \geq 4$, so $\lambda_k = h$, and $C = \binom{2h}{h}$. The lower witness has deficit $(C + o(1))h^n$. Take any sequence of optimal automata and apply lemma 37.1. Along a subsequence with fixed d, r , (32.13) implies

$$\frac{\text{def}}{h^n} \geq (1 - o(1))(C - o(1))^d 2^r. \quad (37.3)$$

For $d \geq 2$ this has lower limit at least $C^2 > C$; for $r \geq 1$ it has lower limit at least $2C > C$. Hence such parameter values cannot occur infinitely often for optimal automata. The only possibility is $d = 1, r = 0$, where the lower limit in (37.3) is C . This proves the even case of theorem 41.1 by comparison with the witness.

More formally, if the asserted liminf bound failed, a sequence violating it by a fixed positive amount would admit the fixed-parameter subsequence above and contradict (37.3). The same argument shows that any sequence whose deficit is asymptotic to the optimum eventually has only spanning coprime two-cycle collision graphs.

37.3 Odd output size: a strict separation of every other shape

Let $k = 2h+1, x = h(h+1), \lambda_k = \sqrt{x}, C = \binom{2h+1}{h}, t_h = (h+1)/h$, and consider $z \mapsto t_h^z + t_h^{-z} = 2 \cosh(z\omega_h)$, strictly increasing on $(0, \infty)$. The lower witness has normalized deficit $2C \cosh(\delta_n \omega_h) + o(1)$. Its coefficient is at most $(17/4)C$, since $1 < t_h \leq 2$ and $\delta_n \leq 2$.

Take a hypothetical infinite sequence of optimal or asymptotically optimal automata with a competing graph. Pass to a subsequence with fixed d, r and fixed residue class of n modulo four; both u, v tend to infinity by lemma 37.1. We now compare its lower-limit coefficient with the witness coefficient.

If $d \geq 2$, the inequality $2 \cosh \geq 2$ and (32.14) give a coefficient at least $(2C)^d (k/\lambda_k)^r$, which is at least $4C^2 > (17/4)C$. If $d = 1, r \geq 2$, the coefficient is at least $2C(k/\lambda_k)^2 \geq 8C > (17/4)C$. These are strict gaps.

Suppose $d = 1, r = 1$. If n is odd, $a + b = n - 1$ is even, so the best possible coefficient is at least $2Ck/\lambda_k$. The witness coefficient for odd n is

$$2C \cosh(\omega_h/2) = C \left(\sqrt{t_h} + \frac{1}{\sqrt{t_h}} \right) = Ck/\lambda_k,$$

exactly half that lower bound. If n is even, $a + b$ is odd, and the coefficient is at least

$$C(k/\lambda_k) \cdot 2 \cosh(\omega_h/2) = C(k/\lambda_k)^2 = C(4 + 1/x).$$

The larger of the even- n witness coefficients is $2C \cosh(2\omega_h) = C(2 + 4/x + 1/x^2)$. Their difference is

$$C(2 - 3/x - 1/x^2) > 0 \quad (x \geq 2). \quad (37.4)$$

Again there is a strict gap, including at $h = 1$.

The only remaining case is $d = 1, r = 0$. Then $\gcd(a, b) = 1$ and $a + b = n$. Writing $z = |b - a|/2$, admissibility implies $z \geq \delta_n$. Hence (32.14) gives a normalized lower limit at least $2C \cosh(\delta_n \omega_h)$. If the split is not the nearest one, the lattice of possible imbalances gives $z \geq \delta_n + 1$. Since $\cosh$ is strictly increasing on positive arguments, this gives another fixed positive gap for the chosen residue class. If z is unbounded the inequality is stronger still.

These comparisons prove the odd case of theorem 41.1. They also prove something stronger: for all sufficiently large n , every optimal automaton must have the nearest coprime two-cycle permutation. Otherwise a subsequence of counterexamples would fall into one of the strictly separated cases. The witness's error tends to zero, whereas the appropriate separation is a fixed positive number for fixed k and residue class.

37.4 From asymptotic separation to exact odd-output optimality

Theorem 37.2 (Eventual exactness for odd output size; manuscript 03). *For each fixed odd $k \geq 3$ there is an integer $N_{\exists}(k)$ such that, for $n \geq N_{\exists}(k)$,*

$$R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n). \quad (37.5)$$

One may take $N_{\exists}(3) = 7$ (theorem 38.1). Manuscript 03 does not compute a numerical $N_{\exists}(k)$ for $k \geq 5$; the proof is an eventual separation argument.

Proof. For an optimal automaton in the surviving graph class, the graph count is exactly $\mathcal{B}_k(a_n, b_n)$. The proper cyclic orbit has size at most $a_n b_n$ when $k \geq 5$, and at most b_n when $k = 3$. Therefore the collision bound gives the exact upper bound $R_2(n, k) \leq k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n)$. The established lower witness (Davies’s Corollary 3, manuscript 03’s Proposition 2.3) attains the right-hand side, and it is minimal by lemma 29.1 once its orbit exceeds k^{n-1} , which holds for all sufficiently large n with k fixed by the estimates above. Theorem 38.1 supplies $N_{\exists}(3) = 7$. $\square$

The argument does not merely infer an exact equality from a first-order asymptotic. It first forces the optimizer into one particular finite-parameter graph class by a strict constant gap; within that class it uses an exact chromatic count and an exact available cyclic correction. Manuscript 03 calls this distinction indispensable. For odd $k \geq 5$ the theorem is now a special case of theorem 33.4 (with $N(k)$ explicit, and every $n \geq \max(7, k + 1)$ when $k = 5, 7$); its proof is a third route.

37.5 Near-optimal structural rigidity

Corollary 37.3 (Manuscript 03). *Fix $k \geq 3$. Suppose a sequence of binary n -state DFAOs, with transitions p_n, s_n and outputs τ_n , satisfies $\text{def}(\tau_n; p_n, s_n)/D_k(n) \rightarrow 1$. For all sufficiently large n , one generator is a permutation with exactly two coprime cycles, and the other is singular and injective on each cycle separately. If k is odd, the two lengths are exactly the nearest split (28.1) for all sufficiently large n .*

Proof. Repeat the constant-gap arguments, which apply equally to a sequence with relative error tending to zero. A selected collision must span the two coprime cycles. Any collision within one of those cycles would give the exponentially more expensive one-cycle obstruction, a contradiction. The odd-output imbalance gap forces the nearest split. No assertion that the same restricted structure suffices for near-optimality is made. $\square$

For even k the corollary does not force the nearest split, and it cannot: theorem 42.1 exhibits a whole interval of asymptotically optimal splits within the witness family. Manuscript 06’s exact rigidity (theorem 39.1) forces the nearest split for exact maximizers at every $k \geq 4$ in its range.

38 A second proof of Part II’s three-output theorem

Manuscript 03’s headline is Part II’s theorem 13.1, proved independently: an analytic argument for $n \geq 32$ and an exact certificate of 3,717 structural inequalities for $7 \leq n \leq 31$. Part II’s proof is analytic with six displayed base cases (section 25). Both are kept; this section is the second proof.

Theorem 38.1 (Exact three-output complexity; manuscript 03). *For every integer $n \geq 7$,*

$$R_2(n, 3) = 3^n - \mathcal{B}_3(a_n, b_n) + b_n, \quad \mathcal{B}_3(a, b) = 3(2^a + 2^b - 2). \quad (38.1)$$

The proof for $n \geq 32$ is mathematical and does not depend on an enumerated range. For $7 \leq n \leq 31$ it uses the complete exact certificate of section 38.3.

This is (13.2): $(a_n, b_n) = (A_n, B_n)$ and $\mathcal{B}_3(a_n, b_n) - b_n = D_n$. The formula for $\mathcal{B}_3$ is Part I's (9.7); manuscript 03 reproves it: if the first part is monochromatic there are $3 \cdot 2^b$ choices; if it uses exactly two colors there are $3(2^a - 2)$ choices, and the second part is forced to use the remaining color.

38.1 A finite structural reduction

Proposition 38.2 (Finite structural reduction for three outputs; manuscript 03). *For a fixed $n \geq 3$, a valid lower bound for every individual three-output deficit is the minimum of the following finite list:*

$$2 \cdot 3^{n-1}, \quad 4 \cdot 3^{n-2} - 1; \quad (38.2)$$

$$(2^{m/d} + (-1)^{m/d} 2)^d 3^{n-m} - H_{n-m}(m), \quad 2 \leq m \leq n, \quad d \mid m, \quad d < m; \quad (38.3)$$

$$\mathcal{B}_3(a/d, b/d)^d 3^r - \begin{cases} \max\{H_r(a), H_r(b)\}, & d = 1, \\ H_r(\text{lcm}(a, b)), & d > 1, \end{cases} \quad (38.4)$$

where $1 \leq a \leq b$, $a + b \leq n$, $d = \gcd(a, b)$, and $r = n - a - b$. Every pair of transformations is covered; no transitivity, rank, or accessibility hypothesis is omitted from this upper-bound reduction.

Proof. The two generators are either both permutations, both singular, or one of each. Apply lemma 29.2 in the first two cases. In the mixed case choose any collision of the singular map, classify it by (9.4) or (9.6), and use the collision obstruction and the three-color period restriction, which is Part I's (9.10): in the two-cycle case with $d = 1$, at least one part of a proper coloring of $K_{a,b}$ is monochromatic, so the period on the occupied vertices divides a or b , and the remaining r vertices contribute the order of a permutation on them; if the initial coloring is not proper, the bound is automatic. The listed parameter ranges contain its type. $\square$

This is Part I's collection (9.11)–(9.15) at $k = 3$, with the two-permutation entry $3^n - M(n, 3)$ replaced by the weaker $2 \cdot 3^{n-1}$ and without the nonsurjective entry, which manuscript 03 does not list (a nonsurjective map has deficit at least $3^n - 2^n$, Part II's lemma 15.1, far above the target). Manuscript 03 remarks that it is not legitimate to replace b by ab in the $d = 1$ entry and expect a sharp three-output answer: the discrepancy is only polynomial, but the exact extremal formula retains precisely that polynomial correction.

38.2 An analytic proof beyond the cutoff

This subsection proves theorem 38.1 for every $n \geq 32$. Set $\Lambda_n = 2^{n/2}$. The desired minimum deficit is $\mathcal{B}_3(a_n, b_n) - b_n$. From (28.1),

$$\mathcal{B}_3(a_n, b_n) = \begin{cases} (9/\sqrt{2})\Lambda_n - 6, & n \text{ odd}, \\ (15/2)\Lambda_n - 6, & n \equiv 0 \pmod{4}, \\ (51/4)\Lambda_n - 6, & n \equiv 2 \pmod{4}. \end{cases} \quad (38.5)$$

We prove that every graph other than a spanning, coprime $K_{a,b}$ forces a deficit strictly larger even than $\mathcal{B}_3(a_n, b_n)$. Throughout, lemma 29.4 gives $\Pi \leq n3^{r/3}$ in the one-cycle case and $\Pi \leq n^2 3^{r/3}$ in the two-cycle case.

Excluding one-cycle collision graphs. Put $\beta_3 = 6^{1/3}$. For every $L \geq 2$,

$$2^L + (-1)^L 2 \geq \beta_3^L. \quad (38.6)$$

For even L this follows from $2 > \beta_3$. For odd $L \geq 3$, $2^L - 2 \geq 2^L(3/4) \geq 2^L(3/4)^{L/3} = 6^{L/3}$. Consequently $P_G(3) \geq \beta_3^{n-r} 3^r$. Since $3^{2/3} > \beta_3$ and $\beta_3^n \geq n$ in the range in question,

$$P_G(3) - \Pi \geq 3^{r/3} \left[\beta_3^n \left(\frac{3^{2/3}}{\beta_3} \right)^r - n \right] \geq \beta_3^n - n. \quad (38.7)$$

For $n \geq 12$,

$$\left(\frac{\beta_3}{\sqrt{2}} \right)^n \geq \left(\frac{\beta_3}{\sqrt{2}} \right)^{12} = \frac{81}{4}, \quad \frac{n}{\Lambda_n} \leq 1.$$

Hence $\beta_3^n - n \geq (77/4)\Lambda_n > (51/4)\Lambda_n \geq \mathcal{B}_3(a_n, b_n)$. This already excludes all one-cycle graphs for $n \geq 12$. (Part II's (16.1) is the same bound $C_L \geq 6^{L/3}$, proved by induction.)

The easy generator bounds also exceed $\mathcal{B}_3(a_n, b_n)$ for $n \geq 32$. For instance, $3/\sqrt{2} > 2$, so each of $2 \cdot 3^{n-1}$ and $4 \cdot 3^{n-2} - 1$ dominates $(51/4)\Lambda_n$ in that range. Their ratios to Λ_n are increasing, and the comparison at $n = 32$ is immediate.

A uniformly small cyclic correction for two-cycle graphs. Write $a = du$, $b = dv$, $r = n - d(u + v)$. For $u, v \geq 1$,

$$\mathcal{B}_3(u, v) = 3(2^u + 2^v - 2) \geq 3 \cdot 2^{(u+v)/2} \geq 2 \cdot 2^{(u+v)/2}. \quad (38.8)$$

Indeed $2^u + 2^v \geq 2 \cdot 2^{(u+v)/2}$ and $2^{(u+v)/2} \geq 2$. Thus

$$P_G(3) \geq 2^d 2^{(n-r)/2} 3^r, \quad \frac{\Pi}{P_G(3)} \leq \frac{n^2}{2^d \Lambda_n} \left(\frac{\sqrt{2}}{3^{2/3}} \right)^r \leq \varepsilon_n := \frac{n^2}{2\Lambda_n}.$$

The sequence ε_n decreases for $n \geq 32$, and $\varepsilon_{32} = 1/128$. It follows that throughout this range

$$P_G(3) - \Pi \geq \frac{127}{128} P_G(3). \quad (38.9)$$

Lemma 38.3 (Manuscript 03). *For $n \geq 32$, a two-cycle graph with $(d, r) \neq (1, 0)$ satisfies*

$$P_G(3) \geq \begin{cases} 9\sqrt{2}\Lambda_n - 18, & n \text{ odd}, \\ (27/2)\Lambda_n - 18, & n \text{ even}. \end{cases} \quad (38.10)$$

Proof. The slightly stronger first inequality in (38.8) gives $P_G(3) \geq 3^d \Lambda_n (3/\sqrt{2})^r$. For $d \geq 3$ the coefficient is at least 27. For $d = 2$, $r \geq 1$ it is at least $27/\sqrt{2}$. For $d = 1$, $r \geq 2$ it is at least $27/2$. Each suffices.

If $d = 2$, $r = 0$, put $w = u + v = n/2 \geq 4$. The arithmetic–geometric mean inequality gives

$$\frac{P_G(3)}{\Lambda_n} = \left(\frac{\mathcal{B}_3(u, v)}{2^{w/2}} \right)^2 \geq (6 - 6/2^{w/2})^2 \geq (9/2)^2 = 81/4,$$

which again suffices. This case has even n .

The remaining case is $d = 1$, $r = 1$. Then $P_G(3) = 9(2^a + 2^b - 2)$ and $a + b = n - 1$. Among integer splits of a fixed sum, $2^a + 2^b$ is smallest at a balanced split. If n is odd this yields $9\sqrt{2}\Lambda_n - 18$; if n is even it yields $(27/2)\Lambda_n - 18$. $\square$

For even n , (38.9) and (38.10) imply a deficit strictly exceeding the largest even target in (38.5), since

$$\frac{127}{128} \left(\frac{27}{2} \Lambda_n - 18 \right) - \left(\frac{51}{4} \Lambda_n - 6 \right) = \frac{165}{256} \Lambda_n - \frac{759}{64} > 0. \quad (38.11)$$

At $\Lambda_n = 2^{16}$ this margin is $2702601/64$; it increases with Λ_n . For odd n , the analogous difference is

$$\frac{127}{128}(9\sqrt{2}\Lambda_n - 18) - \left(\frac{9}{\sqrt{2}}\Lambda_n - 6\right) = \frac{567}{64\sqrt{2}}\Lambda_n - \frac{759}{64} > 0$$

in the same range. All excluded graph types therefore force a deficit larger than $\mathcal{B}_3(a_n, b_n)$, not merely larger than $\mathcal{B}_3(a_n, b_n) - b_n$.

Optimizing the only remaining graph. The sole remaining possibility is $r = 0, d = 1$: two coprime cycles covering all n states, and graph $K_{a,b}$. Assume $a \leq b$. By the three-color period restriction, its deficit is at least $F_n(a, b) = 3(2^a + 2^b - 2) - b$. If $a \geq 2$ and $a \leq b$, moving one vertex from the smaller part to the larger part changes this quantity by

$$F_n(a - 1, b + 1) - F_n(a, b) = 3(2^b - 2^{a-1}) - 1 > 0. \quad (38.12)$$

It is therefore strictly minimized over coprime splits by the closest such split, namely (28.1) (Part II's lemma 18.1 is the same computation). The published lower witness supplies an attaining automaton, and lemma 29.1 applies: the finite certificate and the analytic comparisons give $\mathcal{B}_3(a_n, b_n) - b_n < 2 \cdot 3^{n-1}$ for every $n \geq 7$. This proves theorem 38.1 for every $n \geq 32$.

38.3 The exact finite certificate

The infinite proof leaves exactly the 25 integers $7 \leq n \leq 31$. The certificate is not a sample of automata. It enumerates the entire finite list of proposition 38.2 for each of these integers. The sum of the list lengths is 3,717. Each number is evaluated in integer arithmetic, and in every row the smallest bound is exactly $\mathcal{B}_3(a_n, b_n) - b_n$. Moreover, the only minimizing graph parameters are $r = 0, d = 1, (a, b) = (a_n, b_n)$.

Why the order calculation is exhaustive. The order of a permutation is the least common multiple of its cycle lengths. Enumerating every nondecreasing partition of r into positive integers therefore produces precisely O_r . The producer uses that description. The independent checker instead uses the recurrence

$$O_0 = \{1\}, \quad O_r = \bigcup_{j=1}^r \{\text{lcm}(j, o) : o \in O_{r-j}\}. \quad (38.13)$$

For one inclusion, append a j -cycle to a permutation on $r - j$ points. For the other, remove any cycle from a permutation on r points. Repeated partitions cause no difficulty because the result is a set of orders. Taking the maximum in (9.9) now evaluates the exact period bound, not a numerical approximation to a Landau function.

The independent implementation also changes the chromatic calculation: it reconstructs $\mathcal{B}_3(a, b)$ by inclusion–exclusion for onto maps instead of using the closed form. It enumerates one-cycle cases by component length rather than component count, and two-cycle cases by total occupied size rather than first cycle length. It compares every parameter record and integer value with the producer's certificate. Finally, it finds the optimal coprime split directly, without calling the nearest-split formula.

Table 10: The complete finite range of manuscript 03 (the shipped data/04-deficit-asymptotics-finite_table.tex, input verbatim). Each row records its minimizing deficit $D_3(n) = D_n$ and the number of structural cases checked.

n	(a_n, b_n)	$D_3(n)$	Cases
7	(3,4)	62	23
8	(3,5)	109	30
9	(4,5)	133	36
10	(3,7)	395	44
11	(5,6)	276	50
12	(5,7)	467	61
13	(6,7)	563	68
14	(5,9)	1,617	78
15	(7,8)	1,138	88
16	(7,9)	1,905	100
17	(8,9)	2,289	109
18	(7,11)	6,511	123
19	(9,10)	4,592	133
20	(9,11)	7,663	148
21	(10,11)	9,199	161
22	(9,13)	26,093	175
23	(11,12)	18,414	187
24	(11,13)	30,701	206
25	(12,13)	36,845	220
26	(11,15)	104,427	236
27	(13,14)	73,708	252
28	(13,15)	122,859	271
29	(14,15)	147,435	286
30	(13,17)	417,769	308
31	(15,16)	294,890	324

What the certificate proves, and what it does not. By proposition 38.2, every automaton's deficit is at least one of the listed structural bounds and therefore at least their minimum. The computed minimum is the target, and the established lower witness attains the corresponding number of states. This proves the exact answer for each checked n ; with the analytic tail it proves theorem 38.1 for all $n \geq 7$. This is a computer-assisted finite step in a mathematical proof. It depends on the displayed reduction, the elementary order recurrence, the correctness of the two programs, and their execution. It is not a proof in Lean, and two implementations written during one investigation are not two external referees. The machine-readable certificate (data/04-deficit-asymptotics-finite_certificate.json), producer, independent checker, and their recorded outputs are shipped so these obligations can be examined and rerun. The rows agree with Part II's data/02-three-output-structural_checks.csv and, for $n \leq 30$, with Part I's table (section 35.5).

38.4 Residue formulas and concrete examples

Expanding (38.1) gives manuscript 03's deficit formula

$$D_3(n) = \begin{cases} 9 \cdot 2^{(n-1)/2} - (n+1)/2 - 6, & n \text{ odd}, \\ \frac{15}{2} 2^{n/2} - n/2 - 7, & n \equiv 0 \pmod{4}, \\ \frac{51}{4} 2^{n/2} - n/2 - 8, & n \equiv 2 \pmod{4}, \end{cases} \quad (38.14)$$

which is Part II's (13.4). The linear subtractions are part of the exact theorem; they do not arise from an asymptotic approximation. Manuscript 01 writes the same law on classes modulo four: with $n = 4t + r$,

$$\begin{aligned} D_3(4t) &= \frac{15}{2} 4^t - 2t - 7, & D_3(4t+1) &= 9 \cdot 4^t - 2t - 7, \\ D_3(4t+2) &= \frac{51}{4} 4^t - 2t - 9, & D_3(4t+3) &= 18 \cdot 4^t - 2t - 8. \end{aligned} \quad (38.15)$$

The half-integer coefficients in the even classes cause no integrality problem: the admissible values of t have 4^t divisible by two. Equivalently, on the odd integers, $D_3(2t+1) = 9 \cdot 2^t - t - 7$. (Manuscript 01 states these for D_3 eventually and for D_3^* throughout $n \geq 7$; with Part II they hold for D_3 at every $n \geq 7$. The write phase checked (38.15) and (38.14) against $D_3^*(n)$ for $7 \leq n \leq 199$.) Manuscript 01 adds that the normalized deficit has three limits, $9/\sqrt{2}$, $15/2$ and $51/4$, on odd n , $n \equiv 0 \pmod{4}$ and $n \equiv 2 \pmod{4}$ (Part II's corollary 21.1), so there is no single constant c with $D_3(n) \sim c(\sqrt{2})^n$ along all integers, and replacing the assertion by merely $D_3(n) = \Theta((\sqrt{2})^n)$ would lose genuine arithmetic information.

At $n = 7$, the split is (3, 4), the proper-coloring count is 66, and $R_2(7, 3) = 2187 - 66 + 4 = 2125$. At $n = 8$, the split is (3, 5), giving $R_2(8, 3) = 6561 - 114 + 5 = 6452$. At $n = 10$, the split is (3, 7) rather than (4, 6), because the latter is not coprime; the deficit is 395. At $n = 11$ the deficit drops to 276. Thus $D_3(n)$ itself need not be monotone, despite its exponential scale (as Part II also notes in section 18.1). These values follow from the proved formula, not from interpreting a random search as exhaustive.

38.5 Equality and rigidity for three outputs

Theorem 38.4 (Necessary structure of every optimizer; manuscript 03). *Let $n \geq 7$. If a binary three-output DFAO has reversal complexity $R_2(n, 3)$, then, after possibly interchanging the two input letters:*

- (i) *One letter is a permutation p , and the other is singular, s .*
- (ii) *p has exactly two cycles, of lengths a_n and b_n .*
- (iii) *The restriction of s to each of these two cycles is injective.*
- (iv) *The output coloring is constant on the smaller cycle, uses the other two colors on the larger cycle, and its cyclic word on the larger cycle has least period b_n .*
- (v) *Every coloring that uses some color on both cycles belongs to the reversal orbit.*

Conversely, an accessible automaton with the stated two-cycle permutation, the coloring in (iv), and a singular generator whose orbit contains every coloring in (v), attains $R_2(n, 3)$.

Proof. The bounds for two permutations and two singular maps are strictly larger than the optimal deficit in both the certified and analytic ranges. Thus (i) holds. Choose any collision of s . The only structural parameters that can attain the minimum are a spanning coprime cross graph on the nearest split; this is the strict conclusion of the analytic tail and the recorded uniqueness check in section 38.3. It forces (ii). If s had a collision within one cycle, applying the obstruction to that collision would give a strictly larger deficit. This proves (iii).

For the resulting graph K_{a_n, b_n} , equality in the obstruction requires exactly b_n proper colorings in the orbit and all $3^n - \mathcal{B}_3(a_n, b_n)$ improper colorings in the orbit. In particular τ is proper. A proper surjective three-coloring has a monochromatic part. If the larger part were monochromatic, the cyclic orbit would

have size at most $a_n < b_n$. Thus the smaller part is monochromatic, and the larger has least period b_n . This proves (iv) and (v).

Conversely, (iv) gives b_n distinct proper colorings under powers of p . Condition (v) supplies all the improper ones. Their union has the cardinality in (38.1), which is the universal maximum. The original automaton is minimal by lemma 29.1. $\square$

Manuscript 03 stresses that condition (v) is a reachability condition, not a simple classification of all possible singular generators; injectivity on the two blocks alone is not asserted sufficient, and determining which singular maps satisfy the remaining condition is a genuine next problem (its question 44.7; Part II's question 23.2). Part II's theorem 19.1 is stronger: it adds that s has rank $n - 1$ with exactly one cross double fiber, and its (iv) is (19.1). Manuscript 03's converse (sufficiency given (v)) is the equality case Part II's proof also uses; Part II's example 19.3 and manuscript 06's example 39.3 show that (i)–(iv) alone do not imply (v).

39 Rigidity of maximizing automata

This section and the next are manuscript 06's. Let $\mathcal{E}$ denote the proved exactness range for $k \geq 4$:

$$\mathcal{E} = \{(n, k) : k \geq 4, n \geq N(k) \text{ or } [4 \leq k \leq 7 \text{ and } n \geq \max(7, k + 1)]\}. \quad (39.1)$$

The first part of this range is analytic; the additional finite portions carry the computational status stated in section 34.

Theorem 39.1 (Necessary structure at the maximum; manuscript 06). *Suppose $(n, k) \in \mathcal{E}$ and a binary DFAO attains $R_2(n, k)$. After interchanging the two letters if necessary, its transition maps are p, s with the following properties:*

- (i) p is a permutation with exactly two cycles X, Y , of lengths a_n, b_n ;
- (ii) s has rank $n - 1$, and its unique nontrivial kernel class consists of one point of X and one point of Y ;
- (iii) $\tau(X) \cap \tau(Y) = \emptyset$ and the pure- p orbit of τ has size $a_n b_n$;
- (iv) every coloring improper on K_{a_n, b_n} is reached.

These are necessary conditions. Conditions (i)–(iii) alone are not sufficient.

Proof. The strict exclusions show that exactly one letter is a permutation and one is singular. Select any collision pair of the singular map. The same strict exclusions force its graph to be full, so the permutation has exactly two coprime cycles. Strict balancing forces their lengths to be a_n, b_n , proving (i).

No two points in the same cycle can be identified by s , because selecting such a collision pair would produce a strictly excluded same-cycle graph. Therefore every kernel class has at most two points, and every class of size two has one point in each cycle. Distinct nontrivial classes form a cross matching.

If there were at least two such classes, every word containing s would produce a coloring with at least two disjoint monochromatic cross edges: write the word as $u \circ s \circ p^j$ and pull the two kernel pairs back by p^j . There exist colorings with exactly one monochromatic cross edge. For instance, give one chosen cross pair a common color and use disjoint other palettes on the two remaining sides. This uses at most three colors and is possible since $k \geq 4$. Such a coloring cannot arise from a word containing s . On the other hand, equality in the basic collision bound requires τ to be proper; hence all pure- p colorings are proper and cannot supply this missing improper coloring. The maximum is impossible when there are two or more kernel pairs. Thus there is exactly one, proving (ii).

Finally, in the full graph case the upper bound is $k^n - \mathcal{B}_k(a_n, b_n) + a_n b_n$. To attain it, the proper part must contain all $a_n b_n$ possible pure-orbit states and the improper part must contain all $k^n - \mathcal{B}_k(a_n, b_n)$ improper states. Properness is preserved by p , so τ must be proper. This proves (iii) and (iv). $\square$

This is the $k \geq 4$ counterpart of Part II's theorem 19.1, whose (ii) is proved the same way, with a coloring that has exactly one monochromatic cross pair (compare proposition 19.2); at $k \geq 4$ the proper output has a full period $a_n b_n$ instead of Part II's B_n . Manuscript 01's theorem 36.4 gives (i), the proper-output condition and (iv) for every fixed $k \geq 3$ and large n , and shows that together they are also sufficient; (ii) is where manuscript 06 goes further.

Remark 39.2 (Manuscript 06). The theorem constrains the actual two letter transformations, not merely an abstract monoid isomorphic to their transition monoid. The singular map's unique collision is forced to be a cross-cycle collision. Its image and its action away from the collision still contain information not determined by the theorem.

Example 39.3 (Rank one deficiency and a full output period are not enough; manuscript 06). Use states $0, \dots, 6$ and define, in image-list notation,

$$\begin{aligned} p &= (1, 2, 0, 4, 5, 6, 3), \\ s &= (0, 1, 2, 0, 4, 5, 6), \\ \tau &= (0, 0, 1, 2, 2, 2, 3). \end{aligned}$$

The permutation has cycles $X = \{0, 1, 2\}$ and $Y = \{3, 4, 5, 6\}$. The map s has rank six and its only collision is the cross pair $\{0, 3\}$. The output palettes are disjoint, and the output words on the two cycles have least periods three and four. Thus the pure orbit has size twelve.

Nevertheless, X is invariant under both transformations and s is the identity there. Every reachable coloring restricts to one of the three rotations of $(0, 0, 1)$ on X . Many improper colorings fail this necessary condition and are unreachable. In fact, the supplied literal orbit calculation returns 564 reachable colorings (data/05-quartic-threshold-supplementary.json), far below the optimum 15 472.

This is not an artifact of inaccessible or equivalent original states. Taking initial state 3 gives accessibility: rotate through Y , then apply s to reach X , and rotate through X . States in different blocks are already distinguished by their outputs, and states in the same block are distinguished by powers of p , because their cyclic output words are primitive. Thus the original DFAO is minimal.

The example identifies the missing issue in a complete extremizer classification: rank, cycle shape, and initial coloring do not by themselves guarantee that the singular words cover the entire improper family. It is the $k = 4$ analogue of Part II's example 19.3, with the invariant block on the shorter cycle instead of the longer.

40 A rank-sensitive matching obstruction

The proof of rank rigidity uses only two kernel pairs. Keeping the entire matching produces a stronger family of instance-sensitive upper bounds. This result does not require a large- n hypothesis.

40.1 The matching statistic on output colorings

Suppose p has two coprime cycles X, Y of lengths a, b , and all nontrivial kernel classes of s are disjoint cross pairs. Let their number be $t \geq 1$, so $\text{rank}(s) = a + b - t$. For a coloring c , write

$$a_\gamma = |\{x \in X : c(x) = \gamma\}|, \quad b_\gamma = |\{y \in Y : c(y) = \gamma\}|.$$

The graph of monochromatic cross edges is a disjoint union $\bigsqcup_\gamma K_{a_\gamma, b_\gamma}$. Its maximum matching size is therefore

$$\nu(c) = \sum_{\gamma=1}^k \min(a_\gamma, b_\gamma). \tag{40.1}$$

Define

$$W_{k,t}(a, b) = \#\{c : X \sqcup Y \rightarrow [k] : \nu(c) < t\}. \quad (40.2)$$

The case $t = 1$ is exactly $W_{k,1}(a, b) = \mathcal{B}_k(a, b)$.

Theorem 40.1 (Matching obstruction hierarchy; manuscript 06). *Under the preceding assumptions,*

$$|\tau\langle p, s \rangle| \leq k^{a+b} - W_{k,t}(a, b) + ab. \quad (40.3)$$

Moreover,

$$W_{k,t}(a, b) = a! b! [x^a y^b z^{<t}] \left(\sum_{i,j \geq 0} \frac{x^i y^j}{i! j!} z^{\min(i,j)} \right)^k, \quad (40.4)$$

where $[z^{<t}]$ means summing the coefficients of $z^0, \dots, z^{t-1}$.

Proof. A word containing s can be written $u \circ s \circ p^j$. The t distinct kernel pairs, pulled back by p^j , remain disjoint cross pairs and are monochromatic in the resulting output coloring. Hence that coloring has $\nu \geq t$. All reachable colorings with $\nu < t$ must therefore come from powers of p , contributing at most ab possibilities. This proves (40.3).

For (40.4), a single output color used i times on X and j times on Y contributes $\min(i, j)$ to ν . The two sides are labeled, so the exponential generating factor is $x^i y^j z^{\min(i,j)} / (i! j!)$. Multiply the factors for the k named colors, extract the desired sizes and matching range, and multiply by $a! b!$. Only finitely many terms can affect a specified coefficient, so the formal power-series calculation has no convergence assumption. $\square$

The hierarchy is monotone: increasing t increases the excluded family $\{\nu < t\}$. It uses the actual kernel pattern rather than only one collision. It remains an upper bound, not a claim that every coloring with $\nu \geq t$ is reachable. It carries out, for two coprime cycles, the refinement Part I proposes in section 12 (“using more than one collision”).

40.2 The first exact correction beyond proper colorings

Proposition 40.2 (Manuscript 06). *For $a, b \geq 1$ and $k \geq 2$,*

$$W_{k,2}(a, b) = \mathcal{B}_k(a, b) + \Omega_{k,1}(a, b), \quad (40.5)$$

where

$$\Omega_{k,1}(a, b) = k \sum_{\substack{1 \leq i \leq a, 1 \leq j \leq b \\ \min(i,j)=1}} \binom{a}{i} \binom{b}{j} \mathcal{B}_{k-1}(a-i, b-j). \quad (40.6)$$

In particular,

$$\Omega_{k,1}(a, b) \geq kab \mathcal{B}_{k-1}(a-1, b-1). \quad (40.7)$$

Proof. The condition $\nu = 1$ says that exactly one color is used on both sides, and for that color the smaller of its two multiplicities is one. Choose that color, choose its i vertices on X and j vertices on Y , and then color the remaining graph properly with the other $k-1$ colors. This is a disjoint classification and gives (40.6). The subcase $i = j = 1$ gives (40.7); it counts exactly the colorings with one monochromatic cross edge. $\square$

For an otherwise correctly shaped candidate whose singular map has at least two cross kernel pairs, the loss relative to the full one-collision bound is at least the explicit integer in (40.7). In the proved exactness range with $a = a_n, b = b_n$, this is a quantitative stability statement relative to the actual optimum. At $k = 3$, $kab \mathcal{B}_2(a-1, b-1) = 3ab \cdot 2 = 6ab$ for $a, b \geq 2$, which is Part II’s proposition 19.2.

Remark 40.3 (Different obstructions live on different scales; manuscript 06). The nonfull-graph penalty in (33.3) is on the scale λ_k^n . The one-edge family in (40.7), at near-balanced side sizes, is governed instead by the palette counts for $k - 1$ colors. This distinction suggests more than one asymptotic scale in the near-maximum spectrum. It does not identify the second-largest attainable state complexity: a matching obstruction need not be sharp for any given singular transformation.

41 Exact exponential expansions of the optimum

The formula of theorem 33.4 identifies the optimal deficit from the trivial bound with a finite palette sum minus a quadratic term. It therefore gives considerably more information than its exponential growth rate alone. Throughout this section k is fixed and $n \rightarrow \infty$; the constants in $O_k(\cdot)$ may depend on k , and $\delta_n \in \{\frac{1}{2}, 1, 2\}$ is as in (28.1).

41.1 The first-order law, three times

All three manuscripts prove the sharp first-order deficit for every fixed k ; it is printed once.

Theorem 41.1 (Sharp defect asymptotics; manuscripts 01, 03 and 06). *Fix $k \geq 3$, $h = \lfloor k/2 \rfloor$. If $k = 2h$ is even, then*

$$D_k(n) \sim \binom{2h}{h} h^n. \quad (41.1)$$

If $k = 2h + 1$ is odd, then

$$\frac{D_k(n)}{\lambda_k^n} = 2 \binom{2h+1}{h} \cosh(\delta_n \omega_h) + o(1), \quad (41.2)$$

so for odd k the three residue classes of n (odd, $\equiv 0$, $\equiv 2$ modulo four) have three distinct leading constants. In either case

$$\lim_{n \rightarrow \infty} D_k(n)^{1/n} = \sqrt{\lfloor k^2/4 \rfloor} = \lambda_k, \quad (41.3)$$

and the relative deficit from k^n has exponential base λ_k/k .

Three proofs. (1) Manuscript 01 substitutes the eventual equality $D_k = D_k^*$ (theorem 36.1) into lemma 32.6; the displayed leading constants are positive and bounded above and below independently of n , giving (41.3). In its own normalization the odd law reads $D_k(n)/s_k^n = 2 \binom{k}{m} \cosh(\lambda \delta_n/2) + o(1)$ with $\lambda = \log((m+1)/m)$ and $\delta_n \in \{1, 2, 4\}$. (2) Manuscript 03 obtains the even law from the compactness argument of section 37 (it does not need exactness for even k), and the odd law together with theorem 37.2; in its normalization $D_{2h+1}(n) \sim \binom{2h+1}{h} [h(h+1)]^{n/2} (t_h^{\delta_n} + t_h^{-\delta_n})$. Taking n th roots gives (41.3). (3) Manuscript 06 derives the sharper expansions of theorems 41.2 and 41.4 below from theorem 33.4 (for $k \geq 4$; at $k = 3$ Part II's corollary 21.1 gives $9/\sqrt{2}$, $15/2$, $51/4$, the case $h = 1$ of (41.2)). The three normalizations agree: $2 \binom{k}{h} \cosh(\delta_n \omega_h) = \binom{k}{h} (t_h^{\delta_n} + t_h^{-\delta_n})$ and manuscript 06's $\hat{C} = 2 \binom{k}{h}$ for odd k ; the write phase checked this numerically for $4 \leq k \leq 11$ and all three values of δ_n . For odd k the three constants are distinct: this is an arithmetic oscillation in the leading constant, not an error term caused by rounding a saddle point; it comes from the coprimality requirement on two nearly equal cycle lengths (manuscript 01).

41.2 Even output counts: the residue effect is subleading

Theorem 41.2 (Manuscript 06). *Let $k = 2h \geq 4$, and define*

$$C_0 = \binom{2h}{h}, \quad C_1 = \frac{(2h)!}{(h-1)!(h+1)!}, \quad \mu_1 = \sqrt{h^2 - 1}, \quad \xi_1 = \sqrt{\max\{h(h-1), h^2 - 4\}}.$$

Then $\xi_1 < \mu_1 < h$ and

$$\begin{aligned} k^n - R_2(n, k) &= C_0 h^n + 2C_1 \mu_1^n \cosh\left(\delta_n \log \frac{h+1}{h-1}\right) \\ &\quad - a_n b_n + O_k(\xi_1^n). \end{aligned} \quad (41.4)$$

In particular, $(k^n - R_2(n, k))/h^n \rightarrow C_0$.

Proof. The maximal product in the signed palette formula is h^2 , from (h, h) . The next largest product is $h^2 - 1$, from $(h-1, h+1)$ and its reversal; both terms have coefficient C_1 and positive sign. Their sum is the hyperbolic-cosine term displayed above.

All remaining pairs with total $2h$ have product at most $h^2 - 4$. Pairs with smaller total have product at most $h(h-1)$. The bounded values of δ_n keep every ratio factor uniformly bounded, so the remaining finite sum is $O_k(\xi_1^n)$. Subtract $a_n b_n$ and use eventual exactness. The inequalities between the exponential bases are immediate for $h \geq 2$. $\square$

Thus the mod-four displacement of the nearest coprime split does not change the leading coefficient for even k , but it does affect the next exponential scale. (For even k , $\mu_1 = \mu$ of (32.5); the expansion answers manuscript 03's question on second-order even asymptotics, question 44.5.)

Example 41.3 (A closed formula for four outputs; manuscripts 06 and 01). Specializing the signed palette sum gives the exact identity

$$\mathcal{B}_4(a, b) = 6 \cdot 2^{a+b} + 4(3^a + 3^b) - 12(2^a + 2^b) + 12. \quad (41.5)$$

Therefore for every $n \geq 7$,

$$4^n - R_2(n, 4) = 6 \cdot 2^n + 4(3^{a_n} + 3^{b_n}) - 12(2^{a_n} + 2^{b_n}) + 12 - a_n b_n. \quad (41.6)$$

The first nonconstant residue-dependent term has base $\sqrt{3}$, whereas the leading deficit has base two. This is an exact formula, not a truncated approximation.

Manuscript 01 derives the same identity (41.5) by expanding the three nonzero terms of (32.1), or by onto-map inclusion–exclusion, notes that it is symmetric, as a complete-bipartite coloring count must be, and states (41.6) eventually; with theorem 34.1 it holds for every $n \geq 7$. Its leading constant is always six, with no parity oscillation at the 2^n scale; parity still affects the exponentially smaller terms and the polynomial correction, which is why a leading-term calculation would not have sufficed to identify the exact maximizing coprime split (manuscript 01). (The identity was checked in the write phase for $1 \leq a, b \leq 11$.)

41.3 Odd output counts: three leading constants

Theorem 41.4 (Manuscript 06). *Let $k = 2h + 1 \geq 5$ and $\hat{C} = 2\binom{2h+1}{h}$. Then*

$$k^n - R_2(n, k) = \hat{C} \lambda_k^n \cosh(\delta_n \omega_h) - a_n b_n + O_k((h(h+1) - 2)^{n/2}). \quad (41.7)$$

Consequently, the normalized deficit $(k^n - R_2(n, k))/\lambda_k^n$ has exactly three subsequential limits,

$$\hat{C} \cosh\left(\frac{1}{2}\omega_h\right), \quad \hat{C} \cosh(\omega_h), \quad \hat{C} \cosh(2\omega_h), \quad (41.8)$$

corresponding respectively to odd n , $n \equiv 0 \pmod{4}$, and $n \equiv 2 \pmod{4}$.

Proof. The leading terms and error estimate are lemma 32.2. Eventual exactness permits their direct use for the optimal deficit. The error base is strictly smaller than λ_k , and $a_n b_n / \lambda_k^n \rightarrow 0$. All three residue classes occur infinitely often, and their limiting constants are distinct because $\omega_h > 0$ and $\cosh$ is strictly increasing on positive arguments. There are no other limits because every integer belongs to one of these three classes. $\square$

Manuscript 06 proves (41.3) from these two theorems: the leading coefficients are positive and bounded above and below for fixed k , all errors have smaller exponential base, and the polynomial correction is negligible. The threshold theorem is uniform in k , while these expansion statements are formulated for fixed k . One should not silently replace k by an arbitrary function of n inside an O_k term. A genuinely uniform expansion in a growing-output regime would require tracking its constants and the ratio between successive palette products.

41.4 Eventual recurrences and rational generating functions

This subsection is manuscript 01's. For fixed k , the residue subsequences of the optimal deficit are eventually exponential polynomials, hence satisfy constant-coefficient linear recurrences, which can be written down before any initial values are computed.

Lemma 41.5 (Finite exponential expansion; manuscript 01). *For $a, b \geq 1$,*

$$\mathcal{B}_k(a, b) = \sum_{i=1}^{k-1} \binom{k}{i} (k-i)^b \sum_{j=0}^i (-1)^{i-j} \binom{i}{j} j^a. \quad (41.9)$$

Proof. Choose the exact i -color palette on the left in $\binom{k}{i}$ ways. By inclusion–exclusion, the number of onto maps to that palette is $\sum_{j=0}^i (-1)^{i-j} \binom{i}{j} j^a$. Each right-hand vertex then has $k-i$ choices. Sum over i . $\square$

Define the finite set of positive integer products $\mathcal{P}_k = \{uv : u, v \geq 1, u + v \leq k\}$, and let E be the forward shift on a sequence indexed by t : $(EF)(t) = F(t+1)$.

Theorem 41.6 (An explicit eventual annihilator; manuscript 01). *Fix $k \geq 3$ and $r \in \{0, 1, 2, 3\}$. The sequence $F_r(t) = D_k(4t+r)$ is eventually annihilated by*

$$\mathcal{A}_k(E) = (E-1)^{e_k} \prod_{q \in \mathcal{P}_k \setminus \{1\}} (E - q^2), \quad e_3 = 2, \quad e_k = 3 \quad (k \geq 4). \quad (41.10)$$

The same recurrence holds for $D_k^(4t+r)$ whenever all of the indices involved are in its witness range. Minimality of the displayed annihilator is not asserted for arbitrary k .*

Proof. Within one residue class, a_{4t+r} and b_{4t+r} each increase by two when t increases by one. The nonzero exponential terms in (41.9) have the form $j^a (k-i)^b$, with $j \geq 1$, $k-i \geq 1$, and $j + (k-i) \leq k$. They therefore multiply by $[j(k-i)]^2$ under E . Equal products can be combined; one factor $E - q^2$ kills their combined contribution. The subtracted period is linear in t when $k = 3$, and quadratic when $k \geq 4$. It is killed by $(E-1)^2$ or $(E-1)^3$, respectively; these factors also kill constant exponential terms corresponding to $q = 1$. The eventual equality $D_k = D_k^*$ completes the proof. $\square$

For example, when $k = 3$ the annihilator is $(E-1)^2(E-4) = E^3 - 6E^2 + 9E - 4$, so every residue subsequence eventually satisfies

$$F_r(t+3) = 6F_r(t+2) - 9F_r(t+1) + 4F_r(t). \quad (41.11)$$

With Part II this holds for every t with $4t + r \geq 7$; it is the order-three form, in the step t , of Part II's order-12 recurrence (21.3) (whose characteristic polynomial in z^4 is $(z^4 - 1)^2(z^4 - 4)$). For $k = 4$ the annihilator is $(E - 1)^3(E - 4)(E - 9)(E - 16)$. The code records exact expanded coefficients for $3 \leq k \leq 10$ (data/03-eventual-recurrences.json) and checks 416 recurrence instances on the explicit candidate sequences. This does not supply missing initial values of the actual optimum before the eventual threshold; for $4 \leq k \leq 7$ theorem 34.1 now supplies them from $n = \max(7, k + 1)$.

Corollary 41.7 (Rationality on residue classes; manuscript 01). *For fixed $k \geq 3$ and $r \in \{0, 1, 2, 3\}$, the ordinary generating function of the sequence $D_k(4t + r)$, with any convention for the finitely many indices where $4t + r < k$, is rational. Its denominator divides*

$$(1 - z)^{e_k} \prod_{q \in \mathcal{P}_k \setminus \{1\}} (1 - q^2 z).$$

The generating function of the full sequence $D_k(n)$ is rational as well, with a denominator dividing the expression obtained by replacing z by z^4 .

Proof. Multiply each residue generating function by the displayed denominator. The recurrence makes all sufficiently high coefficients vanish, leaving a polynomial. Changing finitely many initial values changes the generating function by a polynomial, so does not spoil rationality. Interleaving the four residues gives $\sum_{r=0}^3 z^r G_r(z^4)$, again rational. $\square$

This is an existence theorem for a rational generating function, not an explicit numerator for the unknown finite initial segment. It concerns the fixed- k optimum, not a generating function where k varies with n . At $k = 3$ it is Part II's rationality statement in corollary 21.2.

42 Even outputs: an interval of asymptotically optimal proportions

This section is manuscript 03's. For odd k , changing the cycle imbalance changes the leading coefficient. For even k , both balanced palettes have size h , so the main term is $\binom{2h}{h} h^{a+b}$ regardless of the imbalance. Smaller palette terms only become relevant at a specific boundary. This produces an interval, not a unique balanced ratio. Fix $k = 2h \geq 4$ and define

$$\theta_h = \frac{\log((h+1)/h)}{\log((h+1)/(h-1))}, \quad 0 < \theta_h < \frac{1}{2}. \quad (42.1)$$

The upper inequality is equivalent to $h^2 > h^2 - 1$.

Theorem 42.1 (The even-output interval criterion; manuscript 03). *Let $a + b = n \rightarrow \infty$, $1 < a < b$, $\gcd(a, b) = 1$, and use the established $U_{a,b}$ witness with $2h$ outputs. Its deficit is asymptotic to the global optimum $\binom{2h}{h} h^n$ if and only if*

$$\Phi_n = a \log \frac{h}{h-1} - b \log \frac{h+1}{h} \rightarrow +\infty, \quad (42.2)$$

$$\Psi_n = b \log \frac{h}{h-1} - a \log \frac{h+1}{h} \rightarrow +\infty. \quad (42.3)$$

Equivalently,

$$a - \theta_h n \rightarrow +\infty, \quad (1 - \theta_h)n - a \rightarrow +\infty. \quad (42.4)$$

In particular, any sequence of admissible coprime splits with $a/n \rightarrow \theta \in (\theta_h, 1 - \theta_h)$ is asymptotically optimal.

Proof. The witness deficit is $\mathcal{B}_{2h}(a, b) - ab$. The polynomial term is negligible relative to h^n , so it suffices to characterize $\mathcal{B}_{2h}(a, b)/h^n \rightarrow C = \binom{2h}{h}$.

First suppose this convergence holds. The monochromatic-side lower bound $\mathcal{B}_{2h}(a, b) \geq 2h(2h - 1)^{\max(a, b)}$ implies that both a and b are bounded below by a positive multiple of n : indeed $\max(a, b)/n \leq \log h / \log(2h - 1) + o(1) < 1$. The (h, h) term in the positive expansion (32.4) is therefore $(C + o(1))h^n$. The neighboring full-palette terms $(h - 1, h + 1)$ and $(h + 1, h - 1)$, divided by h^n , are positive constant multiples of $(1 + o(1))e^{-\Phi_n}$ and $(1 + o(1))e^{-\Psi_n}$. Since the whole normalized sum tends to C , both must tend to zero. This proves necessity.

Conversely assume (42.2)–(42.3). Both lengths are again linear in n , by (42.4). Consider the logarithm

$$f(i) = a \log i + b \log(2h - i), \quad 0 < i < 2h.$$

It is strictly concave. The two neighboring differences from $f(h)$ are $-\Phi_n$ and $-\Psi_n$. Concavity implies that every other integer $i \neq h$ has $f(i) - f(h) \rightarrow -\infty$: along either direction, successive increments are no larger than the first negative increment. Thus every full-palette term other than (h, h) is $o(h^n)$.

If a term in (32.4) uses fewer than $2h$ colors, increase one of its two palette sizes until the total is $2h$. Because both a, b are linear in n , its pure exponential is smaller than that full-palette exponential by an exponentially vanishing factor. All full-palette exponentials are $O(h^n)$ by the previous paragraph. There are only finitely many palette pairs, so all terms with unused colors are $o(h^n)$ as well. The onto-map errors vanish, leaving precisely Ch^n . Finally, (42.4) follows by expanding Φ_n and Ψ_n with $a + b = n$ and factoring out $\log((h + 1)/(h - 1))$. $\square$

For $k = 4$ the left boundary is $\theta_2 = \log(3/2)/\log 3 \approx 0.369$. The interval is nontrivial and persists as n grows. The theorem is a first-order assertion about the established witness family. It does not imply that every split in the interval is exactly optimal, or that every singular map with the appropriate two-cycle permutation realizes the witness orbit. Indeed, for $4 \leq k \leq 7$ and every $n \geq \max(7, k + 1)$, and for every $k \geq 4$ once $n \geq N(k)$, only the nearest split is exactly optimal (corollary 31.5 with theorem 28.1); manuscript 06's second-order term (41.4) is what separates the splits inside the interval.

Remark 42.2 (Boundary behavior; manuscript 03). If Φ_n tends to a finite real number rather than $+\infty$, the neighboring palette $(h - 1, h + 1)$ contributes a nonvanishing extra constant at scale h^n . If $\Phi_n \rightarrow -\infty$, that contribution dominates the balanced-palette scale. Thus the interval criterion identifies the precise first-order boundary, not merely a convenient sufficient central region.

43 Reproducible checks and their limits

43.1 What each manuscript ran

All programs of the three manuscripts use only the Python standard library, exact integers, and explicit exceptions rather than `assert` statements, so Python's optimization flag does not disable their checks. None performs network access or contains downloaded certificates. Their recorded outputs were produced with Python 3.13.5 (the environment records say so); the write phase reran all of them on scratch copies (section 43.2).

- *Manuscript 06*: three programs with the file prefix `code/05-quartic-threshold-`: the primary `theorem_checks.py`, the independent `independent_check.py`, and `supplementary_checks.py`; $2 \times 9,247,046$ comparisons in 2,217 rows for $k = 4, \dots, 7$, with zero failures and zero unexpected equalities, and the supplementary counts of section 34.
- *Manuscript 01*: the producer `code/03-eventual-verify.py` and the checker `code/03-eventual-check_certificates.py`; 42,960 producer checks, 919/919 certificate rows accepted, eight BFS witnesses with 605,713 states.

- *Manuscript 03*: the producer `code/04-deficit-asymptotics-verify.py` and `code/04-deficit-asymptotics-independent_check.py`; 25 sizes, 3,717 candidates, BFS counts 2125, 6452, 19550, 58654 for $n = 7, \dots, 10$, 12 direct coloring checks, and the cutoff margin 2702601/64.

Manuscript 06. The exactness theorem for $n \geq N(k)$ has a conventional proof written in full, apart from the explicitly imported published lower construction. The extra finite portions for $4 \leq k \leq 7$ are computer-assisted. Neither the conventional proof nor the programs have been checked in Lean, Rocq, or another proof assistant, and the manuscript has not been independently refereed. The infinite theorem does not depend on a generated plot or on numerical floating-point estimates. All mathematical formulas defining the finite obligations appear in the article, and source code and result files are kept separate so that reproduction does not require trusting an existing success summary. The extension to larger k is mathematically permitted, but the required finite computation grows; no unrun finite range is certified.

Manuscript 01. Both programs exit unsuccessfully when a check fails. The independent checker also rejects missing, duplicated, or unexpected certificate rows, and it validates an existing CSV; rerunning the producer regenerates the data before the independent audit. `data/03-eventual-asymptotic_ratios.csv` contains illustrative floating-point ratios of the exact candidate defects to λ_k^n ; these are diagnostic illustrations, not certified interval bounds and not part of any proof. The finite checks do not prove the infinite tail; the tail has a separate proof. Likewise an unspecified tail plus the finite region does not by itself cover any possible intervening gap.

Manuscript 03. The producer reports 25 checked state sizes and 3,717 structural candidates. It separately performs literal breadth-first orbit searches for the three-output witnesses with $n = 7, 8, 9, 10$, obtaining respectively 2125, 6452, 19550 and 58654 states. These four searches are sanity checks of attainment and composition conventions, not the upper-bound proof. The independent checker reconstructs all 3,717 candidate records using the alternative order recurrence (38.13) and coloring calculation, additionally checks 12 small complete-bipartite coloring counts by literal color enumeration, and verifies the rational cutoff margin (38.11). The cutoff inequality’s extension to every larger n is justified in the mathematical proof, not by iterating a finite loop. Both programs were also run with `python -O` (the `optimized_*` consoles). The producer supports exploratory `-max-n` values through 45 and `-bfs-max-n` through 12, but larger explorations are not needed for the proof and change output files. The word “independent” refers to implementation strategy, not to authorship by an independent research team.

For its sanity checks manuscript 03 uses zero-based states: with $a = a_n, b = b_n, p$ is the two increasing cycles on $\{0, \dots, a-1\}$ and $\{a, \dots, n-1\}$; s starts as the identity, with $s(0) = a, s(n-1) = 0$ and, when $a > 2$ is odd, $s(1)$ and $s(2)$ swapped. For three outputs, color 0 is used throughout the first block, color 1 at the first point of the second block, and color 2 on its remaining points; at $n = 7$ this gives Part II’s arrays $p = (1, 2, 0, 4, 5, 6, 3), s = (3, 2, 1, 3, 4, 5, 0), \tau = (0, 0, 0, 1, 2, 2, 2)$ (section 18.1). The pullback action in the code is `tuple(c[q] for q in t)`; starting at τ and repeatedly applying it by p and s enumerates $\tau\langle p, s \rangle$, without an inconsistent forward-composition convention. The package does not infer the all- n generation property from four BFS tests. Its certificate (`data/04-deficit-asymptotics-finite_certificate.json`) is ordinary JSON with exact decimal integers and no executable code: for each n the producer emits the two easy bounds and the records $(m, d, r, P, O, P - O)$ and $(a, b, d, r, P, O, P - O)$ for all parameters of proposition 38.2; the independent checker reconstructs these records, compares the entire sorted record sets, not merely their minima, independently optimizes the coprime-split expression, and raises an exception with a nonzero exit on any mismatch. A future third verifier can ignore the producer, recompute (38.13), evaluate the displayed formulas and compare its own minima. The claims need only the range through 31; expanding the exploratory range does not replace the infinite proof.

43.2 Commands and files

The shipped programs are byte-identical to the delivery and still use the delivered layout: each computes a package root as the parent of its own directory and reads or writes `results/...` there, and `code/05-quartic-threshold-supplementary_checks.py` imports `theorem_checks`, which does not exist under its shipped name. Run in this report's directory they would therefore write stray files into Part I's `results/`, manuscripts 01 and 03 would overwrite each other's `results/verification_summary.json`, the supplementary program would stop at its import, and the two checkers of manuscripts 01 and 03 would fail unless their producers had run there first. The report's README gives a recipe that restores the delivered names in a scratch copy; there the commands are

```
python code/theorem_checks.py          # manuscript 06
python code/independent_check.py
python code/supplementary_checks.py
python code/verify.py                  # manuscript 01
python code/check_certificates.py
python code/verify.py                  # manuscript 03
python code/independent_check.py
```

each group in its own copy. The write phase ran them so on 29 September 2026 (Python 3.14.4; about 30 s, 5 s and 1 s): all passed, every CSV is byte-identical to the shipped one, and every JSON, $\text{\TeX}$ and console record equals the shipped one apart from Windows line endings, except that manuscript 01's `verification_summary.json` (and its identical console copy) records the Python version.

43.3 What the checks do not establish

No finite enumeration proves the universal quantifiers of theorems 28.1, 36.1 and 38.1. They are supplied by the analytic arguments; the finite checks are exact integer comparisons of explicitly reduced obligations, whose sufficiency is proved in the text (section 34 and propositions 35.1 and 38.2). No Lean, Rocq, Isabelle or other proof-assistant verification was carried out for Part III, and none of its statements is formalized anywhere in ProveIt. The published lower theorem is cited, not formally reconstructed. Agreement between two programs written in one investigation is implementation independence, not logical independence of the shared reduction, and not external review. Successful compilation of a PDF is not mathematical verification.

44 Further research questions

The three manuscripts close with ten, ten and nine questions. Where two or three of them ask the same question it is printed once, with every source's comments; questions that another source (or Part II) has since answered are printed with the answer, not as open. Part II's questions in section 23 stay as printed there, with dated pointers where Part III bears on them. Manuscript 01 calls its questions proposed directions arising from the argument, not assertions that each is a previously published open problem; manuscript 03 calls its questions proposals, not hidden assumptions in its proofs; manuscript 06 says they separate exact optimization, structural classification, algorithms, and formalization.

Question 44.1 (The complete two-parameter optimum; manuscripts 06, 01 and 03). Does $R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + G_k(a_n, b_n)$ hold for every $n \geq 7$ and $3 \leq k < n$?

This is Davies's Problem 2 in full, and Part II's question 23.1 for $k \geq 4$. *Status after Part III*: settled for $k = 3$ (Part II), for $4 \leq k \leq 7$ (theorem 34.1), for $k \geq 8$ once $n \geq N(k)$ (theorem 33.4), and for $8 \leq k \leq 12$, $n \leq 100$ (proposition 35.1) and every $k < n \leq 30$ (proposition 10.1). The remaining region is $k \geq 8$ with

$\max(7, k + 1) \leq n < N(k)$ outside those finite ranges. Manuscript 06: a direct target is to replace the palette estimates by a global comparison of every disconnected or isolated biclique graph with the nearest full coprime biclique, including the period correction; finite verification can settle each additional fixed output count, but it cannot by itself close the unbounded two-parameter region. Manuscript 01 (writing before manuscript 06): a concrete route is to replace the limiting palette comparisons in proposition 36.2 by finite inequalities strong enough to cover every admissible collision graph; the certificate bound already provides an exact target for a successful inequality, and small exceptional graph types should be treated separately rather than concealed in an asymptotic error. Manuscript 03 asked the same (“is Davies’s optimized coprime-split lower bound exact for every $n \geq 7$ in its stated range?”).

Question 44.2 (An explicit, and a substantially sharper, threshold; manuscripts 06 and 01). Can one give an explicit, reasonably small upper bound for the least threshold, and how does it grow with k (manuscript 01)? Can one prove an effective linear or near-linear condition in k , rather than the sufficient condition $n \geq 3k^4$ (manuscript 06)?

Manuscript 01’s first part is answered by manuscript 06: $N(k) \leq 3k^4$ is explicit (corollary 33.5); in manuscript 01’s words, even a very large explicit bound turns the theorem into a finite verification program for any fixed k , which is what section 34 carries out for $k \leq 7$. Manuscript 01 had noted that the union bound (32.12) and the elementary order bound are both quantitative, and that the missing work was a usable uniform separation between the finite palette terms, with a controlled treatment of components whose smaller part has not reached the asymptotic regime. The sharper form stays open. [Added 29 September 2026, batch 43: at $k = 4$ Part IV lowers the analytic threshold from $N(4) = 212$ to 26 (remark 50.6); for general k the question stays open.] Manuscript 06: the current losses have identifiable sources, namely the crude multinomial remainder $k^{23k} \mu^n$, the conversion of many small graph components into a factor 2^{d+r} , and the elementary permutation-order bound; sharper palette tails and a direct comparison preserving the component sizes could improve the threshold without changing the collision method. The case k comparable to n is especially important, since fixed- k asymptotics are least informative there.

Question 44.3 (Explicit cutoffs for odd outputs; manuscript 03. Answered). Give an explicit computable bound for $N_{\exists}(k)$ in theorem 37.2, and determine the smallest cutoff for $k = 5$.

Answered by manuscript 06: $N(k)$ of (33.1) is such a bound for every $k \geq 4$, and for $k = 5$ the formula holds for every $n \geq 7$ (theorem 34.1), so no cutoff beyond Davies’s range $n \geq 7$ is needed. Manuscript 03 had observed that its proof already provides exponential upper errors, a uniform period error, and finitely many strict coefficient gaps, that turning the compactness reduction into inequalities with numerical thresholds should produce a rigorous algorithm, and that the exclusion of small reduced parts must be bounded as well as the asymptotic palette errors; manuscript 06’s lemma 33.1 does exactly that.

Question 44.4 (Exact even-output optimality; manuscript 03. Answered eventually). For fixed even $k \geq 4$, is $R_2(n, k) = k^n - \mathcal{B}_k(a_n, b_n) + a_n b_n$ true for every sufficiently large n ?

Answered: yes, by manuscript 01 (theorem 36.1, non-effective) and by manuscript 06 (theorem 33.4, for $n \geq N(k)$). Manuscript 03’s stronger form (every $n \geq 7$) is question 44.1. Manuscript 03 had noted that its argument forces an optimal permutation into two coprime cycles but not their nearest balance for even k , and suggested analyzing the second-largest terms in (32.2), since first-order asymptotics cannot discriminate the interval of theorem 42.1; manuscripts 01 and 06 instead use exact strict balancing.

Question 44.5 (Second-order asymptotics for even outputs; manuscript 03. Answered). After subtracting $\binom{2h}{h} h^n$, what are the next exponential scale and the exact residue-dependent coefficient of $D_{2h}(n)$?

Answered by manuscript 06’s theorem 41.2: the next scale is $(h^2 - 1)^{n/2}$ with coefficient $2C_1 \cosh(\delta_n \log \frac{h+1}{h-1})$, $C_1 = (2h)! / ((h-1)!(h+1)!)$. Manuscript 03 had observed that for a

prescribed near-balanced split the next palette products are $h^2 - 1$, but that an extremal theorem must also show that no different coprime split or singular generator improves the total by changing the relative weights of those terms; exactness (theorem 33.4) supplies this.

Question 44.6 (Rank and kernel structure of an extremal singular letter; manuscript 01. Answered). Must an eventual extremizer's singular generator have rank $n - 1$? Can an extremizer have two or more disjoint cross-cycle kernel pairs?

Answered: rank $n - 1$, one cross pair, at $k = 3$ for every $n \geq 7$ (Part II's theorem 19.1) and at $k \geq 4$ for every $(n, k) \in \mathcal{E}$ (theorem 39.1), hence for every fixed $k \geq 3$ eventually. Manuscript 01 had noted that its theorem 36.4 only rules out within-cycle pairs and kernel classes of size three, and that resolving the question requires exploiting full improper saturation rather than reusing a single-pair chromatic obstruction; both rigidity proofs do exactly that.

Question 44.7 (Which rank- $(n - 1)$ singular letters saturate the bound? Manuscripts 06, 03 and 01). For the forced two-cycle permutation p , characterize those singular maps s for which some proper full-period coloring reaches every improper coloring (manuscript 06); at $k = 3$, characterize the singular s that attain $R_2(n, 3)$ with a permissible output coloring (manuscript 03, Part II's question 23.2). Are all eventual extremal pairs, up to relabeling and generator changes, associated with the monoids U_{a_n, b_n} , or do smaller monoids have the same projected coloring orbit (manuscript 01)?

Manuscript 06: theorem 39.1 gives necessary structure, but example 39.3 rules out sufficiency of cycle shape, rank, and output period. A useful intermediate goal is a classification in terms of the image of s , the location of the omitted point, and how s moves the two cycles. The published lower monoids provide positive examples, not a proof that every extremizer has the same transition monoid. Manuscript 03: theorem 38.4 reduces the problem to maps injective on each block and to the reachability of every improper coloring; a useful answer would replace that reachability condition by conditions on the cross-block matching, image omissions, and the permutation action, and need not require that $\langle p, s \rangle$ equal the entire standard witness monoid. Manuscript 01: the theorem identifies necessary cycle geometry and a necessary and sufficient orbit condition, not a normal form for the semigroup; different transformations can induce the same output coloring, so the projection may erase monoid-level differences.

Question 44.8 (Recognition without enumerating k^n colorings; manuscripts 06 and 01). Can saturation of a two-cycle candidate be recognized in time polynomial in n and the bit length of its description (manuscript 06)? Can the saturation condition of theorem 36.4 be replaced by an efficiently checkable algebraic criterion (manuscript 01)?

This is Part II's question 23.3 for every k . Manuscript 06: a breadth-first search in the coloring space is exponential. The rigidity theorem reduces the input shape, but not automatically the decision complexity. A useful certificate would describe generation of all improper colorings by finitely many local transformations. Conversely, a missing coloring might admit a short invariant-based certificate like the invariant-block obstruction in example 39.3.

Question 44.9 (The second-largest value; manuscript 06). What is the largest reverse complexity strictly below $R_2(n, k)$, and which structural defect produces it?

This is Part II's question 23.4 for $k \geq 4$. The nonfull collision graph and the multiple-kernel-pair obstruction produce different quantitative losses (corollary 33.2 and proposition 40.2), but neither proves that its bound is attained. A complete answer must compare a wrong cycle split, extra kernel pairs, a shortened pure orbit, and incomplete coverage of the improper family even when the coarse structure is correct.

Question 44.10 (Stability of nearly extremal automata; manuscripts 01 and 03). If an orbit misses at most $(1 + \varepsilon)D_k(n)$ colorings, which cycle structures, output periods, and incomplete-saturation patterns are possible, and can exact or asymptotic gap theorems distinguish these mechanisms (manuscript 01)? Find an explicit positive function $c(k)$ such that every sufficiently large automaton outside the required two-cycle structure has deficit at least $(1 + c(k))D_k(n)$, and optimize that constant, with the smallest associated state threshold (manuscript 03).

Manuscript 01: for odd k , the hyperbolic-cosine factor already penalizes imbalance at leading order; for even k , subleading palette terms are essential, and the explicit balancing gap offers a way to quantify that setting. Manuscript 03: its subsequence arguments imply such a separation for the excluded structural types at fixed k , but do not extract its best value or the smallest associated state threshold; odd imbalance gives additional residue-dependent gaps that can be treated separately. *Partial answer (manuscript 06; the reading is the merge's):* for $n \geq N(k)$ every nonfull cross-cycle graph costs at least $(\hat{C}/4 - 1)\lambda_k^n$ more than $D_k^*(n)$ (corollary 33.2), and every same-cycle or unmixed case has deficit above $(3\hat{C}/2)\lambda_k^n$ (section 33). Since $D_k^*(n) < \mathcal{B}_k(a_n, b_n) < (3\hat{C}/2)\lambda_k^n$, these give an explicit $c(k) > 0$ for those structural types, for instance $c(k) = (\hat{C} - 4)/(6\hat{C})$ for nonfull graphs. The wrong-split, extra-kernel-pair and incomplete-saturation mechanisms are not quantified in this form (for extra kernel pairs see proposition 40.2). [Added 29 September 2026, batch 43: at $k = 4$ Part IV quantifies the wrong-split and extra-kernel-pair mechanisms (theorems 53.6 and 53.7); incomplete saturation stays unquantified.]

Question 44.11 (Sharpness of the matching hierarchy; manuscript 06). For fixed a, b, k, t , when can the bound $k^{a+b} - W_{k,t}(a, b) + ab$ be attained?

The matching statistic is necessary because every singular word carries a rotated kernel matching. It forgets the relative displacement pattern of that matching and all later constraints imposed by the word. Refining $W_{k,t}$ by displacement type may reveal additional obstructions, while constructing large orbits from prescribed matchings would test how much information the statistic loses.

Question 44.12 (The attainable near-maximum spectrum; manuscripts 06 and 03). Which integers immediately below the optimum occur as reverse state complexities, and which are absent (manuscript 06)? Which orbit sizes in a neighborhood of $R_2(n, 3)$ are attainable, and does the rigidity theorem imply gaps in the spectrum immediately below the maximum (manuscript 03)?

This is Part II's question 23.5. Manuscript 06: a family of singular maps with controlled defects could connect the matching hierarchy with the spectrum of attainable orbit sizes. This is not determined by knowing the maximum: the orbit cannot generally be modified by adding or deleting one coloring independently of the others. Manuscript 03: a perturbation of the larger cycle's output word changes its period and can lose a polynomial number of proper orbit states; a change in collision graph type loses exponentially many. Understanding the intervening scales could connect extremal structure to the DFAO magic-number question.

Question 44.13 (The diagonal and near-diagonal regimes; manuscripts 06 and 01). What is the exact binary reversal maximum when every original state has a distinct output, $k = n$ (manuscript 06)? How does the extremal defect behave for $k = n$ or $k = n - c$ with c fixed, and can output projection expose a different optimal split than monoid-size maximization in these regimes (manuscript 01)?

Manuscript 06: the lower witness assumes $k < n$, so the exactness theorem does not apply at the boundary. The problem then relates directly to orders of two-generated transformation monoids, a classical direction associated with Holzer and König [2]. It requires a separate attainability analysis, not a substitution of $k = n$ into the formula. Manuscript 01: the witness uses a redundant output in the larger cycle; at $k = n$ that resource disappears completely, and at fixed $n - k$ it is scarce. This is a structural change, not merely a limiting value of the fixed- k formulas. (Part I's section 10 shows the collision bound already has a gap at $(n, k) = (5, 5)$.)

Question 44.14 (Growing output alphabets; manuscripts 01 and 03). For which functions $k = k(n) \rightarrow \infty$ is the closest-coprime two-cycle construction eventually optimal? Is there a transition when k is a positive proportion of n (manuscript 01)? For which growth rates do the fixed-output formulas retain a uniform version, and where does the transition toward the diagonal monoid-size regime occur (manuscript 03)?

This is Part II's question 23.6. *Partial answer (manuscript 06)*: whenever $4 \leq k \leq (n/3)^{1/4}$ the construction is exactly optimal (corollary 33.5), and the extremizer structure of theorem 39.1 persists, and the same holds whenever $n \geq N(k)$; for $k \geq 8$ the range $n < N(k)$, in particular k a positive proportion of n , is open outside the certified finite ranges, and manuscript 06's asymptotic expansions are fixed- k statements. Manuscript 01: the fixed- k proof does not answer this by substitution; onto-map errors, binomial palette multiplicities, and the number of possible palette sizes all change with k , and a uniform occupancy estimate is a natural replacement for the fixed-alphabet palette lemma. Manuscript 03: the number of palette terms, the onto-map approximation, and the strict gap (37.4) all depend on k ; treating these dependencies explicitly is necessary.

Question 44.15 (Minimal recurrences and closed numerators; manuscript 01). What is the minimal recurrence for $D_k(4t + r)$ for general fixed k ? Which bases in $\mathcal{P}_k$ cancel, and can the exact rational generating function, including its finite initial correction, be computed?

The annihilator of theorem 41.6 is explicit but can contain unnecessary factors. Determining cancellations is a finite symbolic problem once the exact threshold question is separated from the tail formula. (For $3 \leq k \leq 7$ the values are now known from $n = \max(7, k + 1)$ on, by Part II and theorem 34.1, so for those k only the minimality question remains; at $k = 3$ Part II's corollary 21.2 gives the denominator $(1 - 4z^4)(1 - z^4)^2$.) [Added 29 September 2026, batch 43: answered at $k = 4$ by Part IV: the order-15 recurrence of theorem 55.2 is minimal in n , and the annihilator of theorem 41.6 is minimal on each residue class (remark 55.4).]

Question 44.16 (Formal verification of the proof chain; manuscripts 06, 01 and 03). Can the graph reduction, strict balancing, effective estimates, finite closure, structural separation and witness saturation be formalized in the repository's Lean development (or Rocq), with a transparent statement of the imported lower-construction theorem?

This is Part II's question 23.7 for $k \geq 4$. Manuscript 06: the proof breaks naturally into finite graph colorings, Stirling-number inequalities, real-power estimates, and exact integer arithmetic; an especially useful first target is a small verified checker for (34.1)–(34.4), together with a proved compression lemma; such a checker should report its imported axioms and its evaluation trust boundary, and replaying a large table without proving that the table covers every graph type would not formalize the theorem. Manuscript 01: the most direct first milestone is a checked finite-certificate theorem, proving proposition 35.1 once and certifying the finite table by a small verified evaluator; the analytic tail can then be added independently, with no reason to delay finite checking until every asymptotic lemma has been formalized. Manuscript 03: formalize proposition 38.2, the order recurrence (38.13) and the certificate evaluation for $7 \leq n \leq 31$; an attractive division is to formalize the symbolic graph classification and let a verified integer evaluator discharge the 3,717 inequalities; the real-power estimates of the analytic tail can be replaced by suitably powered rational comparisons; and the lower construction should retain an explicit dependency declaration until its own generation theorem is formalized.

Question 44.17 (Beyond one cyclic permutation action; manuscripts 06, 01 and 03). Which parts of the argument survive when the available permutation transitions generate a noncyclic group, or when there are several singular letters (manuscript 06)? What exact defect laws arise when permutation letters generate a prescribed abelian or nonabelian group, together with one singular letter, and can orbital chromatic polynomials identify a sharp threshold between near-maximal and full reversal (manuscript 01)? Extend the deficit method when the alphabet contains one permutation and several singular letters, or when all invertible letters generate a fixed abelian group (manuscript 03).

This is Part II's question 23.8. Manuscript 06: the rightmost-singular-letter argument still produces orbit graphs, but those graphs need not be cycles or unions of bicliques; a replacement for the classification would connect group actions, proper-coloring counts, and the number of colorings supplied by the permutation subgroup; the cyclic classification is precisely what makes the present binary problem tractable. Manuscript 01: the single cyclic group is unusually rigid (one pair orbit is a union of cycles or complete bipartite graphs); more general groups introduce orbital graphs whose chromatic behavior may not be controlled by palette splits; this is a natural extension of Part I's abelian-permutation obstruction (section 8), but its exact asymptotics require new graph input. Manuscript 03: the forbidden proper colorings are then governed by a union of edge orbits; an exact graph-class description may lead to new alphabet-versus-output thresholds, and it is not enough merely to count the colors needed for one selected collision graph.

45 Conclusion of Part III

Manuscript 06. The collision-orbit reduction, strict balancing and explicit palette estimates determine $R_2(n, k)$ exactly for every $k \geq 4$ once $n \geq N(k) \leq 3k^4$, uniformly enough to let k grow like $n^{1/4}$, and a finite exact-integer closure extends this to every $n \geq \max(7, k + 1)$ for $4 \leq k \leq 7$. Equality forces one permutation with the two nearest coprime cycles and a singular letter of rank $n - 1$; a matching hierarchy prices additional kernel pairs; and the optimal deficit has one leading constant for even k and three for odd k . The proved results narrow the problem rather than eliminate every remaining direction.

Manuscript 01. The collision-orbit method does more than guarantee that a binary reversal omits some states: with fixed output alphabet size, it identifies the exact eventual maximum. The only competitive permutation geometry is two coprime cycles; strict balancing selects the closest such pair; and the published monoid construction realizes every coloring permitted by the resulting sharp bound. The final defect has exponential base $\sqrt{\lfloor k^2/4 \rfloor}$ and its residue subsequences satisfy explicit eventual linear recurrences. Three points deserve emphasis: the cyclic permutation order is negligible compared with the defect scale only because k is fixed and $\lambda_k > 1$; the strict balancing theorem works at the exact integer level and resolves ties that the even-output leading term cannot see; and a single isolated vertex is not automatically negligible, the positive margin (36.5) being what excludes that possible competitor. The proof also uses the witness's saturation of every improper coloring: a low chromatic count alone is only an upper bound on an arbitrary orbit, and it becomes an exact construction because the imported monoid theorem and the representative-selection argument prove reachability. The remaining gap, as manuscript 01 stated it, was the uniform threshold seven and growing output alphabets; manuscript 06 has since narrowed both.

Manuscript 03. The decisive object for binary reversal is not the entire transformation monoid but the colorings excluded by a collision orbit. With three outputs, properness forces a monochromatic block and sharply restricts the residual cyclic orbit, which turns the repository's finite structural evidence into an exact formula for every $n \geq 7$ (Part II's theorem, proved a second way). For arbitrary fixed output size, balanced disjoint palettes determine the exponential deficit scale, and uniformly controlling the cyclic correction makes this local coloring calculation a global extremal statement. Odd output size supplies a strict imbalance penalty, yielding eventual exact optimality; even output size lacks that penalty at first order and admits a nontrivial interval of asymptotically optimal proportions. The distinction explains the specific obstruction that remained for exact even-output optimality, which manuscripts 01 and 06 remove by exact balancing.

46 The finite verification kernel and threshold data

This section prints manuscript 06's two appendices. The following pseudocode expresses the primary finite comparison, after precomputing chromatic values and residual-order bounds. The implementation

adds typed interfaces, exact cube roots, exception handling, per-row margins, and output records.

```

for n in range(max(7, k + 1), N(k)):
    a, b = nearest_coprime_split(n)
    D = P(k, a, b) - a*b
    check(k**n - k**(n-1) > D)
    check((k-1)**2 * k**(n-2) - 1 > D)

    for component_size in range(2, n + 1):
        u, v = nearest_coprime_split(component_size)
        for d in range(1, n // component_size + 1):
            r = n - d*component_size
            gap = P(k, u, v)**d * k**r - d*u*v*g[r]
            if d == 1 and r == 0:
                check(gap == D)
            else:
                check(gap > D)

    for m in range(2, n + 1):
        r = n - m
        for ell in divisors(m):
            if ell >= 2:
                cycle = (k-1)**ell + (-1)**ell * (k-1)
                gap = cycle**(m//ell) * k**r - m*g[r]
                check(gap > D)

```

(Here P is $\mathcal{B}_k$ and $g[r]$ is γ_r .) The three nontrivial coverage facts are worth restating. First, every mixed pair yields a graph in the classification after choosing a collision pair. Second, fixing the component total and multiplicity permits the nearest-coprime compression because the coloring term and the period term are optimized in the same direction. Third, every same-cycle displacement has a component length dividing its active cycle length. These facts, not just the loop bounds, turn the finite calculation into a universal bound.

The next table records further effective thresholds. It does not claim finite closure below them. In particular, a small value of $N(k)$ relative to $3k^4$ is only an improvement of a sufficient bound.

k	$\hat{H}$	$\hat{C}$	$\hat{L}$	$\hat{S}$	$N(k)$
8	16	70	8	335	2,688
9	20	252	9	206	1,863
10	25	252	9	595	5,364
11	30	924	11	352	3,883
12	36	924	11	952	10,483
13	42	3,432	13	548	7,137
14	49	3,432	13	1,416	18,421
15	56	12,870	15	799	12,000
16	64	12,870	15	1,999	30,000

Table 11: Threshold data beyond the closed slices (manuscript 06; the remaining constants are in data/05-quartic-threshold-thresholds.json).

The difference between odd and even thresholds reflects the gap from the largest to the next palette product: $\hat{H} - 1$ for even k and $\hat{H} - 2$ for odd k . These values are bounds from a deliberately uniform argument, not measurements of when the conjectured formula first becomes true.

47 Sources, dependency boundaries and provenance

This section follows the three manuscripts’ source audits, all dated 28 September 2026 and shipped verbatim:

- manuscript 06: `05-quartic-threshold-SOURCE_AUDIT.md` and `05-quartic-threshold-RESEARCH_STATUS.md`;
- manuscript 01: `03-eventual-SOURCE_AUDIT.md`;
- manuscript 03: `04-deficit-asymptotics-SOURCES.md` and `04-deficit-asymptotics-STATUS.md`.

47.1 Repository evidence the manuscripts inspected

All three read this report through the GitHub connector, in the directory

```
SetTheory/Cardinals/docs/reports/automata-and-formal-languages/
dfao-reversal-coloring-obstruction/
```

The pins in full are

```
e2b1f016a94102663f12b970e6dd434229f90d01      (manuscript 06)
2cdcd74f3abcbff7bd47c6c4265522395def609        (manuscripts 01 and 03)
```

and manuscript 01 records Part I’s blobs as

```
e6dbd47babd24f1b086abf59bca372152bee06c3      (article.tex)
13166e22098bf45e1bdcfaed4801dc83cd8dd16f      (README.md)
```

Manuscript 06, at `e2b1f016a`, inspected the repository and report directory information and the report-collection README, the report’s README.md, its code/reversal.py (collision graphs, chromatic counts, structural upper bounds, the Davies lower expression), and `02-three-output-SOURCE_AUDIT.md`. The last explicitly identifies a complete three-output extension and leaves four and more outputs for further work; the README records finite structural equality over $7 \leq n \leq 30$, $3 \leq k < n$, not a proof of the all- n optimum for arbitrary k . Its bibliography cites the report as *RepoCore* (README and code/reversal.py inspected) and Part II’s audit as *RepoThree*. *Manuscript 01*, at `2cdcd74f3`, inspected the article and README (blobs as above), which it read as proving a proposed nonattainment result, developing collision-orbit bounds and certificates, reporting 372 finite comparisons, and explicitly leaving unrestricted exact optimality unresolved. *Manuscript 03*, at the same commit, inspected the README, RESEARCH_STATUS.md (its explicit non-claim of arbitrary- n exact optimality), and code/reversal.py through the structural-bound and Davies-lower-bound functions, including the exact order sets and $H(r, t)$; the returned file content was truncated later, so this was not a full audit of every function or of the merged article; the general report-collection README and part of the manifest were consulted for topic selection. Its bibliography cites the README, the research-status record and code/reversal.py (structural_bound, H, davies_lower_bound) at that snapshot.

Each manuscript states that these files identify the baseline and that it is not a full audit of every source file, report, formal theorem, or incoming package in ProveIt; that no Lean build was run; and that the report is not treated as formally verified because it lives in ProveIt. Each rederives every upper-bound ingredient it needs, and none uses an unproved repository theorem as an assumption.

Stale text in the shipped audits. Manuscripts 01 and 03 describe the report at `2cdcd74f3`, before Part II: `03-eventual-SOURCE_AUDIT.md` says it “explicitly leaves unrestricted exact optimality unresolved”, and `04-deficit-asymptotics-SOURCES.md` and `..._STATUS.md` describe the finite range $7 \leq n \leq 30$ as the only evidence. Manuscript 03 presents its $k = 3$ theorem as new; it is Part II’s theorem 13.1, placed and written before this addition (section 38). These files are kept verbatim.

47.2 Primary published sources

All three inspected Davies’s arXiv version 2 of 17 October 2017 [1] in HTML and PDF. Manuscript 06: the coloring-orbit formulation (Proposition 4), Theorem 3, Corollary 3, the $k \geq 4$ period contribution, the output-map existence lemma, and the concluding optimality question (Problem 2); printed pages 12 and 17 were inspected as page images. The published statement supplies an accessible (trim) witness; manuscript 06 verifies original-state minimality separately. Manuscript 01: Proposition 4; Section 3’s definition of $U_{a,b}$, its two-generation attribution and the explicit generator; the PDF page with the concluding open problems (printed page 17), where Problem 2 asks about the exact binary lower bound for $n \geq 7$. Manuscript 03: Proposition 4 (printed page 6), the $U_{a,b}$ definition, two-generation attribution and generators (page 7), Theorem 3 (pages 9–12), Corollary 3 (page 12), and Section 5, Question 2 (page 17); the image of page 17 was inspected to check the original question and to distinguish bold exhaustive values from the paper’s other numerical entries, and no numerical entry of that table is treated as a proof.

Holzer and König [2]: manuscript 06 checked the bibliographic record and treats the paper as background for the $k = n$ monoid connection, citing the lower-bound dependency through Davies. Manuscript 01 imports its Theorem 8 (two-generation of $U_{a,b}$) in the form quoted by Davies; the entire original 2004 proof was not independently reconstructed. Manuscript 03 checked the publisher metadata and abstract; the Theorem 8 attribution and its applicability were checked in Davies’s paper, the full proof was not reproduced, and the much stronger extremal-monoid assertions are not imported. Krawetz’s thesis [3] is cited by manuscript 01 for the attribution of the generator through Davies’s account, not as a claim that the thesis was read in full.

47.3 Literature-search scope

Manuscript 06 searched the exact Davies title and combinations of DFAO, reversal, optimality, three and four outputs, and recent-year terms; broad acronym searches returned many unrelated results, which were not used as mathematical evidence. Manuscript 01 searched binary DFAO reversal, the exact lower bound, closest coprime cycles, and eventual or asymptotic optimality. Manuscript 03 searched “DFAO” “reversal” with “optimal”, “asymptotic”, “2025”; the exact Davies title, also with 2025 and 2026; and the exact Holzer–König title. No later solution was located by any of them. None is an exhaustive citation-index review, a survey of theses or unpublished work, or correspondence with specialists; such searches cannot exclude an unindexed paper, thesis, later variant, or equivalent result under different terminology. Non-discovery establishes no priority, and neither absence of search results nor the 2017 paper’s open-problem wording establishes that a problem remained unsolved in later literature. The manuscripts present their proofs for review, without declaring a verified first solution; their extension claims are relative to the inspected sources.

47.4 Dependency ledgers

Manuscript 06 (novelty boundary). The collision-graph mechanism, the orbit formulation, and the published lower construction are antecedents. The results it develops are the all- k effective eventual comparison, the uniform quartic threshold, the four finite-slice closures, the corresponding $k \geq 4$ extremizer constraints, and the matching and asymptotic consequences. Standard identities (Stirling recurrences, Rolle’s theorem, graph coloring, the Chinese remainder criterion, the published lower monoids) are proved or explained, not advertised as new. Its research-status record lists as proved by conventional arguments: strict balancing for $k \geq 3$; nearest-coprime optimization of the Davies expression for every $k \geq 4$, $n \geq 7$, $k < n$, with no asymptotic hypothesis; the matching upper bound for every $k \geq 4$ and $n \geq N(k)$; exactness there with the published lower construction; the necessary extremizer structure, including rank $n - 1$; the matching hierarchy for arbitrary coprime cycle sizes and cross-pair kernels; the first matching correction;

and the even/odd asymptotics. The finite inequalities below $N(k)$ for $k = 4..7$ are computer-assisted and extend both the exact optimum and the rigidity to every $n \geq \max(7, k + 1)$ in these slices.

Manuscript 01.

Ingredient or conclusion	Provenance and role
Reversal as a coloring orbit	Davies; rederived to fix composition and accessibility conventions.
Two-generator monoids $U_{a,b}$	Published Holzer–König theorem; explicit generator recorded by Davies from Krawetz. Imported dependency.
Collision-orbit graphs and chromatic bounds	Inherited from Part I; their proofs and graph classification are supplied.
Strict balancing with an explicit positive gap	Derived and proved, with an elementary proof of the required Stirling positivity.
Exclusion of all non-two-cycle competitors for fixed k	Derived and proved; the odd-output, one-isolated-vertex case is checked explicitly.
Eventual exact optimum and extremizer criterion	Conclusions of the new argument; no explicit threshold and no assertion for growing k .
Parity-sensitive constants and recurrence laws	Consequences derived from the exact eventual formula.
The 919 finite equalities	Exact integer certificates, checked by two implementations; not kernel-checked proofs.

The phrase “derived here” in manuscript 01 describes its own work; it does not establish that the statement has never appeared elsewhere. Manuscript 01 names as the most useful independent checks the strict balancing proof, the one-isolated-vertex inequality for odd k ((36.5)), and the use of the imported $U_{a,b}$ generation theorem.

Manuscript 03 (theorem-by-theorem boundary). (1) Exact three-output optimum, all $n \geq 7$: analytic upper proof for $n \geq 32$, computer-assisted upper step for $7 \leq n \leq 31$, attainment by the imported $U_{a,b}$ witness and Davies’s output construction; two local implementations agree on all 3,717 inequalities. (2) Necessary structure of every three-output optimizer: follows from the strict inequalities and the unique minimizing parameters of the certificate; not a classification of all singular generators. (3) Sharp deficit asymptotics for every fixed $k \geq 3$: mathematical proof, no computational range assumption, fixed k essential. (4) Eventual exact formula for every fixed odd k : strict-separation proof plus the lower witnesses; no numerical $N_{\exists}(k)$ for odd $k \geq 5$. (5) Near-optimal rigidity: necessary structure, not asserted sufficient. (6) Even-output interval criterion: an if-and-only-if criterion for first-order optimality within the $U_{a,b}$ witness family, not an exact even-output optimum. Its new upper arguments require only finite transformations, elementary graph coloring, least common multiples, exact integer arithmetic, and fixed-parameter asymptotic estimates; they do not assume the conjecture that the largest two-generated monoid has a particular form, which is unnecessary because a large coloring orbit is not the same as a large monoid.

47.5 Claims and nonclaims

The principal new contribution of Part III is the exact value of $R_2(n, k)$ for $k \geq 4$ in the range $\mathcal{E}$ of (39.1), with the effective threshold $N(k) \leq 3k^4$ (manuscript 06), together with the non-effective eventual routes, finite certificates, recurrences, sharp asymptotics and the even- k interval criterion of manuscripts 01 and 03. The following are *not* claimed:

- Davies’s Problem 2 in full: the formula for $k \geq 8$ with $\max(7, k + 1) \leq n < N(k)$ outside the certified finite ranges; small n outside the stated ranges; the boundary $k = n$ (manuscript 06; none of the three treats it) and the near-diagonal regime.

- Any other open question of Davies’s Section 5, the largest-two-generated-transformation-monoid problem, or a classification of all extremal transformation monoids (orbit extremality is weaker than monoid-size extremality).
- A necessary-and-sufficient classification of the singular transitions attaining the optimum (manuscripts 06 and 03); example 39.3 is included to prevent an overstatement of theorem 39.1.
- The second-best attainable value or the near-optimal spectrum (manuscript 06).
- Uniform asymptotic error terms for growing k : the main threshold of manuscript 06 is uniform in k , but every displayed asymptotic expansion, and every theorem of manuscripts 01 and 03, is a fixed- k statement; no uniform result is given for $k = k(n) \rightarrow \infty$ beyond $k \leq (n/3)^{1/4}$.
- An exact value or asymptotic estimate for the least threshold: $N(k)$ is a sufficient threshold, and manuscripts 01 and 03 prove only the existence of $N_{\exists}(k)$ (for odd $k \geq 5$ in manuscript 03, with no numerical value).
- Exact even-output optimality from manuscript 03 alone (its plateau theorem is first-order and concerns the witness family; exact even-output optimality comes from manuscripts 01 and 06), and exact optimality of every split in the plateau interval.
- That a finite interval plus an unspecified infinite tail covers every intervening integer (manuscript 01), or that the finite tables and breadth-first searches prove any infinite statement.
- A new proof of Davies’s lower construction or of the Holzer–König two-generation theorem; section 30 proves the saturation step given generation.
- Independent refereeing, worldwide or historical priority beyond the inspected sources, or machine-checked formalization. None of the new material has been checked by a proof assistant or an external referee.
- No novelty for the orbit reduction, the collision-graph method, the lower witness, standard Stirling and palette identities, the small values (the $n = 8$, $k = 5$ value 369020 and the $k = 3$ values are in Part I, Part II or Davies’s experiments), or the finite table beyond extending its range.

47.6 What remains open

The $k \geq 8$ window below $N(k)$; a sharper (near-linear) threshold; exact values for growing k beyond $n^{1/4}$ and the diagonal $k = n$; a classification of saturating singular letters and efficient recognition of saturation; second-best values, stability constants and the near-maximum spectrum; sharpness of the matching hierarchy; minimal recurrences; formal verification; noncyclic permutation groups and several singular letters. Section 44 formulates the corresponding questions. Davies’s Problem 1, the nonattainment question of Part I, is not affected by Part III. [Added 29 September 2026, batch 43: at $k = 4$ Part IV settles the minimal recurrence and quantifies stability (section 59).]

47.7 Where the merge had to choose

- *Placement.* Part III follows Part II’s last section and precedes Part I’s appendices, which contain no numbered equations, tables, figures or theorem-like items; no number of Parts I or II changed (compared in the .aux files of the committed and the new build).
- *Base.* The batch-40 placement named manuscript 01 as the base of its two-manuscript addition; batch 41’s manuscript 06 arrived before the write phase and proves more for $k \geq 4$ (an explicit threshold, the all- n closure for $k \leq 7$, rank rigidity), so it is the base of the three-manuscript Part III (as the batch-41 placement records). Its order of sections is kept; manuscripts 01 and 03 are inserted where their material belongs, and their non-effective eventual proofs (sections 36 and 37) and $k = 3$ proof (section 38) are kept whole as marked second and third routes.
- *Duplicates, printed once.* The orbit reduction, the collision obstruction, the graph classification and the two coloring counts (all three manuscripts; printed in Part I, recalled in section 29.1); the two

unmixed cases (06 and 03, lemma 29.2; 01's weaker variant as remark 29.3); the order bound $3^{r/3}$ (06 and 03, with 03's proof as a second proof); $J(r)$ (01's $A(r)$, Part II's (14.4)); the TP2 inequality (06, with 01's proof as a second route); the balancing theorem (06 qualitative, 01 quantitative, both printed); the nearest-split optimization (06 for $k \geq 4$, 01 for $k \geq 3$, Part II for $k = 3$); the palette formulas (06 and 03, with 03's generating-function proof as a second proof); the two-palette lower bound (all three); the sharpened palette bounds (01 and 03); the first-order asymptotics (all three, theorem 41.1); the $k = 4$ closed form (06 and 01, example 41.3); the $k = 3$ exact formula, its residue forms, the example values and the three limits (03, 01 and Part II; printed once in section 38); the β_k bounds (06, 01 and 03). Proofs that are Part I's or Part II's arguments are not repeated.

- *Questions.* The 29 questions of the three manuscripts are printed as 17, merged where they ask the same thing, with every source's comments kept; five are printed as answered (questions 44.3 to 44.6 fully, the first part of question 44.2), and two carry partial answers (questions 44.10 and 44.14).
- *Order and pieces.* Manuscript 06's status and scope section, its two appendices, and the three source audits are printed as sections 43, 46 and 47; manuscript 01's reproduction appendix and manuscript 03's verification-protocol appendix are merged into section 43; the three title-page status boxes are the status box of section 28.1; the three conclusions are section 45.
- *Notation.* All renamings are in table 6: in particular P_k and $B_k(a, b) \rightarrow \mathcal{B}_k$, $M_k(n)$ and $B_k(n) \rightarrow R_2(n, k)$, T_k and $\gamma_k \rightarrow G_k$, manuscript 06's $D_k(n) \rightarrow D_k^*(n)$, the two thresholds $N(k)$ and $N_\exists(k)$, manuscript 06's constants hatted, manuscript 01's δ_n halved, and $s_k, \alpha_k, \lambda \rightarrow \lambda_k$. No other normalization changed. Where manuscript 01 writes $Q = A \sqcup B$ for the two cycles, and manuscript 06 A, B , Part III writes X, Y (Part II's letters).
- *References to Parts I–II.* The manuscripts cite the report as a repository package; here those citations are internal cross-references, and their bibliography entries for Davies and for Holzer and König are Part I's [1, 2]. Krawetz's thesis is added to the bibliography.
- *Added in the merge.* Remark 33.6 (the $k \geq 4$ case of $U(n, k) = L(n, k)$ on $\mathcal{E}$, with a write-phase check of the nonsurjective entry), the comparisons of section 35.5, the sentences relating the three manuscripts to each other and to Parts I–II (after most results of Part III, and the status paragraphs of the answered questions), and dated pointers in Part I's reading-routes box and sections 1, 10 and 12, and in Part II's sections 13.3 and 27, remark 18.3, and questions 23.1 and 23.6.
- *Reading of the data.* Manuscript 03's finite table (table 10) is input verbatim from the shipped data file named in its caption; its delivered article read it from `results/finite_table.tex`.

Part IV. Four outputs: an all-state route, stability and the missing orbits

Part IV was added on 29 September 2026 in batch 43 of ProveIt's incoming-report intake, from two manuscripts written before Part III was printed. Both prove the four-output formula $R_2(n, 4) = 4^n - \mathcal{B}_4(a_n, b_n) + a_n b_n$ for every $n \geq 7$, which is already Part III's theorem 28.1 at $k = 4$; their proofs of it are printed here as a second route, not as a new theorem. What Part IV adds is a much smaller analytic threshold at $k = 4$ (26, against Part III's $N(4) = 212$), explicit stability theorems, the exact inventory of the missing cyclic orbits, the number of extremal output maps on the standard witness, and a recurrence of proved minimal order 15 for the optimal deficit. Section 48 records the provenance and fixes the notation.

48 Provenance, conventions and the statement

48.1 Two manuscripts, one addition

Part IV merges two AI-assisted research manuscripts prepared for Vladimir Reshetnikov and dated 29 September 2026. They prove the same theorem with the same architecture and in the same order (coloring orbits, collision reduction, four-color counts and balancing, the upper bound, a finite certificate, attainment, rigidity, stability, the missing-orbit inventory, asymptotics and a recurrence), so they form one part. In this report's local file sequence (Part I's unprefix files count as 01, Part II's are 02, Part III's are 03 to 05) they are 06 and 07. Both archives arrived in 06bcc37a8 and were placed in faef2ed2a; despite their similar archive names they are different packages with no file in common.

- *Manuscript 05 of batch 43, the base, called AS (“all states”) below; file prefix 07-all-state-. Exact Binary Reversal with Four Outputs: An all-state formula, quantitative rigidity, and the structure of the missing reversal states, 21 pages as delivered. Archive ProveIt_Four_Output_Reversal (1).zip (inner directory four_output_reversal/). Pinned to ProveIt commit 8315d24e3, with this report's article.tex blob 84923bc5 and the blob f93f98bb of 03-eventual-SOURCE_AUDIT.md.*
- *Manuscript 03 of batch 43, called FO (“four outputs”) below; file prefix 06-four-output-. Exact Four-Output Binary Reversal: Optimal state complexity, collision rigidity, and secondary asymptotics, 21 pages as delivered. Archive ProveIt_Four_Output_Reversal.zip (inner directory of the same name). Pinned to ProveIt commit c130dba62.*

AS is the base because it has the weaker hypothesis and proves more: its theorem covers every $n \geq 4$ (FO's every $n \geq 7$), it proves the recurrence minimal (FO only proves annihilation), and it counts the extremal output maps on the standard transition pair. FO keeps, under its own hypotheses, a second-collision penalty that needs no properness assumption, a stability theorem at the scale $3^{n/2}$ with an explicit imbalance bound, three exact residue formulas, a literal C++ breadth-first witness search through $n = 12$, and a regression sweep through $n = 200$. “Manuscript 03” and “manuscript 06” keep their Part III meanings (batch 40's manuscript 03 and batch 41's manuscript 06); the two batch-43 sources are always called AS and FO.

What the two manuscripts knew. At both pins this report's article.tex had blob 84923bc5: Parts I and II were printed, Part III was not (it was written in 18634ca66, after both pins), although its files were already staged. FO read Part I, Part II (including question 23.1, which it answers for $k = 4$), the README and code/reversal.py. AS read Part I, Part II, the README and Part III's staged 03-eventual-SOURCE_AUDIT.md, which records “no established explicit fixed- k threshold”. Neither knew Part III's explicit threshold $N(k)$ or its all- n closure of $4 \leq k \leq 7$. Both therefore re-derive Part III's $k = 4$ case; this part credits Part III wherever that happens.

Status (both manuscripts). AI-assisted, unrefereed research manuscripts with complete proposed proofs and reproducible finite certificates, not refereed or proof-assistant-checked publications. Nothing in Part IV is checked in Lean, Rocq or any other proof assistant, no Lean or Rocq file is supplied, and the repository’s formal developments in other directories confer no formal status on these results. The upper bound is analytic for $n \geq 26$ and computer-assisted below (AS: 2,082 integer comparisons for $4 \leq n \leq 25$; FO: 2,059 for $7 \leq n \leq 25$), with the ordinary execution and integer-arithmetic trust boundary. Attainment for $n \geq 7$ imports the Holzer–König two-generation theorem in the form stated by Davies; it is cited, not reproved. The claimed resolution is the four-output slice of Davies’s exact binary reversal problem, not the variable-output problem or the largest two-generated transformation-monoid problem. A targeted literature check found no later resolution, but does not establish priority; publication priority has not been established.

48.2 Notation of Part IV

Parts I–III’s symbols keep their meanings; in particular $\mathcal{B}_k(a, b)$, (a_n, b_n) , δ_n , $D_k(n) = k^n - R_2(n, k)$, $D_k^*(n)$ and def are Part III’s (table 6), $P_G(k)$, $C_L(k)$, $H_r(t)$, $M(n, k)$ and $L(n, k)$ are Part I’s, and $J(r)$ and $P_2(d)$ are Part II’s. The two manuscripts use Part II’s letters A, B and D_n for four-output objects, and several letters that Parts I–III already use. Part IV therefore fixes the convention of table 12, used throughout, including in FO’s material. No normalization changed. As in the manuscripts, the output alphabet is $\Delta = \{0, 1, 2, 3\}$ and states are numbered from zero.

Table 12: Symbols of Part IV. The second column gives the manuscripts’ own symbols. The last column prints the tempting false reading.

Symbol here	Manuscripts’ symbols	Meaning	Not to be confused with
$(a_n, b_n), a, b$	AS: (A_n, B_n) , A, B ; FO: (A_n, B_n)	the closest coprime split (28.1) (both manuscripts’ own formula (48.1))	Part II’s (A_n, B_n) is the same split; the manuscripts’ A, B are <i>not</i> the cycles X, Y .
X, Y	AS: X, Y ; FO: L, R	the two cycles (sets) of a two-cycle permutation	Part I’s $L(n, k)$; R_2 .
$\mathcal{B}_4(a, b)$	AS, FO: $\mathcal{B}(a, b)$; FO also $P_{K_{a,b}}(4)$	proper four-colorings of $K_{a,b}$, (41.5)	Part II’s $\mathcal{B} = \mathcal{B}_3$, printed with the same letter.
$D_4^*(n)$	AS, FO: D_n ($n \geq 7$)	$\mathcal{B}_4(a_n, b_n) - a_n b_n$, the witness deficit	Part II’s $D_n = D_3^*(n)$, the <i>three</i> -output deficit.
$D_4(n)$	AS: D_n ($n \geq 4$)	$4^n - R_2(n, 4)$, the optimal deficit; AS defines it by cases, the theorem proves the cases optimal	Part I’s $D(n, k)$, the minimum of its finite collection.
$\text{def}(\tau; s, t)$	FO: $\delta(\tau; s, t)$	$4^n - \tau\langle s, t \rangle $ for one machine	$\delta_n = (b_n - a_n)/2$.
$\tau\langle s, t \rangle$	FO: $\mathcal{O}(\tau; s, t)$; AS: $\tau\langle s, t \rangle$	the reachable coloring set	Part I’s $\mathcal{O}_r$, a set of orders.
δ_n	AS: δ_r ; FO: t (in one proof)	$(b_n - a_n)/2 \in \{\frac{1}{2}, 1, 2\}$	—
$\psi(t)$	AS: $h(t)$; FO: $g(t) = 4\psi(t)$	$3^t - 3 \cdot 2^t$	Part III’s $h = \lfloor k/2 \rfloor$; Landau’s $g(k)$ of theorem 1.1.
$\mathcal{F}_4(a, b)$	AS, FO: $F(a, b)$	$\mathcal{B}_4(a, b) - ab$	Part III’s residue sequences $F_r(t)$.
$\mathcal{K}_n^{\text{AS}}, \mathcal{K}_n^{\text{FO}}$	AS: $\mathcal{L}_n$; FO: a script D_n	the finite certificate lists	Part I’s $L(n, k)$, Part III’s $\mathcal{L}_k(s, \delta)$; deficits D .
$C_L(4)$	AS: $3^L + (-1)^L 3$; FO: C_L	proper four-colorings of an L -cycle (Part I’s (9.3))	residue constants c_n .

Continued on next page

Symbol here	Manuscripts' symbols	Meaning	Not to be confused with
$\Omega_4^e(a, b)$	FO: $E(a, b)$	four-colorings of $K_{a,b}$ with exactly one monochromatic edge	Part III's $\Omega_{4,1}(a, b)$, which counts all colorings with matching number one.
$C_{u,v}$	AS: $E(u, v)$; FO: $uvN_{u,v}$	proper colorings of $K_{a,b}$ with least periods u, v on X, Y	$C_L(k)$; Part III's range $\mathcal{E}$.
$o_{u,v}$	FO: $N_{u,v}$	$C_{u,v}/(uv)$, the number of such p -orbits	Part III's threshold $N(k)$.
$P_q(m)$	AS: $P_q(m)$	words of length m over q letters with least cyclic period m (Part II's P_2)	$P_G(k)$.
$\text{prim}_j(m)$, $\text{sur}_j(m)$	FO: $\text{Prim}_j(m)$, $\text{Sur}_j(m)$	primitive, respectively all, length- m words using every one of j fixed colors (sur_j is Part III's)	—
c_n, d_n, e_n	FO: c_n, d_n, e_n ; AS: $C_r, -E_r, 12 + \delta_r^2$	residue-periodic coefficients (55.6)	Part III's $C_0, C_1, \hat{C}$.
$\chi_4(x)$	AS: $\chi_D(x)$; FO: $\mathcal{A}(E)$	the order-15 characteristic polynomial (55.8)	Part III's $\mathcal{A}_k(E)$, an annihilator in the step t of a residue class; not a chromatic count.
S	AS: S ; FO: E	the forward shift in n	Part III's E , the shift in t ($n = 4t + r$).
$Q_4(z), N_4(z)$	AS: denominator, $N(z)$; FO: $Q(z), N(z)$	denominator and numerator of the tail generating function	the state set Q ; $N(k)$.
κ	FO: C	a positive constant in theorem 53.7	$C_L(k), c_n$.
$\mathbf{1}$	AS: $\mathbf{1}$	indicator	—

Where the manuscripts write M_n for the balanced multinomial coefficient (FO) it is Part I's $M(n, 4)$; AS's residual-order function $H_r(t)$ is Part I's (9.9); both manuscripts' P_G is Part I's $P_G(4)$. AS's letters s, t for an arbitrary pair and p, s for a permutation and a singular map keep their meaning, as in Parts II and III. A *primitive cyclic word* of length m is one whose least positive rotation period is m .

48.3 The statement

Both manuscripts write the split directly: for $n \geq 7$, with $h = \lfloor n/2 \rfloor$,

$$(a_n, b_n) = \begin{cases} (h, h+1), & n = 2h+1, \\ (h-1, h+1), & n = 2h, \text{ } h \text{ even}, \\ (h-2, h+2), & n = 2h, \text{ } h \text{ odd}; \end{cases} \quad (48.1)$$

this is Part III's (28.1) and Part II's (13.1). Then $3 \leq a_n < b_n$, $a_n + b_n = n$, $\gcd(a_n, b_n) = 1$, $b_n \geq 4$, $a_n b_n \geq 12$ and $b_n - a_n \leq 4$. At $k = 4$, Part III's palette formula (41.5) is

$$\mathcal{B}_4(a, b) = 6 \cdot 2^{a+b} + 4(3^a + 3^b) - 12(2^a + 2^b) + 12 \quad (a, b \geq 1), \quad (48.2)$$

and $D_4^*(n) = \mathcal{B}_4(a_n, b_n) - a_n b_n$ for $n \geq 7$.

AS defines $R_2(n, 4)$ as the maximum reversal complexity of a minimal n -state DFAO with at most two input letters and four used output values, FO as the maximum over minimal n -state binary DFAOs with four output symbols. These are the readings of R_2 discussed in Part III's section 28.3; both manuscripts prove the upper bound for the unrestricted maximum of $|\tau\langle s, t \rangle|$ over all transformation pairs and output

maps, without accessibility, minimality or surjectivity, and witnesses that are accessible, minimal and use all four outputs.

Theorem 48.1 (Four outputs at every $n \geq 4$; Part III and Part I, second route by AS and FO). *For every $n \geq 4$, all maps $s, t : Q \rightarrow Q$ and every $\tau : Q \rightarrow \Delta$,*

$$|\tau\langle s, t \rangle| \leq \begin{cases} 176, & n = 4, \\ 826, & n = 5, \\ 3526, & n = 6, \\ 4^n - D_4^*(n), & n \geq 7. \end{cases} \quad (48.3)$$

For each $n \geq 4$ equality is attained by an accessible, minimal n -state DFAO using two input letters and all four output values. Consequently

$$R_2(n, 4) = 4^n - 6 \cdot 2^n - 4(3^{a_n} + 3^{b_n}) + 12(2^{a_n} + 2^{b_n}) - 12 + a_n b_n \quad (n \geq 7), \quad (48.4)$$

and $R_2(4, 4) = 176$, $R_2(5, 4) = 826$, $R_2(6, 4) = 3526$; equivalently $D_4(n) = 80, 198, 570$ for $n = 4, 5, 6$ and $D_4(n) = D_4^(n)$ for $n \geq 7$.*

Status of the statement. For $n \geq 7$ this is Part III's theorem 28.1 at $k = 4$ (analytic for $n \geq N(4) = 212$, computer-assisted for $7 \leq n \leq 211$ in theorem 34.1), with the closed form of example 41.3. For $n = 4, 5, 6$ the three values are Part I's: its table of small cases in section 10 records equality of the structural upper bound and an explicit witness at 176, 826 and 3526, with the witnesses of section A; the value 176 is also among Davies's previously verified maxima (section 9.5). AS's witnesses for $n = 4, 5, 6$ (section 52) are exactly Part I's arrays. What AS and FO add to the theorem is its proof: a second, self-contained route whose analytic part starts at $n = 26$ and whose finite part is a certificate of about two thousand integer comparisons, printed in sections 49 to 52. AS's contribution at $n = 4, 5, 6$ is the uniform certificate, not the numbers.

48.4 What is new and what is re-derived

Item	Status relative to Parts I–III
Orbit reduction, collision graphs, four-color counts, unmixed cases	Part I (and Part III's recollection in section 29.1); reproved by both manuscripts; recalled in section 49.
Balancing at $k = 4$	Part III's theorem 31.3 and corollary 31.5; AS and FO give an elementary $k = 4$ identity, and AS a quantitative gap (lemma 49.4).
The formula for $n \geq 7$	Part III, theorem 28.1 at $k = 4$; second route here.
The values for $n = 4, 5, 6$	Part I's finite table and witnesses.
Analytic threshold 26	New at $k = 4$ (Part III: $N(4) = 212$); like $N(4)$, a sufficient threshold only (section 50).
Finite certificates (2,082 and 2,059 entries)	New as certificates; their range lies inside Part I's $7 \leq n \leq 30$ and Part III's $7 \leq n \leq 211$ (section 51.2).
Attainment	Part III's proposition 30.4 at $k = 4$ (saturation given the imported generation theorem); AS and FO give their own witnesses.
Rigidity (i)–(iv)	Specializes Part III's theorem 39.1; adds the exact $2 + 2$ primitive palettes and the converse (theorem 53.1).
Second-collision penalty	Part III's theorem 40.1 with (40.7) at $k = 4$, $t = 2$; FO proves it under a weaker hypothesis, AS's $24ab$ form is weaker (section 53.1).
Stability (radius $a_n b_n / 2$; scale $3^{n/2}$)	New (theorems 53.6 and 53.7).
Missing-orbit inventory, extremal output count	New at $k = 4$ (Part II has the $k = 3$ inventory, theorem 20.1); section 54.
Residue formulas and asymptotics	Refine Part III's theorem 41.2 at $k = 4$ to exact identities.
Order-15 recurrence, minimal	New (Part III's $k = 4$ annihilator is of order 6 in the step t , order 24 in n , minimality not asserted); section 55.

49 The reduction at four outputs

49.1 What Part IV takes from Parts I–III

Both manuscripts reprove the orbit reduction and the collision machinery, with Part I's composition convention $((u \circ v)(q) = u(v(q)))$, letters act on colorings by $c \mapsto c \circ t$, and the monoid contains the identity). These statements are printed once, in Part I, and recalled with credit.

- *Orbit reduction.* For an accessible DFAO with transitions s, t , initial state q_0 and output τ , the reversal has $|\tau\langle s, t \rangle|$ states (proposition 2.1; Davies's Propositions 3–4). AS (Proposition 2.1) and FO (Proposition 2.1) reprove it by the same argument, to fix the composition convention. FO notes that only the lower witness uses accessibility, and calls a map $Q \rightarrow \Delta$ a coloring without requiring that all four colors occur.
- *Collision obstruction.* For a permutation p , a singular s and a collision pair $\{x, y\}$ of s , let $G = G(p; x, y)$ have edge set $\{\{p^j(x), p^j(y)\} : j \in \mathbb{Z}\}$. Writing a word with its rightmost occurrence of s as $u \circ s \circ p^j$ shows that it identifies $p^{-j}(x), p^{-j}(y)$, the endpoints of an edge of G , so only pure powers of p can reach a proper coloring (lemma 3.3). Since p is an automorphism of G , its pure orbit on τ is either wholly proper or wholly improper, and

$$|\tau\langle p, s \rangle| \leq 4^n - P_G(4) + \begin{cases} |\tau\langle p \rangle|, & \tau \text{ proper for } G, \\ 0, & \tau \text{ improper for } G, \end{cases} \quad (49.1)$$

with $|\tau\langle p \rangle| \leq \text{ord}(p)$ (proposition 9.1; AS's Lemma 2.2, FO's Lemma 3.2). AS stresses that the lemma does *not* assert that every improper coloring is reachable; that is needed for an equality witness and is proved separately (section 52). FO's proof, like AS's, uses the *rightmost* occurrence; FO's audit notes that this direction of composition is essential and that the collapsed edge uses p^{-j} , which is legitimate because G contains all translates.

- *Classification.* A same-cycle pair on an m -cycle with displacement e gives $d = \gcd(m, e)$ cycles of length $L = m/d \geq 2$ (a 2-cycle is one edge) and $r = n - m$ isolates, $P_G(4) = C_L(4)^d 4^r$ with $C_L(4) = 3^L + (-1)^L 3$ ((9.4), (9.3)); a cross pair on cycles of lengths a, b gives $d = \gcd(a, b)$ copies of $K_{a/d, b/d}$ and $r = n - a - b$ isolates, $P_G(4) = \mathcal{B}_4(a/d, b/d)^d 4^r$ ((9.6)). AS (Lemma 2.3) and FO (Lemma 3.3) reprove this with the Chinese remainder criterion. AS proves the cycle count as the trace of the L th power of the 4×4 matrix with zero diagonal and ones elsewhere (eigenvalues $3, -1, -1, -1$), which is Part I's argument; both manuscripts also give the path recurrence of section 56.3, which Part II's section 25 spells out for three colors. FO remarks that the classification does not make the monoid small; it gives a short list of obstructions whose proper-coloring counts can be compared exactly.
- *The four-color biclique count.* Formula (48.2) is Part III's (41.5) (manuscripts 06 and 01 of Part III), and Part I's $B_4(a, b)$. AS (Lemma 3.1) and FO (Lemma 4.1) prove it from the palette sum $\sum_j \binom{4}{j} j! S(a, j) (4 - j)^b$, substituting $2! S(a, 2) = 2^a - 2$ and $3! S(a, 3) = 3^a - 3 \cdot 2^a + 3$; the contributions of $j = 1, 2, 3$ are $4 \cdot 3^b$, $6(2^a - 2)2^b$ and $4(3^a - 3 \cdot 2^a + 3)$, the $j = 4$ term vanishes because $b > 0$, and the result is symmetric in a, b , as it must be. FO notes that it is also the four-color specialization of the count in Davies's Theorem 3.
- *Residual orders.* AS's $H_r(t) = \max_{\lambda \vdash r} \text{lcm}(t, \lambda_1, \dots)$, with $H_0(t) = t$, is Part I's (9.9), an exact maximum; FO uses Part II's product bound $J(r)$ ((14.4), lemma 14.2).

49.2 Unmixed letters and permutation orders

Lemma 49.1 (Unmixed and nonsurjective cases; AS and FO). *Let $n \geq 2$.*

- (i) *If both letters are permutations, the deficit is at least $4^n - M(n, 4) \geq 3 \cdot 4^{n-1}$.*

- (ii) If both letters are singular, the deficit is at least $9 \cdot 4^{n-2} - 1$.
 (iii) (FO) If τ is not surjective, the deficit is at least $4^n - 3^n \geq 4^{n-1}$.

For $n \geq 7$ each of these bounds exceeds $7 \cdot 2^n$ (FO).

These are Part III's lemma 29.2 at $k = 4$ and Part I's (9.11)–(9.13); (iii) is also Part III's remark 29.3. The manuscripts' proofs of the multinomial bound differ from Part III's restriction argument and are kept.

Proofs of the bound $M(n, 4) \leq 4^{n-1}$. AS. Permutations preserve the four color-class sizes, so an orbit lies in one class of size at most $M(n, 4)$; the multinomial recurrence gives $M(n, 4) \leq 4M(n-1, 4)$ and $M(1, 4) = 1$. FO. The maximum occurs when the multiplicities differ by at most one (replacing u, v by $u-1, v+1$ when $u \geq v+2$ reduces the factorial denominator by the factor $(v+1)/u < 1$). Normalize all multinomial coefficients at level n by 4^n : each normalized coefficient is the average of four at level $n-1$ (nonexistent entries read as zero), so the maximum is nonincreasing; it is $1/4$ at level one.

For (ii), choose a collision pair of each letter; a nonidentity word ends on the right with one of them and makes its pair monochromatic, so no proper coloring of the graph formed by these one or two edges is reachable except possibly τ itself; two distinct edges have exactly $9 \cdot 4^{n-2}$ proper four-colorings whether or not they meet, one edge has $12 \cdot 4^{n-2}$. For (iii), a color absent from τ is absent from every orbit element. FO's comparison with $7 \cdot 2^n$: $4^n - 3^n \geq 4^{n-1}$ and $4^n - M(n, 4) \geq 3 \cdot 4^{n-1}$, and all three exceed $7 \cdot 2^n$ at $n = 7$ and remain larger as n grows. $\square$

Lemma 49.2 (Permutation orders). (AS) A permutation on $n \geq 1$ points has order at most 2^{n-1} . (FO) A permutation on r points has order at most $J(r)$, and

$$J(r) \leq 3^{r/3} < (3/2)^r < 2^r \quad (r > 0). \quad (49.2)$$

In the same-cycle case $\text{ord}(p) \leq mJ(r)$, and in the cross-cycle case $\text{ord}(p) \leq \text{lcm}(a, b)J(r) = \frac{ab}{d}J(r)$ (FO); exactly, $\text{ord}(p) \leq H_{n-m}(m)$, respectively $H_{n-a-b}(\text{lcm}(a, b))$ (AS).

Proof. AS. If the cycle lengths are $m_1, \dots, m_c$, the order divides $\prod m_i$, and $m \leq 2^{m-1}$ for every positive integer m by induction; so $\text{ord}(p) \leq 2^{n-c} \leq 2^{n-1}$. FO. The first two inequalities are Part II's lemma 14.2 and Part III's lemma 29.4. FO derives the closed form of J as Part II does and proves $m \leq 3^{m/3}$ by checking $m = 1, \dots, 4$ and using $(m+1)^3/m^3 < 3$ for $m \geq 4$; the strict comparisons follow from $\sqrt[3]{3} < 3/2 < 2$. Keeping the one or two active cycles separate and bounding the order on the remaining r states gives the local bounds. FO calls the Landau function unnecessary here: the simpler $J(r)$ bound makes the proof and the certificate easier to audit. $\square$

49.3 Uniform four-color bounds

Put $\psi(t) = 3^t - 3 \cdot 2^t$ (AS's $h(t)$; FO's $g(t) = 4\psi(t)$). Then

$$\mathcal{B}_4(a, b) = 6 \cdot 2^{a+b} + 4(\psi(a) + \psi(b)) + 12, \quad \psi(1) = \psi(2) = -3, \psi(3) = 3, \quad (49.3)$$

and ψ is increasing from $t = 2$ on, since $\psi(t+1) - \psi(t) = 2 \cdot 3^t - 3 \cdot 2^t > 0$ for $t \geq 2$; in FO's form $\Delta g(t) = 8 \cdot 3^t - 12 \cdot 2^t$ and $\Delta^2 g(t) = 16 \cdot 3^t - 12 \cdot 2^t > 0$, and $\psi \geq -3$ throughout.

Lemma 49.3 (Uniform exponential lower bounds; AS and FO). For positive integers a, b with $w = a + b$ and every $L \geq 2$:

- (a) $\mathcal{B}_4(a, b) \geq 6 \cdot 2^w - 12 \geq 3 \cdot 2^w$ (FO; AS proves the outer bound);
- (b) if $w \geq 5$, then $\mathcal{B}_4(a, b) \geq 6 \cdot 2^w + 12 > 6 \cdot 2^w$ (AS; FO states the strict form);
- (c) $C_L(4) = 3^L + (-1)^L 3 \geq (8/3)^L$ (FO), hence $\geq (5/2)^L$ (AS).

Proof. (a) By (49.3) and $\psi \geq -3$; $w \geq 2$ gives the second inequality. AS's proof of $\mathcal{B}_4 \geq 3 \cdot 2^w$ is different: choose a two-color palette for the left side and the complementary palette for the right; the six choices give $6 \cdot 2^w$ palette-coloring pairs, and a proper coloring occurs in at most two of them (two exactly when both sides are monochromatic). (b) If $w \geq 5$ one side is at least three, so its ψ -value is at least 3 while the other is at least -3 ; hence $\psi(a) + \psi(b) \geq 0$. (c) FO: $C_2 = 12 > (8/3)^2$, and for $L \geq 3$, $C_L \geq 3^L - 3 \geq \frac{8}{9} 3^L \geq (\frac{8}{9})^L 3^L$. AS: $L = 2$ is direct, $3^3 - 3 = 24 > (5/2)^3$, and $3^{L+1} - 3 = 3(3^L - 3) + 6$ gives $3^L - 3 \geq (5/2)^L$ for $L \geq 3$ by induction; the even- L expression is larger still. $\square$

49.4 Balancing at four outputs

The closest coprime split (48.1) is the unique coprime pair $a < b$, $a + b = n$, with least $b - a$: both manuscripts prove this as Part II (lemma 18.1) and Part III (corollary 31.5) do (for odd n consecutive integers; for $n = 2h$ with h even the odd pair $h \pm 1$ has gcd dividing two; for odd $h \geq 5$ that pair is even, and the odd pair $h \pm 2$ has gcd dividing four; the equal split is not coprime).

Lemma 49.4 (Balancing the period-corrected count; AS and FO). *Put $\mathcal{F}_4(a, b) = \mathcal{B}_4(a, b) - ab$. For positive integers $b \geq a + 2$,*

$$\mathcal{F}_4(a, b) - \mathcal{F}_4(a + 1, b - 1) = 8(3^{b-1} - 3^a) - 12(2^{b-1} - 2^a) + (b - a - 1) > 0. \quad (49.4)$$

Hence, among coprime pairs with sum $n \geq 7$, $\mathcal{F}_4$ is uniquely minimized at (a_n, b_n) , with value $D_4^(n)$. (AS) Every other coprime pair $a < b$ with sum n satisfies*

$$\mathcal{F}_4(a, b) > D_4^*(n) + a_n b_n. \quad (49.5)$$

The first assertion is the case $k = 4$ of Part III's corollary 31.5 (via theorem 31.3, or quantitatively theorem 31.4); the manuscripts' proof is a direct $k = 4$ identity.

Proof. Subtract two instances of (48.2); the product term contributes $b - a - 1$. In FO's form the difference is $\Delta g(b - 1) - \Delta g(a) + (b - a - 1)$. The function $8 \cdot 3^t - 12 \cdot 2^t$ is strictly increasing for $t \geq 1$ (its increment is $16 \cdot 3^t - 12 \cdot 2^t > 0$), which gives strict balancing; the intermediate splits need not be coprime, since strict balancing holds for all positive splits. So the closest admissible split is the unique minimum.

The gap (AS). Order the other pair as $a < b$; then $a < a_n$, and the inward moves from (a, b) to (a_n, b_n) include the last move from $(a_n - 1, b_n + 1)$, whose loss is at least

$$16 \cdot 3^{a_n-1} - 12 \cdot 2^{a_n-1} + 1 > 4 \cdot 3^{a_n-1} > (a_n + 2)^2 \geq n^2/4 \geq a_n b_n.$$

The first strict inequality uses $a_n - 1 \geq 2$; the second holds at $a_n = 3$ and persists by induction, since multiplying by three dominates the ratio $(a_n + 3)^2/(a_n + 2)^2$; finally $n \leq 2a_n + 4$. Summing the positive inward losses proves (49.5). $\square$

Both manuscripts explain why four outputs are especially tractable. A proper three-coloring of a biclique is constant on one side (Part II); with four colors both sides can carry primitive nonconstant words on disjoint two-color palettes, so the proper orbit can have length ab rather than $\max(a, b)$ (FO). The leading term of $\mathcal{B}_4(a, b)$ is $6 \cdot 2^{a+b}$, independent of the imbalance of the split; a nonspanning or disconnected graph has a strictly larger leading coefficient, a same-cycle graph a larger exponential base, and the optimal split is selected by a smaller, strictly convex correction (AS and FO).

50 An effective tail: every $n \geq 26$

This section proves the upper bound of theorem 48.1 for $n \geq 26$ twice, by AS's and by FO's elementary estimates. Neither uses asymptotic notation; both reduce to one integer inequality at $n = 26$ and a decreasing ratio.

50.1 AS's exclusion

Lemma 50.1 (An upper estimate on the witness deficit; AS). *For all $n \geq 26$, $D_4^*(n) < 7 \cdot 2^n$.*

Proof. Write $a_n = n/2 - \delta_n$, $b_n = n/2 + \delta_n$ with $0 < \delta_n \leq 2$. Since $a_n b_n \geq 12$,

$$D_4^*(n) \leq 6 \cdot 2^n + 4(3^{\delta_n} + 3^{-\delta_n})3^{n/2} \leq 6 \cdot 2^n + \frac{328}{9}3^{n/2}.$$

At $n = 26$ the last term is strictly less than 2^n , by the integer inequality

$$328 \cdot 3^{13} < 9 \cdot 4^{13}. \quad (50.1)$$

The ratio $3^{n/2}/2^n = (\sqrt{3}/2)^n$ decreases, so the inequality persists. $\square$

Proposition 50.2 (Structural exclusion beyond the threshold; AS). *Let $n \geq 26$. In the complete case split of section 49.1 and lemma 49.1, every case except one spanning coprime biclique has deficit greater than $8 \cdot 2^n$. A spanning coprime biclique with cycle lengths a, b has deficit at least $\mathcal{F}_4(a, b)$. Consequently the upper bound (48.3) holds for every $n \geq 26$, and every structural case other than the spanning biclique with cycle lengths (a_n, b_n) has deficit strictly greater than $D_4^*(n) + a_n b_n$.*

Proof. A cross-cycle collision with $d = \gcd(a, b) \geq 2$ and $r = n - a - b$ has, by lemma 49.3(a), $P_G(4) = \mathcal{B}_4(a/d, b/d)^{d4^r} \geq 3^d 2^{a+b4^r} = 3^d 2^{n+r} \geq 9 \cdot 2^n$; subtracting $\text{ord}(p) \leq 2^{n-1}$ (lemma 49.2) leaves more than $(17/2)2^n > 8 \cdot 2^n$. If $d = 1$ but $r \geq 2$, the same estimate gives $P_G(4) \geq 3 \cdot 2^{n+r} \geq 12 \cdot 2^n$; if $r = 1$, then $a + b = n - 1 \geq 5$ and lemma 49.3(b) gives $P_G(4) \geq (6 \cdot 2^{n-1} + 12)4 > 12 \cdot 2^n$; the deficit is at least $(23/2)2^n$. A same-cycle collision has $P_G(4) \geq (5/2)^{n-r} 4^r \geq (5/2)^n$ by lemma 49.3(c); already at $n = 10$, $(5/4)^n > 17/2$ (that is, $2 \cdot 5^{10} > 17 \cdot 4^{10}$), and this persists, so $P_G(4) - \text{ord}(p) > 8 \cdot 2^n$ for $n \geq 26$. The two bounds of lemma 49.1(i)–(ii) also exceed $8 \cdot 2^n$. This leaves $d = 1, r = 0$: p has exactly two cycles of coprime lengths a, b , its order is ab , and the graph bound is $\mathcal{B}_4(a, b) - ab = \mathcal{F}_4(a, b)$, minimized at (a_n, b_n) by lemma 49.4.

For the margin, the excluded cases have deficit $> 8 \cdot 2^n > D_4^*(n) + 2^n > D_4^*(n) + a_n b_n$, since $a_n b_n \leq n^2/4 < 2^n$ for $n \geq 26$ (true at 26 and preserved by induction); every other spanning coprime split has the margin by (49.5). $\square$

AS: the threshold 26 is sufficient for this particular collection of simple estimates, not a claimed optimal analytic threshold; no conclusion about smaller n is drawn by continuity or extrapolation, and the next section handles them exactly.

50.2 A second route: FO's exclusion

Lemma 50.3 (Eliminating nonoptimal graph types; FO). *For $n \geq 7$, a same-cycle collision forces $\text{def} > 7 \cdot 2^n$, and a cross-cycle collision with $\gcd(a, b) \geq 2$ or $a + b < n$ forces $\text{def} > 8 \cdot 2^n$.*

Proof. For a same-cycle collision, $P_G(4) = C_{m/d}(4)^{d4^r} \geq (8/3)^n$ by lemma 49.3(c), and $\text{ord}(p) < (3/2)^n$ by (49.2), so $\text{def}/2^n > (4/3)^n - (3/4)^n > 7$ for $n \geq 7$; the middle expression increases with n , and at $n = 7$ the claim is the integer inequality $4^{14} - 3^{14} > 7 \cdot 12^7$. For a cross-cycle collision with $d = \gcd(a, b) \geq 2$, $P_G(4) = \mathcal{B}_4(a/d, b/d)^{d4^r} \geq 3^d 2^{n+r} \geq 9 \cdot 2^n$, and $\text{ord}(p) < 2^n$. If $d = 1$ and $r \geq 1$: when $a + b \geq 5$, $P_G(4) > 6 \cdot 2^{a+b4^r} = 6 \cdot 2^{n+r} \geq 12 \cdot 2^n$; when $a + b \leq 4$, $n \geq 7$ forces $r \geq 3$ and $P_G(4) \geq 3 \cdot 2^{n+r} \geq 24 \cdot 2^n$. Again subtracting an order below 2^n suffices. $\square$

Lemma 50.4 (The optimal candidate is below the separating threshold; FO). *For every $n \geq 26$,*

$$D_4^*(n) < 6 \cdot 2^n + 40 \cdot 3^{n/2} < 7 \cdot 2^n. \quad (50.2)$$

Proof. With $a_n = n/2 - \delta_n$, $b_n = n/2 + \delta_n$, $0 \leq \delta_n \leq 2$, one has $3^{\delta_n} + 3^{-\delta_n} \leq 9 + 1/9 < 10$, so $4(3^{a_n} + 3^{b_n}) < 40 \cdot 3^{n/2}$; the remaining correction $-12(2^{a_n} + 2^{b_n}) + 12 - a_n b_n$ is negative. For the second inequality the ratio $40 \cdot 3^{n/2}/2^n$ decreases, and at $n = 26$ the comparison is $40 \cdot 3^{13} = 63,772,920 < 67,108,864 = 2^{26}$. $\square$

Proposition 50.5 (Analytic upper bound and strictness; FO). *For $n \geq 26$, every orbit has deficit at least $D_4^*(n)$. Equality requires one permutation and one singular generator, the permutation having exactly two cycles, of lengths a_n, b_n , and every collision pair of the singular generator joins those two cycles.*

Proof. The easy regimes have deficit greater than $7 \cdot 2^n$ (lemma 49.1), so they cannot attain the candidate of lemma 50.4. In the mixed regime choose any collision pair; lemma 50.3 rules out a same-cycle collision and every cross-cycle graph with $d > 1$ or $r > 0$. Only two full coprime cycles of lengths a, b remain; their order is ab , so $\text{def} \geq \mathcal{B}_4(a, b) - ab \geq D_4^*(n)$, with equality in the second step only at (a_n, b_n) (lemma 49.4). Since the argument applies to *any* collision pair, every collision joins the two cycles in an equality case. $\square$

Remark 50.6 (Comparison with Part III's threshold; added in the merge). Part III's analytic threshold at $k = 4$ is $N(4) = 212$ (equation (33.1), table 7); below it, Part III closes $7 \leq n \leq 211$ by 180,964 exact comparisons in 205 rows per implementation. AS and FO reach 26 because at $k = 4$ the leading term $6 \cdot 2^n$ of every spanning coprime biclique does not depend on the split, so the competing graph types are separated by a whole multiple of 2^n ; Part III's $N(k)$ comes from an argument uniform in k , which the manuscripts' estimates are not. Neither 26 nor 212 is claimed to be least: both are sufficient thresholds, and the formula already holds from $n = 7$ by the finite closures. The first part of question 44.2 (a smaller explicit threshold) is thereby improved at $k = 4$ only. (For comparison: $D_4^*(n) < 7 \cdot 2^n$ first holds at $n = 15$ and fails again at $n = 16, 18$ and 22 ; FO's intermediate bound $6 \cdot 2^n + 40 \cdot 3^{n/2} < 7 \cdot 2^n$ first holds at $n = 26$. The write phase checked both by exact arithmetic.)

51 Finite certificates below the threshold

51.1 Two complete finite lists

For each n both manuscripts form a finite list of integers, one for every structural case left by the reduction, and check that its minimum is the claimed deficit. Neither enumerates transformation monoids or automata; the finite objects are cycle lengths, divisors and residual integer partitions. AS calls this the essential reduction that makes the finite proof small and independently checkable.

AS's list. For $n \geq 4$, $\mathcal{K}_n^{\text{AS}}$ consists of

$$3 \cdot 4^{n-1}, \quad 9 \cdot 4^{n-2} - 1, \quad (51.1)$$

$$C_{m/d}(4)^d 4^{n-m} - H_{n-m}(m), \quad 2 \leq m \leq n, \quad d \mid m, \quad d < m, \quad (51.2)$$

$$\mathcal{B}_4(a/d, b/d)^d 4^{n-a-b} - H_{n-a-b}(\text{lcm}(a, b)), \quad 1 \leq a \leq b, \quad a + b \leq n, \quad d = \gcd(a, b). \quad (51.3)$$

FO's list. For $n \geq 7$, $\mathcal{K}_n^{\text{FO}}$ consists of the three easy entries $4^n - 3^n$, $4^n - M(n, 4)$, $9 \cdot 4^{n-2} - 1$, the same-cycle entries $C_{m/d}(4)^d 4^{n-m} - mJ(n-m)$ over the same range, and the cross-cycle entries $\mathcal{B}_4(a/d, b/d)^d 4^r - \frac{ab}{d}J(r)$, $r = n - a - b$, over the same range. Entries keep their structural tags even when numerical values coincide.

Lemma 51.1 (Completeness; AS and FO). *Every pair of transformations on n points and every output map have deficit at least $\min \mathcal{K}_n^{\text{AS}}$, and at least $\min \mathcal{K}_n^{\text{FO}}$.*

Proof. The easy entries handle two permutations, two singular letters and (FO) a nonsurjective output (lemma 49.1). In the mixed case choose any collision pair. For a same-cycle pair the containing cycle has length m and the other cycles partition $n-m$, so $\text{ord}(p) \leq H_{n-m}(m) \leq mJ(n-m)$; for a cross pair on cycles of lengths a, b the other cycles partition $n-a-b$, so $\text{ord}(p) \leq H_{n-a-b}(\text{lcm}(a, b)) \leq \text{lcm}(a, b)J(n-a-b)$ (lemma 49.2). The classification covers every possible pair, and taking the minimum over the list only weakens the bound. $\square$

Both lists are variants of Part I's structural collection (9.11)–(9.15) at $k = 4$: AS's drops the nonsurjective entry (not needed, since the collision bound holds for every τ) and weakens the multinomial entry to $3 \cdot 4^{n-1}$, keeping Part I's exact order sets; FO's keeps all three easy entries and replaces the exact order sets by Part II's product bound J . At $k = 4$ Part I's period term $T_{r,4}(a, b)$ is $H_r(\text{lcm}(a, b))$ ((9.10)), so AS's cross entries are Part I's.

Proposition 51.2 (Finite certificates; AS and FO). (AS) For every $4 \leq n \leq 25$, $\min \mathcal{K}_n^{\text{AS}} = D_4(n)$ of theorem 48.1, attained by a unique entry: the same-cycle entry $(m, d, r) = (4, 1, 0)$ at $n = 4$, the cross-cycle entry $(a, b, r) = (2, 3, 0)$ at $n = 5$, the same-cycle entry $(6, 2, 0)$ at $n = 6$, and the cross-cycle entry $(a_n, b_n, 0)$ for $7 \leq n \leq 25$, where every other entry exceeds $D_4^*(n) + a_n b_n$. (FO) For every $7 \leq n \leq 25$, the unique minimizing tagged entry of $\mathcal{K}_n^{\text{FO}}$ is the cross-cycle entry $(a, b, d, r) = (a_n, b_n, 1, 0)$, with value $D_4^*(n)$.

Finite computational proofs. AS. Enumerate the ranges and all integer partitions exactly, and evaluate all powers and least common multiples as integers. The resulting 2,082 entries, their parameters and the 22 minima are in `data/07-all-state-finite_certificate.json`; table 13 displays the minimum and the next-smallest entry, so uniqueness and the gap can be checked from the summary (`data/07-all-state-finite_summary.csv`). The producer code `07-all-state-verify.py` enumerates nondecreasing partitions and uses (48.2); the independent checker code `07-all-state-independent_check.py` imports no producer code, enumerates partitions in the reverse direction, evaluates biclique counts by the Stirling-number formula and cycle counts by the path recurrence, and re-enumerates every structural entry instead of trusting the reported minimum. Both completed successfully on all 2,082 entries.

FO. Evaluate the list with the integer formulas (48.2), $C_L(4)$, $M(n, 4)$ and the closed form of J . There are 2,059 entries in total; the complete rows, including the runner-up tag, are in `data/06-four-output-base_certificate.csv`. The primary implementation code `06-four-output-verify.py` uses the closed forms; the independent implementation code `06-four-output-independent_check.py` computes biclique counts by Stirling numbers, cycle counts by the transfer recurrence and J by the dynamic program $J(r) = \max_{1 \leq j \leq r} jJ(r-j)$, enumerates cross-cycle cases by total occupied size rather than by first cycle length, and verifies every delivered table field. Both returned the stated unique minima; all arithmetic is exact, and all checks remain active under Python's optimization flag.

Each is a finite computational proof, with the ordinary execution and integer-arithmetic trust boundary (section 56). $\square$

Table 13: The two finite certificates (AS: $4 \leq n \leq 25$; FO: $7 \leq n \leq 25$). The runner-up is AS's next-smallest entry; the margin is the runner-up minus the minimum, which both manuscripts report and which agrees on the common range; for $n \geq 7$ it exceeds $a_n b_n$. The last two columns count every entry of $\mathcal{K}_n^{\text{AS}}$ and $\mathcal{K}_n^{\text{FO}}$; FO's extra entry is the nonsurjective bound $4^n - 3^n$.

n	(a_n, b_n)	$D_4(n)$	Runner-up	Margin	$ \mathcal{K}_n^{\text{AS}} $	$ \mathcal{K}_n^{\text{FO}} $	$R_2(n, 4)$
4	—	80	93	13	10	—	176
5	—	198	235	37	13	—	826
6	—	570	726	156	19	—	3 526
7	(3, 4)	912	1 346	434	23	24	15 472
8	(3, 5)	2 133	3 684	1 551	30	31	63 403
9	(4, 5)	3 784	8 577	4 793	36	37	258 360
10	(3, 7)	13 359	15 196	1 837	44	45	1 035 217
11	(5, 6)	15 006	19 616	4 610	50	51	4 179 298
12	(5, 7)	32 353	60 114	27 761	61	62	16 744 863
13	(6, 7)	58 482	72 884	14 402	68	69	67 050 382
14	(5, 9)	171 447	234 054	62 607	78	79	268 264 009
15	(7, 8)	226 948	685 923	458 975	88	89	1 073 514 876
16	(7, 9)	472 965	907 960	434 995	100	101	4 294 494 331
17	(8, 9)	882 132	1 017 494	135 362	109	110	17 178 987 052
18	(7, 11)	2 264 023	3 528 744	1 264 721	123	124	68 717 212 713
19	(9, 10)	3 442 146	3 852 836	410 690	133	134	274 874 464 798
20	(9, 11)	7 047 969	12 577 577	5 529 608	148	149	1 099 504 579 807
21	(10, 11)	13 490 734	18 884 980	5 394 246	161	162	4 398 033 020 370
22	(9, 13)	31 517 295	53 963 266	22 445 971	175	176	17 592 154 527 121
23	(11, 12)	53 092 152	56 834 426	3 742 274	187	188	70 368 691 085 512
24	(11, 13)	107 626 165	212 369 004	104 742 839	206	207	281 474 869 084 491
25	(12, 13)	209 682 048	220 945 730	11 263 682	220	221	1 125 899 697 160 576
Total				2 082	2 059		

FO's comments on the table: at $n = 7$ the minimizing entry is $\mathcal{B}_4(3, 4) - 12 = 924 - 12 = 912$ and the closest competitor is $\mathcal{B}_4(2, 5) - 10 = 1346$, giving margin 434; at $n = 10$ the optimum is the less balanced coprime pair (3, 7) and the runner-up is a (4, 5) collision with one isolated state. So the finite check includes the potentially dangerous arithmetic exceptions instead of assuming that all best candidates are balanced full-support bicliques.

Corollary 51.3 (Universal upper bound and structural margin; AS and FO). *The upper bound (48.3) holds for every $n \geq 4$. For every $n \geq 7$, all structural entries other than the spanning coprime biclique with cycle lengths (a_n, b_n) have deficit strictly greater than $D_4^*(n) + a_n b_n$ (AS); in particular, if equality is possible, every selected collision yields the unique minimizing graph type, and the cycle structure and cross-collision restrictions of proposition 50.5 hold for every $n \geq 7$ (FO).*

Proof. Combine proposition 51.2 with proposition 50.2 (AS), respectively with proposition 50.5 (FO); the intervals $4 \leq n \leq 25$ (or $7 \leq n \leq 25$) and $n \geq 26$ cover the domain. $\square$

51.2 Comparison with Parts I and III; added in the merge

The write phase compared the closed form (48.4) with the recorded data of every part, in exact integers: it equals Part III's $4^n - \mathcal{B}_4(a_n, b_n) + a_n b_n$ with the closest-coprime split for $7 \leq n \leq 400$; the upper and lower columns of all 24 rows $k = 4$ ($7 \leq n \leq 30$) of Part I's data/finite_range_bounds.csv; all

94 rows $k = 4$ of `data/03-eventual-finite_certificates.csv`; the split and the target deficit of all 205 rows of `data/05-quartic-threshold-finite_k4.csv` ($7 \leq n \leq 211$); all 213 rows of FO's `data/06-four-output-base_certificate.csv` and `..._structural_sweep.csv`; and AS's 22 rows, including 176, 826 and 3526 at $n = 4, 5, 6$. There were no mismatches. On $7 \leq n \leq 25$ the two certificates have identical margins (FO's margin equals AS's gap); FO has exactly one more entry per n (its nonsurjective entry), 2,059 against AS's 2,040 on the common range. Part III's margins at $k = 4$ are different numbers (its smallest excluded margin is 1265 at $n = 7$), because its list compresses component sizes and bounds residual orders by $\lfloor 3^{r/3} \rfloor$. So the range $7 \leq n \leq 25$ of the two new certificates was already covered by Part I's proposition 10.1 ($n \leq 30$) and by Part III's closure ($n \leq 211$); what is new is that, with the threshold 26, a certificate of about two thousand comparisons suffices for the whole four-output slice.

52 Attainment and the small witnesses

52.1 The imported generation theorem

Both manuscripts import, for attainment only, the Holzer–König two-generation theorem [2, Theorem 8] in the form stated by Davies [1, Section 3, printed p. 7], who attributes the displayed generators to Krawetz [3]. This is Part III's definition 30.1 and the imported theorem of section 30: for $1 < a < b$, $\gcd(a, b) = 1$, $Q = X \sqcup Y$ with $X = \{0, \dots, a-1\}$, $Y = \{a, \dots, n-1\}$ and

$$p = (0 \ 1 \ \dots \ a-1)(a \ a+1 \ \dots \ n-1), \quad s(0) = a, \ s(n-1) = 0, \text{ and, if } a \text{ is odd, } s(1) = 2, \ s(2) = 1, \quad (52.1)$$

with s fixing every other state, the monoid $U_{a,b}$ of powers of p and maps t with $t(X) \cap t(Y) \neq \emptyset$, $Y \not\subseteq t(Q)$ is generated by p and s . (AS calls the parity rule the four-output specialization of Davies's displayed generators; FO writes it as $(s(1), s(2)) = (2, 1)$ for odd a and $(1, 2)$ for even a .) The map s has the unique collision pair $\{0, a\}$, omits $n-1$ from its image and has rank $n-1$ (FO); in the range $a \geq 3$ all listed points exist (AS). Neither manuscript reproves the generation theorem, and both say that finite breadth-first searches do not replace it; AS: all universal lower-bound claims depend on this published theorem, and neither a few searches nor the repository's presence replaces that dependency. The upper bound does not use it (FO).

52.2 Saturating the improper colorings

The saturation step is Part III's lemma 30.2, and the witness Part III's proposition 30.4 at $k = 4$ (which gives X and Y two colors each). AS and FO prove it for their own output words.

Proposition 52.1 (All-state lower witness; AS). *For $n \geq 7$, $a = a_n$, $b = b_n$ and the generators (52.1), the output word $\tau = 0^{a-1}12^{b-1}3$ has orbit size $4^n - \mathcal{B}_4(a, b) + ab$. The DFAO with initial state 0 is accessible and minimal.*

Proof. The graph of the cross collision is $K_{a,b}$, and τ is proper for it (disjoint palettes). Each block word has full cyclic period, because it contains a symbol occurring exactly once, so the combined period is $\text{lcm}(a, b) = ab$ and exactly ab proper colorings lie in $\tau\langle p \rangle$. Let c be any improper coloring, so $c(x) = c(y)$ for some $x \in X$, $y \in Y$. Choose $y_0 \in Y$ labelled 2 by τ ; since $b \geq 4$ this label occurs at another point of Y , so there are representatives $r_i \in Q \setminus \{y_0\}$ with $\tau(r_i) = i$ for every $i \in \Delta$. The map $t(q) = r_{c(q)}$ identifies x, y and omits y_0 , so $t \in U_{a,b}$, and $\tau \circ t = c$. By the imported theorem every improper coloring is reachable; there are $4^n - \mathcal{B}_4(a, b)$ of them, and by (49.1) the only reachable proper colorings are the ab pure powers. The p -cycle reaches X from 0, $s(0) = a$ enters Y , and powers of p reach Y , so the DFAO is accessible. Points on different cycles have different palettes, and two points on one

cycle are separated by a power of p because the cyclic output word is primitive; so the DFAO is minimal, and proposition 2.1 converts the orbit count into reversal complexity. $\square$

Second proof, with another output (FO). For $3 \leq a < b$ coprime, FO takes $\tau|_X = (0, 1, \dots, 1)$ and $\tau|_Y = (2, 3, \dots, 3)$ and proves $|\tau U_{a,b}| = 4^{a+b} - \mathcal{B}_4(a, b) + ab$. Properness and the period ab are as above (a unique occurrence of the first color prevents a nonidentity rotation from fixing a word). Every nonpermutation in $U_{a,b}$ has a cross-block collision, so no other proper coloring occurs. Conversely, for an improper c choose, for each color, one state carrying it under τ , and send each q to the representative of $c(q)$: this map u has $\tau \circ u = c$ and a cross-block collision, and its image contains at most two states of Y (the representatives of colors 2 and 3), so, as $b \geq 4$, it omits a state of Y and lies in $U_{a,b}$. Accessibility and minimality are proved as above. FO remarks that although its count is a specialization of Davies's lower construction, the representative-map proof makes the four-color count and the accessibility requirements explicit.

Example 52.2 (Seven states; FO). The optimal split is $(3, 4)$. In zero-based image-list notation $p = (1, 2, 0, 4, 5, 6, 3)$, $s = (3, 2, 1, 3, 4, 5, 0)$, $\tau = (0, 1, 1, 2, 3, 3, 3)$. Here $\mathcal{B}_4(3, 4) = 924$, $\text{ord}(p) = 12$ and $R_2(7, 4) = 4^7 - 924 + 12 = 15,472$. (These p and s are Part II's arrays of section 18.1.) FO: this lower-bound value already appears in Davies's table (it is also Part I's, section 10); the new assertion is the general upper bound and the all-state theorem.

52.3 The three small cases

For $n = 4, 5, 6$ AS gives the following zero-based arrays (images of $0, \dots, n-1$, not cycle notation); every row uses all four output labels. They are exactly Part I's witnesses of section A.

n	p	s	τ	Reverse states
4	(1, 2, 3, 0)	(0, 0, 3, 2)	(0, 1, 2, 3)	176
5	(1, 0, 3, 4, 2)	(2, 1, 2, 3, 0)	(0, 1, 2, 3, 3)	826
6	(1, 2, 3, 4, 5, 0)	(0, 3, 0, 4, 5, 1)	(0, 0, 1, 1, 2, 3)	3526

At $n = 4$ the collision graph is C_4 , with 84 proper colorings and a four-element proper orbit; at $n = 5$ it is $K_{2,3}$, with 204 and a six-element proper orbit; at $n = 6$ it is *two triangles*, not a single six-cycle: the collision pair $\{0, 2\}$ under the six-cycle has $d = 2$, $L = 3$, giving $24^2 = 576$ proper colorings and a six-element proper orbit (Part I makes the same observation after its table in section 10). These give the upper deficits 80, 198, 570. Full breadth-first enumeration of the coloring orbits gives the counts in the table, once with tuples of colors and once, independently, with integer base-four encodings; both exhaust the reachable set under both transitions. These are finite lower-bound certificates, not samples of words. Accessibility is immediate from the cycles (and the cross transition at $n = 5$); minimality follows from the primitive cyclic words (and the disjoint palettes at $n = 5$) and is also checked by partition refinement. The arrays and results are in `data/07-all-state-bfs_witnesses.json`.

Proof of theorem 48.1. Corollary 51.3 supplies the upper bound, proposition 52.1 (or FO's variant) minimal equality witnesses for $n \geq 7$, and the three arrays above equality for $n = 4, 5, 6$. So both the algebraic orbit maximum and the automaton reversal maximum are $4^n - D_4(n)$ for every admissible n . $\square$

Remark 52.3 (What the small values establish; AS). The values displayed by Davies and the finite repository comparisons are not claimed as new numerical discoveries. Their role is to complete a single rigorously specified all-state statement and to make its lower witnesses directly reproducible. In particular, the special behaviour at $n = 6$ is kept separate rather than forced into the coprime-split formula.

53 Rigidity, a second collision and stability

Throughout this section $n \geq 7$, $a = a_n$, $b = b_n$ and $R = R_2(n, 4)$.

Theorem 53.1 (Necessary structure of every extremizer; AS and FO). *If $|\tau\langle s, t \rangle| = R$ for arbitrary maps s, t and output τ , then after interchanging the letters:*

- (i) *one letter p is a permutation with exactly two cycles X, Y , of lengths a, b ;*
- (ii) *the other letter s has rank $n - 1$; its only nonsingleton fiber has size two, with one point in each cycle (in particular s is injective on each cycle);*
- (iii) *τ is proper for $K_{a,b}$, uses exactly two colors on each cycle, with disjoint palettes, and both cyclic output words are primitive;*
- (iv) *the reachable coloring set consists of every improper coloring of $K_{a,b}$ together with one proper p -orbit of length ab .*

Conversely, (i)–(iv) imply that the orbit has size R (AS). Conditions (i)–(iii) alone are not asserted to imply (iv) for an arbitrary singular generator.

This specializes Part III's theorem 39.1 at $k = 4$, whose (iii) says $\tau(X) \cap \tau(Y) = \emptyset$ with pure orbit of size ab and whose (iv) says every improper coloring is reached; at $k = 4$ these force the $2 + 2$ palette split and the primitive words, and (iv) becomes an equality of sets. It is the four-output counterpart of Part II's theorem 19.1.

Proof (AS). The structural margin of corollary 51.3 rules out two permutations and two singular letters. Write the mixed pair as p, s . Any collision pair of s must give the unique optimal graph entry, so p has precisely two cycles of lengths a, b , and s is injective on each: a same-cycle collision would give a strictly worse bound. Fix a cross-cycle collision. If τ were improper for $K_{a,b}$, (49.1) would give an orbit at most $R - ab$; so τ is proper, its p -orbit has size at most ab , and equality in $4^n - \mathcal{B}_4(a, b) + ab$ forces a full proper orbit of length ab and reachability of every improper coloring. With word periods $u \mid a$, $v \mid b$ (coprime), the combined period is uv , so $u = a$, $v = b$; both words are nonconstant, the palettes are disjoint, and four colors split as two and two. This proves (iii) and (iv), given that there is only one collision pair.

All nonsingleton fibers of s are cross pairs, and distinct ones are disjoint. If there were at least two, consider a coloring with exactly one cross equality: choose $x \in X$, $y \in Y$ and distinct colors α, β, γ , color x, y by α , $X \setminus \{x\}$ by β and $Y \setminus \{y\}$ by γ . A composition with a rightmost s collapses at least two distinct cross pairs after a common rotation by p , so it cannot produce this coloring; nor can a pure power of p , which is proper. This contradicts reachability of every improper coloring. So there is exactly one cross double fiber and s has rank $n - 1$. Finally (iv) supplies $4^n - \mathcal{B}_4(a, b) + ab = R$ distinct reachable colorings, and the upper bound excludes more; this is the converse. $\square$

Second proof (FO). The strict comparisons, including the finite certificate, give (i) and show that every collision of s joins the two cycles, so s is injective on each cycle, every nonsingleton fiber has size two and is a cross pair. If there were two such pairs, proposition 53.4 below would make the orbit strictly smaller than R ; since s is singular there is at least one, which proves (ii). By (49.1) the reachable proper colorings lie in the p -orbit of τ , of size at most ab ; equality requires all improper colorings and exactly ab proper ones, and since p preserves properness, τ is proper with orbit length ab . The periods $u \mid a$, $v \mid b$ are coprime with $uv = ab$, so $u = a$, $v = b$; each primitive word of length at least two uses at least two colors, properness makes the palettes disjoint, and four colors force exactly two each. This proves (iii), and the equality count proves (iv).

AS states the equality conditions in two layers on purpose: the cycle, fiber and output conditions are concrete necessities, while full improper saturation is a separate dynamical property of the generated

monoid. The standard pair of section 52 has that property by the published generation theorem and the representative argument; a complete classification of the other transition pairs with this property is not provided.

Remark 53.2 (Necessary is not sufficient; FO). Statements (i)–(iii) alone do not characterize extremizers. At $n = 7$ take FO's p and τ of example 52.2, but let s send 0 to 3 and fix every other state: s has rank six and the unique cross collision $\{0, 3\}$, the machine is accessible and minimal, and the output words are primitive on disjoint palettes. Yet on the four-state cycle Y , s acts as the identity and p only rotates, so the restriction of every reachable coloring to Y has at most four possibilities, and its restriction to the other three states at most 4^3 ; the orbit has at most $4 \cdot 4^3 = 256$ states, far fewer than 15,472. FO's breadth-first search finds exactly 212 (data/06-four-output-bfs.json); the bound 256 already proves the failure of sufficiency. This parallels Part III's example 39.3 (564 states, with the invariant block on the shorter cycle) and Part II's example 19.3.

53.1 A second collision

Part III prices extra kernel pairs by its matching hierarchy (theorem 40.1): if p has two coprime cycles and the nontrivial kernel classes of s are $t \geq 2$ disjoint cross pairs, then $|\tau\langle p, s \rangle| \leq k^{a+b} - W_{k,2}(a, b) + ab$ with $W_{k,2} = \mathcal{B}_k + \Omega_{k,1}$ and $\Omega_{k,1}(a, b) \geq kab \mathcal{B}_{k-1}(a-1, b-1)$ (proposition 40.2, (40.7)), the last term counting exactly the colorings with one monochromatic cross edge. At $k = 4$ FO proves the same penalty with a direct count, under a weaker hypothesis on s , and AS proves a weaker form.

Lemma 53.3 (Exactly one monochromatic cross edge; FO). *For $a, b \geq 2$, the number of four-colorings of $K_{a,b}$ with exactly one monochromatic edge is*

$$\Omega_4^e(a, b) = 12ab(2^{a-1} + 2^{b-1} - 2) = 4ab \mathcal{B}_3(a-1, b-1). \quad (53.1)$$

Proof. Choose the edge in ab ways and its color in four. No other vertex can have that color (it would create a second monochromatic edge with the opposite chosen endpoint), so the remaining vertices form a proper three-coloring of $K_{a-1, b-1}$. For positive u, v the three-color biclique count is $3(2^u + 2^v - 2)$: if the left side uses one color there are $3 \cdot 2^v$ possibilities, and if it uses two, $3(2^u - 2)$, the right side then being constant in the remaining color (Part II's $\mathcal{B}$). Each coloring has a unique counted edge. $\square$

Proposition 53.4 (A second-collision penalty; FO). *Let p consist of two coprime cycles of lengths $a, b \geq 2$. If s has at least two distinct cross-cycle collision pairs, then for every output map τ ,*

$$|\tau\langle p, s \rangle| \leq 4^{a+b} - \mathcal{B}_4(a, b) - \Omega_4^e(a, b) + ab. \quad (53.2)$$

Proof. A word containing s is $u \circ s \circ p^j$ after isolating its rightmost singular letter. Its coloring has a monochromatic edge for each of the two distinct translated collision pairs, and these remain distinct cross edges. So neither a proper coloring nor a coloring with exactly one monochromatic cross edge arises from such a word. There are $\mathcal{B}_4(a, b) + \Omega_4^e(a, b)$ colorings of those two types, and at most ab of them can be reached by the remaining words, the powers of p . No properness assumption on τ is used. $\square$

This is Part III's bound at $k = 4, t = 2$ with the one-edge term; FO's hypothesis allows s to have further collisions of any kind, whereas theorem 40.1 assumes that all nontrivial kernel classes are disjoint cross pairs.

Proposition 53.5 (A $24ab$ penalty; AS). *Suppose p has cycles X, Y of lengths a, b , s is injective on each cycle and has at least two cross-cycle double fibers, and τ is proper for $K_{a,b}$. Then $|\tau\langle p, s \rangle| \leq R - 24ab$.*

Proof. Use the colorings of the rigidity proof with exactly one cross equality: ab choices of the equal pair and $4 \cdot 3 \cdot 2 = 24$ ordered choices of its three colors give $24ab$ distinct colorings (the equal pair and the colors of the two nonempty complements can be recovered). Each is unreachable by the rightmost-singular-factor argument, and, τ being proper, not a pure power. So at most $4^n - \mathcal{B}_4(a, b) - 24ab$ improper colorings and at most ab proper ones are reachable. $\square$

AS: the properness hypothesis matters, because without it pure permutation powers need not avoid the particular improper family counted, and the proposition is not claimed to be sharp; it supplies a uniform explicit loss sufficient for the stability theorem. (Added in the merge: since $\mathcal{B}_3(a-1, b-1) \geq 6$ for $a, b \geq 2$, $\Omega_4^c(a, b) \geq 24ab$, so proposition 53.4 implies proposition 53.5 without its properness and injectivity hypotheses; the family AS counts consists of the one-edge colorings whose two complements are monochromatic.)

53.2 Stability near the maximum

Theorem 53.6 (Quantitative stability; AS). *If $|\tau\langle s, t \rangle| > R - ab/2$, then after interchanging the letters conditions (i)–(iii) of theorem 53.1 hold: the permutation has the optimal two-cycle split, the singular letter has one cross-cycle double fiber, and the output words are primitive on disjoint two-color palettes. No assertion of complete improper saturation is required or inferred.*

Proof. An excluded letter type or collision graph has deficit $> D_4^*(n) + ab$ (corollary 51.3), hence orbit $< R - ab$. So one letter is a permutation with cycles of lengths a, b , the other is singular, and it has no within-cycle collision. If τ is improper the orbit is at most $R - ab$. For proper τ , a nonfull p -orbit has size a proper divisor of ab , at most $ab/2$, and the chromatic bound gives at most $R - ab/2$; so the period is full, forcing the primitive disjoint $2 + 2$ palettes. If s had more than one cross double fiber, proposition 53.5 would give the loss $24ab$. So it has exactly one, and rank $n - 1$. $\square$

AS: this is a stability statement with a certified, but not necessarily optimal, radius; it concerns the structure of an individual nearly maximal automaton, not merely the asymptotic size of the maximum.

Theorem 53.7 (Stability at the $3^{n/2}$ scale; FO). *Let $n \geq 26$ and let def be an orbit deficit. If $\text{def} < 7 \cdot 2^n$, then, after naming the generators appropriately, one is a permutation p with exactly two full coprime cycles of lengths $a' \leq b'$, $a' + b' = n$, the other is singular, and all its collisions are cross-cycle collisions. Moreover*

$$\text{def} \geq 6 \cdot 2^n + 3^{b'} - \frac{n^2}{4}. \quad (53.3)$$

Consequently, if $\kappa > 0$ and

$$\kappa 3^{n/2} < 2^n, \quad \text{def} \leq 6 \cdot 2^n + \kappa 3^{n/2}, \quad (53.4)$$

then

$$b' - a' \leq 2 \log_3 \left(\kappa + \frac{n^2}{4 \cdot 3^{n/2}} \right). \quad (53.5)$$

For fixed κ the cycle imbalance of such machines is therefore bounded independently of sufficiently large n .

Proof. Lemmas 49.1 and 50.3 eliminate every structure except two full coprime cycles, using only $\text{def} < 7 \cdot 2^n$, and the argument applies to every collision pair. Now $b' \geq n/2 \geq 13$, and for $b' \geq 4$, $4\psi(b') = 4 \cdot 3^{b'} - 12 \cdot 2^{b'} \geq 3^{b'}$ because $(3/2)^{b'} \geq 4$; also $4\psi(a') \geq -12$. So $\mathcal{B}_4(a', b') \geq 6 \cdot 2^n + 3^{b'}$ by (49.3); subtracting the period bound $a'b' \leq n^2/4$ gives (53.3). The hypotheses (53.4) imply $\text{def} < 7 \cdot 2^n$; combining the two estimates gives $3^{b'} \leq \kappa 3^{n/2} + n^2/4$; divide by $3^{n/2}$, take base-three logarithms, and use $b' - n/2 = (b' - a')/2$. $\square$

FO: this is a necessary structural stability statement, not a distance estimate between transition tables; a nearly optimal orbit might still be generated by many different rank- $(n - 1)$ maps on the same two cycles, and describing that freedom is a separate problem.

Relation to Parts II and III (added in the merge). Part III's question 44.10 asks for explicit separations of nearly extremal automata and records that the wrong-split, extra-kernel-pair and incomplete-saturation mechanisms are not quantified there. At $k = 4$, theorem 53.6 quantifies the first two for every $n \geq 7$ (radius $a_nb_n/2$ below the maximum, via the margin $> a_nb_n$ and the $24ab$ loss), and theorem 53.7 bounds the split imbalance on the scale $3^{n/2}$ for $n \geq 26$. Incomplete saturation remains unquantified, and neither manuscript determines the second-largest value (question 44.9, Part II's question 23.4).

54 The missing orbits and the extremal output maps

Theorem 53.1 identifies the missing states set-theoretically: they are the proper colorings of $K_{a,b}$ with one full cyclic orbit removed. This permits an exact orbit inventory, not just a cardinality (AS); an extremizer misses $D_4^*(n)$ colorings, all proper, and their organization into cyclic orbits can be determined exactly (FO). Part II's theorem 20.1 is the three-output counterpart. Throughout, a, b are coprime and p rotates the two blocks X, Y .

54.1 Primitive words and least periods

For $q, m \geq 1$ let $P_q(m) = \sum_{d|m} \mu(m/d)q^d$ (AS), the number of words of length m over a fixed q -letter alphabet with least cyclic period m (Part II's P_2 is (20.1)): every word of length m has a unique least period dividing m , so $q^m = \sum_{d|m} P_q(d)$, and divisor inversion gives the formula. For $m > 1$ such words are nonconstant; a primitive binary word of length greater than one uses both symbols. FO counts instead, for a fixed set of j colors, the words using all of them, $\text{sur}_j(m) = \sum_{i=0}^j (-1)^i \binom{j}{i} (j-i)^m$ (Part III's sur_j), and the primitive ones among them, $\text{prim}_j(m) = \sum_{d|m} \mu(m/d) \text{sur}_j(d)$: every surjective word is a repetition of a unique primitive word with the same palette. For $m > 1$, $\text{prim}_2(m) = P_2(m) = \sum_{d|m} \mu(d)2^{m/d}$.

For $u \mid a$ and $v \mid b$ let $C_{u,v}$ be the number of proper colorings of $K_{a,b}$ whose restrictions to X, Y have least periods u, v , and $o_{u,v} = C_{u,v}/(uv)$.

Lemma 54.1 (Counting exact component periods; AS and FO). *Every coloring counted by $C_{u,v}$ has p -orbit length uv , and*

$$C_{u,v} = \begin{cases} 12, & u = v = 1, \\ 4P_3(v), & u = 1 < v, \\ 4P_3(u), & v = 1 < u, \\ 6P_2(u)P_2(v), & u > 1, v > 1 \end{cases} \quad (\text{AS}), \quad (54.1)$$

$$C_{u,v} = \sum_{d|u} \sum_{e|v} \mu(u/d) \mu(v/e) \mathcal{B}_4(d, e) \quad (\text{AS}), \quad (54.2)$$

$$C_{u,v} = \sum_{\substack{i,j \geq 1 \\ i+j \leq 4}} \frac{4!}{i! j! (4-i-j)!} \text{prim}_i(u) \text{prim}_j(v) \quad (\text{FO}). \quad (54.3)$$

Proof. AS. If both periods are one, choose two distinct constant colors ($4 \cdot 3 = 12$). If only the left side is constant, choose its color in four ways and a primitive length- v word from the remaining three colors; the repetition determines the length- b word. If both sides are nonconstant, properness and the four-color limit force a disjoint $2 + 2$ palette split: six choices of the left palette and $P_2(u)P_2(v)$ primitive words. Since $\gcd(a, b) = 1$, $\gcd(u, v) = 1$ and the simultaneous rotation has period $\text{lcm}(u, v) = uv$. Alternatively,

colorings with periods dividing u, v are the repetitions of proper colorings of $K_{u,v}$, so there are $\mathcal{B}_4(u, v)$ of them; inverting this divisor sum in both variables gives (54.2). *FO*. Choose disjoint used palettes of sizes i, j in $4!/(i! j! (4-i-j)!)$ ways, then a primitive surjective i -color word of length u and a primitive surjective j -color word of length v , and repeat them to lengths a, b ; this counts every proper coloring with the specified periods once. $\square$

Theorem 54.2 (Exact missing-orbit inventory; AS and FO). *For coprime a, b , the proper colorings of $K_{a,b}$ form $o_{u,v}$ orbits of length uv for each $u \mid a, v \mid b$, and*

$$\sum_{u \mid a} \sum_{v \mid b} uv o_{u,v} = \mathcal{B}_4(a, b) \quad (\text{FO}). \quad (54.4)$$

For any extremizer at $n \geq 7$, with $(a, b) = (a_n, b_n)$, the number of missing p -orbits of length uv is $o_{u,v} - \mathbf{1}_{(u,v)=(a,b)}$; every missing orbit occurs this way, and

$$\sum_{u \mid a} \sum_{v \mid b} uv (o_{u,v} - \mathbf{1}_{(u,v)=(a,b)}) = D_4^*(n) \quad (\text{AS}). \quad (54.5)$$

Proof. The proper-coloring set is invariant under p , and lemma 54.1 partitions it by exact component periods, all orbits in one part having size uv ; summing orbit sizes counts all proper colorings. Theorem 53.1(iv) removes exactly one reachable proper orbit, of full size ab , and no others. Since a, b are coprime, the length uv determines the two divisors, so no length is counted ambiguously between different pairs (AS). $\square$

For $a, b \geq 3$ full periods need at least two colors on each side, so only $i = j = 2$ contributes to $o_{a,b}$ (FO):

$$o_{a,b} = \frac{6 P_2(a) P_2(b)}{ab}. \quad (54.6)$$

FO's audit adds that the inventory assumes coprime cycle lengths (so that each joint orbit has size uv), and that the full-period count loses one *orbit*, not one coloring.

Example 54.3 (Seven states; AS and FO). At $n = 7$, $(a, b) = (3, 4)$ and $D_4^*(7) = 912$:

(u, v)	(1, 1)	(1, 2)	(3, 1)	(1, 4)	(3, 2)	(3, 4)	Total
Orbit length uv	1	2	3	4	6	12	
Proper orbits $o_{u,v}$ (FO)	12	12	32	72	12	36	176
Missing orbits (AS)	12	12	32	72	12	35	175

Weighting by orbit lengths gives $12 + 24 + 96 + 288 + 72 + 432 = 924 = \mathcal{B}_4(3, 4)$ proper colorings, of which an extremizer keeps one orbit of length 12 and misses $924 - 12 = 912$. FO: this is more information than the missing-state count alone. AS: complete four-color orbit enumeration at $n = 7$ and $n = 8$ reproduces the corresponding inventories independently of the period-counting formulas (data/07-all-state-bfs_witnesses.json); AS's data/07-all-state-orbit_inventories.json records the algebraic inventories for $7 \leq n \leq 100$, FO's data/06-four-output-inventory.json those for $(a, b) = (3, 4)$ and $(3, 5)$.

54.2 All extremal outputs on the standard pair

Theorem 54.4 (Extremal output maps for the published witness; AS). *Fix the transition pair (52.1) at $n \geq 7$ with $(a, b) = (a_n, b_n)$. Exactly $6P_2(a)P_2(b)$ output maps $\tau : Q \rightarrow \Delta$ attain $R_2(n, 4)$; modulo simultaneous rotation by p there are $6P_2(a)P_2(b)/(ab)$ of them.*

Proof. Necessity follows from theorem 53.1: the palettes split as $2 + 2$ and both block words are primitive, giving six left palettes and $P_2(a)P_2(b)$ pairs of words. Conversely, such a map is proper, surjective and has full p -period ab ; since $b \geq 4$ and only two colors are used on Y , some color on Y has at least two representatives, so a redundant point $y_0 \in Y$ can be chosen, and the representative argument of proposition 52.1 (Part III's lemma 30.2) realizes every improper coloring by a map in $U_{a,b}$ omitting y_0 , which the imported theorem makes reachable by the fixed pair. The action on these maps has full orbit size ab . $\square$

FO makes the same observation: there are $6P_2(a)P_2(b)$ full-period proper output maps on the fixed labelled cycles, and in the monoid $U_{a,b}$ each attains the same orbit size as its witness, because each uses two colors on Y , whose size exceeds two. At $n = 7$ there are $6 \cdot 6 \cdot 12 = 432$ such maps in 36 rotation classes, the 36 full-period orbits of example 54.3. AS: the theorem classifies output maps for one fixed, known saturating pair; it does not classify all singular generators that work with the optimal permutation, nor identify output maps under arbitrary state or color relabelings, which are different equivalence problems.

55 Exact sequences, asymptotics and a minimal recurrence

The following are consequences of the all-state formula, not independent claims that residue-class asymptotics or primitive-word techniques are new methods (AS). For $n \geq 7$ they concern $D_4(n) = D_4^*(n)$.

55.1 Exact residue formulas

Substituting (48.1) into $D_4^*(n)$ gives, with $h = \lfloor n/2 \rfloor$ (FO),

$$D_4^*(n) = \begin{cases} 6 \cdot 2^n + 16 \cdot 3^h - 36 \cdot 2^h + 12 - h(h+1), & n = 2h + 1, \\ 6 \cdot 2^n + 40 \cdot 3^{h-1} - 60 \cdot 2^{h-1} + 13 - h^2, & n = 2h, h \text{ even}, \\ 6 \cdot 2^n + 328 \cdot 3^{h-2} - 204 \cdot 2^{h-2} + 16 - h^2, & n = 2h, h \text{ odd}, \end{cases} \quad (55.1)$$

and, in AS's form without square roots, for $4j + r \geq 7$,

$$D_4^*(4j) = 6 \cdot 16^j + \frac{40}{3}9^j - 30 \cdot 4^j + 13 - 4j^2, \quad (55.2)$$

$$D_4^*(4j+1) = 12 \cdot 16^j + 16 \cdot 9^j - 36 \cdot 4^j + 12 - 4j^2 - 2j, \quad (55.3)$$

$$D_4^*(4j+2) = 24 \cdot 16^j + \frac{328}{3}9^j - 102 \cdot 4^j + 15 - 4j^2 - 4j, \quad (55.4)$$

$$D_4^*(4j+3) = 48 \cdot 16^j + 48 \cdot 9^j - 72 \cdot 4^j + 10 - 4j^2 - 6j. \quad (55.5)$$

These are exact identities for $n \geq 7$, not asymptotic expansions (FO); the write phase checked all seven against the closed form for $7 \leq n \leq 399$.

Proposition 55.1 (Exact separated-scale expression; AS and FO). *Let c_n, d_n, e_n have period four, given in residue order 0, 1, 2, 3 by*

$$(c_n) = \left(\frac{40}{3}, \frac{16}{\sqrt{3}}, \frac{328}{9}, \frac{16}{\sqrt{3}} \right), \quad (d_n) = (-30, -18\sqrt{2}, -51, -18\sqrt{2}), \quad (e_n) = \left(13, \frac{49}{4}, 16, \frac{49}{4} \right).$$

Then for every $n \geq 7$

$$D_4^*(n) = 6 \cdot 2^n + c_n 3^{n/2} + d_n 2^{n/2} - \frac{n^2}{4} + e_n, \quad (55.6)$$

with $c_n = 4(3^{\delta_n} + 3^{-\delta_n})$, $d_n = -12(2^{\delta_n} + 2^{-\delta_n})$ and $e_n = 12 + \delta_n^2$. Consequently

$$\begin{aligned} \frac{D_4(n)}{2^n} &\rightarrow 6, & \frac{R_2(n, 4)}{4^n} &= 1 - 6 \cdot 2^{-n} + O((\sqrt{3}/4)^n), \\ \frac{D_4(n) - 6 \cdot 2^n}{3^{n/2}} &\rightarrow \begin{cases} 16/\sqrt{3}, & n \text{ odd}, \\ 40/3, & n \equiv 0 \pmod{4}, \\ 328/9, & n \equiv 2 \pmod{4}. \end{cases} \end{aligned} \quad (55.7)$$

Proof. Substitute $a_n = n/2 - \delta_n$, $b_n = n/2 + \delta_n$ into $D_4^*(n)$, using $a_n b_n = n^2/4 - \delta_n^2$ (AS's constants $C_r = c_n$, $E_r = -d_n$). Divide by the indicated scales; the remaining exponential bases and polynomial terms are smaller, and $3^{n/2} = o(2^n)$. $\square$

In particular the leading deficit is $6 \cdot 2^n$ with no leading residue-class oscillation; the oscillation first appears at the smaller scale $3^{n/2}$, and the fraction of the coloring space that is missed is asymptotic to $6 \cdot 2^{-n}$ (AS). The oscillation is arithmetic, not a numerical artifact: odd n permits adjacent cycle lengths, while for even n the closest coprime split alternates between offsets one and two from the midpoint, and these bounded offsets have no effect on $6 \cdot 2^n$ but survive in the $3^{n/2}$ correction (FO). Despite the radicals in (55.6), the integer formulas are preferable for exact computation (FO).

Relation to Part III (added in the merge). This is Part III's theorem 41.2 at $h = 2$, made exact: there $C_0 = 6$, $C_1 = 4$, $\mu_1 = \sqrt{3}$ and $2C_1 \cosh(\delta_n \log 3) = 4(3^{\delta_n} + 3^{-\delta_n}) = c_n$, with an error $O(\xi_1^n) = O(2^{n/2})$, which (55.6) replaces by the explicit term $d_n 2^{n/2} - n^2/4 + e_n$. Part III's example 41.3 already notes that the first residue-dependent term has base $\sqrt{3}$.

55.2 A minimal recurrence and the generating function

Let S be the forward shift in n and

$$\chi_4(x) = (x-2)(x^4-9)(x^4-4)(x-1)^2(x^4-1) = (x-2)(x^4-9)(x^4-4)(x-1)^3(x+1)(x^2+1). \quad (55.8)$$

The first form is AS's χ_D , the second FO's operator $\mathcal{A}(E)$; they are the same polynomial of degree 15 (checked in the write phase).

Theorem 55.2 (Minimal constant-coefficient recurrence; AS, annihilation also FO). *The sequence $(D_4(n))_{n \geq 7}$ satisfies $\chi_4(S)D_4 = 0$ from every starting index $n \geq 7$, and χ_4 is its minimal characteristic polynomial: the minimal order is 15. The tail $(R_2(n, 4))_{n \geq 7}$ has minimal characteristic polynomial $(x-4)\chi_4(x)$ and minimal order 16.*

Proof. Annihilation (AS and FO). In (55.6), 2^n is annihilated by $S-2$, a period-four coefficient times $3^{n/2}$ by S^4-9 , and times $2^{n/2}$ by S^4-4 ; the quadratic polynomial and the remaining period-four term are jointly annihilated by $(S-1)^2(S^4-1)$, which has a triple root at 1 and simple roots $-1, \pm i$ (FO: a quadratic by $(S-1)^3$, a period-four sequence by $S^4-1 = (S-1)(S+1)(S^2+1)$). Since $R_2(n, 4) = 4^n - D_4(n)$, one more factor $S-4$ suffices.

Minimality (AS). Expand each period-four coefficient in its four Fourier modes. For a coefficient vector (c_0, c_1, c_2, c_3) the two imaginary-frequency coefficients equal $(c_0 - c_2)/4$, the alternating one is $(c_0 + c_2 - 2c_1)/4$ and the constant one $(c_0 + c_2 + 2c_1)/4$. For both (c_n) and (d_n) all four are nonzero: for c the alternating numerator is $448/9 - 32/\sqrt{3} > 0$ and $40/3 \neq 328/9$; for $-d$ it is $81 - 36\sqrt{2} > 0$ and $30 \neq 51$; the constant coefficients are positive. So all four roots of x^4-9 and of x^4-4 are needed. The period-four vector $\delta_n^2 = (1, \frac{1}{4}, 4, \frac{1}{4})$ has alternating coefficient $9/8$ and imaginary-frequency coefficients $-3/4$, all nonzero; the term $-n^2/4$ needs multiplicity three at the root 1; the coefficient of 2^n is $6 \neq 0$. Exponential-polynomial sequences with distinct bases are linearly independent, also on an infinite tail

(apply shift differences $S - \lambda$ to eliminate all but one base, then use the nonvanishing leading coefficient of the remaining polynomial), so no factor or multiplicity can be removed. The base 4 enters $R_2(n, 4)$ with coefficient one and needs exactly one more factor. $\square$

FO proves the annihilation and says that no minimal-order claim is needed; its audit states that no minimal recurrence order is claimed. Its supplementary checks use exact integers. The write phase confirmed, independently of both manuscripts, that $\chi_4(S)$ annihilates the closed form from every starting index $7 \leq n \leq 384$ and that removing any single linear factor of χ_4 (over $\mathbb{C}$) destroys the annihilation.

Corollary 55.3 (Rational generating function; AS and FO).

$$\sum_{n \geq 7} D_4(n) z^{n-7} = \frac{N_4(z)}{Q_4(z)}, \quad Q_4(z) = (1 - 2z)(1 - 9z^4)(1 - 4z^4)(1 - z)^2(1 - z^4), \quad (55.9)$$

where

$$\begin{aligned} N_4(z) = & 912 - 1515z - 188z^2 + 7064z^3 - 36544z^4 + 52766z^5 - 19986z^6 - 29624z^7 \\ & + 146416z^8 - 210575z^9 + 89558z^{10} + 22752z^{11} - 111168z^{12} + 159804z^{13} - 69624z^{14}. \end{aligned}$$

The displayed denominator is reduced (AS); FO shows that the denominator divides $Q_4(z)$, written as $(1 - 2z)(1 - 9z^4)(1 - 4z^4)(1 - z)^3(1 + z)(1 + z^2)$.

Proof. The recurrence gives a denominator of this form, and multiplication by the initial series determines N_4 : with the ascending coefficients

$$(q_0, \dots, q_{15}) = (1, -4, 5, -2, -14, 56, -70, 28, 49, -196, 245, -98, -36, 144, -180, 72)$$

of Q_4 , $n_i = \sum_{j \leq i} q_j D_4(7 + i - j)$ for $0 \leq i < 15$, and all later coefficients vanish by the recurrence (FO). If numerator and denominator shared a nonconstant factor, cancellation would give a smaller-order eventual recurrence, contrary to theorem 55.2 (AS). $\square$

Explicitly, $\sum_{j=0}^{15} q_j D_4(n - j) = 0$ for $n \geq 22$ (AS), equivalently $\sum_{j=0}^{15} \alpha_j D_4(n + j) = 0$ for $n \geq 7$ with FO's advance coefficients

$$(\alpha_0, \dots, \alpha_{15}) = (72, -180, 144, -36, -98, 245, -196, 49, 28, -70, 56, -14, -2, 5, -4, 1),$$

the q_j in reverse order (FO's a_j , renamed because of a_n). The first fifteen values, from $D_4(7)$, are

$$\begin{aligned} & 912, 2133, 3784, 13359, 15006, 32353, 58482, 171447, 226948, 472965, \\ & 882132, 2264023, 3442146, 7047969, 13490734. \end{aligned}$$

These coefficients and values are also in `data/07-all-state-recurrence.json` and `data/06-four-output-recurrence.json`; they can be recomputed from the closed formula, and were not used to infer the theorem from a guessed recurrence (AS). Providing a factored operator and exact coefficients avoids any reliance on numerical root identification (FO). FO adds that the recurrence permits evaluation modulo a supplied integer with $O(\log n)$ ring operations by polynomial powering; this is not a claim of logarithmic bit complexity for the exact integer $D_4(n)$, whose binary length grows linearly with n .

Remark 55.4 (Residue classes; added in the merge). Part III's theorem 41.6 annihilates each residue sequence $F_r(t) = D_4(4t + r)$ by $(E - 1)^3(E - 4)(E - 9)(E - 16)$ in the step t , and does not assert minimality. By (55.2)–(55.5), every residue sequence is a combination of 16^t , 9^t , 4^t and a quadratic in t with leading coefficient -4 , all with nonzero coefficients; by the linear-independence argument above, that annihilator is therefore minimal for each of the four residue classes, from the first t with $4t + r \geq 7$. (The write phase also checked this by exact arithmetic.) Its order-24 interleaving in n is not minimal: χ_4 has order 15. This answers Part III's question 44.15 at $k = 4$.

56 Reproducible checks and their limits

56.1 What each manuscript ran

Both packages use Python’s standard library with exact integer arithmetic; FO adds a standalone C++17 program. The recorded runs of FO used Python 3.13.5. Both packages ship their recorded outputs, not merely code that has not been run (AS).

AS. The checks have different logical roles:

Check	Executed range	Role in the argument
Complete structural collection	$4 \leq n \leq 25$; 2,082 entries	Finite upper-bound proof and strict structural margin; re-enumerated by an independent implementation.
Reachable coloring orbits	$4 \leq n \leq 9$ (producer); $4 \leq n \leq 8$ (checker)	Finite lower-bound proof for $n = 4, 5, 6$; regression tests at larger sizes.
Accessibility and minimality	six witnesses, $4 \leq n \leq 9$	Confirms the original machines have the stated number of states.
Missing-state orbit inventory	complete coloring space, $n = 7, 8$	Checks the orbit formulas against actual reachable and missing sets.
Primitive-period formulas	$7 \leq n \leq 100$	Compares the palette formula with double Möbius inversion and checks all weighted inventory sums.
Order-15 recurrence	$22 \leq n \leq 500$	Regression test, not the proof of the recurrence or of its minimality.
Threshold inequalities	exact integers at 26 and 10	Checks the numerical anchors of the infinite-tail induction.

The producer and checker use different data representations and independent implementations of the principal counting formulas; that protects against coding errors, but both rely on the same graph classification, so their agreement is not logical independence of two proofs, nor a replacement for auditing lemma 51.1 (AS).

FO. The following checks completed successfully:

Check	Completed scope
Finite proof certificate	$n = 7, \dots, 25$; 2,059 entries
Primary structural regression	$n = 7, \dots, 200$; 752,291 entries
Independent structural implementation	$n = 7, \dots, 100$; 101,568 entries
Brute-force graph-coloring checks	17 graphs; 180,128 colorings
Inventory and unique-edge counts	(3, 4), (3, 5); 81,920 colorings
Explicit witness accessibility and minimality	$n = 7, \dots, 200$
Split and residue-formula comparisons	$n = 7, \dots, 1000$
Order-15 recurrence checks	starting indices $n = 7, \dots, 300$
Full witness-orbit breadth-first searches	$n = 7, \dots, 12$

The largest complete breadth-first search reached 16,744,863 colorings; the six exact witness counts were

n	7	8	9	10	11	12
$ \tau\langle p, s \rangle $	15 472	63 403	258 360	1 035 217	4 179 298	16 744 863

(data/06-four-output-bfs.json, with the 212-state non-extremal example of remark 53.2). They also check the interpretation of composition and the odd- a swap in the published witness; they do not

verify its generation theorem for unbounded a, b . (FO's regression count 752,291 is the size of the structural enumeration for $7 \leq n \leq 200$, which does not depend on the number of outputs; Part II's verifier reports the same count at $k = 3$.) FO on independence: two programs derived from the same recurrence are not two proofs of it; here the two certificate implementations share the mathematically proved reduction but use distinct formulas for each principal count (the symmetric closed form, the cycle chromatic formula and the explicit J , against Stirling surjection counting, the two-coordinate transfer recurrence and a partition-product dynamic program), and the independent code imports no primary code. This detects transcription and implementation errors; it does not certify the graph reduction, which is argued in section 49, and the agreement of the residue formulas and of the recurrence checks is arithmetic consistency, not the extremal theorem from which the sequence is derived. The Python checks use explicit error conditions, not removable `assert` statements, and neither program downloads code or data. The C++ program restricts its endpoint to at most 12 (its memory guard); the largest run needs roughly 85 MB for its coloring mask and queue.

56.2 The dependency boundary

AS: the graph classification, coloring inequalities, balancing theorem, threshold-26 exclusion, rigidity, orbit inventories and recurrence arguments are proved in the manuscript. The finite comparison is a computer-assisted proof with an explicitly bounded domain; trusting it requires the correctness of the short enumeration algorithms, the Python runtime and integer arithmetic, and the recorded execution. The lower construction for all $n \geq 7$ additionally uses the published two-generator theorem, which the package does not prove. No independent referee has reviewed the arguments. These qualifications do not alter the mathematical scope of the proposed theorem: the proof addresses every $n \geq 4$, with no unbounded computational search left implicit. FO: the certificate is part of the proof for nineteen specified indices; the larger sweeps and breadth-first searches are supplementary checks, not arguments for an infinite range.

56.3 The finite algorithms

This subsection prints AS's appendix on finite algorithms and FO's compact enumerator.

Partitions and residual orders (AS). To enumerate nondecreasing partitions of r , choose a first part i between a lower bound and r and recurse on partitions of $r - i$ with parts at least i ; $r = 0$ contributes the empty partition. Every partition has a unique first part and a unique remaining partition with this constraint, so induction proves completeness and absence of duplicates. Least common multiples of the parts are exactly the possible orders of permutations on r points; repeated values may be deduplicated because only their maximum after combination with t is needed. The independent checker uses nonincreasing partitions (choose the largest part up to a bound and recurse with that bound equal to the chosen part; the same induction proves completeness) and computes $\text{lcm}(t, \lambda_1, \dots, \lambda_j)$ directly for each partition.

The structural enumerator (AS). All variables are integers and all endpoints inclusive:

```

For n = 4, ..., 25:
  Insert the two homogeneous-letter bounds.
  For m = 2, ..., n:
    For each proper positive divisor d of m:
      Insert the same-cycle bound with r = n-m.
  For a = 1, ..., n-1:
    For b = a, ..., n-a:
      Insert the cross-cycle bound with d = gcd(a,b).
  Compare every entry with the proposed minimum.
  Check uniqueness and, for n >= 7, the strict AB margin.
```

The checker instead groups cases by the residual size r , then enumerates $m = n - r$, divisors $L \mid m$ and splits $a + b = m$, and compares the entire multiset of triples (case type, parameters, integer value) with the supplied certificate. This detects missing, extra and repeated cases and incorrect integer evaluations, not just an incorrect displayed minimum.

A second derivation of the cycle count (AS and FO). Fix the first color of a path, and let S_L and T_L count proper length- L words ending, respectively, in the first color and in a different color. Then $(S_1, T_1) = (1, 0)$, $S_{L+1} = T_L$, $T_{L+1} = 3S_L + 2T_L$; closing the path into a cycle requires a different last color, so $C_L(4) = 4T_L$. Solving this two-term transfer recurrence gives $3^L + (-1)^L 3$ (FO, whose x_L, y_L are S_L, T_L); it is the independent formula used by both checkers.

Breadth-first orbit closure (AS). Starting with $\{\tau\}$, repeatedly add $c \circ p$ and $c \circ s$ for each newly discovered coloring c . Every inserted coloring is reachable by induction on its discovery, and every reachable coloring is eventually inserted by induction on word length; since there are only 4^n colorings the algorithm terminates with exactly the orbit. The tuple implementation stores $(c(0), \dots, c(n-1))$; the independent one stores $\sum_i c(i)4^i$, extracts colors by bit shifts and marks visited integers in a byte array. Both use the same mathematical transition; the different encodings protect against implementation mistakes but are not different mathematics.

FO's compact enumerator. FO prints the whole finite comparison in deliberately small form (its production checker additionally writes the full CSV, reports coverage and performs the supplementary checks). The code keeps FO's variable names (A is a_n , $B(a, b)$ is $\mathcal{B}_4(a, b)$):

```
from math import gcd, factorial

def B(a, b):
    return (6*2**(a+b) + 4*(3**a + 3**b)
           - 12*(2**a + 2**b) + 12)

def J(r):
    if r < 2:
        return 1
    q, t = divmod(r, 3)
    if t == 0:
        return 3**q
    return 4*3**(q-1) if t == 1 else 2*3**q

for n in range(7, 26):
    A = max(a for a in range(1, n//2 + 1)
            if gcd(a, n-a) == 1)
    target = B(A, n-A) - A*(n-A)
    q, z = divmod(n, 4)
    M = factorial(n)//(factorial(q)**(4-z)
                     * factorial(q+1)**z)
    entries = [(4**n-3**n, None), (4**n-M, None),
              (9*4**(n-2)-1, None)]
    for m in range(2, n+1):
        for d in range(1, m):
            if m % d == 0:
                L, r = m//d, n-m
                P = (3**L + (-1)**L*3)**d * 4**r
                entries.append((P-m*J(r), None))
    for a in range(1, n):
        for b in range(a, n-a+1):
            d, r = gcd(a, b), n-a-b
            P = B(a//d, b//d)**d * 4**r
            entries.append((P-(a*b//d)*J(r),
```

```

        (a, b, d, r)))
    least = min(value for value, tag in entries)
    tags = [tag for value, tag in entries
            if value == least]
    if least != target or tags != [(A, n-A, 1, 0)]:
        raise RuntimeError((n, least, target, tags))
print("PASS: all nineteen finite comparisons")

```

The loops exhaust $\mathcal{K}_n^{\text{FO}}$, including two-vertex edges, singleton cycles, arbitrary residual states and $J(0) = 1$; none of these boundary cases may be dropped (FO).

56.4 Commands and files

The shipped programs are byte-identical to the deliveries and still use the delivered layout: each computes its package root as the parent of its own directory and reads and writes `data/...` there under the delivered, unprefix names. Run in this report's directory, `code/07-all-state-verify.py` would therefore *overwrite* Part I's `data/bfs_witnesses.json` and `data/verification_summary.json`, and `code/07-all-state-independent_check.py` would then read Part I's file; `code/06-four-output-verify.py` would write five stray files, and `code/06-four-output-independent_check.py` would fail on the missing `data/base_certificate.csv`; both Makefiles name the delivered scripts, which do not exist here, and their `pdf` and `clean` targets act on the merged `article.tex` in place. The report's README gives a recipe that restores the delivered names in a scratch copy; there the commands are

```

python code/verify.py           # AS (07-all-state-)
python code/independent_check.py
python -O code/verify.py        # FO (06-four-output-)
python -O code/independent_check.py
g++ -O3 -std=c++17 -Wall -Wextra -pedantic code/witness_bfs.cpp -o witness_bfs
./witness_bfs 12 > data/bfs.json

```

each group in its own copy (FO's scripts are meant to run under `python -O`; they use no assertions, so the flag does not disable any check). The write phase ran them so on 29 September 2026 (Python 3.14.4, and the WinLibs g++; about 2 s each): all passed. Every CSV is byte-identical to the shipped one; every JSON record equals the shipped one apart from Windows line endings, except that FO's `verification.json` and `independent_check.json` record the Python version (3.14.4 instead of the delivered 3.13.5); the C++ output equals `data/06-four-output-bfs.json` apart from line endings.

56.5 What the checks do not establish

No finite enumeration proves the universal statements; they rest on the analytic tail of section 50, whose sufficiency for the finite reduction is proved in the text. Breadth-first searches are not exhaustive enumerations of all n -state automata; universality comes from the structural bounds (AS). Agreement between two programs written in one investigation is implementation independence, not logical independence of the shared reduction, and not external review. No Lean, Rocq or other proof-assistant verification was carried out for Part IV, and none of its statements is formalized anywhere in ProveIt.

56.6 Routes to formal certification

AS. A formalization decomposes into a small finite part and a reusable graph-theoretic part: define coloring orbits on a finite state type and prove the rightmost-singular-factor lemma by induction on words; prove the

two graph descriptions with finite cyclic groups and the generalized Chinese remainder theorem; check the four-color formulas by surjection counting or short recurrences. For the finite proof no sophisticated solver is needed: a verified integer-partition enumerator, a verified divisor enumerator and closed arithmetic evaluation recompute the 2,082 inequalities, and the same framework would serve larger output alphabets. The small lower witnesses can be certified by a verified reachability closure on $\{0, 1, 2, 3\}^Q$: a list of states closed under both transitions, with predecessor words, supplies both directions of the equality between the certificate set and the reachable set. The infinite tail needs only elementary inequalities for powers, finite permutation orders and exact polynomial identities. The substantial remaining external formalization is the Holzer–König theorem; a useful intermediate formal theorem would expose it as an explicit hypothesis while proving all upper bounds, finite cases and conditional attainment without further assumptions.

FO. The proof separates into finite graph theory, integer arithmetic and the external monoid-generation input. A Lean or Rocq development could first encode transformations, the rightmost-singular factorization and the two collision-graph forms; then prove the coloring counts by finite combinatorial sums, including the special two-vertex cycle convention; the whole small-state certificate can then be a finite reflection theorem over $\mathcal{K}_n^{\text{FO}}$, with an explicit checker whose soundness follows from the reduction. The analytic comparison avoids most real analysis: the threshold is one integer inequality propagated by a fixed ratio, balancing is a strict integer finite difference, period estimates can be expressed with cubed inequalities, and parity splitting removes the square roots of the presentation. One can either formalize the $U_{a,b}$ generation theorem or expose it as a named dependency while proving all upper-bound and conditional-attainment statements. No such formalization is supplied, and no kernel-checked status is claimed.

57 Further research questions

AS and FO close with eight questions each. Where both ask the same question it is printed once, with both sources' comments; where Part III (or Part II) already asks it, the counterpart is named; questions that Part III has answered are printed with the answer, not as open. AS states its questions as directions arising from its manuscript (apart from Davies's cited variable-output problem), not as claims that a comprehensive search has verified their open status; FO says the first three concern the remaining mathematical obstruction rather than a computational limit.

Question 57.1 (The next fixed output alphabet; AS and FO. Answered by Part III). Can the exact binary reversal maximum with five outputs be determined for all $n \geq 7$, and can non-effective fixed- k thresholds be replaced by explicit thresholds and bounded certificates (AS)? For every fixed $k \geq 5$, is Davies's coprime two-cycle lower bound optimal for all $n \geq 7$ with $k < n$, and can equality be proved uniformly in n and k (FO)?

Answered, except for the uniform part: Part III proves the formula for $k = 5, 6, 7$ and every $n \geq \max(7, k+1)$ (theorem 34.1) and, for every $k \geq 4$, with the explicit threshold $N(k) \leq 3k^4$ (theorem 33.4); the full two-parameter statement is question 44.1, open for $k \geq 8$ below $N(k)$ outside the certified finite ranges. AS had observed that for four outputs the dominant palette split $2 + 2$ contributes the split-independent $6 \cdot 2^n$, whereas for five outputs the competing $2 + 3$ and $3 + 2$ terms depend on the imbalance of the cycles, so the leading palette terms should be separated from a uniformly controlled remainder before residual vertices and multiple components are treated, without assuming that the $k = 4$ estimates transfer unchanged. FO had observed that its proof works because the biclique count splits into a split-independent leading term and one strictly convex correction, that for larger k several palette splits compete, and that a first step is a uniform balancing inequality for $\mathcal{B}_k(a, b) - ab$ with structural gaps for repeated bicliques and isolated cycles; Part III's theorem 31.3 and corollary 31.5 supply that balancing.

Question 57.2 (Uniformity when the number of outputs grows; AS). For which functions $k = k(n)$ can the published coprime two-cycle construction be proved optimal, and what inequalities would settle Davies's full Problem 2?

This is Part III's question 44.14, partly answered there ($4 \leq k \leq (n/3)^{1/4}$). AS: it needs uniform control of proper-coloring counts and available cyclic periods; an estimate valid for each fixed k may lose all strength as k grows, so explicit dependence on k is an essential part of the question, not an afterthought.

Question 57.3 (Which singular generators saturate the improper set? AS and FO). Fix the optimal two-cycle permutation and a singular map with exactly one cross-cycle double fiber; characterize the maps for which every improper four-coloring is reachable from a primitive proper $2 + 2$ output map (AS). For the optimal cycles and a primitive disjoint two-palette output map, which rank- $(n - 1)$ cross-collision transformations reach every improper coloring (FO)?

This is Part III's question 44.7 at $k = 4$. AS: a complete answer would upgrade the necessary conditions to a classification of transition pairs; it is not enough to classify kernels, since the placement of the image and the permutations induced away from the double fiber affect the generated monoid. FO: theorem 53.1 reduces the problem to this reachability question and remark 53.2 shows why the reduction is not yet a classification; one could first classify these transformations modulo the centralizer of the two-cycle permutation, then seek a graph or semigroup criterion; requiring the monoid to equal $U_{a,b}$ is sufficient but may be stronger than necessary for its four-color projection.

Question 57.4 (Stability radius, second maximum and the singular generator; AS and FO). What is the smallest deficit below $R_2(n, 4)$ that can occur without the structure of theorem 53.6, and what is the exact second-largest orbit size (AS)? Does a deficit within $o(2^n)$ of the optimum force the singular generator to have exactly one cross collision, and under stronger near-optimality can one bound a natural distance from the family of extremizing transition tables (FO)?

These are the $k = 4$ cases of Part III's questions 44.9 and 44.10 (Part II's question 23.4). AS: the radius $a_n b_n / 2$ separates structural failures from saturation failures but is not proved best possible; determining the first failed saturation examples among pairs satisfying (i)–(iii) could be more informative than optimizing the exponential estimates. FO: the penalty $\Omega_4^c(a, b)$ is of order $ab 2^{\max(a,b)}$ for balanced parts, so it isolates one intermediate scale but does not settle an $o(2^n)$ criterion; a sharper hierarchy of obstruction counts may distinguish the number and arrangement of several collisions.

Question 57.5 (Several collisions; AS and FO). If the singular letter is injective on both optimal cycles and has $q \geq 2$ cross-cycle double fibers, what is the sharp loss in reachable colorings (AS)? For $r \geq 2$ distinct cross collisions, what is the sharp number of colorings excluded beyond the one-collision obstruction (FO)?

Part III's matching hierarchy (theorem 40.1) already gives, for t disjoint cross kernel pairs, the upper bound $4^{a+b} - W_{4,t}(a, b) + ab$ with the exact count (40.4); its sharpness is question 44.11. AS: the $24ab$ obstruction counts only colorings with exactly one equal cross pair; counting colorings whose equality pattern is a small bipartite graph could give penalties depending on q and on the relative rotation positions of the pairs, and the rightmost-singular-factor argument gives the precise compatibility condition to exploit. FO: counting colorings with fewer than r monochromatic cross edges gives an immediate lower bound, but the orbit of the whole kernel pattern under p keeps positional information that an edge count loses; exact formulas for small kernel patterns could improve both stability and classification.

Question 57.6 (Constructive word bounds; AS). For the standard optimal pair, how long a word is needed to reach an arbitrary improper coloring? Can such a word be found without exploring 4^n colorings or the entire transition monoid?

AS: the representative proof gives a target transformation in $U_{a,b}$ but no bound on its word length in the two generators; a constructive refinement of the generation theorem would turn saturation into an efficient synthesis algorithm and give stronger finite certificates. (Compare Part III's question 44.8.)

Question 57.7 (Enumerating inequivalent extremizers; AS). How many extremal transition-output triples exist modulo state relabeling, input-letter interchange and output relabeling?

AS: [theorem 54.4](#) counts output maps on one fixed pair only; a classification of saturating singular generators and their centralizer actions would be needed before an orbit-counting formula could be applied.

Question 57.8 (Removing the computational boundary; AS and FO). Can the structural comparisons be proved uniformly already for $n \geq 7$, leaving only the three small equality witnesses, or can the certificate be checked by a small proof-assistant kernel (AS)? Can the inequalities of [section 50](#) be strengthened enough to cover $7 \leq n \leq 25$ without a finite certificate (FO)?

AS: either outcome would strengthen confidence; the first needs tighter small- n inequalities, the second is directly actionable because the certificate has just 2,082 integer comparisons, not a search over automata. FO: this would simplify the trust boundary rather than change the theorem; the difficulty is genuine, since small $n \equiv 2 \pmod{4}$ have an optimal split farther from the midpoint and a graph with isolated states can become the closest competitor, and the recorded runner-up tags identify the cases a uniform proof must dominate.

Question 57.9 (Four-color projections versus whole transformation monoids; FO). Can a two-generated monoid different from U_{a_n, b_n} attain the maximum four-color orbit, and how small can such a monoid be?

This is the last part of Part III's [question 44.7](#) (manuscript 01) at $k = 4$. FO: a positive answer would show explicitly that extremal coloring orbits and extremal monoid sizes are different optimization problems; neither [theorem 48.1](#) nor [theorem 53.1](#) settles the largest-two-generated-transformation-monoid question.

Question 57.10 (Attainable near-maximum sizes; FO). Which values immediately below $R_2(n, 4)$ occur as reversal complexities of minimal n -state binary DFAOs with four outputs?

This is Part III's [question 44.12](#) at $k = 4$. FO: the missing proper-orbit inventory describes one structured source of gaps, incomplete reachability among improper colorings another; it would be valuable to distinguish the two mechanisms and decide whether they force intervals of unattainable sizes near the maximum.

Question 57.11 (A complete formal theorem; FO). Can the collision classification, certificate checker, strict comparison and the two-generator monoid theorem be formalized together, so that both sides of (48.4) are kernel-checked without admitted inputs?

This is Part III's [question 44.16](#) at $k = 4$; the two manuscripts' plans are in [section 56.6](#). FO: the short upper-bound reduction is a natural first milestone, and the published generation theorem likely the more substantial algebraic component; separating the tasks allows a formally verified universal upper bound before the attainment formalization is complete.

58 Conclusion of Part IV

AS. The four-output problem separates completely into a chromatic upper bound and an existing monoid construction that attains it. The identity for $\mathcal{B}_4(a, b)$ makes the universal comparison explicit: beyond $n = 25$ every collision structure except a spanning coprime biclique has a strictly larger exponential deficit, and strict balancing selects the closest coprime split; the finite certificate covers every smaller state count. Equality forces not only the best cycle sizes but also a single cross collision and primitive disjoint binary output words. The result is more than a collection of matching numerical values: an all-state formula, an explicit proof boundary, a quantitative structural theorem for near-extremizers and an exact description of

the missing states. Its claims are confined to four outputs; the uniform variable-output conjecture and the classification of all saturating transition pairs remain separate tasks.

FO. The four-output problem has an exact all-state answer because one collision already forces a very restricted graph, and four-color biclique counts separate its possible shapes sharply enough; after this reduction only the closest coprime two-cycle split survives. Published generation results attain the bound, while strictness yields necessary extremizer rigidity and a meaningful stability theorem. The result resolves the four-output slice of the optimality problem, relative to the proof and finite certificate given; it does not resolve the variable-output question or classify every extremizing singular map, and these remaining issues are isolated as explicit structural and reachability problems rather than an undifferentiated search over automata.

The merge. Both conclusions were written before Part III: the four-output slice of Davies’s Problem 2 is Part III’s (theorem 28.1), and what Part IV contributes to it is the second route with threshold 26, the stability theorems, the orbit inventory, the output count and the minimal recurrence.

59 Sources, dependency boundaries and provenance

This section follows the two manuscripts’ audits, dated 29 September 2026 and shipped verbatim: AS’s 07-all-state-SOURCE_AUDIT.md and 07-all-state-PROOF_AUDIT.md (delivered in notes/), and FO’s 06-four-output-SOURCE_AUDIT.md and 06-four-output-PROOF_AUDIT.md.

59.1 Repository evidence the manuscripts inspected

Both read this report through the GitHub connector, at the pins

8315d24e33a0f4901ca85f0e330ed1d5e19b1ae2	(AS)
c130dba623c420551d90db41dae7b3d9cff72dfd	(FO)

At both, `article.tex` had blob 84923bc50f6233d63fd1a7341a0e16801cd715b8 (AS records it and checked it explicitly at its pin) and `README.md` blob 1716ad6b: the text of Parts I and II, with Part III’s files staged but Part III not yet written. AS inspected the article (orbit reduction, rightmost-singular argument, graph forms, structural collection, three-output extension), the README (the exact three-output theorem, the finite-range result and the article’s explicit limits), 03-eventual-SOURCE_AUDIT.md (blob f93f98bb9d992ffc7cbc5396c9e06cbba287dde7), and the repository root README and research-report manifest, used to choose the topic; unrelated claims there are not dependencies. It inspected the eventual audit, not every proof of its companion manuscript, and does not rely on that eventual theorem. FO inspected the repository overview and research-report manifest, then the README, the full article source and code/`reversal.py`; a GitHub code search for “four-output” in ProveIt returned one unrelated Hilbert-tenth-problem file and no relevant reversal continuation. Both say that the inspected repository papers are research drafts, not formal proof certificates, and both reprove every part of the upper-bound argument they use instead of treating an unrefereed repository statement as an axiom.

Stale text in the shipped audits and the unshipped manuscripts. FO writes that Part II “explicitly leaves the four-or-more-output cases open” and that the contribution relative to the snapshot is the all-state four-output upper bound; its source audit lists the exact result for $k \geq 4$ as “explicitly left open there”. AS writes that the eventual audit records “no established explicit fixed- k threshold” and that its conclusions “make [the repository’s] fixed-four-output eventual analysis effective at every state count”. Both were true of the text at the pins; Part III, written in 18634ca66 after both pins, proves the four-output formula for every $n \geq 7$ with an explicit threshold (theorems 28.1 and 34.1), so in the current report the four-output formula is Part III’s and AS’s and FO’s proofs are a second route (section 48.3). The shipped audits are kept verbatim.

59.2 Primary published sources

Both inspected Davies’s arXiv version 2 of 17 October 2017 [1]. AS: HTML and PDF, with the relevant PDF pages, including the small-value table and the concluding problems, visually inspected; Propositions 3–4, Section 3 ($U_{a,b}$, the two-generation theorem, the generators attributed to Krawetz), Theorem 3 and Corollary 3, Section 5 Problem 2 (printed p. 17; only the four-output slice is claimed), and Table 3 (prior small values, and the distinction between established and candidate optima; small numerical values are not new). FO: the full parsed article and the displayed open-question page; printed pp. 5–6 (Propositions 3–4), p. 7 ($U_{a,b}$ and the parity-adjusted generators), p. 9 (Theorem 3), p. 12 (Corollary 3) and p. 17 (Problem 2 and the small-value table); the version inspected has 18 printed pages. Neither claims the generator, the orbit interpretation or the small lower-bound numbers as new.

Holzer and König [2]: both import Theorem 8 through Davies’s statement; the original 2004 proof was not reconstructed or independently audited, and FO checked the official publisher metadata. The larger question of the maximum size of an arbitrary two-generated transformation monoid is neither assumed nor resolved (FO). Davies’s attribution of the explicit generators to Krawetz [3] is retained; the thesis was not read in full (AS).

59.3 Literature-search scope

AS searched DFAO reversal, Davies’s paper, the binary exact maximum and later resolutions, and searched the repository for four-output follow-up work. FO searched the full Davies title, “DFAO reversal four”, and combinations of “Davies”, “reversal”, “optimal” and “output”. Neither located a later resolution. Neither is a systematic citation-index review, a survey of all theses or unindexed manuscripts, or correspondence with the authors; a missing search hit does not establish priority or prove the absence of a later result, and neither manuscript makes a verified global-priority claim. FO’s claim of extension is specifically relative to the inspected repository version.

59.4 Dependency ledgers

AS (contribution claimed by its package): the exact four-output optimum for every $n \geq 4$, with the formula for $n \geq 7$; an explicit elementary structural exclusion for $n \geq 26$ plus a complete certificate for $4 \leq n \leq 25$; a strict structural margin greater than ab , an $ab/2$ stability radius and the specialized $24ab$ penalty; the exact missing-orbit inventory and the extremal output-map count on the standard transition pair; specialized exact sequence consequences, including the minimal order-15 recurrence. General ideas of asymptotics, recurrence extraction and primitive-word enumeration are not claimed as new methods; the repository’s finite comparisons including four outputs are not claimed as new computations of previously unknown numbers. AS’s proof audit lists its boundary checks: the closest-split formula is used only for $n \geq 7$; the $n = 6$ witness has two triangular components, not one C_6 ; proper-orbit bounds do not imply generic improper saturation; the $24ab$ penalty assumes injectivity on both cycles and a proper initial output and is not claimed sharp; the count $6P_2(a)P_2(b)$ concerns one fixed transition pair; a reverse-state search is not an enumeration of all automata; and the two implementations are independent only at the implementation level.

FO (boundary against the predecessor): already present at its pin were the coloring-orbit reduction, the collision-graph method and classification, a universal structural upper bound, finite equality with Davies’s construction for $7 \leq n \leq 30$, $3 \leq k < n$, and the all-state three-output result with its rigidity. FO’s contribution relative to that snapshot: the all-state four-output upper bound, the strict comparison selecting the nearest coprime split, four-output rigidity, the four-color second-collision penalty, secondary-scale stability, and the derived orbit inventory and exact asymptotics. Graph coloring, the orbit reduction, primitive-word counting, the biclique chromatic polynomial and the lower witness are not claimed as new;

the three-output proof is not used as an admitted premise. FO's proof audit records that the certificate subtracts $mJ(r)$ or $\text{lcm}(a, b)J(r)$, conservative upper bounds on the permutation order, not a product of unrelated coloring periods, and that $J(0) = J(1) = 1$.

59.5 Claims and nonclaims

Part IV's new contributions are the threshold-26 route with its 2,082- and 2,059-entry certificates, the structural margin, the two stability theorems, the missing-orbit inventory, the count of extremal output maps, and the minimal order-15 recurrence with its reduced generating function. The following are *not* claimed:

- The four-output formula as a new theorem: it is Part III's theorem 28.1 at $k = 4$ for $n \geq 7$, and Part I's finite table for $n = 4, 5, 6$; AS and FO give a second proof.
- Anything for five or more outputs, the variable-output problem, growing k , or the diagonal: both manuscripts' claims are confined to four outputs.
- A least threshold: 26 is sufficient for the particular estimates, not optimal, and no conclusion below it is drawn by continuity or extrapolation (AS).
- A new proof of Davies's lower construction or of the Holzer–König two-generation theorem; finite breadth-first searches do not replace it.
- A classification of extremizers: theorem 53.1 gives necessary conditions, (i)–(iii) are not sufficient (remark 53.2); no classification of saturating singular generators or of extremal monoids (orbit extremality is weaker than monoid-size extremality); theorem 54.4 concerns one fixed transition pair and no other equivalence problem.
- Sharpness of the penalties or of the stability radius; a second-largest value; a distance estimate between transition tables (FO); the near-maximum spectrum.
- Novelty for the orbit reduction, the collision-graph method, the lower witness, primitive-word counting, standard Stirling and palette identities, or the small values 176, 826, 3526 and 15,472 (Part I's tables, Davies's Table 3).
- That the finite tables, sweeps or breadth-first searches prove any infinite statement.
- Independent refereeing, priority, or machine-checked formalization.

59.6 What remains open

At $k = 4$: the least analytic threshold and a computation-free proof for $7 \leq n \leq 25$ (question 57.8); a classification of saturating singular letters (questions 57.3 and 57.9); the optimal stability radius, the second-largest value and the near-maximum spectrum (questions 57.4 and 57.10); sharp penalties for several collisions (question 57.5); word-length bounds for saturation and the enumeration of inequivalent extremizers (questions 57.6 and 57.7); formal verification (question 57.11). Beyond $k = 4$ the open questions are Part III's.

59.7 Where the merge had to choose

- *Role*. Both manuscripts re-derive Part III's $k = 4$ case, which neither had read. Part IV therefore prints the formula as Part III's (and Part I's for $n \leq 6$) and the manuscripts' proofs of it as a second route (theorem 48.1 and section 48.3), instead of as a new theorem; the batch-43 placement (faef2ed2a) records the same decision.
- *Placement*. Part IV follows Part III's last section and precedes Part I's appendices, which contain no numbered equations, tables, figures or theorem-like items; no number of Parts I–III changed (compared in the .aux files of the committed and the new build).

- *Base.* AS (weaker hypothesis $n \geq 4$, proved minimality of the recurrence, the output count); its order of sections is kept, and FO's material is inserted where it belongs. FO's proofs that differ from AS's are kept as marked second routes (lemmas 50.3 and 50.4 and proposition 50.5, the second proof of theorem 53.1, FO's witness); FO-only results keep FO's hypotheses (proposition 53.4 without properness, theorem 53.7 for $n \geq 26$). FO's proofs were re-read for hidden uses of properness, the hypothesis AS's penalty needs: none was found; its penalty uses only the two cross pairs, and its stability theorem only $\text{def} < 7 \cdot 2^n$.
- *Duplicates, printed once.* The orbit reduction, the collision obstruction, the classification, the cycle and biclique counts (both manuscripts; Part I, Part III); the unmixed cases (both; Part I, Part III's lemma 29.2); the closest split and balancing (both; Part III's corollary 31.5); the upper bound for $n \geq 7$ and its witness (both; Part III); the rigidity theorem (both, and Part III's theorem 39.1 in weaker form); the one-edge count (FO; Part III); the inventory (AS's and FO's formulas, one theorem); the output-map count (AS's theorem, FO's remark); the residue asymptotics (both; Part III's theorem 41.2 as an expansion); the order-15 operator (both: AS proves minimality, FO annihilation); the generating-function data (both, identical numbers).
- *Tables.* AS's and FO's certificate tables are printed as one (table 13; the margins agree, the entry counts differ by FO's nonsurjective entry); the column $R_2(n, 4)$ is AS's maximum field of data/07-all-state-finite_summary.csv. AS's and FO's $n = 7$ inventories are printed as one table (example 54.3).
- *Questions.* The 16 questions are printed as 11, merged where they ask the same thing, with both sources' comments; question 57.1 is printed as answered by Part III (except its uniform part), and each question names its Part III counterpart.
- *Order and pieces.* AS's verification section and appendix on finite algorithms, and FO's reproducibility section and compact enumerator, are merged into section 56; the two formalization plans are section 56.6; FO's literature check and the four audits are this section; AS's appendix on generating-function data is in section 55; the two title-page status statements are the status box of section 48.1; the two conclusions are section 58.
- *Notation.* All renamings are in table 12; in particular $A, B \rightarrow a_n, b_n$ (or a, b), $D_n \rightarrow D_4^*(n)$ and $D_4(n)$ (Part II's D_n is the three-output deficit), $\mathcal{B}(a, b) \rightarrow \mathcal{B}_4(a, b)$, FO's cycles $L, R \rightarrow X, Y$, $h(t), g(t) \rightarrow \psi(t), 4\psi(t)$, FO's $E(a, b) \rightarrow \Omega_4^e$ and AS's $E(u, v) \rightarrow C_{u,v}$ (the two manuscripts' E collide with each other), FO's $N_{u,v} \rightarrow o_{u,v}, \chi_D, \mathcal{A}(E) \rightarrow \chi_4(\mathbf{S})$, FO's advance coefficients $a_j \rightarrow \alpha_j$, FO's constant $C \rightarrow \kappa$. No normalization changed.
- *References to Parts I–III.* The manuscripts cite the report as a repository package (AS: *RepoMain*, *RepoEventual*; FO: *ProveIt*); here those citations are internal cross-references, and their bibliography entries for Davies and for Holzer and König are Part I's [1, 2].
- *Added in the merge.* The status paragraph after theorem 48.1, section 48.4, remark 50.6, section 51.2, the relation paragraphs after theorems 53.1 and 53.7 and propositions 53.4, 53.5 and 55.1, remark 55.4, the counterparts named in section 57, and dated pointers in Part I's reading-routes box and in Part III's section 28.3, questions 44.2, 44.10 and 44.15 and its list of open problems.

A Explicit small witnesses

The following arrays use state labels $0, \dots, n-1$. Except for the $(5, 4)$ witness, the permutation is the full cycle $p = (1, 2, \dots, n-1, 0)$. The singular arrays used are

n	s
3	(0, 0, 2)
4	(0, 0, 3, 2)
5	(0, 0, 3, 4, 2)
6	(0, 3, 0, 4, 5, 1).

Suitable output arrays are

n	k	τ	$ \tau\langle p, s \rangle $
3	3	(0, 1, 2)	24
4	3	(0, 1, 0, 2)	67
4	4	(0, 1, 2, 3)	176
5	3	(0, 1, 0, 1, 2)	218
5	5	(0, 1, 2, 3, 4)	2110
6	3	(0, 0, 1, 1, 2, 2)	699
6	4	(0, 0, 1, 1, 2, 3)	3526
6	5	(0, 0, 1, 2, 3, 4)	12031
6	6	(0, 1, 2, 3, 4, 5)	32262.

For $(n, k) = (5, 4)$ use instead

$$p = (1, 0, 3, 4, 2), \quad s = (2, 1, 2, 3, 0), \quad \tau = (0, 1, 2, 3, 3),$$

whose orbit has size 826. Every listed machine is accessible from state 0 and is output-minimal; both properties are checked in the supplied code. The arrays are explicit witnesses only. Their use does not assert that every machine with the same collision graph has the same orbit size.

B Evaluating the finite bounds

The Stirling numbers needed in (9.5) are computed by the recurrence

$$\begin{Bmatrix} u \\ i \end{Bmatrix} = i \begin{Bmatrix} u-1 \\ i \end{Bmatrix} + \begin{Bmatrix} u-1 \\ i-1 \end{Bmatrix},$$

with $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} = 1$ and all impossible indices zero. This follows by considering whether the last element joins one of i existing blocks or forms a new singleton block. Thus no symbolic algebra package is needed.

The possible permutation orders can be enumerated recursively through nondecreasing cycle lengths. If the remaining number of points is r and all remaining parts must be at least j , choose a first part $i \geq j$ and combine i by least common multiple with an order from a partition of $r - i$ having parts at least i . Retaining only distinct resulting orders substantially reduces repeated work. The base case for zero remaining points is the singleton set $\{1\}$.

A direct evaluator then performs the following operations:

```
orders[r] = orders obtained from every partition of r
H(r, t)   = max(lcm(t, o) for o in orders[r])

candidates = [nonsurjective_gap, permutation_gap,
              two_singular_gap]

for m in 2..n:
    for each proper positive divisor d of m:
        add C_(m/d)(k)^d * k^(n-m) - H(n-m, m)

for a in 1..n-1:
    for b in a..n-a:
        d = gcd(a, b)
        r = n-a-b
        period = H(r, lcm(a,b))
        if k == 3 and d == 1:
```

```

    period = max(H(r,a), H(r,b))
    add B_k(a/d,b/d)^d * k^r - period

upper = k^n - min(candidates)

```

The lower evaluator independently scans coprime splits $1 < \ell < m$ with $\ell + m = n$ and maximizes (10.1). All arithmetic is exact. The data file retains both results rather than storing only a Boolean equality flag. It therefore records the actual extremal values, including numbers far larger than those feasible for literal coloring-orbit enumeration.

C A complete three-state reachability certificate

For the maps of (9.8), a displayed word $x_1 \cdots x_r$ denotes successive pullbacks, so the resulting coloring is $\tau \circ x_1 \circ \cdots \circ x_r$. The empty word is ε . Each row can be checked by direct tuple substitution; no search program is needed to verify the certificate once this table is given. This is the one place in the article where an exact orbit size is established entirely by hand.

Coloring	Word	Coloring	Word
(0, 0, 0)	<i>bab</i>	(1, 1, 2)	<i>aabaab</i>
(0, 0, 1)	<i>abaab</i>	(1, 2, 0)	<i>a</i>
(0, 0, 2)	<i>b</i>	(1, 2, 1)	<i>aabaaba</i>
(0, 1, 0)	<i>abaaba</i>	(1, 2, 2)	<i>aabaa</i>
(0, 1, 1)	<i>abaa</i>	(2, 0, 0)	<i>baa</i>
(0, 1, 2)	ε	(2, 0, 1)	<i>aa</i>
(0, 2, 0)	<i>ba</i>	(2, 0, 2)	<i>baaba</i>
(0, 2, 2)	<i>baabaa</i>	(2, 1, 1)	<i>aabaabaa</i>
(1, 0, 0)	<i>abaabaa</i>	(2, 1, 2)	<i>aaba</i>
(1, 0, 1)	<i>aba</i>	(2, 2, 0)	<i>baab</i>
(1, 1, 0)	<i>ab</i>	(2, 2, 1)	<i>aab</i>
(1, 1, 1)	<i>abab</i>	(2, 2, 2)	<i>aabab</i>

There are 24 distinct colorings in the table. Combined with the triangle obstruction excluding the other three, this proves the exact orbit size without relying on a claim that a program has examined every word. The same table is stored in `results/sharp_n3_orbit.json` as 24 pairs of a coloring and a word.

D Proof audit: points where the argument could go wrong

The main proof has a short dependency chain:

collision pair $\implies$ three-colorable orbital graph $\implies$ surjective proper k -coloring,
 fiber-partition stabilizer in a cyclic group $\implies |\text{Orb}_\tau \cap \mathcal{R}(c)| \leq g(k)$,
 rightmost singular factor $\implies$ proper reachable states use only the permutation,
 these three statements $\implies k! - g(k)$ missing states.

The cases with two singular or two invertible generators are handled separately and do not require the cyclic graph argument. The following points are stated explicitly because omitting any one of them can make an attractive informal argument incomplete or incorrect. This list merges two separately written

audit lists, and every item of each is retained: the two overlap on the rightmost-factor and surjectivity points, but each carried items the other lacked, so a shorter list would silently drop a genuine warning.

The side of composition. A word containing a singular map must be factored as $u \circ s \circ p^j$, using its *rightmost* singular factor. The collapsed input pair is then $p^{-j}\{x, y\}$. Factoring on the other side without adjusting the argument would not identify a fixed graph edge.

Surjectivity in the relabeling argument. From $\sigma \circ c = \rho \circ c$ one can conclude $\sigma = \rho$ only on the image of c . The construction therefore refines a proper coloring until it uses every output label before invoking the symmetric group; this is also what makes the $k!$ relabelings of that coloring pairwise distinct, so that the counting argument is not silently overcounting.

The fiber partition is stabilized setwise, not blockwise. In the formulation of [remark 5.2](#), H consists of the group elements that permute the fibers of c among themselves. Requiring each block to be preserved individually would give a smaller subgroup, a trivial induced color permutation, and a bound that proves nothing. Conversely, allowing arbitrary set maps rather than elements of the acting group would give no subgroup at all.

A cyclic component of size two is a single edge. When the quotient length is 2, the component of the collision-orbit graph consists of one undirected edge, not a two-edge multigraph on the same pair. The chromatic factor is $k(k-1)$, which is what (9.3) returns at $L=2$; treating it as a genuine 2-cycle with two parallel edges would be a modeling error even though the count happens to agree.

Different actions commute; arbitrary output permutations do not. Equation (2.1) is associativity of postcomposition and precomposition. The contradiction is that putting two noncommuting output permutations in one cyclic input orbit would force them to commute on a surjective coloring. No commutativity of all function composition is assumed.

Distinct colorings versus distinct words. The orbit size counts functions $Q \rightarrow \Delta$, not products or input words. The subgroup-image proof and all BFS implementations deduplicate colorings explicitly.

Proper colorings need not be surjective in the counting bound. The chromatic polynomial counts all proper maps into a fixed k -symbol alphabet. This is exactly the set excluded by singular-containing words. Surjectivity is needed for the factorial relabeling argument, not for (9.1).

Inaccessible proper colorings do not make improper ones accessible. The argument shows only that proper colorings are unreachable through words containing the singular letter. It says nothing about which improper colorings occur, and in particular does not license the converse reading that everything not excluded by the graph is reached. That converse is the saturation property discussed in [section 9](#), and it has to be established separately for any claimed equality.

A graph upper bound does not create a witness. Knowing that at most $P_G(k) - T$ colorings are lost gives only an upper bound on orbit size. Lower witnesses and their reachability must be supplied separately. The finite optimality statements explicitly combine a universal upper calculation with a lower construction or a complete orbit search of an explicit machine.

References

- [1] Sylvie Davies. *State Complexity of Reversals of Deterministic Finite Automata with Output*. arXiv:1705.07150v2, 17 October 2017. <https://arxiv.org/abs/1705.07150>. Version-specific text: <https://arxiv.org/html/1705.07150v2>. The selected open problem is Section 5, Problem 1, on printed page 17; the orbit characterization is Proposition 4, ternary attainability is Theorem 1 and Corollary 2, the small maxima used as regression checks are Table 3, and the lower construction used in the finite comparison is Corollary 3. A conference version is associated with CIAA 2018, doi:10.1007/978-3-319-94812-6_12; all numbering and page references here are to the arXiv version actually consulted.

- [2] Markus Holzer and Barbara König. *On deterministic finite automata and syntactic monoid size*. Theoretical Computer Science **327**(3), 319–347, 2004. doi:10.1016/j.tcs.2004.04.010. Background on the two-generated transformation-monoid problem; not required as a black-box theorem for the principal obstruction proof. Theorem 13 contains the classical bound $n^n - n! + g(n)$, to which the present main bound specializes at $k = n$; that special case is not claimed as new. Sections 4–5 use graph colorings and chromatic polynomials in the transformation-monoid setting, so graph coloring as a method is likewise not claimed here as an invention. [Added 29 September 2026, batches 40–41: Part III’s manuscript 01 imports its Theorem 8, the two-generation of the monoid $U_{a,b}$, in the form quoted by Davies (section 30).]
- [3] B. Krawetz. *Monoids and the State Complexity of the Operation $\text{root}(L)$* . Master’s thesis, University of Waterloo, 2003. <https://cs.uwaterloo.ca/~shallit/krawetz.pdf>. Cited in Part III, after manuscript 01, for the attribution of the $U_{a,b}$ generator through Davies’s account; not read in full by that manuscript.
- [4] Marc Deléglise and Jean-Louis Nicolas. *The Landau Function and the Riemann Hypothesis*. arXiv:1907.07664, 2019. <https://arxiv.org/abs/1907.07664>. Cited only for the standard name and definition of $g(k)$; no analytic result from this work is used.